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(54) **METHODS AND COMPOSITIONS FOR
MODULATING A GENOME**

(71) Applicant: **Flagship Pioneering Innovations VI,
LLC**, Cambridge, MA (US)

(72) Inventors: **Jacob Rosenblum Rubens**, Cambridge,
MA (US); **Geoffrey A. von Maltzahn**,
Somerville, MA (US); **Robert James
Citorik**, Somerville, MA (US); **Barrett
Ethan Steinberg**, Somerville, MA
(US); **Donghui Li**, Somerville, MA
(US); **William Edward Salomon**, West
Roxbury, MA (US)

(73) Assignee: **Flagship Pioneering Innovations VI,
LLC**, Cambridge, MA (US)

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(56) **References Cited**

U.S. PATENT DOCUMENTS

6,693,086 B1 2/2004 Dow et al.
8,889,418 B2 11/2014 Zhang et al.

9,267,932 B2 2/2016 Boeke et al.
10,113,163 B2 10/2018 Liu et al.
12,024,728 B2 7/2024 Altshuler et al.
12,031,162 B2 7/2024 Altshuler et al.
12,037,602 B2 7/2024 Cleaver et al.
12,037,617 B2 7/2024 Altshuler et al.
12,065,669 B2 8/2024 Cleaver et al.
12,123,034 B2 10/2024 Altshuler et al.
12,157,898 B2 12/2024 Bothmer et al.
2003/0121063 A1 6/2003 Kazazian et al.
2007/0037759 A1 2/2007 Deininger et al.
2010/0154070 A1 6/2010 Xu et al.
2011/0045591 A1 2/2011 Schumann et al.
2011/0059502 A1 3/2011 Chalasani
2011/0171729 A1 7/2011 Wang et al.
2014/0004569 A1 1/2014 Lambowitz et al.
2014/0113375 A1 4/2014 Liu et al.
2014/0273234 A1 9/2014 Zhang et al.
2014/0315985 A1 10/2014 May et al.
2014/0349400 A1 11/2014 Jakimo et al.
2017/0275665 A1 9/2017 Silas et al.
2018/0298391 A1 10/2018 Jakimo et al.
2019/0055543 A1 2/2019 Tran et al.
2019/0177735 A1 6/2019 Sederoff et al.
2020/0109398 A1 4/2020 Rubens et al.
2020/0248155 A1 8/2020 Halperin et al.
2022/0396813 A1 12/2022 Feala et al.
2023/0131847 A1 4/2023 Rubens et al.
2023/0235358 A1 7/2023 Citorik et al.
2023/0242899 A1 8/2023 Steinberg et al.
2023/0272430 A1 8/2023 Bothmer et al.
2023/0332184 A1 10/2023 Rubens et al.
2023/0348939 A1 11/2023 Bothmer et al.
2024/0002822 A1 1/2024 Altshuler et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 1342758 A 4/2002
CN 105308184 A 2/2016

(Continued)

OTHER PUBLICATIONS

U.S. Appl. No. 18/282,644, filed Sep. 18, 2023.
U.S. Appl. No. 18/280,749, filed Sep. 7, 2023.
U.S. Appl. No. 18/563,127, filed Nov. 21, 2023.
U.S. Appl. No. 18/690,134, filed Mar. 7, 2024.
U.S. Appl. No. 18/595,904, filed Mar. 5, 2024.
U.S. Appl. No. 18/590,275, filed Feb. 28, 2024.
U.S. Appl. No. 18/690,120, filed Mar. 7, 2024.
Fujimoto et al., "Integration of the 5' end of the retrotransposon,
R2Bm, can be complemented by homologous recombination,"
Nucleic Acids Research (2004) vol. 32, No. 4, pp. 1555-1565.

(Continued)

Primary Examiner — Ganapathirama Raghu

(74) Attorney, Agent, or Firm — K&L Gates LLP

(57)

ABSTRACT

Methods and compositions for modulating a target genome
are disclosed. The composition may comprise a first RNA
encoding a polypeptide comprising a retrotransposase
reverse transcriptase domain and a retrotransposase endo-
nuclease domain. The composition may also comprise a
second RNA comprising a sequence that binds the polypep-
tide and a heterologous object sequence. The composition
may insert the sequence of the heterologous object sequence
into a target DNA.

18 Claims, 24 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited**U.S. PATENT DOCUMENTS**

2024/0002886 A1 1/2024 Altshuler et al.
 2024/0018551 A1 1/2024 Altshuler et al.
 2024/0026324 A1 1/2024 Altshuler et al.
 2024/0035049 A1 2/2024 Bothmer et al.
 2024/0042058 A1 2/2024 Citorik et al.
 2024/0076638 A1 3/2024 Altshuler et al.
 2024/0076698 A1 3/2024 Cleaver et al.
 2024/0082429 A1 3/2024 Altshuler et al.
 2024/0084333 A1 3/2024 Cleaver et al.
 2024/0084334 A1 3/2024 Altshuler et al.
 2024/0093186 A1 3/2024 Altshuler et al.
 2024/0200104 A1 6/2024 Citorik et al.
 2024/0247243 A1 7/2024 Altshuler et al.
 2024/0252682 A1 8/2024 Altshuler et al.
 2024/0263153 A1 8/2024 Rubens et al.
 2024/0318170 A1 9/2024 Rubens et al.
 2024/0374759 A1 11/2024 Altshuler et al.
 2024/0417757 A1 12/2024 Bothmer et al.

FOREIGN PATENT DOCUMENTS

CN 107208078 A 9/2017
 EP 1700914 A1 9/2006
 EP 3448990 B1 6/2021
 WO 2003064644 A1 8/2003
 WO 2005049789 A2 6/2005
 WO 2008072540 A1 6/2008
 WO 2011064750 A1 6/2011
 WO 2014093718 A1 6/2014
 WO 2014150624 A1 9/2014
 WO 2015035139 A2 3/2015
 WO 2015191693 A2 12/2015
 WO 2016028843 A2 2/2016
 WO 2016036754 A1 3/2016
 WO 2016065364 A1 4/2016
 WO 2016201047 A1 12/2016
 WO 2016205728 A1 12/2016
 WO 2017059241 A1 4/2017
 WO 2017123609 A1 7/2017
 WO 2017137768 A1 8/2017
 WO 2017151719 A1 9/2017
 WO 2017173004 A1 10/2017
 WO 2017173054 A1 10/2017
 WO 2017180711 A1 10/2017
 WO 2017186550 A1 11/2017
 WO 2017197238 A1 11/2017
 WO 2018027078 A1 2/2018
 WO 2018049168 A1 3/2018
 WO 2018071663 A1 4/2018
 WO 2018089860 A1 5/2018
 WO 2018165629 A1 9/2018
 WO 2018176009 A1 9/2018
 WO 2018218166 A1 11/2018
 WO 2019051097 A1 3/2019
 WO 2020014528 A1 1/2020
 WO 2020047124 A1 3/2020
 WO 2020082076 A1 4/2020
 WO 2020186262 A1 9/2020
 WO 2020191233 A1 9/2020
 WO 2020191234 A1 9/2020
 WO 2020191239 A1 9/2020
 WO 2020191241 A1 9/2020
 WO 2020191242 A1 9/2020
 WO 2020191243 A1 9/2020
 WO 2020191245 A1 9/2020
 WO 2020191246 A1 9/2020
 WO 2020191248 A1 9/2020
 WO 2020191249 A1 9/2020
 WO 2020191171 A9 10/2020
 WO 2020191153 A9 11/2020
 WO 2020252361 A1 12/2020
 WO 2021016075 A1 1/2021
 WO 2021080922 A1 4/2021
 WO 2021102390 A1 5/2021
 WO 2021178709 A1 9/2021

WO 2021178717 A2 9/2021
 WO 2021178720 A2 9/2021
 WO 2021178898 A9 11/2021
 WO 2021248102 A1 12/2021

OTHER PUBLICATIONS

Fujiwara H., Site-specific non-LTR retrotransposons. ASMscience.org/MicrobiolSpectrum, 2015, pp. 1-16. (Year: 2015).
 Galantou RN., Cellular factors controlling human L1 retrotransposition. Ph.D. Dissertation, 2017, Universite Cote D'Azur, France, pp. 1-175. (Year: 2017).
 Garcia-rodriguez et al., Use of the computer-retargeted group II intron RmInt1 of Sinorhizobium meliloti for gene targeting, RNA Biology (2014) vol. 11, No. 4, pp. 391-401.
 Garneau et al., "The CRISPR-Cas bacterial immune system cleaves bacteriophage and plasmid DNA," Nature (2010) vol. 468, pp. 67-71.
 George et al., "Analysis of the 5' Junctions of R2 Insertions With the 28S Gene-Implications for Non-LTR Retrotransposition," Genetics (1996) vol. 142, pp. 853-863.
 Gilbert et al., "Multiple Fates of L1 Retrotransposition Intermediates in Cultured Human Cells," Molecular and Cellular Biology (2005) vol. 25, No. 17, pp. 7780-7795.
 Gladyshev-Rotifer et al., "rDNA-specific R9 retrotransposable elements generate an exceptionally long target site duplication upon insertion," Gene (2009) vol. 448, pp. 145-150.
 Goodier et al., "Restricting retrotransposons—a review," Mobile DNA (2016) vol. 7, Article 16, 30 pages.
 Govindaraju et al., "Endonuclease domain of non-LTR retrotransposons: loss-of-function mutants and modeling of the R2Bm endonuclease," Nucleic Acids Research (2016) vol. 44, No. 7, pp. 3276-3287.
 Govindaraju A., Protein/Nucleic-Acid Structure and Sequence Requirements for Specifying Second-Strand Cleavage and Second-Strand Synthesis During the Integration of the Site-Specific Line R2Bm. Ph.D. Dissertation, The Univ. Texas, Arlington, 2017, pp. 1-101. (Year: 2017).
 Ha et al., "Exosomes as therapeutic drug carriers and delivery vehicles across biological membranes: current perspectives and future challenges," Acta Pharmaceutica Sinica B (2016) vol. 6, Issue 4, pp. 287-296.
 Haack et al., "Cryo-EM Structures of a Group II Intron Reverse Splicing into DNA," Cell (2019) vol. 178, pp. 612-623.
 Halperin et al., "CRISPR-guided DNA polymerases enable diversification of all nucleotides in a tunable window," Nature (2018) vol. 560, No. 7717, pp. 248-252.
 Han et al., "Circular retrotransposition products generated by a LINE retrotransposon," Nucleic Acids Research (2012) vol. 40, No. 21, pp. 10866-10877.
 Han, "Non-long terminal repeat (non-LTR) retrotransposons-mechanisms, recent developments, and unanswered questions," Mobile DNA (2010) vol. 1, Article 15, 12 pages.
 Herschhorn et al., "Retroviral reverse transcriptases," Cellular and Molecular Life Sciences (2010) vol. 67, pp. 2717-2747.
 Hockemeyer et al., "Efficient targeting of expressed and silent genes in human ESCs and iPSCs using zinc-finger nucleases," Nature Biotechnology 27(9): 851-857, 2009.
 Holstein et al., Efficient Non-viral Gene Delivery into Human Hematopoietic Stem Cells by Minicircle Sleeping Beauty Transposon Vectors. Mol. Therapy., 2018, vol. 26(4): 1137-1153. (Year: 2018).
 Hwang et al., "Web-based design and analysis tools for CRISPR base editing," BMC Bioinformatics (2018) vol. 19, Article 542, 7 pages.
 International Search Report and Written Opinion issued in PCT/US2019/048607 mailed on Jan. 15, 2020, 15 pages.
 Ivancevic et al., "LINEs between Species-Evolutionary Dynamics of LINE-1 Retrotransposons across the Eukaryotic Tree of Life," Genome Biology and Evolution (2016) vol. 8, No. 11, pp. 3301-3322.
 Ivics et al., "Molecular Reconstruction of Sleeping Beauty, a Tc1-like Transposon from Fish, and Its Transposition in Human Cells," Cell 1997, vol. 91, pp. 501-510.

(56)

References Cited**OTHER PUBLICATIONS**

- Jamburuthugoda et al., "The Reverse Transcriptase Encoded by the Non-LTR Retrotransposon R2 Is as Error-Prone as That Encoded by HIV-1," *Journal of Molecular Biology* (2011).
- Jamburuthugoda et al., Identification of RNA binding motifs in the R2 retrotransposon-encoded reverse transcriptase, *Nuc Acids Res* (2014) vol. 42, No. 13, pp. 8405-8415.
- Jinek et al., "A Programmable Dual-RNA-Guided DNA Endonuclease in Adaptive Bacterial Immunity," *Science* (2012) vol. 337, pp. 816-821 and Supplementary Materials.
- Jurka et al., "Sequence patterns indicate an enzymatic involvement in integration of mammalian retroposons," *Proceedings of the National Academy of Sciences* (1997) vol. 94, pp. 1872-1877.
- Kajikawa et al., "A new mechanism to ensure integration during LINE retrotransposition—A suggestion from analyses of the 5' extra nucleotides," *Gene* (2012) vol. 505, pp. 345-351.
- Kawashima et al., "A novel target-specific gene delivery system combining baculovirus and sequence-specific long interspersed nuclear elements," *Virus Research* (2007) vol. 127, pp. 49-60.
- Kelley et al., "The Phyre2 web portal for protein modeling, prediction and analysis," *Nature Protocols* (2015) vol. 10, No. 6, pp. 845-858.
- Kierzek et al., "Secondary structures for 5' regions of R2 retrotransposon RNAs reveal a novel conserved pseudoknot and regions that evolve under different constraints," *J Mol Biol* (2009) vol. 390, No. 3, pp. 428-442.
- Kleinstiver et al., "Engineered CRISPR-Cas9 nucleases with altered PAM specificities," *Nature* (2015) vol. 523, pp. 481-485.
- Klompe et al., "Transposon-encoded CRISPR-Cas systems direct RNA-guided DNA integration," *Nature* (2019) vol. 571, No. 7764, pp. 219-225.
- Kojima et al., "The Wide Distribution and Change of Target Specificity of R2 Non-LTR Retrotransposons in Animals," *PLOS One* (2016) vol. 11, No. 9, Article e0163496, 16 pages.
- Kojima et al., Recent Expansion of a New Ingi-Related Clade of Vingi non-LTR Retrotransposons in Hedgehogs, *Molecular Biology and Evolution* (2011) vol. 28, No. 1, pp. 17-20.
- Komor et al., "Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage," *Nature* (2016) vol. 523, pp. 420-424.
- Krzywkowski et al., "Limited reverse transcriptase activity of phi29 DNA polymerase," *Nucleic Acids Research* (2018) vol. 46, No. 7, pp. 3625-3632.
- Kuroki-Kami et al., "Targeted gene knockin in zebrafish using the 28S rDNA-specific non-LTR-retrotransposon R201," *Mobile DNA* (2019) vol. 10, Article 23, 12 pages.
- Kurzynska-Kokorniak et al., "DNA-directed DNA Polymerase and Strand Displacement Activity of the Reverse Transcriptase Encoded by the R2 Retrotransposon," *Journal of Molecular Biology* (2007) vol. 374, pp. 322-333.
- Landrum et al., "ClinVar: public archive of interpretations of clinically relevant variants," *Nucleic Acids Research* (2015) vol. 44, pp. D862-D868.
- Larson et al., "Spliced integrated retrotransposed element (SpIRE) formation in the human genome," *PLoS Biology* (2018) vol. 16, No. 3, Article e2003067, 37 pages.
- Lathe et al., "A single lineage of r2 retrotransposable elements is an active, evolutionarily stable component of the *Drosophila* rDNA locus," *Molecular Biology and Evolution* (1997) vol. 14, No. 12, pp. 1232-1241.
- Letunic et al., "20 Years of the SMART protein domain annotation resource," *Nucleic Acids Research* (2018) vol. 46, pp. D493-D496.
- Li et al., "A Review of the Structure, Preparation, and Application of NLCs, PNPs, and PLNs," *Nanomaterials* 7, 122, 2017.
- Lim et al., "Specific insertions of zinc finger domains into Gag-Pol yield engineered retroviral vectors with selective integration properties," *PNAS* (2010) vol. 107, No. 28, pp. 12475-12480.
- Liu et al., "Computational approaches for effective CRISPR guide RNA design and evaluation," *Computational and Structural Biotechnology Journal* (2020): vol. 18, pp. 35-44.
- Longworth et al., "Expanding the Enzymatic Activity of the programmable Endonuclease Cas9 in Zebrafish," *Rice University Master of Arts Thesis* (2018) 42 pages.
- Luan et al., "RNA Template Requirements for Target DNA-Primed Reverse Transcription by the R2 Retrotransposable Element," *Molecular and Cellular Biology* (1995) vol. 15, No. 7, pp. 3882-3891.
- Lubelsky et al., "Sequences enriched in Alu repeats drive nuclear localization of long RNAs in human cells," *Nature* 555 (107-111), 2018.
- Mahbub et al., "Globular domain structure and function of restriction-like-endonuclease LINEs—similarities to eukaryotic splicing factor Prp8," *Mobile DNA* (2017) vol. 8, Article 16, 15 pages.
- Templeton et al., "Improved DNA: liposome complexes for increased systemic delivery and gene expression," *Nature Biotech*, 15:647-652, 1997.
- Thompson et al., "Independently derived targeting of the 28S rDNA by A- and D-clade R2 retrotransposons," *Mobile Genetic Elements* (2011) vol. 1, pp. 29-37.
- Trevino et al., "Genome Editing Using Cas9 Nickases," *Methods in Enzymology*, Eds. Doudna et al. (2014) vol. 546, Chapter 8, pp. 161-174.
- Truong et al., "Retrohoming of a Mobile Group II Intron in Human Cells Suggests How Eukaryotes Limit Group II Intron Proliferation," *PLoS Genetics* (2015) vol. 11, No. 8, Article e1005422, 35 pages.
- U.S. Appl. No. 62/263,223, filed Oct. 2, 2015.
- U.S. Appl. No. 62/332,099, filed Apr. 13, 2016.
- Ustyantsev et al., "Convergence of retrotransposons in oomycetes and plants," *Mobile DNA* (2017) vol. 8, Article 4, 11 pages.
- Wagstaff et al., "Molecular reconstruction of extinct LINE-1 elements and their interaction with nonautonomous elements," *Molecular Biology and Evolution* (2013) 30(1): 88-99.
- Wallace et al., "L1 mobile element expression causes multiple types of toxicity," *Gene* (2008) vol. 419, pp. 75-81.
- Wicker et al., "A unified classification system for eukaryotic transposable elements," *Nature Reviews Genetics* (2007) vol. 8, pp. 973-982.
- Xiong et al., "Functional expression of a sequence-specific endonuclease encoded by the retrotransposon R2Bm," *Cell* (1988) vol. 55, pp. 235-246.
- Xiong et al., "Origin and evolution of retroelements based upon their reverse transcriptase sequences," *The EMBO Journal* (1990) vol. 9, No. 10, pp. 3353-3362.
- Xu et al., "Accuracy and efficiency define Bxb1 integrase as the best of fifteen candidate serine recombinases for the integration of DNA into the human genome," *BMC Biotechnology* (2013) vol. 13, Article 87, 17 pages.
- Yang et al., "Identification and characterization of nuclear and nucleolar localization signals in 58-kDa microspherule protein (MSP58)," *Journal of Biomedical Science* (2015) vol. 22, Article 33, 15 pages.
- Yang et al., "Identification of the endonuclease domain encoded by R2 and other site-specific, non-long terminal repeat retrotransposable elements," *PNAS* (1999) vol. 96, pp. 7487-7492.
- Yasukawa et al., "Next-generation sequencing-based analysis of reverse transcriptase fidelity," *Biochemical and Biophysical Research Communications* (2017) vol. 492, pp. 147-153.
- Zhang et al., "A novel RNA motif mediates the strict nuclear localization of a long noncoding RNA," *Molecular and Cellular Biology* 34, 2318-2329 (2014).
- Zhang et al., "Genome Editing with mRNA Encoding ZFN, TALEN, and Cas9," *Molecular Therapy* (2019) vol. 27, No. 4, pp. 735-746.
- Zhao et al., "An ultra-processive, accurate reverse transcriptase encoded by a metazoan group II intron," *RNA* (2018) 24: 183-195.
- Zhao et al., "Crystal structures of a group II intron maturase reveal a missing link in spliceosome evolution," *Nature Structural & Molecular Biology* (2016) vol. 23, No. 6, pp. 558-567.
- Zhu et al., "The iCRISPR Platform for Rapid Genome Editing in Human Pluripotent Stem Cells," *Methods in Enzymology*, Eds. Doudna et al. (2014) vol. 546, Chapter 11, pp. 215-250.
- Zimmerly et al., "A Group II Intron RNA Is a Catalytic Component of a DNA Endonuclease Involved in Intron Mobility," *Cell* (1995) vol. 83, pp. 529-538.

(56)

References Cited**OTHER PUBLICATIONS**

- Zimmerly et al., "An Unexplored Diversity of Reverse Transcriptases in Bacteria," *Microbiology Spectrum* (2015) vol. 3, No. 2, Article MDNA-0058-2014, 16 pages.
- Zimmerly et al., "Evolution of group II introns," *Mobile DNA* (2015) vol. 6, Article 7, 19 pages.
- Zimmerman et al., "A Completely Reimplemented MPI Bioinformatics Toolkit with a New Hhpred Server at its Core," *J Mol Biol* (2018) vol. 430, pp. 2237-2243.
- Zingler et al., "Analysis of 5' junctions of human LINE-1 and Alu retrotransposons suggests an alternative model for 5'-end attachment requiring microhomology-mediated end-joining," *Genome Research* (2005) vol. 15, pp. 780-789.
- 63rd Fujihara Seminar, "A new horizon of retroposon research," Jul. 31-Aug. 3, 2012, Kyoto, Japan, program booklet with abstracts, 90 pages.
- [No Author Listed] Invitrogen—"ProQuest Two-Hybrid System: A sensitive method for detecting protein-protein interactions," User Manual, Version A (2005) 116 pages.
- [No Author Listed] Rice University Events Listing for Aug. 3, 2018: Thesis Defense by Gia Longworth, "Expanding the Enzymatic Activity of the Programmable Endonuclease Cas9 in Zebrafish," with Abstract, first posted on Aug. 3, 2018.
- Ade et al., "Evaluating different DNA binding domains to modulate L1 ORF2p-driven site-specific retrotransposition events in human cells," *Gene* (2018) vol. 642, pp. 188-198.
- Altenbuchner, "Editing of the *Bacillus subtilis* Genome by the CRISPR-Cas9 System," *Applied and Environmental Microbiology* (2016) vol. 82, No. 17, pp. 5421-5427.
- An et al., "Plug and play modular strategies for synthetic retrotransposons," *Methods* (2009) vol. 49, pp. 227-235.
- Anand et al., "Structure based design of protein linkers for zinc finger nuclease," *FEBS Letters*, 587:19, 2013.
- Anzalone et al., "Search-and-replace genome editing without double-strand breaks or donor DNA," *Nature* (2019) vol. 576, No. 7785, pp. 149-157.
- Babushok et al., "Progress in understanding the biology of the human mutagen LINE-1," *Human Mutation* (2007) vol. 28, No. 6, pp. 527-539.
- Bailey et al., "The MEME Suite," *Nucleic Acids Research* (2015) vol. 43, pp. W39-W49.
- Bao et al., "Rebase Update, a database of repetitive elements in eukaryotic genomes," *Mobile DNA* (2015) vol. 6, Article 11, 6 pages.
- Barnes et al., "Recombinant R2 Retrotransposon for Targeted Integration of Large Genetic Cassettes into the Human Genome," *Molecular Therapy* (2019) vol. 27, No. 4S1, Abstract 216, p. 109.
- Bateman et al., "UniProt—the universal protein knowledgebase," *Nucleic Acids Research* (2017) vol. 45, pp. D158-D169.
- Belfort et al., "Group II Intron RNPs and Reverse Transcriptases—From Retroelements to Research Tools," *Cold Spring Harbor Perspectives in Biology* (2019) 11:a032375, 17 pages.
- Benitez-Guijarro et al., "RNase H2, mutated in Aicardi-Goutieres syndrome, promotes LINE-1 retrotransposition," *The EMBO Journal* (2018) vol. 37, Article e98506, 22 pages.
- Bibillo et al., "High Processivity of the Reverse Transcriptase from a Non-long Terminal Repeat Retrotransposon," *J Biol Chem* (2002) vol. 277, No. 38, pp. 34836-34845.
- Bibillo et al., "The Reverse Transcriptase of the R2 Non-LTR Retrotransposon: Continuous Synthesis of cDNA on Non-continuous RNA Templates," *J Mol Biol* (2002) vol. 316, pp. 459-473.
- Birbach et al., "Cytosolic, nuclear and nucleolar localization signals determine subcellular distribution and activity of the NF-kappaB inducing kinase NIK," *Journal of Cell Science*, 117 (3615-3624), 2004.
- Bodea et al., "Retrotransposon-induced mosaicism in the neural genome," *Open Biology* (2018) vol. 8, No. 7, Article 180074, 17 pages.
- Boissinot et al., "L1 (LINE-1) Retrotransposon Evolution and Amplification in Recent Human History," *Molecular Biology and Evolution* 2000, 915-928.
- Bono et al., "Connecting genotypes, phenotypes and fitness: harnessing the power of CRISPR/Cas9 genome editing," *Molecular Ecology* (2015) vol. 24, pp. 3810-3822.
- Burke et al., "R4, a non-LTR retrotransposon specific to the large subunit rRNA genes of nematodes," *Nucleic Acids Res.* 23, 4628-34 (1995).
- Burke et al., "Sequence relationship of retrotransposable elements R1 and R2 within and between divergent insect species," *Molecular Biology and Evolution* (1993) vol. 10, No. 1, pp. 163-185.
- Burke et al., "The domain structure and retrotransposition mechanism of R2 elements are conserved throughout arthropods," *Molecular Biology and Evolution* (1999) vol. 16, No. 4, pp. 502-511.
- Candales et al., "Database for bacterial group II introns," *Nucleic Acids Research* (2012) vol. 40, pp. D187-D190.
- Chan et al., "Crystal structure of a group II intron in the pre-catalytic state," *Nature Structural & Molecular Biology* (2012) vol. 19, No. 5, pp. 555-557.
- Chen et al., "Fusion Protein Linkers: Property, Design and Functionality," *Adv Drug Deliv Rev* (2013) 65(10): 1357-1369.
- Chillón et al., "In vitro characterization of the splicing efficiency and fidelity of the RmInt1 group II intron as a means of controlling the dispersion of its host mobile element," *RNA* (2014) vol. 20, No. 12, pp. 2000-2010.
- Choi et al., "Interplay between RNASEH2 and MOV10 controls LINE-1 retrotransposition," *Nucleic Acids Research* (2018) doi: 10/1093/nar/gkx1312, 15 pages.
- Choudhuri, "Bioinformatics for Beginners: Genes, Genomes, Molecular Evolution, Databases and Analytical Tools," Elsevier (2014) p. 64.
- Christensen et al., "Footprint of the Retrotransposon R2Bm Protein on its Target Site Before and After Cleavage," *Journal of Molecular Biology* (2004) vol. 336, No. 5, pp. 1035-1045.
- Christensen et al., "R2 Target-Primed Reverse Transcription-Ordered Cleavage and Polymerization Steps by Protein Subunits Asymmetrically Bound to the Target DNA," *Molecular and Cellular Biology* (2005) vol. 25, No. 15, pp. 6617-6628.
- Christensen et al., "RNA from the 5' end of the R2 retrotransposon controls R2 protein binding to and cleavage of its DNA target site," *Proceedings of the National Academy of Sciences* (2006) vol. 103, No. 47, pp. 17602-17607.
- Christensen et al., "Role of the *Bombyx mori* R2 element N-terminal domain in the target-primed reverse transcription (TPRT) reaction," *Nucleic Acids Research* (2005) vol. 33, No. 20, pp. 6461-6468.
- Cordaux et al., "The impact of retrotransposons on human genome evolution," *Nature Reviews Genetics* (2009) *Nature Reviews* vol. 10, pp. 691-703.
- Cost et al., "Targeting of Human Retrotransposon Integration Is Directed by the Specificity of the L1 Endonuclease for Regions of Unusual DNA Structure," *Biochemistry* (1998) vol. 37, pp. 18081-18093.
- Craig et al., Editors, "Mobile Dna III," ASM Press (2015) pp. 1-1346.
- Denli et al., "Primate-Specific ORF0 Contributes to Retrotransposon-Mediated Diversity," *Cell* (2015) vol. 163, No. 3, pp. 583-593.
- Dewannieux et al., "Role of poly(A) tail length in Alu retrotransposition," *Genomics* (2005) vol. 86, pp. 378-381.
- Doucet et al., "A 3' Poly(A) Tract Is Required for LINE-1 Retrotransposition," *Molecular Cell* (2015) vol. 60, pp. 728-741.
- Eickbush et al., "Integration of *Bombyx mori* R2 Sequences into the 28S Ribosomal RNA Genes of *Drosophila melanogaster*," *Molecular and Cellular Biology* (2000) vol. 20, No. 1, pp. 213-223.
- Eickbush et al., "Integration, Regulation, and Long-Term Stability of R2 Retrotransposons," *Microbiology Spectrum* (2015) vol. 3, No. 2, MDNA3-011, 20 pages.
- Eickbush et al., "R2 and R2-R1 hybrid non-autonomous retrotransposons derived by internal deletions of full-length elements," *Mobile DNA* (2012) vol. 3, Article 10, 15 pages.

(56)

References Cited**OTHER PUBLICATIONS**

- Eickbush et al., "R2 Retrotransposons Encode a Self-Cleaving Ribozyme for Processing from an rRNA Cotranscript," *Molecular and Cellular Biology* (2010) vol. 30, No. 13, pp. 3142-3150.
- Ellefson et al., "Synthetic evolutionary origin of a proofreading reverse transcriptase," *Science* (2016) vol. 352, Issue 6293, pp. 1590-1593.
- Enyeart et al., "Biotechnological applications of mobile group II introns and their reverse transcriptases-gene targeting, RNA-seq, and non-coding RNA analysis," *Mobile DNA* (2014) vol. 5, Article 2, 19 pages.
- Faure et al., "CRISPR-Cas in mobile genetic elements: counter-defence and beyond," *Nat Rev Microbiol* (2019) vol. 17, No. 8, pp. 513-525.
- Finn et al., "InterPro in 2017—beyond protein family and domain annotations," *Nucleic Acids Res* (2017) vol. 45, No. D1, pp. D190-D199.
- Finnegan, "Transposable elements—How non-LTR retrotransposons do it," *Current Biology* (1997) vol. 7, pp. R245-R248.
- Flasch et al., "Genome-wide de novo L1 Retrotransposition Connects Endonuclease Activity with Replication," *Cell* (2019) vol. 177, p. 877-851.
- Maita et al., "Crystal Structure of the Endonuclease Domain Encoded by the Telomere-specific Long Interspersed Nuclear Element, TRAS1," *Journal of Biological Chemistry* (2004) vol. 279, No. 39, p. 41607-41076.
- Malik et al., "Ribonuclease H evolution in retrotransposable elements," *Cytogenetic and Genome Research* (2005) vol. 110, pp. 392-401.
- Mastroianni et al., "Group II Intron-Based Gene Targeting Reactions in Eukaryotes," *PLoS ONE* (2008) vol. 3, Issue 9, Article e3121, 15 pages.
- Meers et al., "DNA repair by RNA: Templated, or not templated, that is the question," *DNA Repair* (2016) vol. 44, pp. 17-21.
- Mills et al., "Which transposable elements are active in the human genome," *Trends in Genetics* (2007) vol. 23, No. 4, pp. 183-191.
- Mita et al., "LINE-1 protein localization and functional dynamics during the cell cycle," *eLife* (2018) vol. 7, Article e30058, 35 pages.
- Miyagawa et al., "Identification of cis- and trans-acting factors involved in the localization of MALAT-1 noncoding RNA to nuclear speckles," *RNA* (2012) vol. 18, pp. 738-751.
- Miyoshi et al., "Poly(ADP-Ribose) Polymerase 2 Recruits Replication Protein A to Sites of LINE-1 Integration to Facilitate Retrotransposition," *Molecular Cell* (2019) vol. 75, pp. 1286-1298.
- Mohr et al., "A Reverse Transcriptase-Cas1 Fusion Protein Contains a Cas6 Domain Required for Both CRISPR RNA Biogenesis and RNA Spacer Acquisition," *Molecular Cell* (2018) vol. 72 pp. 700-714.
- Moran et al., "High Frequency Retrotransposition in Cultured Mammalian Cells," *Cell* (1996) vol. 87, pp. 917-927.
- Moss et al., "The R2 retrotransposon RNA families," *RNA Biology* (2011) vol. 8, No. 5, pp. 714-718.
- Mukha et al., "Endonuclease domain of the *Drosophila melanogaster* R2 non-LTR retrotransposon and related retroelements—a new model for transposition," *Frontiers in Genetics* (2013) vol. 4, Article 63, 15 pages.
- Naldini et al., "In Vivo Gene Delivery and Stable Transduction of Nondividing Cells by a Lentiviral Vector" *Science* (1996) vol. 272 pp. 263-267.
- Nami et al., "Strategies for In Vivo Genome Editing in Nondividing Cells," *Trends in Biotechnology* (2018) vol. 36, No. 8, pp. 770-786.
- Nichuguti et al., "Both the Exact Target Site Sequence and a Long Poly(A) Tail Are Required for Precise Insertion of the 18S Ribosomal DNA-Specific Non-Long Terminal Repeat Retrotransposon R7Ag," *Molecular and Cellular Biology* (2016) vol. 36, No. 10, pp. 1494-1508.
- Núñez et al., "Cas1-Cas2 complex formation mediates spacer acquisition during CRISPR-Cas adaptive immunity," *Nature Structural & Molecular Biology* (2014) vol. 21, No. 6, pp. 528-534.
- Opposition Brief filed in European Patent No. 3448990, mailed Mar. 15, 2022, 66 pages.
- Osanaï et al., "Essential Motifs in the 3' Untranslated Region Required for Retrotransposition and the Precise Start of Reverse Transcription in Non-Long-Terminal-Repeat Retrotransposon SART1," *Molecular and Cellular Biology* (2004) vol. 24, No. 18, pp. 7902-7913.
- Ostertag et al., "Biology of Mammalian L1 Retrotransposons" *Annual Review of Genetics* (2001) vol. 35, pp. 501-538.
- Ostertag et al., "Twin Priming—A Proposed Mechanism for the Creation of Inversions in L1 Retrotransposition," *Genome Research* (2001) vol. 11, pp. 2059-2065.
- Paix et al., "Scalable and Versatile Genome Editing Using Linear DNAs with Microhomology to Cas9 Sites in *Caenorhabditis elegans*," *Genetics* (2014) vol. 198, pp. 1347-1356.
- Pei et al., "PROMALS3D—a tool for multiple protein sequence and structure alignments," *Nucleic Acids Research* (2008) vol. 36, No. 7, pp. 2295-2300.
- Pellenz et al., "New human chromosomal safe harbor sites for genome engineering with CRISPR/Cas9, TAL effector and homing endonucleases," *Human Gene Therapy* (2018) doi: 10.1101/396390, 39 pages.
- Peters et al., "Recruitment of CRISPR-Cas systems by Tn7-like transposons," *PNAS* (2017) vol. 114, No. 35, pp. E7358-E7366.
- Piskareva et al., "The carboxy-terminal segment of the human LINE-1 ORF2 protein is involved in RNA binding," *FEBS Open Bio* (2013) vol. 3, pp. 433-437.
- Qu et al., "Structure of a group II intron in complex with its reverse transcriptase," *Nature Structural & Molecular Biology* (2016) vol. 23, No. 6, pp. 549-557.
- Ran et al., "Genome engineering using the CRISPR/Cas9 system," *Nature Protocols* (2013) vol. 8, No. 11, pp. 2281-2308.
- Ruminski et al., "Processing and Translation Initiation of Non-long Terminal Repeat Retrotransposons by Hepatitis Delta Virus (HDV)-like Self-cleaving Ribozymes," *J Biol Chem* (2011) vol. 286, No. 48, pp. 41286-41295.
- Ruschak et al., "Secondary structure models of the 3' untranslated regions of diverse R2 RNAs" *RNA* (2004) vol. 10(6): 978-987.
- San Filippo et al., "Characterization of the C-Terminal DNA-binding-DNA Endonuclease Region of a Group II Intron-encoded Protein," *Journal of Molecular Biology* (2002) vol. 324, pp. 933-951.
- Sano et al., "Mutations to create thermostable reverse transcriptase with bacterial family A DNA polymerase from *Thermotoga petrophila* K4," *Journal of Bioscience and Bioengineering* (2012) vol. 113, No. 3, pp. 315-321.
- Sharma et al., "In vivo genome editing of the albumin locus as a platform for protein; replacement therapy," *Blood* (2015) vol. 126, No. 15: 1777-1784.
- Sharon et al., "Functional Genetic Variants Revealed by Massively Parallel Precise Genome Editing," *Cell* (2018) vol. 175, pp. 544-557.
- Shivram et al., "Targeting novel sites: The N-terminal DNA binding domain of non-LTR retrotransposons is an adaptable module that is implicated in changing site specificities," *Mobile Genetic Elements* (2011) vol. 1, No. 3, pp. 169-178.
- Shukla et al., "High-throughput identification of RNA nuclear enrichment sequences," *The EMBO Journal* (2018) vol. 37, Article e98452, 11 pages.
- Silas et al., "Direct CRISPR spacer acquisition from RNA by a natural reverse-transcriptase-Cas1 fusion protein," *Science* (2016) vol. 351, No. 6276, Article aad4234, 31 pages.
- Simon et al., "Retrons and their applications in genome engineering," *Nucleic Acids Research* (2019) doi: 10.1093/nar/gkz865, 13 pages.
- Smyshlyaev et al., "Acquisition of an Archaea-like ribonuclease H domain by plant L1 retrotransposons supports modular evolution," *Proceedings of the National Academy of Sciences* (2013) vol. 110, No. 50, p. 20140-20145.
- Spuch and Navarro, "Liposomes for Targeted Delivery of Active Agents against Neurodegenerative Diseases (Alzheimer's Disease and Parkinson's Disease)," *Journal of Drug Delivery*, vol. 2011, Article ID 469679, 12 p. 2011. doi:10.1155/2011/469679 for review.

(56)

References Cited

OTHER PUBLICATIONS

Stamos et al., "Structure of a Thermostable Group II Intron Reverse Transcriptase with Template-Primer and Its Functional and Evolutionary Implications," *Molecular Cell* (2017) vol. 68, pp. 926-939.

Strecker et al., "RNA-guided DNA insertion with CRISPR-associated transposases," *Science* (2019) vol. 365, No. 6448, pp. 48-53.

Su et al., "Human DNA polymerase [eta] has reverse transcriptase activity in cellular environments," *J Biol Chem* (2019) vol. 294, No. 15, pp. 6073-6081.

Su et al., "Sequence-specific retrotransposition of 28S rDNA-specific Line R2OI in human cells" *RNA* (2019) vol. 25, pp. 1432-1438.

Sultana et al., "Integration site selection by retroviruses and transposable elements in eukaryotes," *Nature Reviews Genetics* (2017) doi: 10.1038/nrg.2017.7, 17 pages.

Sultana et al., "The Landscape of L1 Retrotransposons in the Human Genome Is Shaped by Pre-insertion Sequence Biases and

Post-insertion Selection," *Molecular Cell* (2019) vol. 74, pp. 555-570.

Sultana T., Influence of the genomic context on integration site selection by human L1 retrotransposons. Ph.D. Dissertation, 2016, Universite Cote D'Azur, France, pp. 1-184. (Year: 2016).

Takahashi et al., "Transplantation of target site specificity by swapping the endonuclease domains of two LINES," *The EMBO Journal* (2002) vol. 21, No. 3, pp. 408-417.

Takasu et al., "Targeted mutagenesis in the silkworm *Bombyx mori* using zinc finger nuclease mRNA injection," *Insect Biochemistry and Molecular Biology* 40(10): 759-765, 2010.

Tang et al., "Rewritable multi-event analog recording in bacterial and mammalian cells," *Science* (2018) vol. 360, No. 6385, eaap8992.

Taylor et al., "Affinity Proteomics Reveals Human Host Factors Implicated in Discrete Stages of LINE-1 Retrotransposition," *Cell* (2013) vol. 155, pp. 1034-1048.

Luo et al., "Comparative analysis of chimeric ZFP-, TALE- and Cas9-piggyBac transposases for integration into a single locus in human cells," *Nucleic Acids Research* (2017) vol. 45, No. 14, pp. 8411-8422.

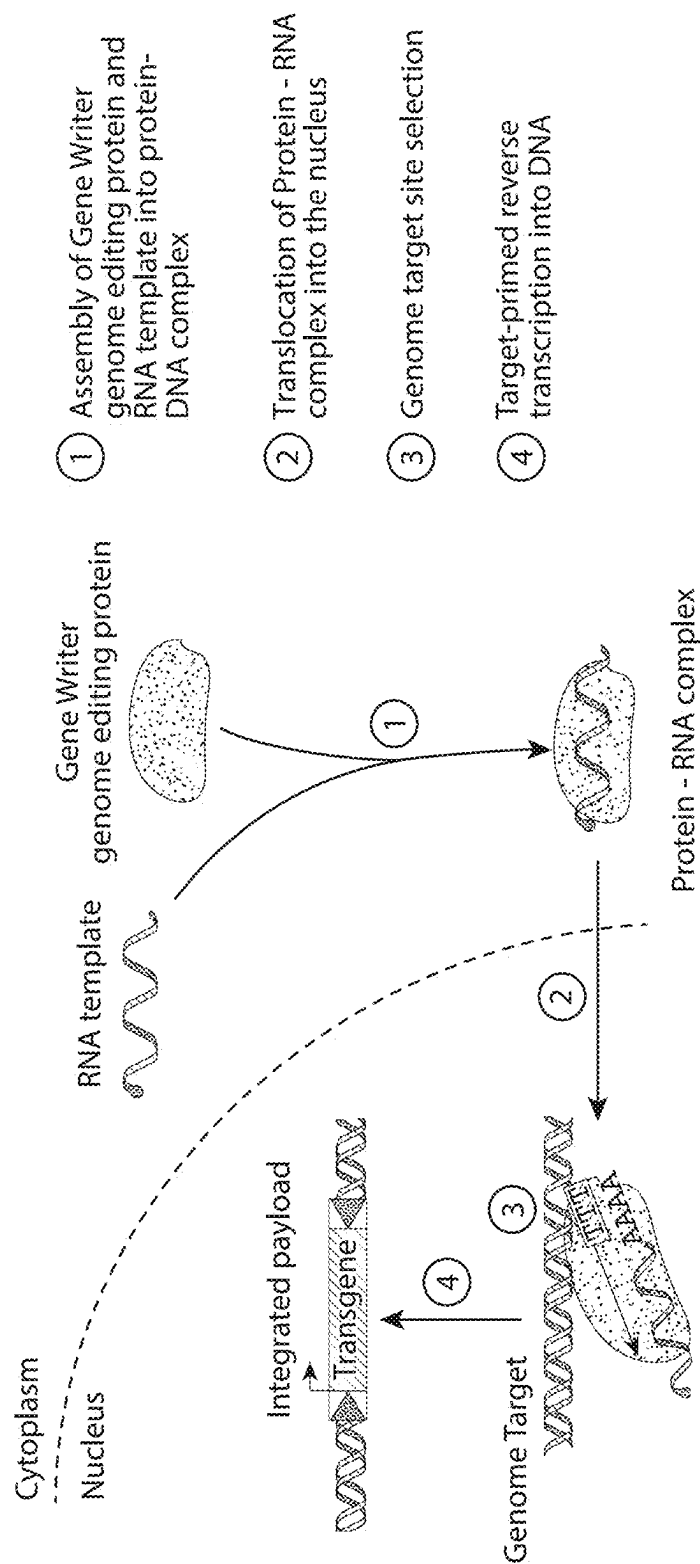


Figure 1

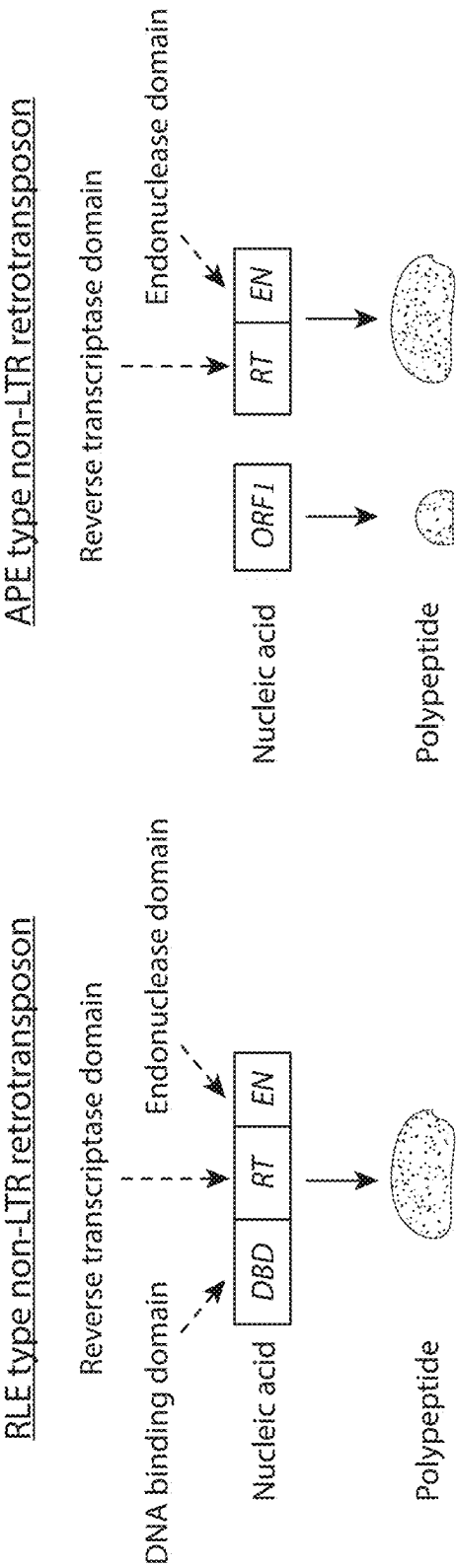


Figure 2

Gene Writer genome editor proteins with different DNA binding domains target different DNA sites for Gene Writing genome editing

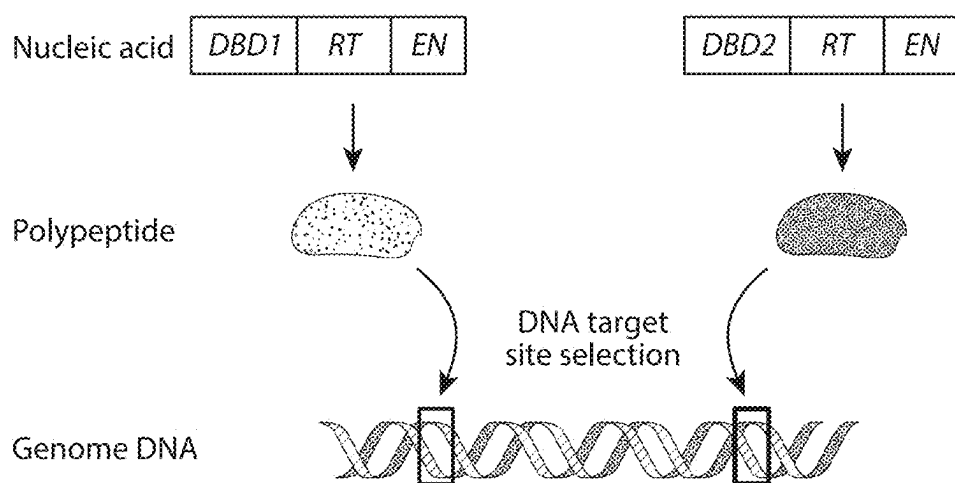


Figure 3

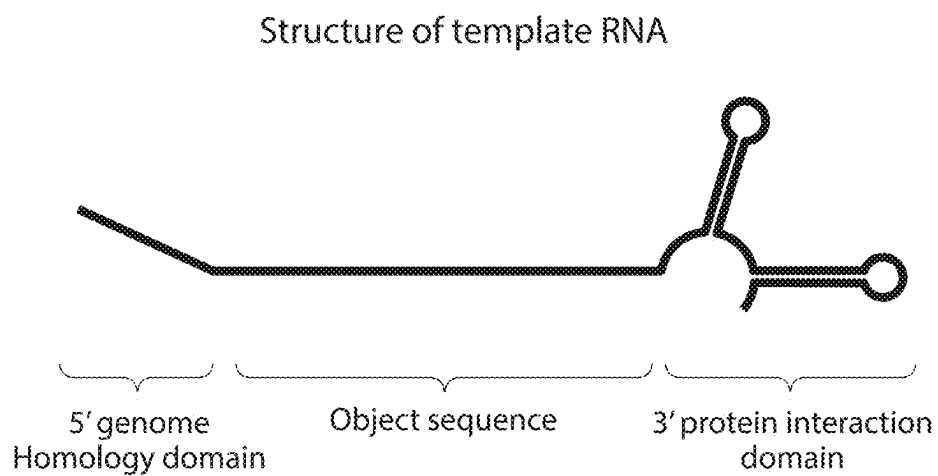


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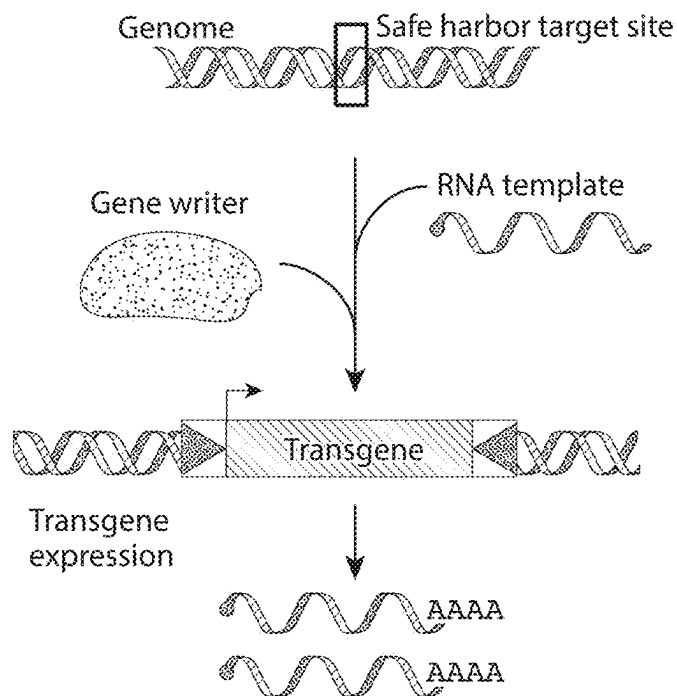


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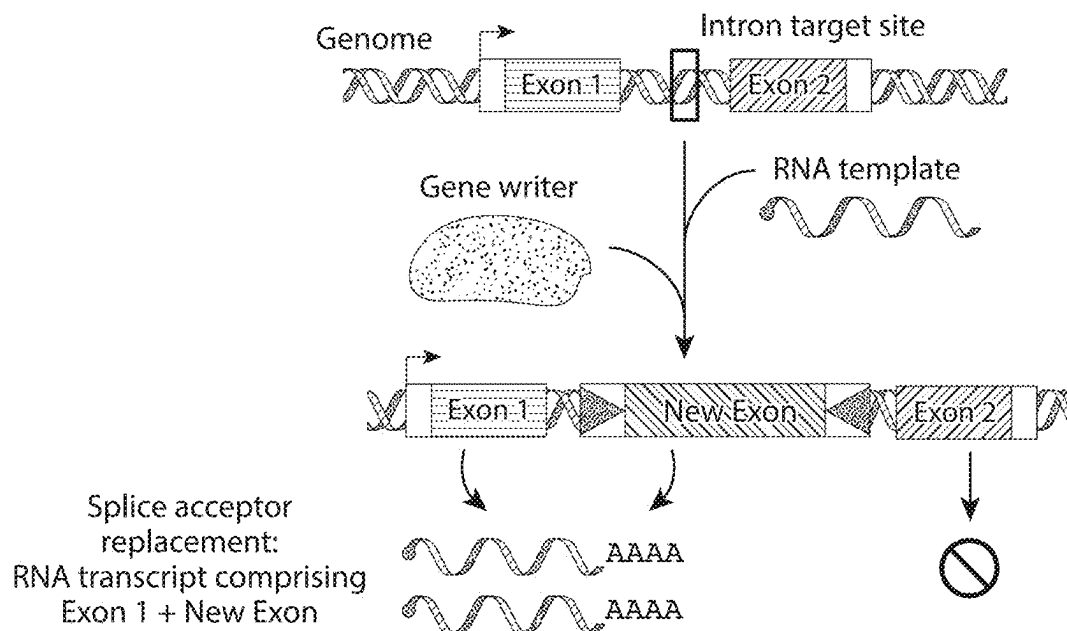


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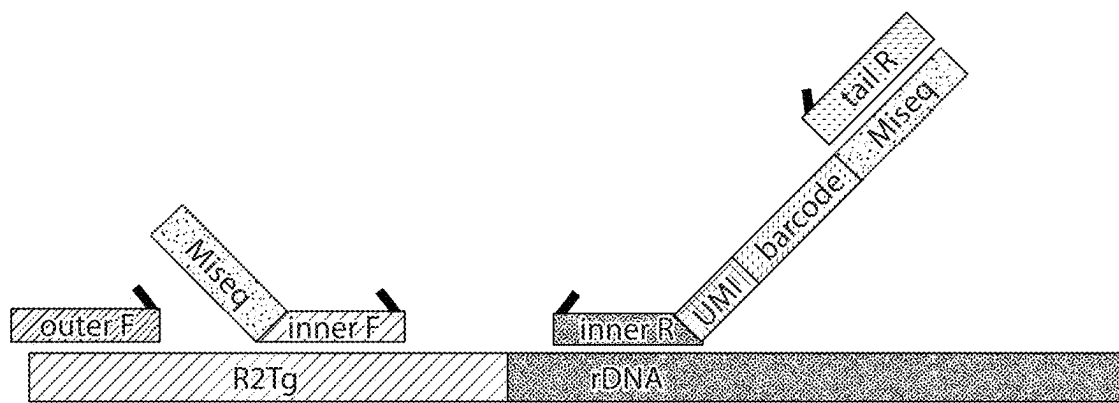


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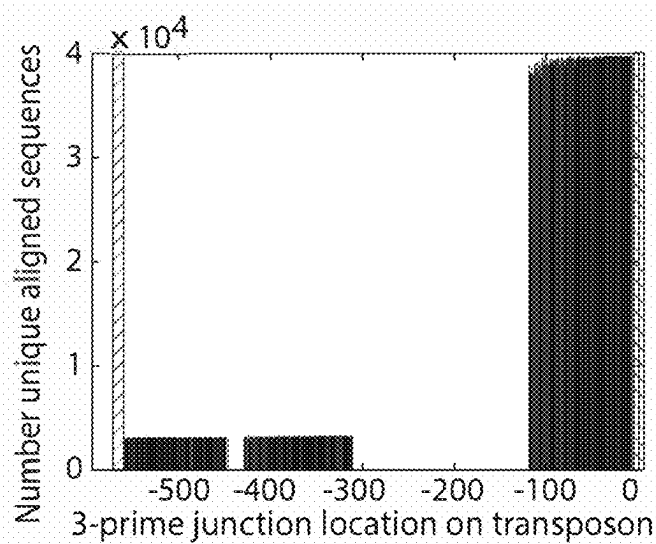


Figure 8A

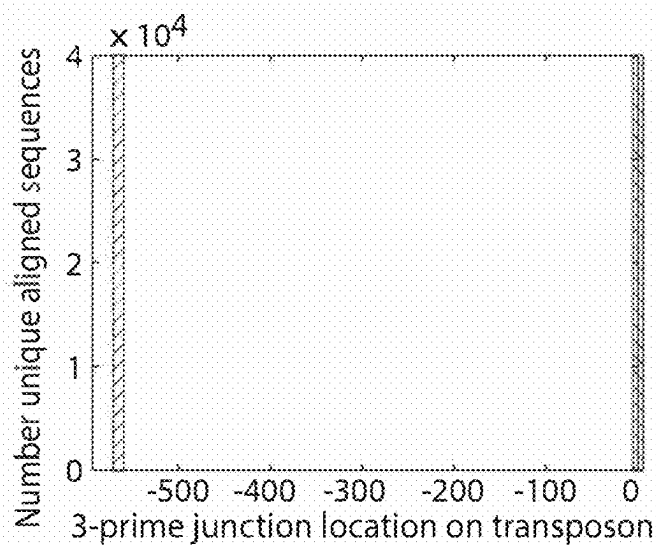


Figure 8B

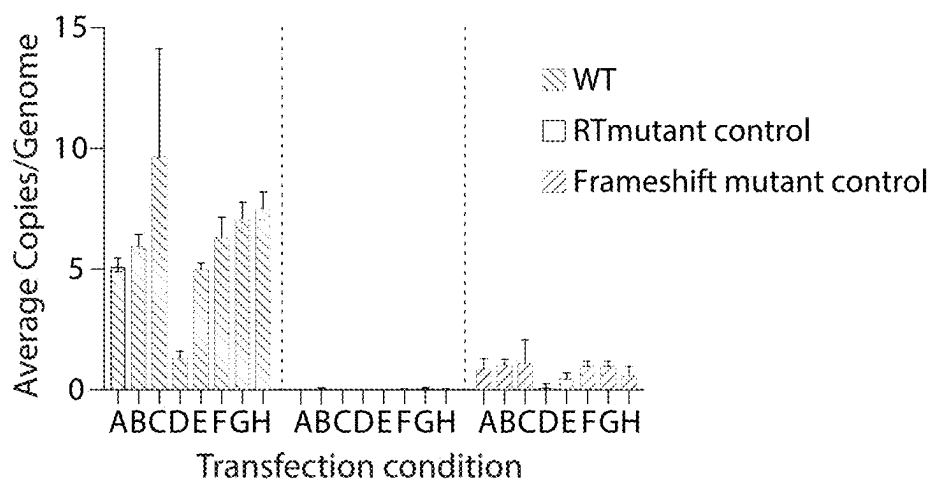


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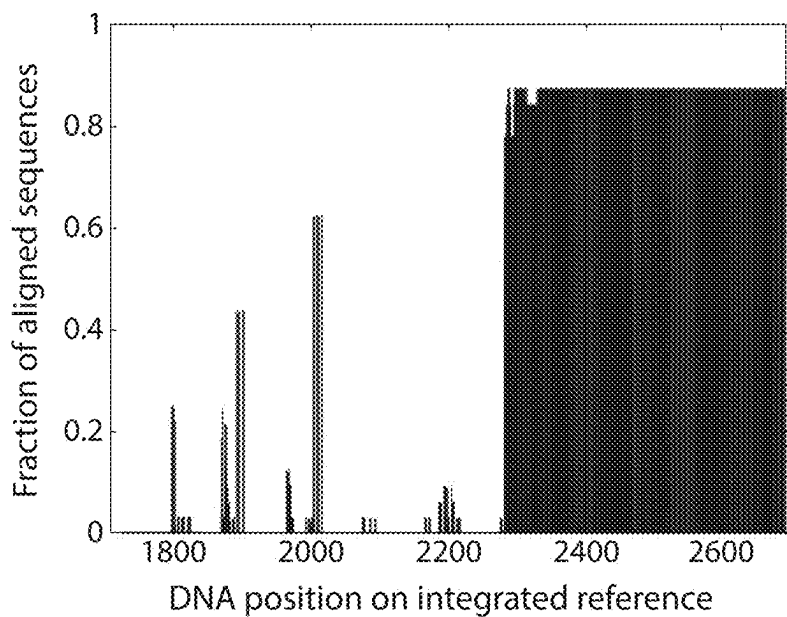


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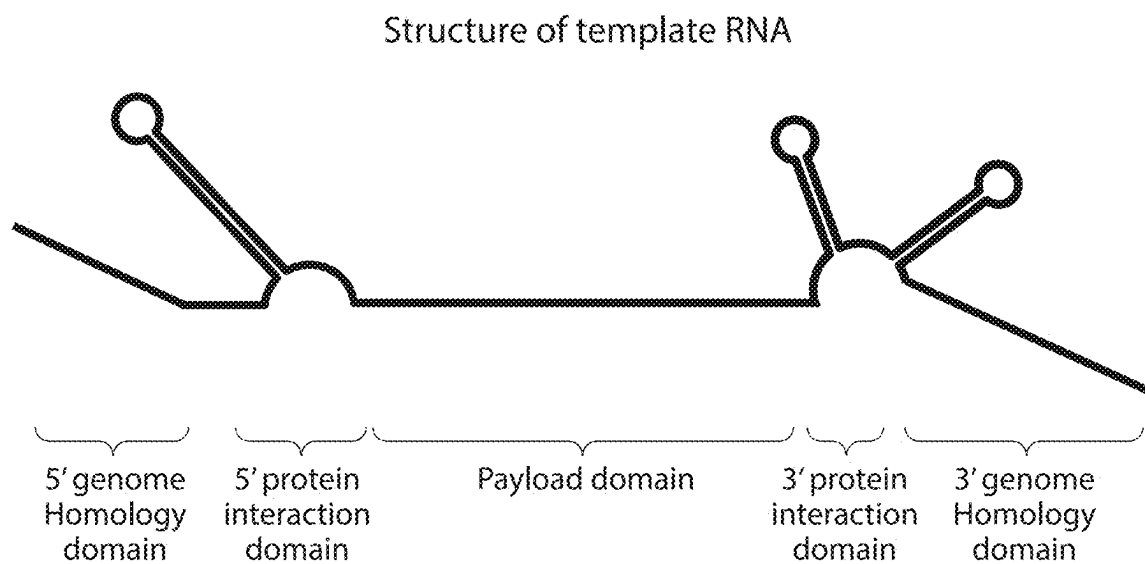


Figure 11

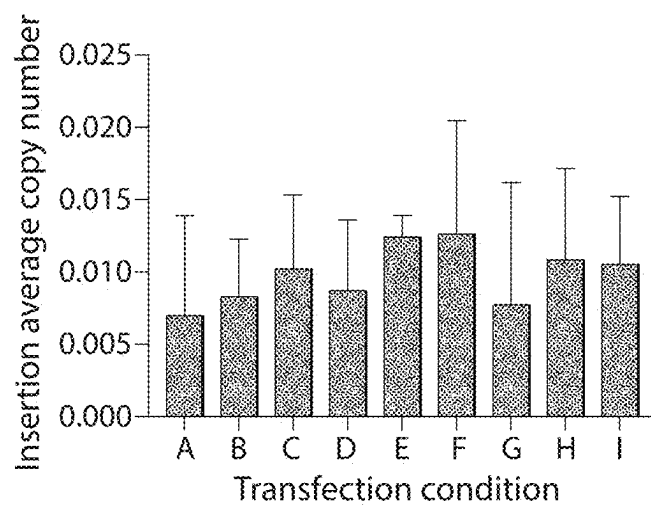


Figure 12

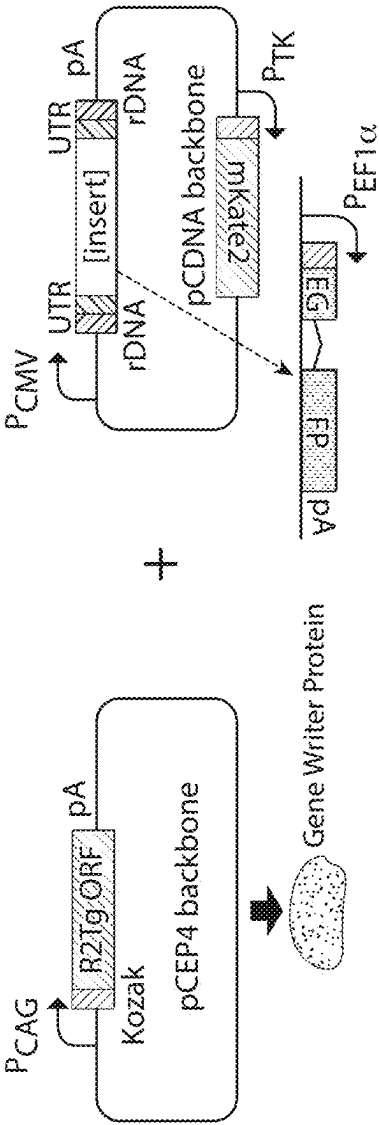


Figure 13

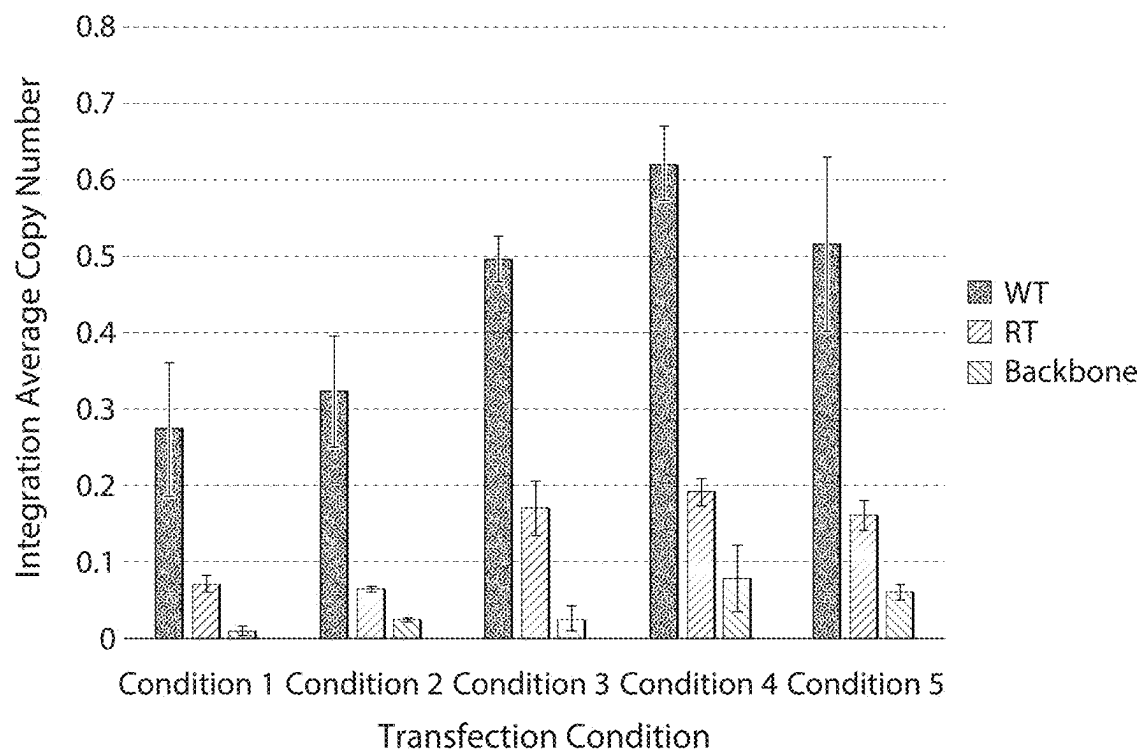


Figure 14

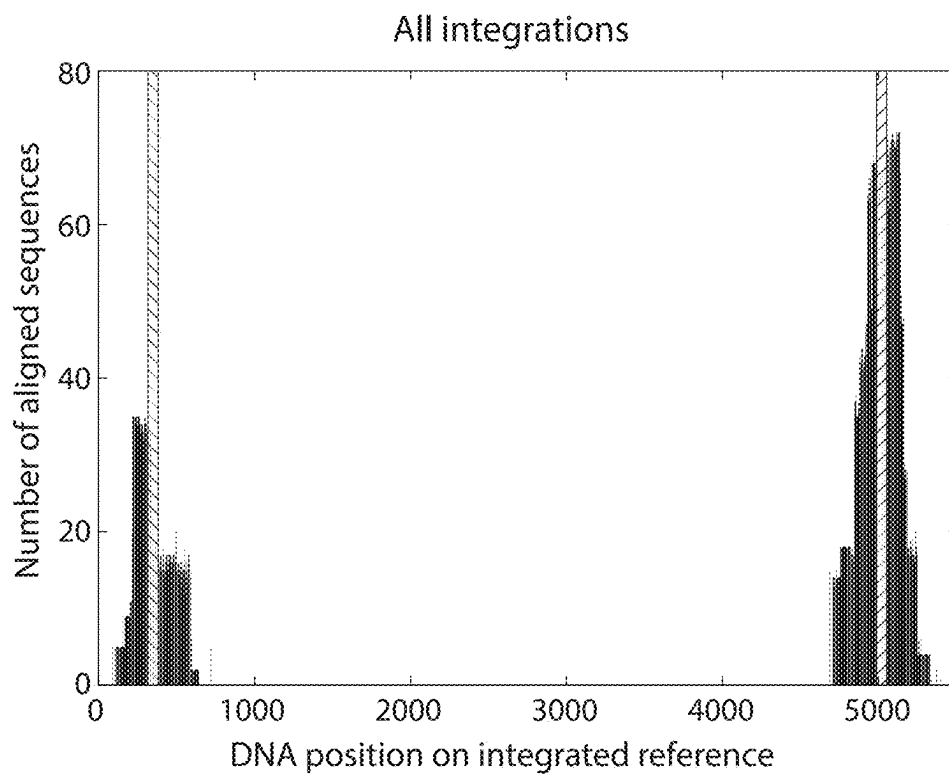


Figure 15A

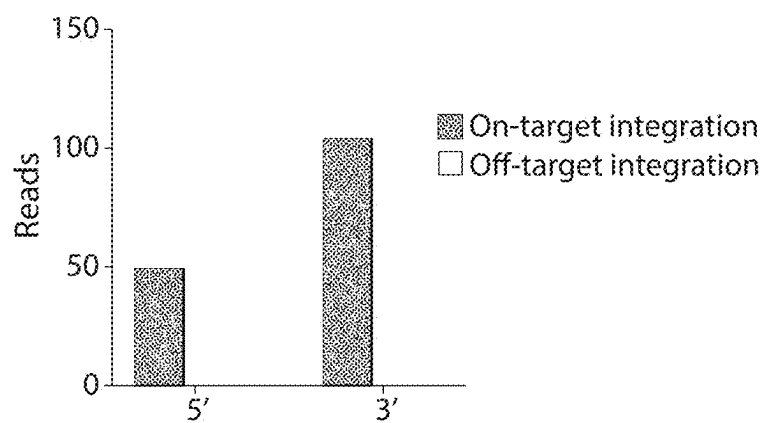


Figure 15B

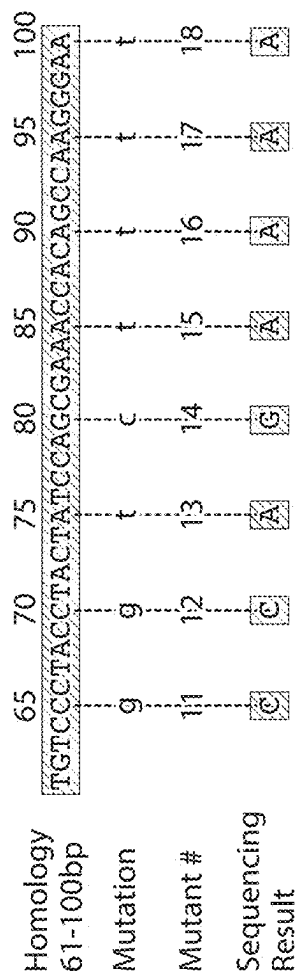
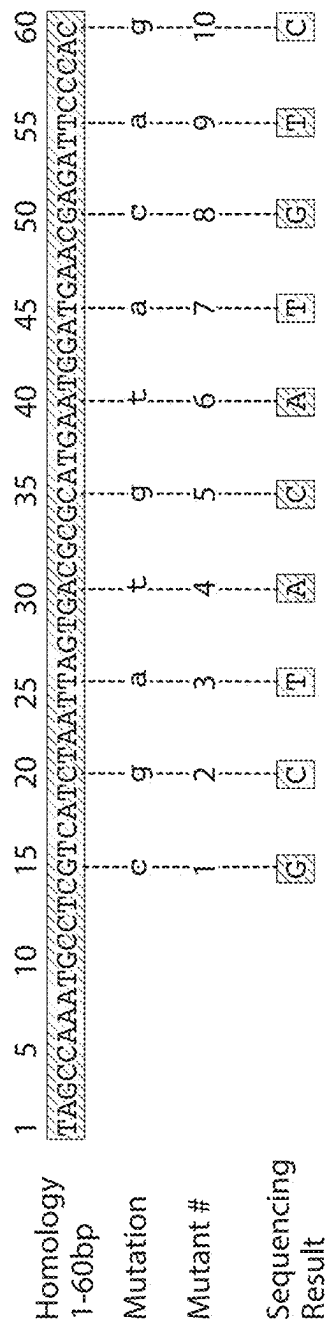


Figure 16

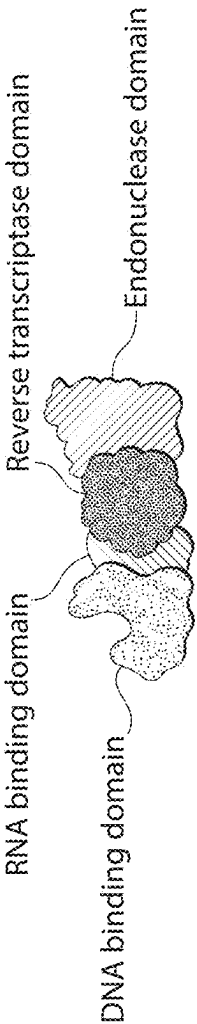


Figure 17A



Figure 17B

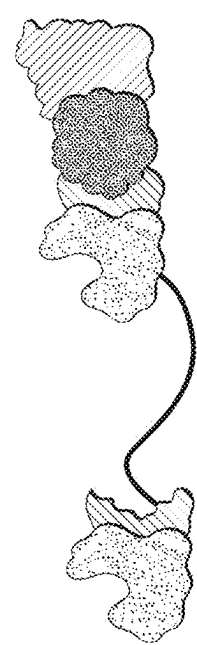


Figure 17C



Figure 17D



Figure 17E



Figure 17F

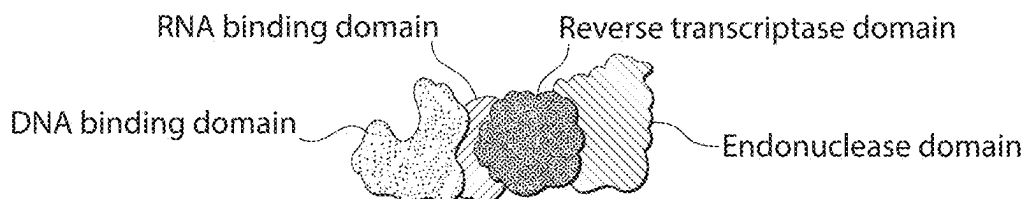
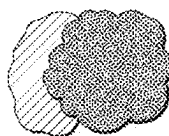


Figure 18A



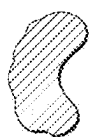
DNA binding
Zinc fingers
Cas9
Transcription factors

Figure 18B



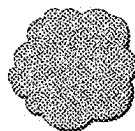
Full-length Reverse transcriptase
Different full-length RT from other R2
Different full-length RT combined with additional RNA binding domain

Figure 18C



RNA binding
B-box protein
MS2 coat protein
dCas protein
UTR binding protein

Figure 18D



RT without RNA binding
Truncated RT from R2
HIV RT
AMV RT

Figure 18E



Endonuclease
Cas9 nickase
Cas ortholog
FokI
Restriction enzyme

Figure 18F

FIG. 19

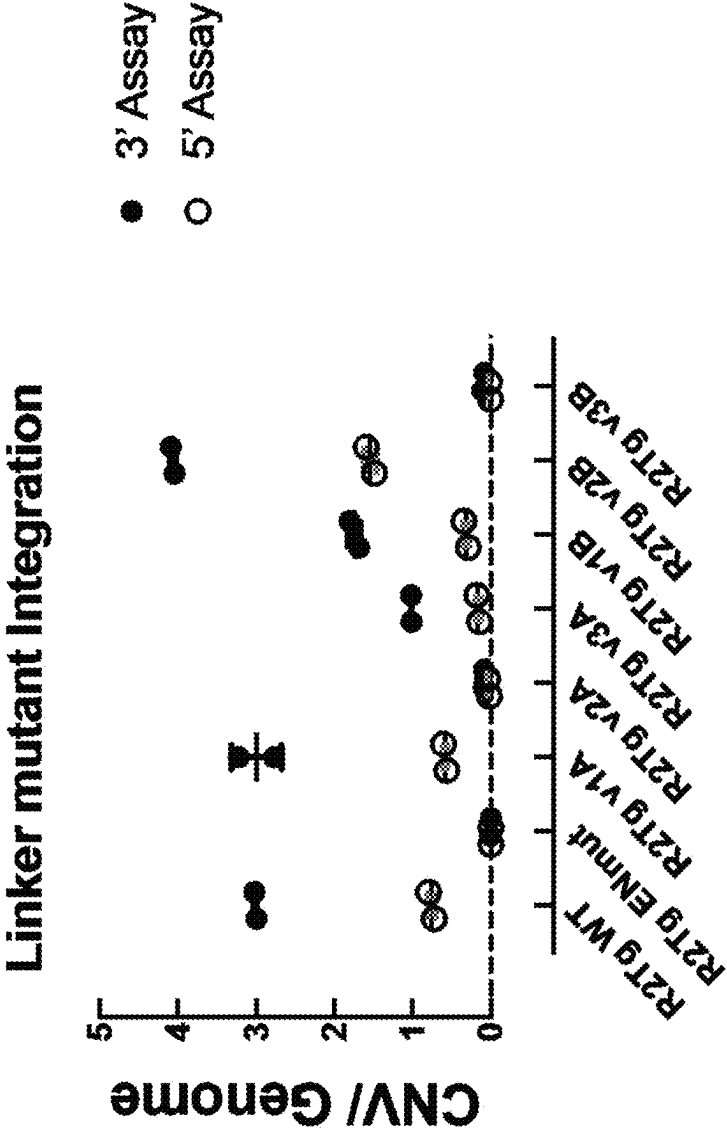


FIG. 20A

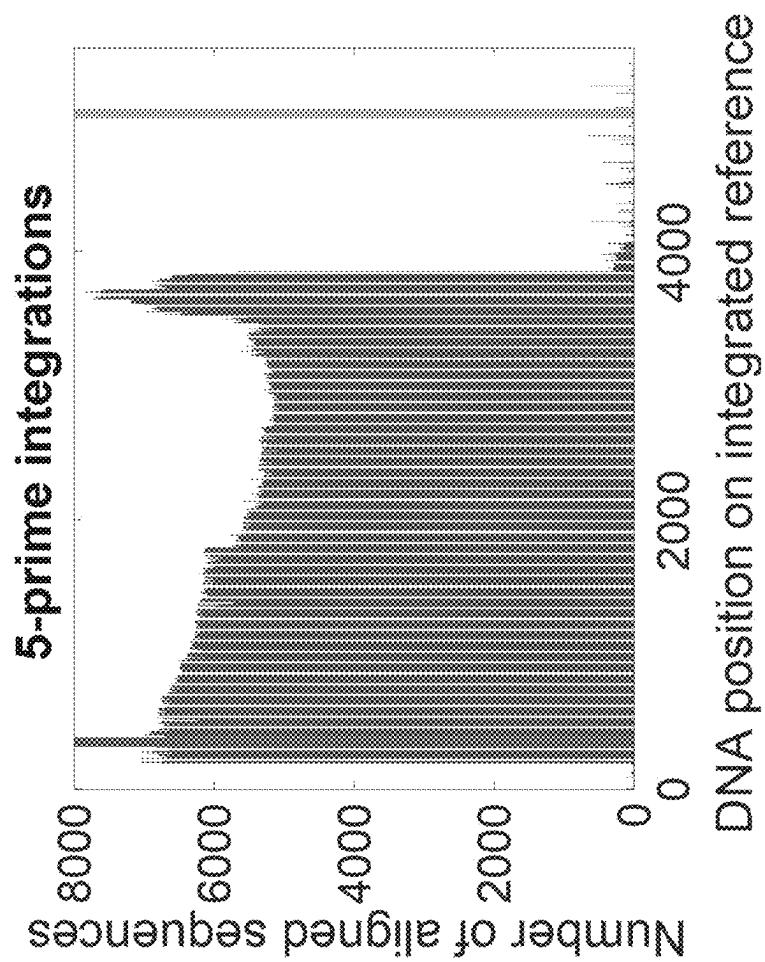


FIG. 20B

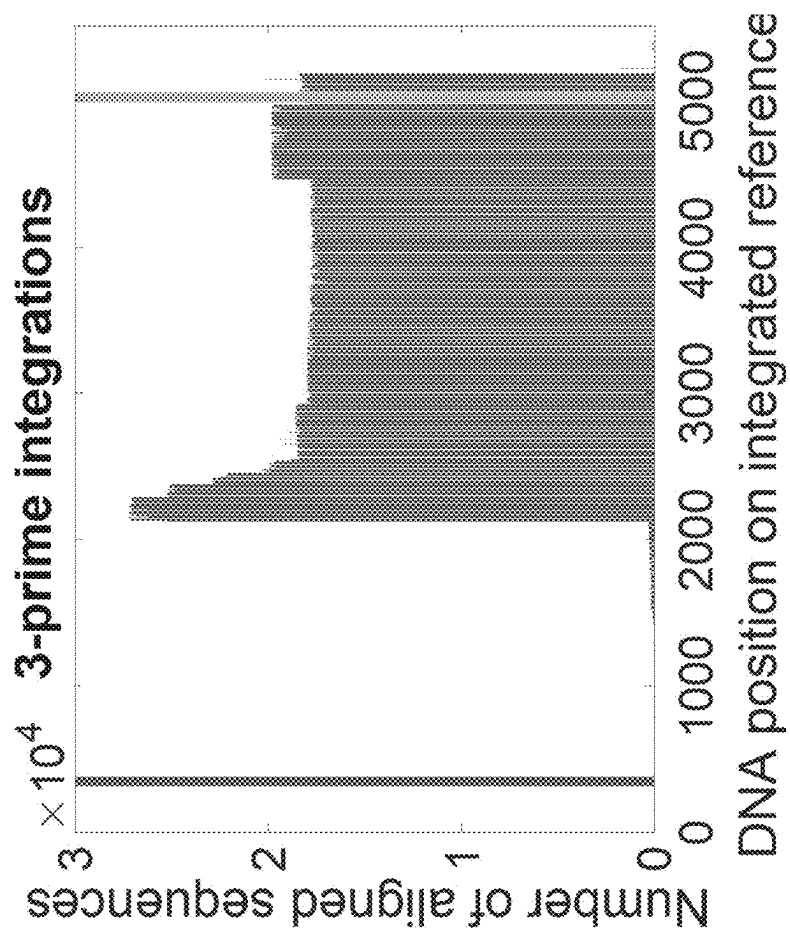


FIG. 21A

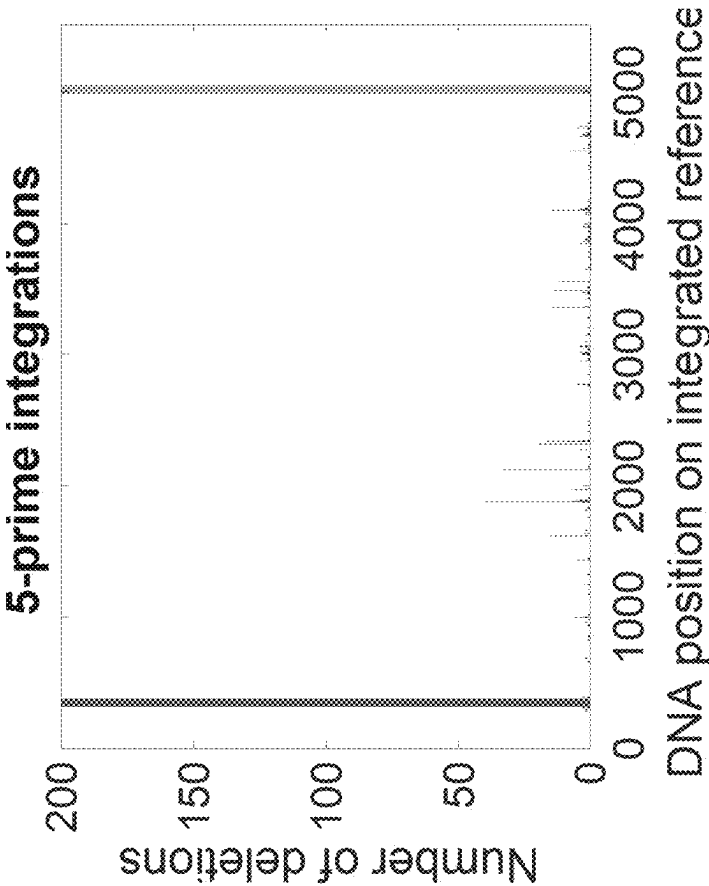


FIG. 21B

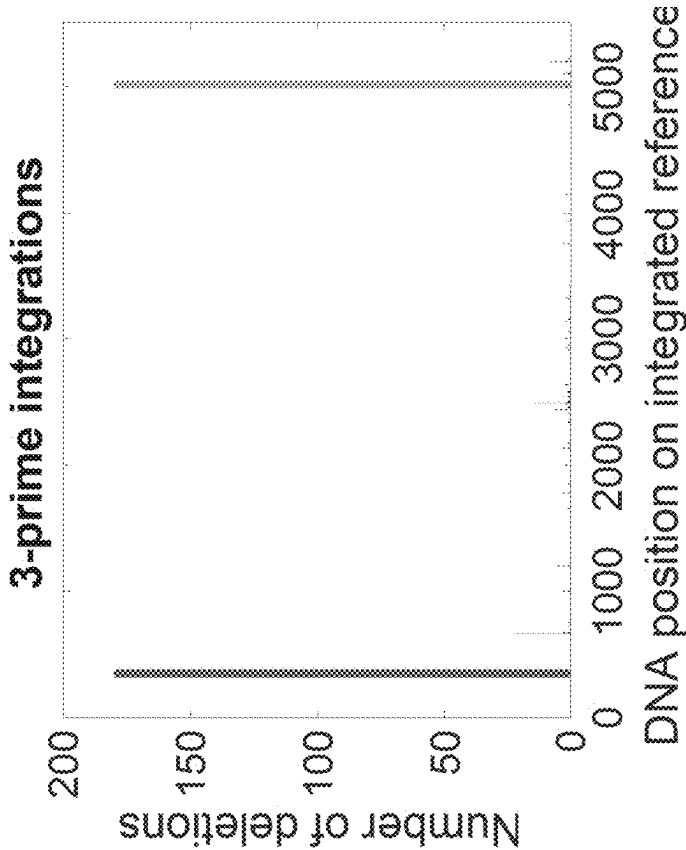


FIG. 23

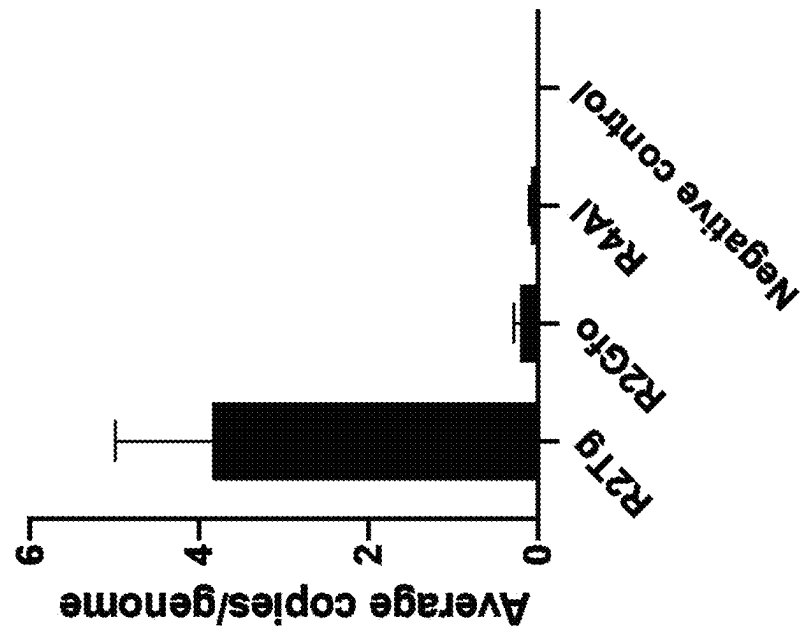


FIG. 24

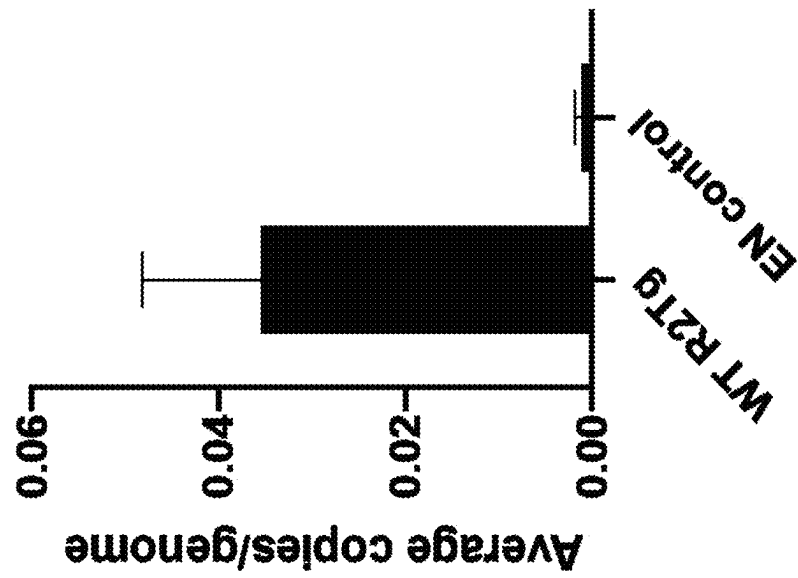
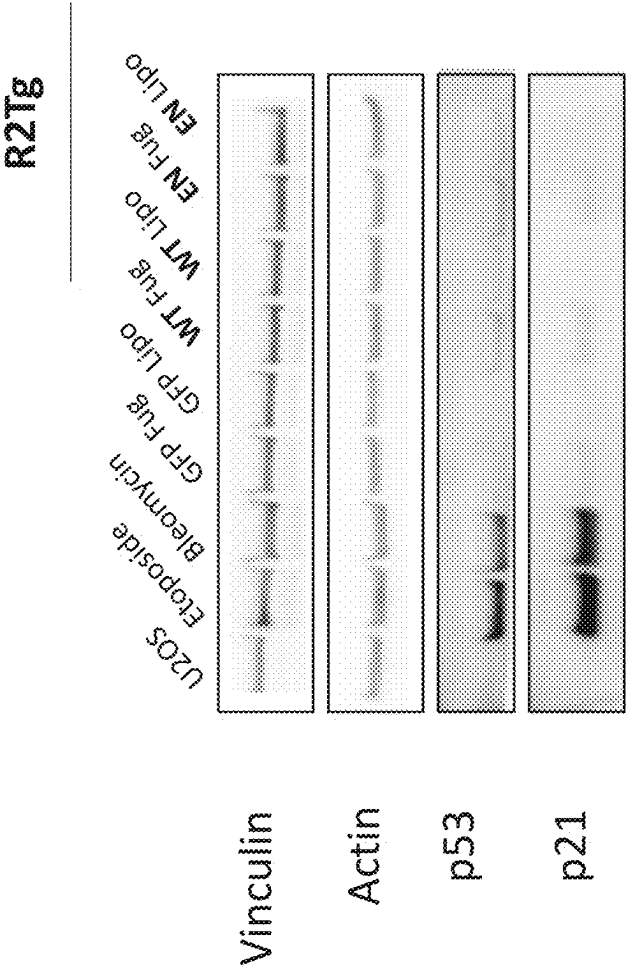


FIG. 25



1

METHODS AND COMPOSITIONS FOR MODULATING A GENOME

This application is a Divisional of U.S. application Ser. No. 16/706,448, filed Dec. 6, 2019, now U.S. Pat. No. 12,031,129, which is a Continuation of International Application No. PCT/US2019/048607, filed Aug. 28, 2019, which claims priority to U.S. Ser. No. 62/723,886 filed Aug. 28, 2018, U.S. Ser. No. 62/725,778 filed Aug. 31, 2018, U.S. Ser. No. 62/850,883 filed May 21, 2019, and U.S. Ser. No. 62/864,924 filed Jun. 21, 2019, the entire contents of each of which is incorporated herein by reference.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on May 16, 2024, is named V2065-700040_SL.xml and is 3,505,256 bytes in size.

BACKGROUND

Integration of a nucleic acid of interest into a genome occurs at low frequency and with little site specificity, in the absence of a specialized protein to promote the insertion event. Some existing approaches, like CRISPR/Cas9, are more suited for small edits and are less effective at integrating longer sequences. Other existing approaches, like Cre/loxP, require a first step of inserting a loxP site into the genome and then a second step of inserting a sequence of interest into the loxP site. There is a need in the art for improved proteins for inserting sequences of interest into a genome.

SUMMARY OF THE INVENTION

This disclosure relates to novel compositions, systems and methods for altering a genome at one or more locations in a host cell, tissue or subject, in vivo or in vitro. In particular, the invention features compositions, systems and methods for the introduction of exogenous genetic elements into a host genome.

Features of the compositions or methods can include one or more of the following enumerated embodiments.

1. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence that encodes a therapeutic polypeptide or that encodes a mammalian (e.g., human) polypeptide, or a fragment or variant thereof.

2. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence, wherein one or more of:
 - i. the heterologous object sequence encodes a protein, e.g. an enzyme (e.g., a lysosomal enzyme) or a blood factor (e.g., Factor I, II, V, VII, X, XI, XII or XIII);

2

- ii. the heterologous object sequence comprises a tissue specific promoter or enhancer;
- iii. the heterologous object sequence encodes a polypeptide of greater than 250, 300, 400, 500, or 1,000 amino acids, and optionally up to 7,500 amino acids;
- iv. the heterologous object sequence encodes a fragment of a mammalian gene but does not encode the full mammalian gene, e.g., encodes one or more exons but does not encode a full-length protein;
- v. the heterologous object sequence encodes one or more introns;
- vi. the heterologous object sequence is other than a GFP, e.g., is other than a fluorescent protein or is other than a reporter protein.
- vii. the heterologous object sequence is other than a T cell chimeric antigen receptor

3. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

4. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a target DNA binding domain, (ii) a reverse transcriptase domain and (iii) an endonuclease domain; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

5. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain, wherein one or both of (i) or (ii) are derived from an avian retrotransposase, e.g., have a sequence of Table 2 or 3 or at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

6. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain, wherein the polypeptide has an activity at 37° C. that is no less than 70%, 75%, 80%, 85%, 90%, or 95% of its activity at 25° C. under otherwise similar conditions; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

7. The system of embodiment 6, wherein the polypeptide is derived from an avian retrotransposase, e.g., an avian retrotransposase of column 8 of Table 3, or a sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto.

8. The system of embodiment 6, wherein the avian retrotransposase is a retrotransposase from *Taeniopygia guttata*, *Geospiza fortis*, *Zonotrichia albicollis*, or *Tinamus guttatus*, or a sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto.

9. The system of embodiment 6, wherein the polypeptide is derived from a retrotransposase of column 8 of Table 3, or a sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto.

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10. The system of any of the preceding embodiments, wherein the template RNA comprises a sequence of Table 3 (e.g., one or both of a 5' untranslated region of column 6 of Table 3 and a 3' untranslated region of column 7 of Table 3), or a sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto.

11. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence, wherein one or more of:
 - i. the nucleic acid encoding the polypeptide and the template RNA or a nucleic acid encoding the template RNA are separate nucleic acids;
 - ii. the template RNA does not encode an active reverse transcriptase, e.g., comprises an inactivated mutant reverse transcriptase, e.g., as described in Examples 1-2, or does not comprise a reverse transcriptase sequence; or
 - iii. the template RNA does not encode an active endonuclease, e.g., comprises an inactivated endonuclease or does not comprise an endonuclease; or
 - iv. the template RNA comprises one or more chemical modifications.

12. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a 5' untranslated sequence that binds the polypeptide, (ii) a 3' untranslated sequence that binds the polypeptide, (iii) a heterologous object sequence, and (iv) a promoter operably linked to the heterologous object sequence, wherein the promoter is disposed between the 5' untranslated sequence that binds the polypeptide and the heterologous sequence, or wherein the promoter is disposed between the 3' untranslated sequence that binds the polypeptide and the heterologous sequence.

13. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a 5' untranslated sequence that binds the polypeptide, (ii) a 3' untranslated sequence that binds the polypeptide, and (iii) a heterologous object sequence, and wherein the heterologous object sequence comprises an open reading frame (or the reverse complement thereof) in a 5' to 3' orientation on the template RNA; or wherein the heterologous object sequence comprises an open reading frame (or the reverse complement thereof) in a 3' to 5' orientation on the template RNA.

14. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain, wherein at least one of (i) or (ii) is heterologous, and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

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15. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a target DNA binding domain, (i) a reverse transcriptase domain and (iii) an endonuclease domain, wherein at least one of (i), (ii) or (iii) is heterologous, and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

16. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a sequence at least 80% identical (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identical) to a reverse transcriptase domain of a purinic/aprimidinic endonuclease (APE)-type non-LTR retrotransposon and (ii) a sequence at least 80% identical (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identical) to an endonuclease domain of an APE-type non-LTR retrotransposon; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

17. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a sequence at least 80% identical (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identical) to a reverse transcriptase domain of a restriction enzyme-like endonuclease (RLE)-type non-LTR retrotransposon, (ii) a sequence at least 80% identical (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identical) to an endonuclease domain of a RLE-type non-LTR retrotransposon, and (iii) a heterologous target DNA binding domain (e.g., a heterologous zinc-finger DNA binding domain); and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

18. The system of any of the preceding embodiments, wherein the template RNA comprises (iii) a promoter operably linked to the heterologous object sequence.

19. The system of any of the preceding embodiments, wherein the polypeptide further comprises (iii) a DNA-binding domain.

20. The system of embodiment 17, wherein the polypeptide comprises a sequence at least 80% identical (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identical) to the sequence of SEQ ID NO: 1016.

21. The system of any of the preceding embodiments, wherein the polypeptide comprises a sequence at least 80% identical (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identical) to a sequence in column 8 of Table 3.

22. The system of any of the preceding embodiments, wherein the nucleic acid encoding the polypeptide and the template RNA or the nucleic acid encoding the template RNA are covalently linked, e.g., are part of a fusion nucleic acid.

23. The system of embodiment 22, wherein the fusion nucleic acid comprises RNA.

24. The system of embodiment 22, wherein the fusion nucleic acid comprises DNA.

25. The system of any of the preceding embodiments, wherein (b) comprises template RNA.

26. The system of embodiment 25, wherein the template RNA further comprises a nuclear localization signal.

27. The system of any of the preceding embodiments, wherein (a) comprises RNA encoding the polypeptide.

28. The system of embodiment 27, wherein the RNA of (a) and the RNA of (b) are separate RNA molecules.

29. The system of embodiment 28, wherein the RNA of (a) and the RNA of (b) are present at a ratio of between 10:1 and 5:1, 5:1 and 2:1, 2:1 and 1:1, 1:1 and 1:2, 1:2 and 1:5, or 1:5 and 1:10.

30. The system of embodiment 28, wherein the RNA of (a) does not comprise a nuclear localization signal.

31. The system of any of the preceding embodiments, wherein the polypeptide further comprises a nuclear localization signal and/or a nucleolar localization signal.

32. The system of any of the preceding embodiments, wherein (a) comprises an RNA that encodes: (i) the polypeptide and (ii) a nuclear localization signal and/or a nucleolar localization signal.

33. The system of any of the preceding embodiments, wherein the RNA comprises a pseudoknot sequence, e.g., 5' of the heterologous object sequence.

34. The system of embodiment 33, wherein the RNA comprises a stem-loop sequence or a helix, 5' of the pseudoknot sequence.

35. The system of embodiment 33 or 34, wherein the RNA comprises one or more (e.g., 2, 3, or more) stem-loop sequences or helices 3' of the pseudoknot sequence, e.g. 3' of the pseudoknot sequence and 5' of the heterologous object sequence.

36. The system of any of embodiments 33-35, wherein the template RNA comprising the pseudoknot has catalytic activity, e.g., RNA-cleaving activity, e.g. cis-RNA-cleaving activity.

37. The system of any of the preceding embodiments, wherein the RNA comprises at least one stem-loop sequence or helix, e.g., 3' of the heterologous object sequence, e.g. 1, 2, 3, 4, 5 or more stem-loop sequences, hairpins or helices sequences.

38. Any above-numbered system, wherein the polypeptide comprises a sequence of at least 50 amino acids (e.g., at least 100, 150, 200, 300, 500 amino acids) having at least 80% identity (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identity) to a sequence of a polypeptide listed in Table 1, or a reverse transcriptase domain or endonuclease domain thereof.

39. Any above-numbered system, wherein the polypeptide comprises a sequence of at least 50 amino acids (e.g., at least 100, 150, 200, 300, 500 amino acids) having at least 80% identity (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identity) to a sequence of a polypeptide listed in any of Tables 2-3 or a reverse transcriptase domain, endonuclease domain, or DNA binding domain thereof.

40. Any above-numbered system, wherein the polypeptide comprises a sequence of at least 50 amino acids (e.g., at least 100, 150, 200, 300, 500 amino acids) having at least 80% identity (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identity) to the amino acid sequence of column 8 of Table 3, or a reverse transcriptase domain, endonuclease domain, or DNA binding domain thereof.

41. Any above-numbered system, wherein the template RNA comprises a sequence of Table 3 (e.g., one or both of a 5' untranslated region of column 6 of Table 3 and a 3' untranslated region of column 7 of Table 3), or a sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto.

42. The system of embodiment 41, wherein the template RNA comprises a sequence of about 100-125 bp from a 3' untranslated region of column 7 of Table 3, e.g., wherein the sequence comprises nucleotides 1-100, 101-200, or 201-325 of the 3' untranslated region of column 7 of Table 3, or a

sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto.

43. Any above-numbered system, wherein (a) comprises RNA and (b) comprises RNA.

44. Any above-numbered system, which comprises only RNA, or which comprises more RNA than DNA by an RNA:DNA ratio of at least 10:1, 20:1, 30:1, 40:1, 50:1, 60:1, 70:1, 80:1, 90:1, or 100:1.

45. Any above-numbered system, which does not comprise DNA, or which does not comprise more than 10%, 5%, 4%, 3%, 2%, or 1% DNA by mass or by molar amount.

46. Any above-numbered system, which is capable of modifying DNA by insertion of the heterologous object sequence without an intervening DNA-dependent RNA polymerization of (b).

47. Any above-numbered system, which is capable of modifying DNA by insertion of a heterologous object sequence in the presence of an inhibitor of a DNA repair pathway (e.g., SCR7, a PARP inhibitor), or in a cell line deficient for a DNA repair pathway (e.g., a cell line deficient for the nucleotide excision repair pathway or the homology-directed repair pathway).

48. Any above-numbered system, which does not cause formation of a detectable level of double stranded breaks in a target cell.

49. Any above-numbered system, which is capable of modifying DNA using reverse transcriptase activity, and optionally in the absence of homologous recombination activity.

50. Any above-numbered system, wherein the template RNA has been treated to reduce secondary structure, e.g., was heated, e.g., to a temperature that reduces secondary structure, e.g., to at least 70, 75, 80, 85, 90, or 95 C.

51. The system of embodiment 50, wherein the template RNA was subsequently cooled, e.g., to a temperature that allows for secondary structure, e.g., to less than or equal to 30, 25, or 20 C.

52. A system for modifying DNA comprising:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain; and
- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide, (ii) a heterologous object sequence, (iii) a first homology domain having at least 10 bases of 100% identity to a target DNA strand, at the 5' end of the template RNA, and (iv) a second homology domain having at least 10 bases of 100% identity to a target DNA strand, 5' end of the template RNA.

53. The system of any of the preceding embodiments, wherein (a) and (b) are part of the same nucleic acid.

54. The system of any of embodiments 1-52, wherein (a) and (b) are separate nucleic acids.

55. The system of any of the preceding embodiments, wherein the template RNA comprises at least 10 bases of 100% identity to a target DNA strand (e.g., wherein the target DNA strand is a human DNA sequence), at the 5' end of the template RNA.

56. The system of any of the preceding embodiments, wherein the template RNA comprises at least 10 bases of 100% identity to a target DNA strand (e.g., wherein the target DNA strand is a human DNA sequence), at the 3' end of the template RNA.

57. A host cell (e.g., a mammalian cell, e.g., a human cell) comprising any preceding numbered system.

58. A method of modifying a target DNA strand in a cell, tissue or subject, comprising administering any preceding numbered system to the cell, tissue or subject, wherein the

system reverse transcribes the template RNA sequence into the target DNA strand, thereby modifying the target DNA strand.

59. The method of embodiment 58, wherein the cell, tissue or subject is a mammalian (e.g., human) cell, tissue or subject.

60. The method of any of the preceding embodiments, wherein the cell is a fibroblast.

61. The method of any of the preceding embodiments, wherein the cell is a primary cell.

62. The method of any of the preceding embodiments, where in the cell is not immortalized.

63. A method of modifying the genome of a mammalian cell, comprising contacting the cell with:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain, (ii) an endonuclease domain, and optionally (iii) a DNA-binding domain; and

- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence.

64. The method of embodiment 63, wherein the polypeptide does not comprise a target DNA binding domain.

65. The method of embodiment 63, wherein the polypeptide is derived from an APE-type transposon reverse transcriptase.

66. The method of embodiment 63, wherein the (i) a reverse transcriptase domain (ii) an endonuclease domain, or both of (i) and (ii), have a sequence of Table 1 or a sequence having at least 80%, 85%, 90%, 95%, 97%, 98%, 99%, 100% identity thereto.

67. The method of embodiment 63, wherein the polypeptide further comprises a target DNA binding domain.

68. A method of modifying the genome of a mammalian cell, comprising contacting the cell with:

- (a) an RNA encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain, (ii) an endonuclease domain, and optionally (iii) a DNA-binding domain; and

- (b) a template RNA comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence, wherein the method does not comprise contacting the mammalian cell with DNA, or wherein the compositions of (a) and (b) do not comprise more than 1%, 0.5%, 0.2%, 0.1%, 0.05%, 0.02%, or 0.01% DNA by mass or by molar amount of nucleic acid.

69. The method of embodiment 68, which results in the addition of at least 5, 10, 20, 50, 100, 200, 500, 1,000, 2,000, or 5,000 base pairs of exogenous DNA sequence to the genome of the mammalian cell.

70. The method of embodiment 68 or 69, which results in the addition of a protein coding sequence to the genome of the mammalian cell.

71. A method of inserting DNA into the genome of a mammalian cell, comprising contacting the cell with an RNA composition, wherein the RNA composition comprises:

- (a) a first RNA that directs insertion of a template RNA into the genome, and

- (b) a template RNA comprising a heterologous sequence, wherein the method does not comprise contacting the mammalian cell with DNA, or wherein the compositions of (a) and (b) do not comprise more than 1%, 0.5%, 0.2%, 0.1%, 0.05%, 0.02%, or 0.01% DNA by mass or by molar amount of nucleic acid,

wherein the method results in the addition of at least 5, 10, 20, 50, 100, 200, 500, 1,000, 2,000, or 5,000 base pairs

of DNA (e.g., exogenous DNA) sequence to the genome of the mammalian cell.

72. The method of embodiment 71, wherein the first RNA encodes a polypeptide (e.g., a polypeptide of any of Tables 1, 2, or 3 herein), wherein the polypeptide directs insertion of the template RNA into the genome.

73. The method of embodiments 72, wherein the template RNA further comprises a sequence that binds the polypeptide.

74. A method of adding at least 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 150, 200, 250, 300, 350, 400, 500, 1000 bp of exogenous DNA to the genome of a mammalian cell, without delivery of DNA to the cell.

75. A method of adding at least 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 150, 200, 250, 300, 350, 400, 500, 1000 bp of exogenous DNA to the genome of a mammalian cell, wherein the method does not comprise contacting the mammalian cell with DNA, or wherein the method comprises contacting the mammalian cell with a composition comprising less than 1%, 0.5%, 0.2%, 0.1%, 0.05%, 0.02%, or 0.01% DNA by mass or by molar amount of nucleic acid.

76. A method of adding at least 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 150, 200, 250, 300, 350, 400, 500, 1000 bp of exogenous DNA to the genome of a mammalian cell, comprising delivering only RNA to the mammalian cell.

77. A method of adding at least 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 150, 200, 250, 300, 350, 400, 500, 1000 bp of exogenous DNA to the genome of a mammalian cell, comprising delivering RNA and protein to the mammalian cell.

78. The method of any one of embodiments 68-77, wherein the template RNA serves as the template for insertion of the exogenous DNA.

79. The method of any one of embodiments 68-78, which does not comprise DNA-dependent RNA polymerization of exogenous DNA.

80. The method of any of embodiments 58-79, which results in the addition of at least 5, 10, 20, 50, 100, 200, 500, 1,000, 2,000, or 5,000 base pairs of DNA to the genome of the mammalian cell.

81. The methods of any of embodiments 68-80, wherein the RNA of (a) and the RNA of (b) are covalently linked, e.g., are part of the same transcript.

82. The methods of any of embodiments 68-80, wherein the RNA of (a) and the RNA of (b) are separate RNAs.

83. The method of any of embodiments 58-82, which does not comprise contacting the mammalian cell with a template DNA.

84. A method of modifying the genome of a human cell, comprising contacting the cell with:

- (a) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain, (ii) an endonuclease domain, and optionally (iii) a DNA-binding domain; and

- (b) a template RNA (or DNA encoding the template RNA) comprising (i) a sequence that binds the polypeptide and (ii) a heterologous object sequence,

wherein the method results in insertion of the heterologous object sequence into the human cell's genome,

wherein the human cell does not show upregulation of any DNA repair genes and/or tumor suppressor genes, or wherein no DNA repair gene and/or tumor suppressor gene is upregulated by more than 10%, 5%, 2%, or 1%, e.g., wherein upregulation is measured by RNA-seq, e.g., as described in Example 14.

85. A method of adding an exogenous coding region to the genome of a cell (e.g., a mammalian cell), comprising contacting the cell with an RNA comprising the non-coding strand of the exogenous coding region, wherein optionally the RNA does not comprise a coding strand of the exogenous coding region, wherein optionally the delivery comprises non-viral delivery.

86. A method of expressing a polypeptide in a cell (e.g., a mammalian cell), comprising contacting the cell with an RNA, wherein the RNA comprises a non-coding strand that is the reverse complement of a sequence that would encoding the polypeptide, wherein optionally the RNA does not comprise a coding strand encoding the polypeptide, wherein optionally the delivery comprises non-viral delivery.

87. The method of any of embodiments 58-86, wherein the sequence that is inserted into the mammalian genome is a sequence that is exogenous to the mammalian genome.

88. The method of any of embodiments 58-87, which operates independently of a DNA template.

89. The method of any of embodiments 58-88, wherein the cell is part of a tissue.

90. The method of any of embodiments 58-89, wherein the mammalian cell is euploid, is not immortalized, is part of an organism, is a primary cell, is non-dividing, is a hepatocyte, or is from a subject having a genetic disease.

91. The method of any of embodiments 58-90, wherein the contacting comprises contacting the cell with a plasmid, virus, viral-like particle, virosome, liposome, vesicle, exosome, or lipid nanoparticle.

92. The method of any of embodiments 58-91, wherein the contacting comprises using non-viral delivery.

93. The method of any of embodiments 58-92, which comprises contacting the cell with the template RNA (or DNA encoding the template RNA), wherein the template RNA comprises the non-coding strand of an exogenous coding region, wherein optionally the template RNA does not comprise a coding strand of the exogenous coding region, wherein optionally the delivery comprises non-viral delivery, thereby adding the exogenous coding region to the genome of the cell.

94. The method of any of embodiments 58-93, which comprises contacting the cell with the template RNA (or DNA encoding the template RNA), wherein the template RNA comprises a non-coding strand that is the reverse complement of a sequence that would encoding the polypeptide, wherein optionally the template RNA does not comprise a coding strand encoding the polypeptide, wherein optionally the delivery comprises non-viral delivery, thereby expressing the polypeptide in the cell.

95. The method of any of embodiments 63-94, wherein the contacting comprises administering (a) and (b) to a subject, e.g., intravenously.

96. The method of any of embodiments 63-95, wherein the contacting comprises administering a dose of (a) and (b) to a subject at least twice.

97. The method of any of embodiments 63-96, wherein the polypeptide reverse transcribes the template RNA sequence into the target DNA strand, thereby modifying the target DNA strand.

98. The method of any embodiments 63-97, wherein (a) and (b) are administered separately.

99. The method of any of embodiments 63-97, wherein (a) and (b) are administered together.

100. The method of any of embodiments 63-99, wherein the nucleic acid of (a) is not integrated into the genome of the host cell.

101. Any preceding numbered method, wherein the sequence that binds the polypeptide has one or more of the following characteristics:

- (a) is at the 3' end of the template RNA;
- (b) is at the 5' end of the template RNA;
- (b) is a non-coding sequence;
- (c) is a structured RNA; or
- (d) forms at least 1 hairpin loop structures.

102. Any preceding numbered method, wherein the template RNA further comprises a sequence comprising at least 20 nucleotides of at least 80% identity (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identity) to a target DNA strand.

103. Any preceding numbered method, wherein the template RNA further comprises a sequence comprising at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150 nucleotides of at least 80% identity (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identity) to a target DNA strand.

104. Any preceding numbered method, wherein the sequence comprising at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150 nucleotides, or about: 2-10, 10-20, 20-30, 30-40, 40-50, 50-60, 60-70, 70-80, 80-90, 90-100, 10-100, or 2-100 nucleotides, of at least 80% identity to a target DNA strand is at the 3' end of the template RNA.

105. Any preceding numbered method, wherein the template RNA further comprises a sequence comprising at least 100 nucleotides of at least 80% identity (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identity) to a target DNA strand, e.g., at the 3' end of the template RNA.

106. The method of embodiment 104 or 105, wherein the site in the target DNA strand to which the sequence comprises at least 80% identity is proximal to (e.g., within about: 0-10, 10-20, 20-30, 30-50, or 50-100 nucleotides of) a target site on the target DNA strand that is recognized (e.g., bound and/or cleaved) by the polypeptide comprising the endonuclease.

107. Any preceding numbered method, wherein the sequence comprising at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150 nucleotides, or about: 2-10, 10-20, 20-30, 30-40, 40-50, 50-60, 60-70, 70-80, 80-90, 90-100, 10-100, or 2-100 nucleotides, of at least 80% identity to a target DNA strand is at the 3' end of the template RNA;

optionally wherein the site in the target DNA strand to which the sequence comprises at least 80% identity is proximal to (e.g., within about: 0-10, 10-20, or 20-30 nucleotides of) a target site on the target DNA strand that is recognized (e.g., bound and/or cleaved) by the polypeptide comprising the endonuclease.

108. The method of embodiment 107, wherein the target site is the site in the human genome that has the closest identity to a native target site of the polypeptide comprising the endonuclease, e.g., wherein the target site in the human genome has at least about: 16, 17, 18, 19, or 20 nucleotides identical to the native target site.

109. Any preceding numbered method, wherein the template RNA has at least 3, 4, 5, 6, 7, 8, 9, or 10 bases of 100% identity to the target DNA strand.

110. Any preceding numbered method, wherein the at least 3, 4, 5, 6, 7, 8, 9, or 10 bases of 100% identity to the target DNA strand are at the 3' end of the template RNA.

111. Any preceding numbered method, wherein the at least 3, 4, 5, 6, 7, 8, 9, or 10 bases of 100% identity to the target DNA strand are at the 5' end of the template RNA.

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112. Any preceding numbered method, wherein the template RNA comprises at least 3, 4, 5, 6, 7, 8, 9, or 10 bases of 100% identity to the target DNA strand at the 5' end of the template RNA and at least 3, 4, 5, 6, 7, 8, 9, or 10 bases of 100% identity to the target DNA strand at the 3' end of the template RNA.

113. Any preceding numbered method, wherein the heterologous object sequence is between 50-50,000 base pairs (e.g., between 50-40,000 bp, between 500-30,000 bp between 500-20,000 bp, between 100-15,000 bp, between 500-10,000 bp, between 50-10,000 bp, between 50-5,000 bp).

114. Any preceding numbered method, wherein the heterologous object sequence is at least 10, 25, 50, 100, 150, 200, 250, 300, 400, 500, 600, or 700 bp.

115. Any preceding numbered method, wherein the heterologous object sequence is at least 715, 750, 800, 950, 1,000, 2,000, 3,000, or 4,000 bp.

116. Any preceding numbered method, wherein the heterologous object sequence is less than 5,000, 10,000, 15,000, 20,000, 30,000, or 40,000 bp.

117. Any preceding numbered method, wherein the heterologous object sequence is less than 700, 600, 500, 400, 300, 200, 150, or 100 bp.

118. Any preceding numbered method, wherein the heterologous object sequence comprises:

- (a) an open reading frame, e.g., a sequence encoding a polypeptide, e.g., an enzyme (e.g., a lysosomal enzyme), a membrane protein, a blood factor, an exon, an intracellular protein (e.g., a cytoplasmic protein, a nuclear protein, an organellar protein such as a mitochondrial protein or lysosomal protein), an extracellular protein, a structural protein, a signaling protein, a regulatory protein, a transport protein, a sensory protein, a motor protein, a defense protein, or a storage protein;
- (b) a non-coding and/or regulatory sequence, e.g., a sequence that binds a transcriptional modulator, e.g., a promoter, an enhancer, an insulator;
- (c) a splice acceptor site;
- (d) a polyA site;
- (e) an epigenetic modification site; or
- (f) a gene expression unit.

119. Any preceding numbered method, wherein the target DNA is a genomic safe harbor (GSH) site.

120. Any preceding numbered method, wherein the target DNA is a genomic NATURAL HARBOR™ site.

121. Any preceding numbered method, which results in insertion of the heterologous object sequence into the a target site in the genome at an average copy number of at least 0.01, 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2, 2.5, 3, 4, or 5 copies per genome.

122. Any preceding numbered method, which results in about 25-100%, 50-100%, 60-100%, 70-100%, 75-95%, 80%-90%, of integrants into a target site in the genome being non-truncated, as measured by an assay described herein, e.g., an assay of Example 6.

123. Any preceding numbered method, which results in insertion of the heterologous object sequence only at one target site in the genome of the cell.

124. Any preceding numbered method, which results in insertion of the heterologous object sequence into a target site in a cell, wherein the inserted heterologous sequence comprises less than 10%, 5%, 2%, 1%, 0.5%, 0.2%, or 0.1% mutations (e.g., SNPs or one or more deletions, e.g., trun-

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cations or internal deletions) relative to the heterologous sequence prior to insertion, e.g., as measured by an assay of Example 12.

125. Any preceding numbered method, which results in insertion of the heterologous object sequence into a target site in a plurality of cells, wherein less than 10%, 5%, 2%, or 1% of copies of the inserted heterologous sequence comprise a mutation (e.g., a SNP or a deletion, e.g., a truncation or an internal deletion), e.g., as measured by an assay of Example 12.

126. Any preceding numbered method, which results in insertion of the heterologous object sequence into a target cell genome, and wherein the target cell does not show upregulation of p53, or shows upregulation of p53 by less than 10%, 5%, 2%, or 1%, e.g., wherein upregulation of p53 is measured by p53 protein level, e.g., according to the method described in Example 30, or by the level of p53 phosphorylated at Ser15 and Ser20.

127. Any preceding numbered method, which results in insertion of the heterologous object sequence into a target cell genome, and wherein the target cell does not show upregulation of any DNA repair genes and/or tumor suppressor genes, or wherein no DNA repair gene and/or tumor suppressor gene is upregulated by more than 10%, 5%, 2%, or 1%, e.g., wherein upregulation is measured by RNA-seq, e.g., as described in Example 14.

128. Any preceding numbered method, which results in insertion of the heterologous object sequence into the target site (e.g., at a copy number of 1 insertion or more than one insertion) in about 1-80% of cells in a population of cells contacted with the system, e.g., about: 1-10%, 10-20%, 20-30%, 30-40%, 40-50%, 50-60%, 60-70%, or 70-80% of cells, e.g., as measured using single cell ddPCR, e.g., as described in Example 17.

129. Any preceding numbered method, which results in insertion of the heterologous object sequence into the target site at a copy number of 1 insertion in about 1-80% of cells in a population of cells contacted with the system, e.g., about: 1-10%, 10-20%, 20-30%, 30-40%, 40-50%, 50-60%, 60-70%, or 70-80% of cells, e.g., as measured using colony isolation and ddPCR, e.g., as described in Example 18.

130. Any preceding numbered method, which results in insertion of the heterologous object sequence into the target site (on-target insertions) at a higher rate that insertion into a non-target site (off-target insertions) in a population of cells, wherein the ratio of on-target insertions to off-target insertions is greater than 10:1, 20:1, 30:1, 40:1, 50:1, 60:1, 70:1, 80:1, 90:1, 100:1, 200:1, 500:1, or 1,000:1, e.g., using an assay of Example 11.

131. Any above-numbered method, results in insertion of a heterologous object sequence in the presence of an inhibitor of a DNA repair pathway (e.g., SCR7, a PARP inhibitor), or in a cell line deficient for a DNA repair pathway (e.g., a cell line deficient for the nucleotide excision repair pathway or the homology-directed repair pathway).

132. Any preceding numbered system, formulated as a pharmaceutical composition.

133. Any preceding numbered system, disposed in a pharmaceutically acceptable carrier (e.g., a vesicle, a liposome, a natural or synthetic lipid bilayer, a lipid nanoparticle, an exosome).

134. A method of making a system for modifying the genome of a mammalian cell, comprising:

- a) providing a template RNA as described in any of the preceding embodiments, e.g., wherein the template RNA comprises (i) a sequence that binds a polypeptide

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comprising a reverse transcriptase domain and an endonuclease domain, and (ii) a heterologous object sequence; and

b) treating the template RNA to reduce secondary structure, e.g., heating the template RNA, e.g., to at least 70, 75, 80, 85, 90, or 95 C, and

c) subsequently cooling the template RNA, e.g., to a temperature that allows for secondary structure, e.g., to less than or equal to 30, 25, or 20 C.

135. The method of embodiment 134, which further comprises contacting the template RNA with a polypeptide that comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain, or with a nucleic acid (e.g., RNA) encoding the polypeptide.

136. The method of embodiment 134 or 135, which further comprises contacting the template RNA with a cell.

137. The system or method of any of the preceding embodiments, wherein the heterologous object sequence encodes a therapeutic polypeptide.

138. The system or method of any of the preceding embodiments, wherein the heterologous object sequence encodes a mammalian (e.g., human) polypeptide, or a fragment or variant thereof.

139. The system or method of any of the preceding embodiments, wherein the heterologous object sequence encodes an enzyme (e.g., a lysosomal enzyme), a blood factor (e.g., Factor I, II, V, VII, X, XI, XII or XIII), a membrane protein, an exon, an intracellular protein (e.g., a cytoplasmic protein, a nuclear protein, an organellar protein such as a mitochondrial protein or lysosomal protein), an extracellular protein, a structural protein, a signaling protein, a regulatory protein, a transport protein, a sensory protein, a motor protein, a defense protein, or a storage protein.

140. The system or method of any of the preceding embodiments, wherein the heterologous object sequence comprises a tissue specific promoter or enhancer.

141. The system or method of any of the preceding embodiments, wherein the heterologous object sequence encodes a polypeptide of greater than 250, 300, 400, 500, or 1,000 amino acids, and optionally up to 1300 amino acids.

142. The system or method of any of the preceding embodiments, wherein the heterologous object sequence encodes a fragment of a mammalian gene but does not encode the full mammalian gene, e.g., encodes one or more exons but does not encode a full-length protein.

143. The system or method of any of the preceding embodiments, wherein the heterologous object sequence encodes one or more introns.

144. The system or method of any of the preceding embodiments, wherein the heterologous object sequence is other than a GFP, e.g., is other than a fluorescent protein or is other than a reporter protein.

145. The system or method of any of the preceding embodiments, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain, wherein one or both of (i) or (ii) are derived from an avian retrotransposase, e.g., have a sequence of Table 2 or 3 or at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto.

146. The system or method of any of the preceding embodiments, wherein the polypeptide has an activity at 37° C. that is no less than 70%, 75%, 80%, 85%, 90%, or 95% of its activity at 25° C. under otherwise similar conditions.

147. The system or method of any of the preceding embodiments, wherein the nucleic acid encoding the polypeptide and the template RNA or a nucleic acid encoding the template RNA are separate nucleic acids.

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148. The system or method of any of the preceding embodiments, wherein the template RNA does not encode an active reverse transcriptase, e.g., comprises an inactivated mutant reverse transcriptase, e.g., as described in Example 1 or 2, or does not comprise a reverse transcriptase sequence.

149. The system or method of any of the preceding embodiments, wherein the template RNA comprises one or more chemical modifications.

150. The system or method of any of the preceding embodiments, wherein the heterologous object sequence is disposed between the promoter and the sequence that binds the polypeptide.

151. The system or method of any of the preceding embodiments, wherein the promoter is disposed between the heterologous object sequence and the sequence that binds the polypeptide.

152. The system or method of any of the preceding embodiments, wherein the heterologous object sequence comprises an open reading frame (or the reverse complement thereof) in a 5' to 3' orientation on the template RNA.

153. The system or method of any of the preceding embodiments, wherein the heterologous object sequence comprises an open reading frame (or the reverse complement thereof) in a 3' to 5' orientation on the template RNA.

154. The system or method of any of the preceding embodiments, wherein the polypeptide comprises (a) a reverse transcriptase domain and (b) an endonuclease domain, wherein at least one of (a) or (b) is heterologous.

155. The system or method of any of the preceding embodiments, wherein the polypeptide comprises (a) a target DNA binding domain, (b) a reverse transcriptase domain and (c) an endonuclease domain, wherein at least one of (a), (b) or (c) is heterologous.

156. A substantially pure polypeptide comprising (a) a reverse transcriptase domain and (b) a heterologous endonuclease domain.

157. A substantially pure polypeptide comprising (a) a target DNA binding domain, (b) a reverse transcriptase domain and (c) an endonuclease domain, wherein at least one of (a), (b) or (c) is heterologous.

158. A substantially pure polypeptide comprising (a) a reverse transcriptase domain, (b) an endonuclease domain, and (c) a heterologous target DNA binding domain.

159. A polypeptide or a nucleic acid encoding the polypeptide, wherein the polypeptide comprises (a) a reverse transcriptase domain and (b) an endonuclease domain, wherein at least one of (a) or (b) is heterologous to the other.

160. A polypeptide or a nucleic acid encoding the polypeptide, wherein the polypeptide comprises (a) a target DNA binding domain, (b) a reverse transcriptase domain and (c) an endonuclease domain, wherein at least one of (a), (b) or (c) is heterologous to the other.

161. Any polypeptide of numbered embodiments 156-160, wherein the reverse transcriptase domain has at least 80% identity (e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identity) to a reverse transcriptase domain of an APE-type or RLE-type non-LTR retrotransposon listed in any of Tables 1-3.

162. Any polypeptide of numbered embodiments 156-161, wherein the endonuclease domain has at least 80% identity e.g., at least 85%, 90%, 95%, 97%, 98%, 99%, 100% identity, to a endonuclease domain of an APE-type or RLE-type non-LTR retrotransposon listed in any of Tables 1-3.

163. Any polypeptide of numbered embodiments 156-162 or any preceding numbered method, wherein the DNA binding domain has at least 80% identity e.g., at least 85%, 90%,

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95%, 97%, 98%, 99%, 100% identity, to a DNA binding domain of a sequence listed in Table 1, 2, or 3.

164. A nucleic acid encoding the polypeptide of any preceding numbered embodiment.

165. A vector comprising the nucleic acid of numbered embodiment 164.

166. A host cell comprising the nucleic acid of numbered embodiment 164.

167. A host cell comprising the polypeptide of any preceding numbered embodiment.

168. A host cell comprising the vector of numbered embodiment 165.

169. A host cell (e.g., a human cell) comprising: (i) a heterologous object sequence (e.g., a sequence encoding a therapeutic polypeptide) at a target site in a chromosome, and (ii) one or both of an untranslated region (e.g., a retrotransposon untranslated sequence, e.g., a sequence of column 6 of Table 3) on one side (e.g., upstream) of the heterologous object sequence, and an untranslated region (e.g., a retrotransposon untranslated sequence, e.g., a sequence of column 7 of Table 3) on the other side (e.g., downstream) of the heterologous object sequence.

170. A host cell (e.g., a human cell) comprising: (i) a heterologous object sequence (e.g., a sequence encoding a therapeutic polypeptide) at a target site in a chromosome, wherein the target locus is a NATURAL HARBOR™ site, e.g., a site of Table 4 herein.

171. The host cell of embodiment 170, which further comprises (ii) one or both of an untranslated region 5' of the heterologous object sequence, and an untranslated region 3' of the heterologous object sequence.

172. The host cell of embodiment 170, which further comprises (ii) one or both of an untranslated region (e.g., a retrotransposon untranslated sequence, e.g., a sequence of column 6 of Table 3) on one side (e.g., upstream) of the heterologous object sequence, and an untranslated region (e.g., a retrotransposon untranslated sequence, e.g., a sequence of column 7 of Table 3) on the other side (e.g., downstream) of the heterologous object sequence.

173. The host cell of any of embodiments 169-173, which comprises heterologous object sequence at only the target site.

174. A pharmaceutical composition, comprising any preceding numbered system, nucleic acid, polypeptide, or vector; and a pharmaceutically acceptable excipient or carrier.

175. The pharmaceutical composition of embodiment 174, wherein the pharmaceutically acceptable excipient or carrier is selected from a vector (e.g., a viral or plasmid vector), a vesicle (e.g., a liposome, an exosome, a natural or synthetic lipid bilayer), a lipid nanoparticle.

176. A polypeptide of any of the preceding embodiments, wherein the polypeptide further comprises a nuclear localization sequence.

177. A method of modifying a target DNA strand in a cell, tissue or subject, comprising administering any preceding numbered system to the cell, tissue or subject, thereby modifying the target DNA strand.

178. Any preceding numbered embodiment, wherein the polypeptide comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to an amino acid sequence listed in Table 5 (e.g., any one of SEQ ID NOs: 1017-1022), or a functional fragment thereof.

179. Any preceding numbered embodiment, wherein the reverse transcriptase domain comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the

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reverse transcriptase domain of an amino acid sequence listed in Table 5 (e.g., any one of SEQ ID NOs: 1017-1022), or a functional fragment thereof.

180. Any preceding numbered embodiment, wherein the retrotransposase comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to an amino acid sequence listed in Table 5 (e.g., any one of SEQ ID NOs: 1017-1022), or a functional fragment thereof.

181. Any preceding numbered embodiment, wherein the polypeptide comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence SGSETPGTSESATPES (SEQ ID NO: 1023) or GGGS (SEQ ID NO: 1024).

182. Any preceding numbered embodiment, wherein the reverse transcriptase domain comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence SGSETPGTSESATPES (SEQ ID NO: 1023) or GGGS (SEQ ID NO: 1024).

183. Any preceding numbered embodiment, wherein the retrotransposase comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence SGSETPGTSESATPES (SEQ ID NO: 1023) or GGGS (SEQ ID NO: 1024).

184. Any preceding numbered embodiment, wherein the polypeptide, reverse transcriptase domain, or retrotransposase comprises a linker comprising an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence SGSETPGTSESATPES (SEQ ID NO: 1023) or GGGS (SEQ ID NO: 1024).

185. Any preceding numbered embodiment, wherein the polypeptide comprises a DNA binding domain covalently attached to the remainder of the polypeptide by a linker, e.g., a linker comprising at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 200, 300, 400, or 500 amino acids.

186. Numbered embodiment 185, wherein the linker is attached to the remainder of the polypeptide at a position in the DNA binding domain, RNA binding domain, reverse transcriptase domain, or endonuclease domain (e.g., as shown in any of FIGS. 17A-17F).

187. Numbered embodiment 185 or 186, wherein the linker is attached to the remainder of the polypeptide at a position in the N-terminal side of an alpha helical region of the polypeptide, e.g., at a position corresponding to version v1 as described in Example 26.

188. Numbered embodiment 185 or 186, wherein the linker is attached to the remainder of the polypeptide at a position in the C-terminal side of an alpha helical region of the polypeptide, e.g., preceding an RNA binding motif (e.g., a -1 RNA binding motif), e.g., at a position corresponding to version v2 as described in Example 26.

189. Numbered embodiment 185 or 186, wherein the linker is attached to the remainder of the polypeptide at a position in the C-terminal side of a random coil region of the polypeptide, e.g., N-terminal relative to a DNA binding motif (e.g., a c-myb DNA binding motif), e.g., at a position corresponding to version v3 as described in Example 26.

190. Any one of numbered embodiments 185-189, wherein the linker comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or

100% sequence identity to the amino acid sequence SGSETPGTSESATPES (SEQ ID NO: 1023) or GGGS (SEQ ID NO: 1024).

191. Any preceding numbered embodiment, wherein a polynucleotide sequence comprising at least about 500, 1000, 2000, 3000, 3500, 3600, 3700, 3800, 3900, or 4000 contiguous nucleotides from the 5' end of the template RNA sequence are integrated into a target cell genome.

192. Any preceding numbered embodiment, wherein a polynucleotide sequence comprising at least about 500, 1000, 2000, 2500, 2600, 2700, 2800, 2900, or 3000 contiguous nucleotides from the 3' end of the template RNA sequence are integrated into a target cell genome.

193. Any preceding numbered embodiment, wherein the nucleic acid sequence of the template RNA, or a portion thereof (e.g., a portion comprising at least about 100, 200, 300, 400, 500, 1000, 2000, 2500, 3000, 3500, or 4000 nucleotides) integrates into the genomes of a population of target cells at a copy number of at least about 0.21, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, or 1.0 integrants/genome.

194. Any preceding numbered embodiment, wherein the nucleic acid sequence of the template RNA, or a portion thereof (e.g., a portion comprising at least about 100, 200, 300, 400, 500, 1000, 2000, 2500, 3000, 3500, or 4000 nucleotides) integrates into the genomes of a population of target cells at a copy number of at least about 0.085, 0.09, 0.1, 0.15, or 0.2 integrants/genome.

195. Any preceding numbered embodiment, wherein the nucleic acid sequence of the template RNA, or a portion thereof (e.g., a portion comprising at least about 100, 200, 300, 400, 500, 1000, 2000, 2500, 3000, 3500, or 4000 nucleotides) integrates into the genomes of a population of target cells at a copy number of at least about 0.036, 0.04, 0.05, 0.06, 0.07, or 0.08 integrants/genome.

196. Any preceding numbered embodiment, wherein the polypeptide comprises a functional endonuclease domain (e.g., wherein the endonuclease domain does not comprise a mutation that abolishes endonuclease activity, e.g., as described herein).

197. Any preceding numbered embodiment, wherein the polypeptide comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the R2 polypeptide from a medium ground finch, e.g., *Geospiza fortis* (e.g., as described herein, e.g., R2-1_GFo), or a functional fragment thereof.

198. Any preceding numbered embodiment, wherein the reverse transcriptase domain comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the R2 polypeptide from a medium ground finch, e.g., *Geospiza fortis* (e.g., as described herein, e.g., R2-1_GFo), or a functional fragment thereof.

199. Any preceding numbered embodiment, wherein the retrotransposase comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the R2 polypeptide from a medium ground finch, e.g., *Geospiza fortis* (e.g., as described herein, e.g., R2-1_GFo), or a functional fragment thereof.

200. Any one of numbered embodiments 197-199, wherein the nucleic acid sequence of the template RNA, or a portion thereof (e.g., a portion comprising at least about 100, 200, 300, 400, 500, 1000, 2000, 2500, 3000, 3500, or 4000 nucleotides) integrates into the genomes of a population of target cells at a copy number of at least about 0.21 integrants/genome.

201. Any preceding numbered embodiment, wherein the polypeptide comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the R4 polypeptide from a large roundworm, e.g., *Ascaris lumbricoides* (e.g., as described herein, e.g., R4_AL), or a functional fragment thereof.

202. Any preceding numbered embodiment, wherein the reverse transcriptase domain comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the R4 polypeptide from a large roundworm, e.g., *Ascaris lumbricoides* (e.g., as described herein, e.g., R4_AL), or a functional fragment thereof.

203. Any preceding numbered embodiment, wherein the retrotransposase comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the R4 polypeptide from a large roundworm, e.g., *Ascaris lumbricoides* (e.g., as described herein, e.g., R4_AL), or a functional fragment thereof.

204. Any one of numbered embodiments 201-203, wherein the nucleic acid sequence of the template RNA, or a portion thereof (e.g., a portion comprising at least about 100, 200, 300, 400, 500, 1000, 2000, 2500, 3000, 3500, or 4000 nucleotides) integrates into the genomes of a population of target cells at a copy number of at least about 0.085 integrants/genome.

205. Any preceding numbered embodiment, wherein introduction of the system into a target cell does not result in alteration (e.g., upregulation) of p53 and/or p21 protein levels, H2AX phosphorylation (e.g., gamma H2AX), ATM phosphorylation, ATR phosphorylation, Chk1 phosphorylation, Chk2 phosphorylation, and/or p53 phosphorylation.

206. Any preceding numbered embodiment, wherein introduction of the system into a target cell results in upregulation of p53 protein level in the target cell to a level that is less than about 0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 30%, 40%, 45%, 50%, 55%, 60%, 70%, 80%, or 90% of the p53 protein level induced by introducing a site-specific nuclease, e.g., Cas9, that targets the same genomic site as said system.

207. Numbered embodiment 205 or 206, wherein the p53 protein level is determined according to the method described in Example 30.

208. Any preceding numbered embodiment, wherein introduction of the system into a target cell results in upregulation of p53 phosphorylation level in the target cell to a level that is less than about 0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 30%, 40%, 45%, 50%, 55%, 60%, 70%, 80%, or 90% of the p53 phosphorylation level induced by introducing a site-specific nuclease, e.g., Cas9, that targets the same genomic site as said system.

209. Any preceding numbered embodiment, wherein introduction of the system into a target cell results in upregulation of p21 protein level in the target cell to a level that is less than about 0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 30%, 40%, 45%, 50%, 55%, 60%, 70%, 80%, or 90% of the p53 protein level induced by introducing a site-specific nuclease, e.g., Cas9, that targets the same genomic site as said system.

210. Numbered embodiment 205 or 209, wherein the p21 protein level is determined according to the method described in Example 30.

211. Any preceding numbered embodiment, wherein introduction of the system into a target cell results in upregulation of H2AX phosphorylation level in the target cell to a level that is less than about 0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 30%, 40%, 45%, 50%, 55%, 60%, 70%, 80%, or 90% of the H2AX phosphorylation level induced by introducing a site-specific nuclease, e.g., Cas9, that targets the same genomic site as said system.

212. Any preceding numbered embodiment, wherein introduction of the system into a target cell results in upregulation of ATM phosphorylation level in the target cell to a level that is less than about 0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 30%, 40%, 45%, 50%, 55%, 60%, 70%, 80%, or 90% of the ATM phosphorylation level induced by introducing a site-specific nuclease, e.g., Cas9, that targets the same genomic site as said system.

213. Any preceding numbered embodiment, wherein introduction of the system into a target cell results in upregulation of ATR phosphorylation level in the target cell to a level that is less than about 0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 30%, 40%, 45%, 50%, 55%, 60%, 70%, 80%, or 90% of the ATR phosphorylation level induced by introducing a site-specific nuclease, e.g., Cas9, that targets the same genomic site as said system.

214. Any preceding numbered embodiment, wherein introduction of the system into a target cell results in upregulation of Chk1 phosphorylation level in the target cell to a level that is less than about 0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 30%, 40%, 45%, 50%, 55%, 60%, 70%, 80%, or 90% of the Chk1 phosphorylation level induced by introducing a site-specific nuclease, e.g., Cas9, that targets the same genomic site as said system.

215. Any preceding numbered embodiment, wherein introduction of the system into a target cell results in upregulation of Chk2 phosphorylation level in the target cell to a level that is less than about 0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 30%, 40%, 45%, 50%, 55%, 60%, 70%, 80%, or 90% of the Chk2 phosphorylation level induced by introducing a site-specific nuclease, e.g., Cas9, that targets the same genomic site as said system.

Definitions

Domain: The term “domain” as used herein refers to a structure of a biomolecule that contributes to a specified function of the biomolecule. A domain may comprise a contiguous region (e.g., a contiguous sequence) or distinct, non-contiguous regions (e.g., non-contiguous sequences) of a biomolecule. Examples of protein domains include, but are not limited to, an endonuclease domain, a DNA binding domain, a reverse transcription domain; an example of a domain of a nucleic acid is a regulatory domain, such as a transcription factor binding domain.

Exogenous: As used herein, the term exogenous, when used with reference to a biomolecule (such as a nucleic acid sequence or polypeptide) means that the biomolecule was introduced into a host genome, cell or organism by the hand of man. For example, a nucleic acid that is as added into an existing genome, cell, tissue or subject using recombinant DNA techniques or other methods is exogenous to the existing nucleic acid sequence, cell, tissue or subject.

Genomic safe harbor site (GSH site): A genomic safe harbor site is a site in a host genome that is able to accommodate the integration of new genetic material, e.g., such that the inserted genetic element does not cause significant alterations of the host genome posing a risk to the host cell or organism. A GSH site generally meets 1, 2, 3, 4, 5, 6, 7, 8 or 9 of the following criteria: (i) is located >300 kb from a cancer-related gene; (ii) is >300 kb from a miRNA/other functional small RNA; (iii) is >50 kb from a 5' gene end; (iv) is >50 kb from a replication origin; (v) is >50 kb away from any ultraconserved element; (vi) has low transcriptional activity (i.e. no mRNA+/-25 kb); (vii) is not in copy number variable region; (viii) is in open chromatin; and/or (ix) is unique, with 1 copy in the human genome. Examples of GSH sites in the human genome that meet some or all of these criteria include (i) the adeno-associated virus site 1 (AAVS1), a naturally occurring site of integration of AAV virus on chromosome 19; (ii) the chemokine (C—C motif) receptor 5 (CCR5) gene, a chemokine receptor gene known as an HIV-1 coreceptor; (iii) the human ortholog of the mouse Rosa26 locus; (iv) the rDNA locus. Additional GSH sites are known and described, e.g., in Pellenz et al. *epub* Aug. 20, 2018 (doi.org/10.1101/396390).

Heterologous: The term heterologous, when used to describe a first element in reference to a second element means that the first element and second element do not exist in nature disposed as described. For example, a heterologous polypeptide, nucleic acid molecule, construct or sequence refers to (a) a polypeptide, nucleic acid molecule or portion of a polypeptide or nucleic acid molecule sequence that is not native to a cell in which it is expressed, (b) a polypeptide or nucleic acid molecule or portion of a polypeptide or nucleic acid molecule that has been altered or mutated relative to its native state, or (c) a polypeptide or nucleic acid molecule with an altered expression as compared to the native expression levels under similar conditions. For example, a heterologous regulatory sequence (e.g., promoter, enhancer) may be used to regulate expression of a gene or a nucleic acid molecule in a way that is different than the gene or a nucleic acid molecule is normally expressed in nature. In another example, a heterologous domain of a polypeptide or nucleic acid sequence (e.g., a DNA binding domain of a polypeptide or nucleic acid encoding a DNA binding domain of a polypeptide) may be disposed relative to other domains or may be a different sequence or from a different source, relative to other domains or portions of a polypeptide or its encoding nucleic acid. In certain embodiments, a heterologous nucleic acid molecule may exist in a native host cell genome, but may have an altered expression level or have a different sequence or both. In other embodiments, heterologous nucleic acid molecules may not be endogenous to a host cell or host genome but instead may have been introduced into a host cell by transformation (e.g., transfection, electroporation), wherein the added molecule may integrate into the host genome or can exist as extra-chromosomal genetic material either transiently (e.g., mRNA) or semi-stably for more than one generation (e.g., episomal viral vector, plasmid or other self-replicating vector).

Mutation or Mutated: The term “mutated” when applied to nucleic acid sequences means that nucleotides in a nucleic acid sequence may be inserted, deleted or changed compared to a reference (e.g., native) nucleic acid sequence. A single alteration may be made at a locus (a point mutation) or multiple nucleotides may be inserted, deleted or changed at a single locus. In addition, one or more alterations may be

made at any number of loci within a nucleic acid sequence. A nucleic acid sequence may be mutated by any method known in the art.

Nucleic acid molecule: Nucleic acid molecule refers to both RNA and DNA molecules including, without limitation, cDNA, genomic DNA and mRNA, and also includes synthetic nucleic acid molecules, such as those that are chemically synthesized or recombinantly produced, such as RNA templates, as described herein. The nucleic acid molecule can be double-stranded or single-stranded, circular or linear. If single-stranded, the nucleic acid molecule can be the sense strand or the antisense strand. Unless otherwise indicated, and as an example for all sequences described herein under the general format "SEQ. ID NO:," "nucleic acid comprising SEQ. ID NO:1" refers to a nucleic acid, at least a portion which has either (i) the sequence of SEQ. ID NO: 1, or (ii) a sequence complimentary to SEQ. ID NO:1. The choice between the two is dictated by the context in which SEQ. ID NO:1 is used. For instance, if the nucleic acid is used as a probe, the choice between the two is dictated by the requirement that the probe be complimentary to the desired target. Nucleic acid sequences of the present disclosure may be modified chemically or biochemically or may contain non-natural or derivatized nucleotide bases, as will be readily appreciated by those of skill in the art. Such modifications include, for example, labels, methylation, substitution of one or more naturally occurring nucleotides with an analog, inter-nucleotide modifications such as uncharged linkages (for example, methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, etc.), charged linkages (for example, phosphorothioates, phosphorodithioates, etc.), pendant moieties, (for example, polypeptides), intercalators (for example, acridine, psoralen, etc.), chelators, alkylators, and modified linkages (for example, alpha anomeric nucleic acids, etc.). Also included are synthetic molecules that mimic polynucleotides in their ability to bind to a designated sequence via hydrogen bonding and other chemical interactions. Such molecules are known in the art and include, for example, those in which peptide linkages substitute for phosphate linkages in the backbone of a molecule. Other modifications can include, for example, analogs in which the ribose ring contains a bridging moiety or other structure such as modifications found in "locked" nucleic acids.

Gene expression unit: a gene expression unit is a nucleic acid sequence comprising at least one regulatory nucleic acid sequence operably linked to at least one effector sequence. A first nucleic acid sequence is operably linked with a second nucleic acid sequence when the first nucleic acid sequence is placed in a functional relationship with the second nucleic acid sequence. For instance, a promoter or enhancer is operably linked to a coding sequence if the promoter or enhancer affects the transcription or expression of the coding sequence. Operably linked DNA sequences may be contiguous or non-contiguous. Where necessary to join two protein-coding regions, operably linked sequences may be in the same reading frame.

Host: The terms host genome or host cell, as used herein, refer to a cell and/or its genome into which protein and/or genetic material has been introduced. It should be understood that such terms are intended to refer not only to the particular subject cell and/or genome, but to the progeny of such a cell and/or the genome of the progeny of such a cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not, in fact, be identical to the parent cell, but are still included within the scope of the term

"host cell" as used herein. A host genome or host cell may be an isolated cell or cell line grown in culture, or genomic material isolated from such a cell or cell line, or may be a host cell or host genome which composing living tissue or an organism. In some instances, a host cell may be an animal cell or a plant cell, e.g., as described herein. In certain instances, a host cell may be a bovine cell, horse cell, pig cell, goat cell, sheep cell, chicken cell, or turkey cell. In certain instances, a host cell may be a corn cell, soy cell, wheat cell, or rice cell.

Pseudoknot: A "pseudoknot sequence" sequence, as used herein, refers to a nucleic acid (e.g., RNA) having a sequence with suitable self-complementarity to form a pseudoknot structure, e.g., having: a first segment, a second segment between the first segment and a third segment, wherein the third segment is complementary to the first segment, and a fourth segment, wherein the fourth segment is complementary to the second segment. The pseudoknot may optionally have additional secondary structure, e.g., a stem loop disposed in the second segment, a stem-loop disposed between the second segment and third segment, sequence before the first segment, or sequence after the fourth segment. The pseudoknot may have additional sequence between the first and second segments, between the second and third segments, or between the third and fourth segments. In some embodiments, the segments are arranged, from 5' to 3': first, second, third, and fourth. In some embodiments, the first and third segments comprise five base pairs of perfect complementarity. In some embodiments, the second and fourth segments comprise 10 base pairs, optionally with one or more (e.g., two) bulges. In some embodiments, the second segment comprises one or more unpaired nucleotides, e.g., forming a loop. In some embodiments, the third segment comprises one or more unpaired nucleotides, e.g., forming a loop.

Stem-loop sequence: As used herein, a "stem-loop sequence" refers to a nucleic acid sequence (e.g., RNA sequence) with sufficient self-complementarity to form a stem-loop, e.g., having a stem comprising at least two (e.g., 3, 4, 5, 6, 7, 8, 9, or 10) base pairs, and a loop with at least three (e.g., four) base pairs. The stem may comprise mismatches or bulges.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic of the GENE WRITING™ genome editing system.

FIG. 2 is a schematic of the structure of the GENE WRITER™ genome editor polypeptide.

FIG. 3 is a schematic of a GENE WRITER™ genome editor polypeptide comprising a heterologous DNA binding domain designed to target different sites of the genome.

FIG. 4 is a schematic of the structure of GENE WRITER™ genome editor template RNA.

FIG. 5 is a schematic showing the GENE WRITING™ genome editing system to add a gene expression unit into a safe harbor site in the genome.

FIG. 6 is a schematic showing GENE WRITING™ genome editing to add a new exon into an specific intron in the genome and replace downstream exons.

FIG. 7 illustrates a schematic of MISEQ™ library construction. Nested PCR was performed across the R2Tg-rDNA junction using (1) outer forward primer and tailed inner reverse primer followed by (2) tailed inner forward primer and tail reverse primer. The inner reverse primer contains a 1-4 base stagger, an 8-nucleotide randomized

UMI, and a multiplexing barcode. The UMI allows for counting of individual amplification events to eliminate PCR bias.

FIGS. 8A-8B: Results of MISEQ™ and MATLAB™ analysis of DNA-mediated R2Tg integration into Hek293T cells. Each graph shows analysis of (FIG. 8A) the experimental R2Tg (FIG. 8B) and 1 bp deletion negative control. The y-axis indicates aligned counts of unique sequences determined via unique UMIs found via MATLAB™. The X-axis indicates the sequence position of sequence coverage. The vertical gray line at the left of the graph indicates the position of the forward primer, while the vertical gray line at the right of the graph indicates the expected Tg-rDNA junction site. Bars at the right end of the graph indicate insertion without a truncation, and bars at the left end of the graph indicate truncation. FIG. 8A shows that most sequences show high alignment to the expected integration product.

FIG. 9 shows a ddPCR evaluation of copy number variation of the R2Tg-rDNA junction in human cells across transfection conditions. Forward primer and probe were expected to bind to the 3' UTR of the R2Tg, while reverse primer was targeted to the human rDNA. The resulting ddPCR signal was normalized to that of reference assay RPP30 to determine copy number. Significantly higher average copies per genome were found with the wildtype (WT, left set of bars) R2Tg as compared to genetic control altering translation with a 1-bp deletion (Frameshift mutant control, right set of bars).

FIG. 10 illustrates the sequence alignment and coverage of TOPO cloning the nested PCR product from Example 7. The gray line at the right edge of the graph indicates the expected transgene-rDNA junction. Most sequences showed high alignment to the expected integrated product.

FIG. 11 is a schematic of an exemplary template RNA. It comprises a payload domain in the center (e.g., a heterologous object sequence, e.g., comprising a promoter and a protein-coding sequence). The payload domain is flanked by 5' and 3' protein interaction domains, e.g., sequences capable of binding the GENE WRITER™ polypeptide, e.g., 5' and 3' UTR sequences shown in Table 3. Flanking the protein interaction domains are 5' and 3' homology domains, which have homology to the desired insertion region in the genome.

FIG. 12 is a graph showing retrotransposition efficiency measured by ddPCR (digital droplet PCR) using different transfection conditions. Bars A-C represent samples that were transfected using 0.15 µl Lipofectamine™ RNAiMAX with 100 ng, 250 ng, or 500 ng respectively. Bars D-F represent samples that were transfected using 0.3 µl Lipofectamine™ RNAiMAX with 100 ng, 250 ng, or 500 ng respectively. Bars G-I represent samples that were transfected using 1 µl TransIT®-mRNA transfection kit with 100 ng, 250 ng, or 500 ng respectively.

FIG. 13. Schematic of trans-transgene delivery machinery. This schematic illustrates a driver plasmid (left) with a pCEP4 backbone, which encodes the reverse transcriptase R2Tg, with a promoter and Kozak sequence upstream, and a polyadenylation signal downstream. The driver plasmid can drive expression of the GENE WRITER™ protein. The transgene plasmid (right), with a pCDNA backbone, comprises (in order) a CMV promoter, an rDNA homology sequence, a 5' UTR, an antisense-orientation insert, a 3' UTR, a second rDNA homology sequence, a second polyadenylation signal, and a TK promoter driving a mKate2 marker. The antisense-orientation insert comprises an EF1α promoter, a coding region for EGFP that comprises an

intron, and a polyadenylation signal. Use of the CMV promoter in the transgene plasmid drives expression of a template RNA comprising the rDNA homology regions, the UTRs, and the antisense-orientation insert.

FIG. 14 shows ddPCR evaluation of copy number variation of the transgene-rDNA junction in human cells across transfection conditions. Forward primer and probe were designed to bind to the 3' UTR of the R2Tg, while reverse primer was targeted to the human rDNA. The resulting ddPCR signal was normalized to that of reference assay RPP30 to determine copy number. Significantly higher average copies per genome were found with the wildtype (WT) R2Tg as compared to backbone construct with no R2Tg sequence involved. Condition 1 denotes a driver plasmid: transgene plasmid molar ratio of 9:1; condition 2 denotes the ratio is 4:1, condition 3 denotes the ratio is 1:1, condition 4 denotes the ratio is 1:4, and condition 5 denotes the ratio is 1:9.

FIGS. 15A and 15B. FIG. 15A: Hybrid capture of R2Tg identified on-target integrations in the human genome. The read coverage as aligned to the expected target integration in the R2 ribosomal site is indicated on the y-axis. The 5' junction between rDNA and R2Tg is indicated by the left vertical line, while the 3' junction is indicated by the right vertical line. Next-generation sequencing identifies reads spanning the expected junctions. FIG. 15B shows the number of reads from this experiment categorized as on-target integration or off-target integration at the 5' end and 3' end of the integrated sequence.

FIG. 16. Sanger sequencing result of the 3' junction nested PCR. Lowercase nucleotides represent the designed SNP. Shaded uppercase nucleotides represent WT sequence. FIG. 16 discloses SEQ ID NO: 1538.

FIGS. 17A-17F are schematic diagrams depicting various covalently dimerized GENE WRITER™ protein configurations. The proteins depicted are. FIG. 17A: a wild-type full length enzyme. FIG. 17B, two full-length enzymes (each comprising a DNA-binding domain, an RNA-binding domain, a reverse transcriptase domain, and an endonuclease domain) connected by a linker. FIG. 17C, a DNA binding domain and an RNA binding domain connected by a linker to a full-length enzyme. FIG. 17D, a DNA-binding domain and an RNA-binding domain connected by a linker to an RNA-binding domain, a reverse transcriptase domain, and an endonuclease domain. FIG. 17E, a DNA-binding domain connected by a first linker to an RNA-binding domain, which is connected by a second linker to a second RNA-binding domain, a reverse transcriptase domain, and an endonuclease domain. FIG. 17F, a DNA-binding domain connected by a first linker to an RNA-binding domain, which is connected by a second linker to a plurality of RNA-binding domains (in this figure, the molecule comprises three RNA-binding domains), which are connected by a linker to a reverse transcriptase domain and an endonuclease domain. In some embodiments, each R2 binds UTRs in the template RNA. In some embodiments, at least one module comprises a reverse transcriptase domain and an endonuclease domain. In some embodiments, the protein comprises a plurality of RNA-binding domains. In some embodiments, the modular system is split and is only active when it binds on DNA where the system uses two different DNA binding modules, e.g., a first protein comprising a first DNA binding module that is fused to an RNA binding module that recruits the RNA template for target primed reverse transcription, and second protein that comprises a second DNA binding module that binds at the site of integration and is fused to the reverse transcription and

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endonuclease modules. In some embodiments, the nucleic acid encoding the GENE WRITER™ comprises an intein such that the GENE WRITER™ protein is expressed from two separate genes and is fused by protein splicing after being translated. In some embodiments, the GENE WRITER™ is derived from a non-LTR protein, e.g., an R2 protein.

FIGS. 18A-18F are a schematic diagram showing different modular components of a GENE WRITER™ protein. The proteins depicted are: FIG. 18A: a wild-type full length enzyme. FIG. 18B: the DNA-binding domain of a GENE WRITER™ may comprise zinc fingers, Cas9, or a transcription factor, or a fragment or variant of any of the foregoing. FIG. 18C: the reverse transcriptase domain and RNA-binding domain together may comprise a reverse transcriptase domain (e.g., from an R2 protein) that is heterologous to one or more other domains of the protein, and may optionally further comprise one or more additional RNA binding domains, or a fragment or variant of any of the foregoing. FIG. 18D: the RNA binding domain may comprise, e.g., a B-box protein, an MS2 coat protein, a dCas protein, or a UTR binding protein, or a fragment or variant of any of the foregoing. FIG. 18E: the reverse transcriptase domain may comprise, e.g., a truncated reverse transcriptase domain, e.g., from an R2 protein; a reverse transcriptase domain from a virus (e.g., HIV), or a reverse transcriptase domain from AMV (avian myeloblastosis virus), or a fragment or variant of any of the foregoing. FIG. 18F: the endonuclease domain can comprise, e.g., a Cas9 nickase, a Cas ortholog, Fok I, or a restriction enzyme, or a fragment or variant of any of the foregoing. In some embodiments, a separate DNA binding domain can be attached to a polypeptide described herein (e.g., a DNA binding domain having stronger affinity for the target DNA sequence than an existing or prior DNA binding domain of the polypeptide, or a DNA binding sequence that binds to a different target DNA sequence than the existing or prior DNA binding domain of the polypeptide). In some embodiments, DNA binding domain mutants can be generated, e.g., having increased affinity to the target DNA sequence. In embodiments, the DNA binding domain comprises a zinc finger. In embodiments, the DNA binding domain is attached to the polypeptide (e.g., at the N-terminal or C-terminal ends) via a linker, e.g., as described herein. In embodiments, a zinc finger is attached to a DNA binding domain mutant (e.g., as described herein), such that the polypeptide exhibits increased binding to the target DNA sequence (e.g., as dictated by the zinc finger) without competition with the rDNA.

FIG. 19 is a graph showing linker mutant integration into the genome of HEK293T cells, assessed by a ddPCR assay evaluating copy number of R2Tg integration per genome. In v1 mutants, an insertion is located at the N-terminal side of an alpha helical region of R2Tg that preceded the predicted -1 RNA binding motif, in v2 mutants, an insertion is located at the C-terminal side of an alpha helical region of R2Tg that preceded the predicted -1 RNA binding motif, and in v3 mutants, an insertion is located C-terminal to a random coil region that came after the predicted c-myc DNA binding motif of R2Tg.

FIGS. 20A-20B are a series of graphs showing long-read sequencing confirming fidelity of R2Tg cis integration. Unique sequence coverage, as determined by UMI, is graphed across the expected reference sequence. The left vertical bar indicates expected 5' junction of the rDNA and R2Tg, while the right vertical bar indicates the 3' junction. Two separate amplicons spanning the 5' junction and 3' junction are shown.

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FIGS. 21A-21B are a series of graphs showing long-read sequencing confirming fidelity of R2Tg cis integration. Unique sequence deletions (>3 bp) as determined by UMI is graphed across the expected reference sequence. The left vertical bar indicates expected 5' junction of the rDNA and R2Tg, while the right vertical bar indicates the 3' junction. Two separate amplicons spanning the 5' junction and 3' junction are shown.

FIG. 22 is a diagram showing exemplary plasmid map PLV033 for cis integration of R2Gfo.

FIG. 23 is a graph showing integration of R2Gfo, R4A1, and R2Tg in cis in HEK293T cells. The mean of four replicates is shown; error bars indicate standard deviation.

FIG. 24 is a graph showing that R2Tg integrates into human fibroblasts in cis. Integration efficiency of the wild-type (WT) and endonuclease (EN) control R2Tg were plotted over four replicate experiments as measured via ddPCR at the 3' junction of R2Tg and the rDNA target.

FIG. 25 is a diagram showing Western Blot analysis for p53, p21, Actin, and Vinculin. U2OS cells were treated with the indicated compound or plasmid: GFP, R2Tg-WT (wild-type), or R2Tg-EN (endonuclease domain mutant). Plasmid transfections were performed with either lipofectamine 3000 (Lipo) or Eugene HD (Fug). Cells were analyzed 24 hours after treatment or transfection.

DETAILED DESCRIPTION

This disclosure relates to compositions, systems and methods for targeting, editing, modifying or manipulating a DNA sequence (e.g., inserting a heterologous object DNA sequence into a target site of a mammalian genome) at one or more locations in a DNA sequence in a cell, tissue or subject, e.g., in vivo or in vitro. The object DNA sequence may include, e.g., a coding sequence, a regulatory sequence, a gene expression unit.

More specifically, the disclosure provides retrotransposon-based systems for inserting a sequence of interest into the genome. This disclosure is based, in part, on a bioinformatic analysis to identify retrotransposase sequences and the associated 5' UTR and 3' UTR from a variety of organisms (see Table 3). While not wishing to be bound by theory, in some embodiments, retrotransposases identified in homeothermic (warm blooded) species, like birds, may have improved thermostability relative to some other enzymes that evolved at lower temperatures, and the thermostable retrotransposases may therefore be better suited for use in human cells. The disclosure also provides experimental evidence that several retrotransposases from different species, e.g., different species of animal and/or different species and clade of retrotransposon (e.g., as grouped by reverse transcriptase phylogeny, e.g., as described in Su et al. (2019) RNA; incorporated herein by reference in its entirety), can be used to catalyze DNA insertion into a target site in human cells (see Examples 7 and Example 28).

In some embodiments, systems described herein can have a number of advantages relative to various earlier systems. For instance, the disclosure describes retrotransposases capable of inserting long sequences (e.g., over 3000 nucleotides) of heterologous nucleic acid into a genome (see, e.g., FIG. 20A). In addition, retrotransposases described herein can insert heterologous nucleic acid in an endogenous site in the genome, such as the rDNA locus (see, e.g., Example 7). This is in contrast to Cre/loxP systems which require a first step of inserting an exogenous loxP site before a second step of inserting a sequence of interest into the loxP site.

GENE WRITER™ Genome Editors

Non-long terminal repeat (LTR) retrotransposons are a type of mobile genetic elements that are widespread in eukaryotic genomes. They include two classes: the apurinic/apyrimidinic endonuclease (APE)-type and the restriction enzyme-like endonuclease (RLE)-type. The APE class retrotransposons are comprised of two functional domains: an endonuclease/DNA binding domain, and a reverse transcriptase domain. The RLE class are comprised of three functional domains: a DNA binding domain, a reverse transcription domain, and an endonuclease domain. The reverse transcriptase domain of non-LTR retrotransposon functions by binding an RNA sequence template and reverse transcribing it into the host genome's target DNA. The RNA sequence template has a 3' untranslated region which is specifically bound to the transposase, and a variable 5' region generally having Open Reading Frame(s) ("ORF") encoding transposase proteins. The RNA sequence template may also comprise a 5' untranslated region which specifically binds the retrotransposase.

The inventors have found that, surprisingly, the elements of such non-LTR retrotransposons can be functionally modularized and/or modified to target, edit, modify or manipulate a target DNA sequence, e.g., to insert an object (e.g., heterologous) nucleic acid sequence into a target genome, e.g., a mammalian genome, by reverse transcription. Such modularized and modified nucleic acids, polypeptide compositions and systems are described herein and are referred to as GENE WRITER™ gene editors. A GENE WRITER™ gene editor system comprises: (A) a polypeptide or a nucleic acid encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain, and either (x) an endonuclease domain that contains DNA binding functionality or (y) an endonuclease domain and separate DNA binding domain; and (B) a template RNA comprising (i) a sequence that binds the polypeptide and (ii) a heterologous insert sequence. For example, the GENE WRITER™ genome editor protein may comprise a DNA-binding domain, a reverse transcriptase domain, and an endonuclease domain. In other embodiments, the GENE WRITER™ genome editor protein may comprise a reverse transcriptase domain and an endonuclease domain. In certain embodiments, the elements of the GENE WRITER™ gene editor polypeptide can be derived from sequences of non-LTR retrotransposons, e.g., APE-type or RLE-type retrotransposons or portions or domains thereof. In some embodiments the RLE-type non-LTR retrotransposon is from the R2, NeSL, HERO, R4, or CRE clade. In some embodiments the GENE WRITER™ genome editor is derived from R4 element X4_Line, which is found in the human genome. In some embodiments the APE-type non-LTR retrotransposon is from the R1, or Tx1 clade. In some embodiments the GENE WRITER™ genome editor is derived from Tx1 element Mare6, which is found in the human genome. The RNA template element of a GENE WRITER™ gene editor system is typically heterologous to the polypeptide element and provides an object sequence to be inserted (reverse transcribed) into the host genome. In some embodiments the GENE WRITER™ genome editor protein is capable of target primed reverse transcription.

In some embodiments the GENE WRITER™ genome editor is combined with a second polypeptide. In some embodiments the second polypeptide is derived from an APE-type non-LTR retrotransposon. In some embodiments the second polypeptide has a zinc knuckle-like motif. In some embodiments the second polypeptide is a homolog of Gag proteins.

Polypeptide Component of GENE WRITER™ Gene Editor System

RT Domain:

In certain aspects of the present invention, the reverse transcriptase domain of the GENE WRITER™ system is based on a reverse transcriptase domain of an APE-type or RLE-type non-LTR retrotransposon. A wild-type reverse transcriptase domain of an APE-type or RLE-type non-LTR retrotransposon can be used in a GENE WRITER™ system or can be modified (e.g., by insertion, deletion, or substitution of one or more residues) to alter the reverse transcriptase activity for target DNA sequences. In some embodiments the reverse transcriptase is altered from its natural sequence to have altered codon usage, e.g. improved for human cells. In some embodiments the reverse transcriptase domain is a heterologous reverse transcriptase from a different retrovirus, LTR-retrotransposon, or non-LTR retrotransposon. In certain embodiments, a GENE WRITER™ system includes a polypeptide that comprises a reverse transcriptase domain of an RLE-type non-LTR retrotransposon from the R2, NeSL, HERO, R4, or CRE clade, or of an APE-type non-LTR retrotransposon from the R1, or Tx1 clade. In certain embodiments, a GENE WRITER™ system includes a polypeptide that comprises a reverse transcriptase domain of a retrotransposon listed in Table 1, Table 2, or Table 3. In embodiments, the amino acid sequence of the reverse transcriptase domain of a GENE WRITER™ system is at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99% identical to the amino acid sequence of a reverse transcriptase domain of a retrotransposon whose DNA sequence is referenced in Table 1, Table 2, or Table 3. A person having ordinary skill in the art is capable of identifying reverse transcription domains based upon homology to other known reverse transcription domains using routine tools as Basic Local Alignment Search Tool (BLAST). In some embodiments, reverse transcriptase domains are modified, for example by site-specific mutation. In embodiments, the reverse transcriptase domain is engineered to bind a heterologous template RNA.

Endonuclease Domain:

In certain embodiments, the endonuclease/DNA binding domain of an APE-type retrotransposon or the endonuclease domain of an RLE-type retrotransposon can be used or can be modified (e.g., by insertion, deletion, or substitution of one or more residues) in a GENE WRITER™ system described herein. In some embodiments the endonuclease domain or endonuclease/DNA binding domain is altered from its natural sequence to have altered codon usage, e.g. improved for human cells. In some embodiments the endonuclease element is a heterologous endonuclease element, such as FokI nuclease, a type-II restriction I-like endonuclease (RLE-type nuclease), or another RLE-type endonuclease (also known as REL). In some embodiments the heterologous endonuclease activity has nickase activity and does not form double stranded breaks. The amino acid sequence of an endonuclease domain of a GENE WRITER™ system described herein may be at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99% identical to the amino acid sequence of an endonuclease domain of a retrotransposon whose DNA sequence is referenced in Table 1, 2, or 3. A person having ordinary skill in the art is capable of identifying endonuclease

clease domains based upon homology to other known endonuclease domains using tools as Basic Local Alignment Search Tool (BLAST). In certain embodiments, the heterologous endonuclease is FokI or a functional fragment thereof. In certain embodiments, the heterologous endonuclease is a Holliday junction resolvase or homolog thereof, such as the Holliday junction resolving enzyme from *Sulfolobus solfataricus*-SsoI Hje (Govindaraju et al., *Nucleic Acids Research* 44:7, 2016). In certain embodiments, the heterologous endonuclease is the endonuclease of the large fragment of a spliceosomal protein, such as Prp8 (Mahbub et al., *Mobile DNA* 8:16, 2017). For example, a GENE WRITER™ polypeptide described herein may comprise a reverse transcriptase domain from an APE- or RLE-type retrotransposon and an endonuclease domain that comprises FokI or a functional fragment thereof. In still other embodiments, homologous endonuclease domains are modified, for example by site-specific mutation, to alter DNA endonuclease activity. In still other embodiments, endonuclease domains are modified to remove any latent DNA-sequence specificity.

DNA Binding Domain:

In certain aspects, the DNA-binding domain of a GENE WRITER™ polypeptide described herein is selected, designed, or constructed for binding to a desired host DNA target sequence. In certain embodiments, the DNA-binding domain of the engineered RLE is a heterologous DNA-binding protein or domain relative to a native retrotransposon sequence. In some embodiments the heterologous DNA binding element is a zinc-finger element or a TAL effector element, e.g., a zinc-finger or TAL polypeptide or functional fragment thereof. In some embodiments the heterologous DNA binding element is a sequence-guided DNA binding element, such as Cas9, Cpf1, or other CRISPR-related protein that has been altered to have no endonuclease activity. In some embodiments the heterologous DNA binding element retains endonuclease activity. In some embodiments the heterologous DNA binding element replaces the endonuclease element of the polypeptide. In specific embodiments, the heterologous DNA-binding domain can be any one or more of Cas9, TAL domain, ZF domain, Myb domain, combinations thereof, or multiples thereof. In certain embodiments, the heterologous DNA-binding domain is a DNA binding domain of a retrotransposon described in Table 1, Table 2, or Table 3. A person having ordinary skill in the art is capable of identifying DNA binding domains based upon homology to other known DNA binding domains using tools as Basic Local Alignment Search Tool (BLAST). In still other embodiments, DNA-binding domains are modified, for example by site-specific mutation, increasing or decreasing DNA-binding elements (for example, number and/or specificity of zinc fingers), etc., to alter DNA-binding specificity and affinity. In some embodiments the DNA binding domain is altered from its natural sequence to have altered codon usage, e.g. improved for human cells

In certain aspects of the present invention, the host DNA-binding site integrated into by the GENE WRITER™ system can be in a gene, in an intron, in an exon, an ORF, outside of a coding region of any gene, in a regulatory region of a gene, or outside of a regulatory region of a gene. In other aspects, the engineered RLE may bind to one or more than one host DNA sequence.

In certain embodiments, a GENE WRITER™ gene editor system RNA further comprises an intracellular localization sequence, e.g., a nuclear localization sequence. The nuclear localization sequence may be an RNA sequence that pro-

motes the import of the RNA into the nucleus. In certain embodiments the nuclear localization signal is located on the template RNA. In certain embodiments, the retrotransposase polypeptide is encoded on a first RNA, and the template RNA is a second, separate, RNA, and the nuclear localization signal is located on the template RNA and not on an RNA encoding the retrotransposase polypeptide. While not wishing to be bound by theory, in some embodiments, the RNA encoding the retrotransposase is targeted primarily to the cytoplasm to promote its translation, while the template RNA is targeted primarily to the nucleus to promote its retrotransposition into the genome. In some embodiments the nuclear localization signal is at the 3' end, 5' end, or in an internal region of the template RNA. In some embodiments the nuclear localization signal is 3' of the heterologous sequence (e.g., is directly 3' of the heterologous sequence) or is 5' of the heterologous sequence (e.g., is directly 5' of the heterologous sequence). In some embodiments the nuclear localization signal is placed outside of the 5' UTR or outside of the 3' UTR of the template RNA. In some embodiments the nuclear localization signal is placed between the 5' UTR and the 3' UTR, wherein optionally the nuclear localization signal is not transcribed with the transgene (e.g., the nuclear localization signal is an anti-sense orientation or is downstream of a transcriptional termination signal or polyadenylation signal). In some embodiments the nuclear localization sequence is situated inside of an intron. In some embodiments a plurality of the same or different nuclear localization signals are in the RNA, e.g., in the template RNA. In some embodiments the nuclear localization signal is less than 5, 10, 25, 50, 75, 100, 150, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900 or 1000 bp in length. Various RNA nuclear localization sequences can be used. For example, Lubelsky and Ulitsky, *Nature* 555 (107-111), 2018 describe RNA sequences which drive RNA localization into the nucleus. In some embodiments, the nuclear localization signal is a SINE-derived nuclear RNA localization (SIRLOIN) signal. In some embodiments the nuclear localization signal binds a nuclear-enriched protein. In some embodiments the nuclear localization signal binds the HNRNPK protein. In some embodiments the nuclear localization signal is rich in pyrimidines, e.g., is a C/T rich, C/U rich, C rich, T rich, or U rich region. In some embodiments the nuclear localization signal is derived from a long non-coding RNA. In some embodiments the nuclear localization signal is derived from MALAT1 long non-coding RNA or is the 600 nucleotide M region of MALAT1 (described in Miyagawa et al., *RNA* 18, (738-751), 2012). In some embodiments the nuclear localization signal is derived from BORG long non-coding RNA or is a AGCCC motif (described in Zhang et al., *Molecular and Cellular Biology* 34, 2318-2329 (2014). In some embodiments the nuclear localization sequence is described in Shukla et al., *The EMBO Journal* e98452 (2018). In some embodiments the nuclear localization signal is derived from a non-LTR retrotransposon, an LTR retrotransposon, retrovirus, or an endogenous retrovirus.

In certain embodiments, a GENE WRITER™ gene editor system polypeptide further comprises an intracellular localization sequence, e.g., a nuclear localization sequence and/or a nucleolar localization sequence. The nuclear localization sequence and/or nucleolar localization sequence may be amino acid sequences that promote the import of the protein into the nucleus and/or nucleolus, where it can promote integration of heterologous sequence into the genome. In certain embodiments, a GENE WRITER™ gene editor system polypeptide (e.g., a retrotransposase, e.g., a poly-

peptide according to any of Tables 1, 2, or 3 herein) further comprises a nucleolar localization sequence. In certain embodiments, the retrotransposase polypeptide is encoded on a first RNA, and the template RNA is a second, separate, RNA, and the nucleolar localization signal is encoded on the RNA encoding the retrotransposase polypeptide and not on the template RNA. In some embodiments, the nucleolar localization signal is located at the N-terminus, C-terminus, or in an internal region of the polypeptide. In some embodiments, a plurality of the same or different nucleolar localization signals are used. In some embodiments, the nuclear localization signal is less than 5, 10, 25, 50, 75, or 100 amino acids in length. Various polypeptide nucleolar localization signals can be used. For example, Yang et al., *Journal of Biomedical Science* 22, 33 (2015), describe a nuclear localization signal that also functions as a nucleolar localization signal. In some embodiments, the nucleolar localization signal may also be a nuclear localization signal. In some embodiments, the nucleolar localization signal may overlap with a nuclear localization signal. In some embodiments, the nucleolar localization signal may comprise a stretch of basic residues. In some embodiments, the nucleolar localization signal may be rich in arginine and lysine residues. In some embodiments, the nucleolar localization signal may be derived from a protein that is enriched in the nucleolus. In some embodiments, the nucleolar localization signal may be derived from a protein enriched at ribosomal RNA loci. In some embodiments, the nucleolar localization signal may be derived from a protein that binds rRNA. In some embodiments, the nucleolar localization signal may be derived from MSP58. In some embodiments, the nucleolar localization signal may be a monopartite motif. In some embodiments, the nucleolar localization signal may be a bipartite motif. In some embodiments, the nucleolar localization signal may consist of a multiple monopartite or bipartite motifs. In some embodiments, the nucleolar localization signal may consist of a mix of monopartite and bipartite motifs. In some

embodiments, the nucleolar localization signal may be a dual bipartite motif. In some embodiments, the nucleolar localization motif may be a KRASSQALG-TIPKRRSSSRFIKRKK (SEQ ID NO: 1530). In some embodiments, the nucleolar localization signal may be derived from nuclear factor-KB-inducing kinase. In some embodiments, the nucleolar localization signal may be an RKKRKKK motif (SEQ ID NO: 1531) (described in Birbach et al., *Journal of Cell Science*, 117 (3615-3624), 2004).

In some embodiments, a nucleic acid described herein (e.g., an RNA encoding a GENE WRITER™ polypeptide, or a DNA encoding the RNA) comprises a microRNA binding site. In some embodiments, the microRNA binding site is used to increase the target-cell specificity of a GENE WRITER™ system. For instance, the microRNA binding site can be chosen on the basis that it is recognized by a miRNA that is present in a non-target cell type, but that is not present (or is present at a reduced level relative to the non-target cell) in a target cell type. Thus, when the RNA encoding the GENE WRITER™ polypeptide is present in a non-target cell, it would be bound by the miRNA, and when the RNA encoding the GENE WRITER™ polypeptide is present in a target cell, it would not be bound by the miRNA (or bound but at reduced levels relative to the non-target cell). While not wishing to be bound by theory, binding of the miRNA to the RNA encoding the GENE WRITER™ polypeptide may reduce production of the GENE WRITER™ polypeptide, e.g., by degrading the mRNA encoding the polypeptide or by interfering with translation. Accordingly, the heterologous object sequence would be inserted into the genome of target cells more efficiently than into the genome of non-target cells. A system having a microRNA binding site in the RNA encoding the GENE WRITER™ polypeptide (or encoded in the DNA encoding the RNA) may also be used in combination with a template RNA that is regulated by a second microRNA binding site, e.g., as described herein in the section entitled "Template RNA component of GENE WRITER™ gene editor system."

TABLE 1

TABLE 1: APE-type non-LTR retrotransposon elements				
Family	Sequence Accession	Mobile Element	Name	Organism
Dewa	AB097143	ORF2	<i>Danio rerio</i> retrotransposon DewaDr1 DNA, complete sequence	<i>Danio rerio</i>
HeT-A	KJ081250	non-LTR retrotransposon: HeT-A	<i>Drosophila melanogaster</i> non-LTR retrotransposon HeT-A, partial sequence	<i>Drosophila melanogaster</i>
Keno	AB111948	ORF2	<i>Tetraodon nigroviridis</i> retrotransposon KenoTn1 DNA, partial sequence	<i>Tetraodon nigroviridis</i>
KenoDr1; Keno	AB097144	ORF2	<i>Danio rerio</i> retrotransposon KenoDr1 DNA, complete sequence	<i>Danio rerio</i>
KenoFr1; Keno	AB111947	ORF2	<i>Takifugu rubripes</i> retrotransposon KenoFr1 DNA, complete sequence	<i>Takifugu rubripes</i>
Kibi	AB097139	ORF2	<i>Danio rerio</i> retrotransposon KibiDr2 DNA, complete sequence	<i>Danio rerio</i>
Kibi	AB097138	ORF2	<i>Danio rerio</i> retrotransposon KibiDr1 DNA, complete sequence	<i>Danio rerio</i>
Kibi	AB097137	ORF2	<i>Tetraodon nigroviridis</i> retrotransposon KibiTn1 DNA, complete sequence	<i>Tetraodon nigroviridis</i>
Kibi	AB097136	ORF2	<i>Takifugu rubripes</i> retrotransposon KibiFr1 DNA, complete sequence	<i>Takifugu rubripes</i>
KoshiTn1	AB097135	ORF2	<i>Tetraodon nigroviridis</i> retrotransposon KoshiTn1 DNA, complete sequence	<i>Tetraodon nigroviridis</i>
Mutsu	AB097142	ORF2	<i>Danio rerio</i> retrotransposon MutuDr3 DNA, partial sequence	<i>Danio rerio</i>

TABLE 1-continued

TABLE 1: APE-type non-LTR retrotransposon elements				
Family	Sequence Accession	Mobile Element	Name	Organism
Mutsu	AB097141	ORF2	<i>Danio rerio</i> retrotransposon MutsuDr2 DNA, partial sequence	<i>Danio rerio</i>
Mutsu	AB097140	ORF2	<i>Danio rerio</i> retrotransposon MutsuDr1 DNA, complete sequence	<i>Danio rerio</i>
R1	HQ284568	non-LTR retrotransposon: R1-like	<i>Trilocha</i> sp. GAS-2011 isolate TrilSp.6 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Trilocha</i> sp. GAS-2011
R1	HQ284534	non-LTR retrotransposon: R1-like	<i>Scopula ornata</i> isolate ScoOrn.6 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Scopula ornata</i>
R1	HQ284496	non-LTR retrotransposon: R1-like	<i>Perigonia ilus</i> isolate PerIlus.31 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Perigonia ilus</i>
R1	HQ284489	non-LTR retrotransposon: R1-like	<i>Oxytenis modestia</i> isolate OxyMod.2_3_4_7_9 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Oxytenis modestia</i>
R1	HQ284488	non-LTR retrotransposon: R1-like	<i>Oxytenis modestia</i> isolate OxyMod.1 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Oxytenis modestia</i>
R1	HQ284476	non-LTR retrotransposon: R1-like	<i>Oeneis magna dubia</i> isolate OenMag.26 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Oeneis magna dubia</i>
R1	HQ284437	non-LTR retrotransposon: R1-like	<i>Lymantria dispar</i> isolate LymDis.2 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Lymantria dispar</i>
R1	HQ284435	non-LTR retrotransposon: R1-like	<i>Lymantria dispar</i> isolate LymDis.1 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Lymantria dispar</i>
R1	HQ284432	non-LTR retrotransposon: R1-like	<i>Janiodes laverna</i> isolate JanLav.911 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Janiodes laverna</i>
R1	HQ284431	non-LTR retrotransposon: R1-like	<i>Janiodes laverna</i> isolate JanLav.811 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Janiodes laverna</i>
R1	HQ284430	non-LTR retrotransposon: R1-like	<i>Janiodes laverna</i> isolate JanLav.5 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Janiodes laverna</i>
R1	HQ284428	non-LTR retrotransposon: R1-like	<i>Janiodes laverna</i> isolate JanLav.411 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Janiodes laverna</i>
R1	HQ284426	non-LTR retrotransposon: R1-like	<i>Janiodes laverna</i> isolate JanLav.211 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Janiodes laverna</i>
R1	HQ284421	non-LTR retrotransposon: R1-like	<i>Heteropterus morpheus</i> isolate HetMor.3 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Heteropterus morpheus</i>
R1	HQ284402	non-LTR retrotransposon: R1-like	<i>Erinnyis ello</i> isolate EriEllo.22 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Erinnyis ello</i>
R1	HQ284399	non-LTR retrotransposon: R1-like	<i>Erebia theano</i> isolate EreThe.29 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Erebia theano</i>
R1	HQ284398	non-LTR retrotransposon: R1-like	<i>Erebia theano</i> isolate EreThe.28 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Erebia theano</i>
R1	HQ284397	non-LTR retrotransposon: R1-like	<i>Erebia theano</i> isolate EreThe.27 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Erebia theano</i>

TABLE 1-continued

TABLE 1: APE-type non-LTR retrotransposon elements				
Family	Sequence Accession	Mobile Element	Name	Organism
R1	HQ284391	non-LTR retrotransposon: R1-like	<i>Emesis lucinda</i> isolate EmeLuc.23 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Emesis lucinda</i>
R1	HQ284390	non-LTR retrotransposon: R1-like	<i>Emesis lucinda</i> isolate EmeLuc.2 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Emesis lucinda</i>
R1	HQ284364	non-LTR retrotransposon: R1-like	<i>Coenonympha glycerion</i> isolate CoeGly.9 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Coenonympha glycerion</i>
R1	HQ284363	non-LTR retrotransposon: R1-like	<i>Coenonympha glycerion</i> isolate CoeGly.8 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Coenonympha glycerion</i>
R1	HQ284362	non-LTR retrotransposon: R1-like	<i>Coenonympha glycerion</i> isolate CoeGly.7 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Coenonympha glycerion</i>
R1	HQ284361	non-LTR retrotransposon: R1-like	<i>Coenonympha glycerion</i> isolate CoeGly.5 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Coenonympha glycerion</i>
R1	HQ284357	non-LTR retrotransposon: R1-like	<i>Coenonympha glycerion</i> isolate CoeGly.13 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Coenonympha glycerion</i>
R1	HQ284356	non-LTR retrotransposon: R1-like	<i>Coenonympha glycerion</i> isolate CoeGly.11 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Coenonympha glycerion</i>
R1	HQ284350	non-LTR retrotransposon: R1-like	<i>Catocyclotis adelina</i> isolate CatAde.18 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Catocyclotis adelina</i>
R1	HQ284340	non-LTR retrotransposon: R1-like	<i>Caria rhacotis</i> isolate CarRha.11 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Caria rhacotis</i>
R1	HQ284339	non-LTR retrotransposon: R1-like	<i>Caria rhacotis</i> isolate CarRha.1 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Caria rhacotis</i>
R1	HQ284319	non-LTR retrotransposon: R1-like	<i>Archiearis parthenias</i> isolate BrePar.1 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Archiearis parthenias</i>
R1	HQ284318	non-LTR retrotransposon: R1-like	<i>Brangas neora</i> isolate BraNeo.32 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Brangas neora</i>
R1	HQ284292	non-LTR retrotransposon: R1-like	<i>Araschnia levana</i> isolate AraLev.31 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Araschnia levana</i>
R1	HQ284286	non-LTR retrotransposon: R1-like	<i>Araschnia levana</i> isolate AraLev.1 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Araschnia levana</i>
R1	HQ284280	non-LTR retrotransposon: R1-like	<i>Anteros formosus</i> isolate AntForm.34 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Anteros formosus</i>
R1	HQ284279	non-LTR retrotransposon: R1-like	<i>Anteros formosus</i> isolate AntForm.32 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Anteros formosus</i>
R1	HQ284278	non-LTR retrotransposon: R1-like	<i>Anteros formosus</i> isolate AntForm.31 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Anteros formosus</i>
R1	HQ284270	non-LTR retrotransposon: R1-like	<i>Agrotis exclamationis</i> isolate AgrExcl.27 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Agrotis exclamationis</i>

TABLE 1-continued

TABLE 1: APE-type non-LTR retrotransposon elements				
Family	Sequence Accession	Mobile Element	Name	Organism
R1	HQ284267	non-LTR retrotransposon: R1-like	<i>Agrius cingulata</i> isolate AgrCing.36_39 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Agrius cingulata</i>
R1	HQ284266	non-LTR retrotransposon: R1-like	<i>Agrius cingulata</i> isolate AgrCing.3 non-LTR retrotransposon R1-like reverse transcriptase-like gene, partial sequence	<i>Agrius cingulata</i>
R1	HQ284263	non-LTR retrotransposon: R1-like	<i>Aglia tau</i> isolate AglTau.8 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Aglia tau</i>
R1	HQ284262	non-LTR retrotransposon: R1-like	<i>Aglia tau</i> isolate AglTau.7 non-LTR retrotransposon R1-like reverse transcriptase gene, partial cds	<i>Aglia tau</i>
R1	DQ836362	MalR1	<i>Maculineaalcon</i> R1-like non-LTR retrotransposon R1 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengarisalcon</i>
R1	DQ836391	MnaR1	<i>Maculinea nausithous</i> R1-like non-LTR retrotransposon R1 reverse transcriptase (RT) gene, partial cds	<i>Phengaris nausithous</i>
R1	KU543683	non-LTR retrotransposon and non-LTR retrovirus reverse transcriptase; Region: RT_nLTR	<i>Bactrocera tryoni</i> clone Btry_5404 non-LTR retrotransposon R1, complete sequence	<i>Bactrocera tryoni</i>
R1	KU543682	non-LTR retrotransposon and non-LTR retrovirus reverse transcriptase; Region: RT_nLTR	<i>Bactrocera tryoni</i> clone Btry_5167 non-LTR retrotransposon R1, complete sequence	<i>Bactrocera tryoni</i>
R1	KU543679	non-LTR retrotransposon and non-LTR retrovirus reverse transcriptase; Region: RT_nLTR	<i>Bactrocera tryoni</i> clone Btry_4956 non-LTR retrotransposon R1, complete sequence	<i>Bactrocera tryoni</i>
R1	KU543678	non-LTR retrotransposon and non-LTR retrovirus reverse transcriptase; Region: RT_nLTR	<i>Bactrocera tryoni</i> clone Btry_5979 non-LTR retrotransposon R1, complete sequence	<i>Bactrocera tryoni</i>
R1	AB078933	ORF1	<i>Papilio xuthus</i> non-LTR retrotransposon gene for gag-like protein, partial cds, clone: SARTPx2-2	<i>Papilio xuthus</i>
R1	AB078932	ORF1	<i>Papilio xuthus</i> non-LTR retrotransposon gene for gag-like protein, partial cds, clone: SARTPx2-1	<i>Papilio xuthus</i>
R1	AB078936	ORF2	<i>Papilio xuthus</i> non-LTR retrotransposon genes for gag-like protein, reverse transcriptase, partial cds, clone: SARTPx4-N18	<i>Papilio xuthus</i>
R1	AB078935	ORF2	<i>Papilio xuthus</i> non-LTR retrotransposon genes for gag-like protein, reverse transcriptase, partial and complete cds, clone: SARTPx3-N7	<i>Papilio xuthus</i>
R1	AB078934	ORF2	<i>Papilio xuthus</i> non-LTR retrotransposon genes for gag-like protein, reverse transcriptase, partial cds, clone: SARTPx3-N3	<i>Papilio xuthus</i>
R1	AB078931	ORF2	<i>Papilio xuthus</i> non-LTR retrotransposon genes for gag-like protein, reverse transcriptase, partial and complete cds, clone: SARTPx1-N14	<i>Papilio xuthus</i>
R1	AB078930	ORF2	<i>Papilio xuthus</i> non-LTR retrotransposon genes for gag-like protein, reverse transcriptase, partial and complete cds, clone: SARTPx1-N5	<i>Papilio xuthus</i>

TABLE 1-continued

TABLE 1: APE-type non-LTR retrotransposon elements				
Family	Sequence Accession	Mobile Element	Name	Organism
R1	AB078929	ORF2	<i>Papilio xuthus</i> non-LTR retrotransposon genes for gag-like protein, reverse transcriptase, partial and complete cds, clone: SARTPx1-N4	<i>Papilio xuthus</i>
R1	AB078928	ORF2	<i>Papilio xuthus</i> non-LTR retrotransposon gene for gag-like protein, reverse transcriptase, complete and partial cds, clone: SARTPx1-3	<i>Papilio xuthus</i>
R1	KP771712	ORF2; contains endonuclease, reverse transcriptase and RNaseH	<i>Blattella germanica</i> non-LTR retrotransposon TRAS-like 2, complete sequence	<i>Blattella germanica</i>
R1	KP771711	ORF2; contains endonuclease, reverse transcriptase and RNaseH	<i>Blattella germanica</i> non-LTR retrotransposon TRAS-like 1, complete sequence	<i>Blattella germanica</i>
R1	AF015813	R1 ORF	<i>Dugesia</i> sp. retrotransposon R1 reverse transcriptase gene, partial cds	<i>Aphonopelma</i> sp. WDB-1998
R1	AF015489	R1 ORF	<i>Dugesia</i> sp. retrotransposon R1 reverse transcriptase gene, partial cds	<i>Aphonopelma</i> sp. WDB-1998
R1Bm	AB182560	non-LTR retrotransposon R1Bmks ORF2	<i>Bombyx mori</i> genes for non-LTR retrotransposon R1Bmks ORF1 protein, non-LTR retrotransposon R1Bmks ORF2 protein, complete cds	<i>Bombyx mori</i>
R6	AB090819	ORF2	<i>Anopheles gambiae</i> retrotransposon R6Ag3 DNA, complete sequence	<i>Anopheles gambiae</i>
R6	AB090818	ORF2	<i>Anopheles gambiae</i> retrotransposon R6Ag2 DNA, complete sequence	<i>Anopheles gambiae</i>
R6	AB090817	ORF2	<i>Anopheles gambiae</i> retrotransposon R6Ag1 DNA, complete sequence	<i>Anopheles gambiae</i>
R6	KJ958615	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reVIR6, partial sequence	<i>Bacillus rossius</i>
R6	KJ958596	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reBER6, partial sequence	<i>Bacillus rossius</i>
R6	AF352480	transposon: NLRCth1-like non-LTR retrotransposon	<i>Chironomus circumdatus</i> clone cir6 transposon NLRCth1-like non-LTR retrotransposon reverse transcriptase gene, partial cds	<i>Chironomus circumdatus</i>
R6	AF373367	transposon: non-LTR retrotransposon LINE2	<i>Clelia rustica</i> clone CR6 non-LTR retrotransposon LINE2 reverse transcriptase pseudogene, partial sequence	<i>Paraphimophis rusticus</i>
R7	AB090820	ORF2	<i>Anopheles gambiae</i> retrotransposon R7Ag1 DNA, complete sequence	<i>Anopheles gambiae</i>
R7	AB090821	ORF2	<i>Anopheles gambiae</i> retrotransposon R7Ag2 DNA, complete sequence	<i>Anopheles gambiae</i>
R7	KJ958622	R2	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR7, partial sequence	<i>Bacillus rossius</i>
R7	KJ958616	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reVIR7, partial sequence	<i>Bacillus rossius</i>
R7	KJ958597	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reBER7, partial sequence	<i>Bacillus rossius</i>
R7	AF352514	transposon: NLRCth1-like non-LTR retrotransposon	<i>Chironomus circumdatus</i> clone cir7 transposon NLRCth1-like non-LTR retrotransposon reverse transcriptase pseudogene, partial sequence	<i>Chironomus circumdatus</i>
Rt2	AY379084	truncated; similar to reverse transcriptase	<i>Leptocheirus plumulosus</i> retrotransposon LpRt2, partial sequence	<i>Leptocheirus plumulosus</i>
Rt2	MSQRT2RET		<i>Anopheles gambiae</i> retrotransposon RT2, complete sequence	<i>Anopheles gambiae</i>
RTAg4	AB090813	ORF2	<i>Anopheles gambiae</i> retrotransposon RTAg4 DNA, complete sequence	<i>Anopheles gambiae</i>

TABLE 1-continued

TABLE 1: APE-type non-LTR retrotransposon elements				
Family	Sequence Accession	Mobile Element	Name	Organism
TRAS1	BMOTRAS1	DNA binding domain at AA1103-1120.	<i>Bombyx mori</i> gene, complete sequence of retrotransposon TRAS1	<i>Bombyx mori</i>
TRAS3	JX875955	similar to reverse transcriptases transposon:	<i>Acyrtosiphon pisum</i> clone LSR1 non-LTR retrotransposon TRAS3, complete sequence	<i>Acyrtosiphon pisum</i>
Tx1	AJ621359	non-LTR retrotransposon TX1-1__Tet	<i>Tetraodon nigroviridis</i> non-LTR retrotransposon TX1-1__Tet, complete sequence	<i>Tetraodon nigroviridis</i>
Tx1	AJ621360	transposon: non-LTR retrotransposon TX1-2__Tet	<i>Tetraodon nigroviridis</i> partial non-LTR retrotransposon TX1-2__Tet	<i>Tetraodon nigroviridis</i>
Tx1	AJ621361	transposon: non-LTR retrotransposon TX1-3__Tet	<i>Tetraodon nigroviridis</i> partial non-LTR retrotransposon TX1-3__Tet	<i>Tetraodon nigroviridis</i>
Tx1	AJ621362	transposon: non-LTR retrotransposon TX1-4__Tet	<i>Tetraodon nigroviridis</i> partial non-LTR retrotransposon TX1-4__Tet	<i>Tetraodon nigroviridis</i>
Tx1	DQ118004	transposon: Tx1-like retrotransposon Tx1Aru	<i>Acipenser ruthenus</i> clone dg194 transposon Tx1-like retrotransposon Tx1Aru reverse transcriptase-like gene, partial sequence	<i>Acipenser ruthenus</i>
Tx1	AB097134	ORF2	<i>Takifugu rubripes</i> retrotransposon KoshiFr1 DNA, complete sequence	<i>Takifugu rubripes</i>
Tx1	AB090816	ORF2	<i>Anopheles gambiae</i> retrotransposon MinoAg1 DNA, complete sequence	<i>Anopheles gambiae</i>
Tx1	AB090812	ORF2	<i>Anopheles gambiae</i> retrotransposon RTAg3 DNA, complete sequence	<i>Anopheles gambiae</i>
Waldo	AH009917	non-LTR retrotransposon: Waldo-A	<i>Drosophila melanogaster</i> Waldo-A non-LTR retrotransposon, 5' sequence	<i>Drosophila melanogaster</i>
Waldo	AH009916	non-LTR retrotransposon: Waldo-A	<i>Drosophila melanogaster</i> clone CBE9 Waldo-A non-LTR retrotransposon, 5' sequence	<i>Drosophila melanogaster</i>
Waldo	AH009915	non-LTR retrotransposon: Waldo-A	<i>Drosophila melanogaster</i> Waldo-A non-LTR retrotransposon, 5' sequence	<i>Drosophila melanogaster</i>
Waldo	AH009914	non-LTR retrotransposon: Waldo-A	<i>Drosophila melanogaster</i> Waldo-A non-LTR retrotransposon	<i>Drosophila melanogaster</i>
Waldo	AH009920	non-LTR retrotransposon: Waldo-B	<i>Drosophila melanogaster</i> Waldo-B non-LTR retrotransposon, 5' sequence	<i>Drosophila melanogaster</i>
Waldo	AH009919	non-LTR retrotransposon: Waldo-B		<i>Drosophila melanogaster</i>
Waldo	AH009918	non-LTR retrotransposon: Waldo-B		<i>Drosophila melanogaster</i>
Waldo	AB090815	ORF2	<i>Anopheles gambiae</i> retrotransposon WaldoAg2 DNA, complete sequence	<i>Anopheles gambiae</i>
Waldo	AB090814	ORF2	<i>Anopheles gambiae</i> retrotransposon WaldoAg1 DNA, complete sequence	<i>Anopheles gambiae</i>
Waldo	AB078939	ORF2	<i>Forficula scudderii</i> non-LTR retrotransposon pseudogene for reverse transcriptase, clone: WaldoFs1-26	<i>Forficula scudderii</i>
Waldo	AB078938	ORF2	<i>Forficula scudderii</i> non-LTR retrotransposon pseudogene for reverse transcriptase, clone: WaldoFs1-2	<i>Forficula scudderii</i>
Waldo	AB078937	ORF2	<i>Forficula scudderii</i> non-LTR retrotransposon pseudogene for reverse transcriptase, clone: WaldoFs1-1	<i>Forficula scudderii</i>

TABLE 2

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
CRE	EF067892		<i>Colletotrichum cereale</i> clone 9F8-1558 Cret3 non-LTR retrotransposon, partial sequence	<i>Colletotrichum cereale</i>
CRE	EF067894		<i>Colletotrichum cereale</i> clone 9F8-2137 Cret3 non-LTR retrotransposon, partial sequence	<i>Colletotrichum cereale</i>
CRE	MG028000	non-LTR retrotransposon: Rex3	<i>Characidium gomesi</i> voucher MNRJ20998 non-LTR retrotransposon Rex3, partial sequence	<i>Characidium gomesi</i>
CRE	KY566213	non-LTR retrotransposon: Rex3	<i>Characidium gomesi</i> non-LTR retrotransposon Rex3, partial sequence	<i>Characidium gomesi</i>
CRE	GU949558		<i>Kaloterms flavicollis</i> isolate Crete non-LTR retrotransposon R2, complete sequence; and R2 protein gene, complete cds	<i>Kaloterms flavicollis</i>
CRE; CRE2	CFU19151	poly dA tracts in 5' and 3' UTRs	<i>Crithidia fasciculata</i> retrotransposon CRE2 in mini-exon gene, putative reverse transcriptase gene, complete cds	<i>Crithidia fasciculata</i>
CZAR	BR000987	pol	TPA_inf: <i>Capsaspora owczarzaki</i> DNA, non-LTR retrotransposon CoL4, complete sequence, strain: ATCC 30864	<i>Capsaspora owczarzaki</i>
CZAR	BR000986	pol	TPA_inf: <i>Capsaspora owczarzaki</i> DNA, non-LTR retrotransposon CoL3, complete sequence, strain: ATCC 30864	<i>Capsaspora owczarzaki</i>
CZAR	BR000985	pol	TPA_inf: <i>Capsaspora owczarzaki</i> DNA, non-LTR retrotransposon CoL2, complete sequence, strain: ATCC 30864	<i>Capsaspora owczarzaki</i>
CZAR	BR000984	pol	TPA_inf: <i>Capsaspora owczarzaki</i> DNA, non-LTR retrotransposon CoL1, complete sequence, strain: ATCC 30864	<i>Capsaspora owczarzaki</i>
DongAG; Dong	AB097127	rt	<i>Anopheles gambiae</i> retrotransposon DongAg DNA, partial sequence	<i>Anopheles gambiae</i>
EhRLE2	AB097128	rt	<i>Entamoeba histolytica</i> retrotransposon EhRLE2 DNA, complete sequence	<i>Entamoeba histolytica</i>
EhRLE3	AB097129	rt	<i>Entamoeba histolytica</i> retrotransposon EhRLE3 DNA, complete sequence	<i>Entamoeba histolytica</i>
Genie	AF440196	endonuclease	<i>Giardia intestinalis</i> non-LTR retrotransposon GENIE 1 pol polyprotein gene, complete cds	<i>Giardia intestinalis</i>
Genie	BK000097	endonuclease	TPA_exp: <i>Giardia intestinalis</i> non-LTR retrotransposon Genie 1A gene, partial sequence	<i>Giardia intestinalis</i>
Genie	BK000095	endonuclease	TPA_exp: <i>Giardia intestinalis</i> non-LTR retrotransposon Genie 1 gene, partial sequence	<i>Giardia intestinalis</i>
Genie	BK000096	insertion site for non-LTR retrotransposon Genie 1	TPA_exp: <i>Giardia intestinalis</i> non-LTR retrotransposon Genie 1 target site sequence	<i>Giardia intestinalis</i>
Genie	AY216701	non-experimental evidence, no additional details recorded	<i>Girardia tigrina</i> GENIE retrotransposon, complete sequence	<i>Girardia tigrina</i>
Genie	BK000098	similar to endonuclease	TPA_exp: <i>Giardia intestinalis</i> non-LTR retrotransposon Genie 2 gene, complete sequence	<i>Giardia intestinalis</i>
GilD	AF433877	(tca)n (SEQ ID NO: 1532) or (tga)n (SEQ ID NO: 1533), n = 2-4	<i>Giardia intestinalis</i> inactive non-LTR retrotransposon GilD, consensus sequence	<i>Giardia intestinalis</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
GilM	AF433875	poly(dA) tract	<i>Giardia intestinalis</i> non-LTR LINE-like retrotransposon GilM, complete sequence	<i>Giardia intestinalis</i>
Hero	AB097132	rt	<i>Danio rerio</i> retrotransposon HERODr DNA, complete sequence	<i>Danio rerio</i>
Hero	AB097130	rt	<i>Takifugu rubripes</i> retrotransposon HEROFr DNA, complete sequence	<i>Takifugu rubripes</i>
HEROTn	AB097131	rt	<i>Tetraodon nigroviridis</i> retrotransposon HEROTn DNA, complete sequence	<i>Tetraodon nigroviridis</i>
NeSL_3_135_68117	FJ905846	non-LTR retrotransposon: NeSL_3_135_68117	<i>Daphnia pulex</i> non-LTR retrotransposon NeSL_3_135_68117, complete sequence	<i>Daphnia pulex</i>
NeSL; NeSL-1	DQ099731	target site duplication	<i>Caenorhabditis briggsae</i> transposon NeSL-1-like non-LTR retrotransposon NeSL-1Cb reverse transcriptase (pol) gene, complete cds	<i>Caenorhabditis briggsae</i>
PERERE-9	BN000800		TPA_exp: <i>Schistosoma mansoni</i> Perere-9 non-LTR retrotransposon	<i>Schistosoma mansoni</i>
R2	AF015814	R2	<i>Limulus polyphemus</i> retrotransposon R2, complete sequence	<i>Limulus polyphemus</i>
R2	AF090145	R2	<i>Nasonia vitripennis</i> R2 non-LTR retrotransposable element reverse transcriptase gene, partial cds	<i>Nasonia vitripennis</i>
R2	AF015818	R2	<i>Porcellio scaber</i> retrotransposon R2, complete sequence	<i>Porcellio scaber</i>
R2	AF015815	R2	<i>Anurida maritima</i> retrotransposon R2, complete sequence	<i>Anurida maritima</i>
R2	M16558	R2	<i>Bombyx mori</i> rDNA insertion element R2 (typeII), complete cds	<i>Bombyx mori</i>
R2	AF015819	R2	<i>Forficula auricularia</i> retrotransposon R2, complete sequence.	<i>Forficula auricularia</i>
R2	EU854578	R2	<i>Triops cancriformis</i> non-LTR retrotransposon R2 reverse transcriptase gene, complete cds	<i>Triops cancriformis</i>
R2	GU949555	R2	<i>Reticulitermes lucifugus</i> non-LTR retrotransposon R2, complete sequence; and R2 protein gene, complete cds	<i>Reticulitermes lucifugus</i>
R2	AB097123	rt	<i>Ciona intestinalis</i> retrotransposon R2Ci-C DNA, partial sequence	<i>Ciona intestinalis</i>
R2	AB097124	rt	<i>Ciona intestinalis</i> retrotransposon R2Ci-D DNA, partial sequence	<i>Ciona intestinalis</i>
R2	FJ461304	R2	<i>Rhynchosciara americana</i> non-LTR retrotransposon RaR2 reverse transcriptase gene, complete cds	<i>Rhynchosciara americana</i>
R2	AB097121	rt	<i>Ciona intestinalis</i> retrotransposon R2Ci-A DNA, complete sequence	<i>Ciona intestinalis</i>
R2	KP657892	R2	<i>Bacillus rossius</i> isolate roCAP(full).9 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657890	R2	<i>Bacillus rossius</i> isolate roCAP(full).7 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657888	R2	<i>Bacillus rossius</i> isolate roCAP(full).5 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	KP657870	R2	<i>Bacillus rossius</i> isolate roCAP(full).1 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657833	R2	<i>Bacillus rossius</i> isolate roANZ(full).13 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657832	R2	<i>Bacillus rossius</i> isolate roANZ(full).12 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657830	R2	<i>Bacillus rossius</i> isolate roANZ(full).10 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657807	R2	<i>Bacillus rossius</i> isolate roANZ(-101).8 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657806	R2	<i>Bacillus rossius</i> isolate roANZ(-101).7 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657805	R2	<i>Bacillus rossius</i> isolate roANZ(-101).6 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657802	R2	<i>Bacillus rossius</i> isolate roANZ(-101).3 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657799	R2	<i>Bacillus rossius</i> isolate roANZ(-101).1 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	FJ461304	R2	<i>Rhynchosciara americana</i> non-LTR retrotransposon RaR2 reverse transcriptase gene, complete cds	<i>Rhynchosciara americana</i>
R2	JQ082370	polyA_signal_sequence	<i>Eyprepocnemis plorans</i> non-LTR retrotransposon R2 R2 protein gene, complete cds	<i>Eyprepocnemis plorans</i>
R2	KJ958672	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCUR4_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958671	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCUR3_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958670	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCUR2_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958669	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCUR1_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958668	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCDF6_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958667	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCDF5_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958666	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCDF4_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958665	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCDF3_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958664	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCDF2_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958663	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCDF1_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958662	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM10_deg, partial sequence	<i>Bacillus rossius</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	KJ958661	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM5_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958660	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM7_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958659	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM3_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958658	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM2_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958657	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reMSN6_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958656	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reMSN5_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958655	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reMSN4_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958654	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reMSN3_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958653	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reMSN2_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958652	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reMSN1_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958651	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT8_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958650	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT7_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958649	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT6_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958648	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT5_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958647	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT4_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958646	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT3_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958645	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT2_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958644	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT1_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958643	R2	<i>Bacillus rossius</i> non-LTR retrotransposon rePAT9_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958642	R2	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR2_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958641	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB9_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958640	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB8_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958639	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB7_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958638	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB6_deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958637	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB5_deg, partial sequence	<i>Bacillus rossius</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	KJ958636	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB4__deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958635	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB3__deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958634	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB2__deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958633	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB1__deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958632	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS1__deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958631	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS11__deg, partial sequence	<i>Bacillus rossius</i>
R2	KJ958629	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM8, partial sequence	<i>Bacillus rossius</i>
R2	KJ958628	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM6, partial sequence	<i>Bacillus rossius</i>
R2	KJ958627	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM4, partial sequence	<i>Bacillus rossius</i>
R2	KJ958626	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCOM1, partial sequence	<i>Bacillus rossius</i>
R2	KJ958624	R2	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR10, partial sequence	<i>Bacillus rossius</i>
R2	KJ958623	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reGAB10, partial sequence	<i>Bacillus rossius</i>
R2	KJ958619	R2	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR4, partial sequence	<i>Bacillus rossius</i>
R2	KJ958618	R2	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR3, partial sequence	<i>Bacillus rossius</i>
R2	KJ958617	R2	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR1, partial sequence	<i>Bacillus rossius</i>
R2	KJ958613	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reVIR4, partial sequence	<i>Bacillus rossius</i>
R2	KJ958612	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reVIR3, partial sequence	<i>Bacillus rossius</i>
R2	KJ958611	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reVIR2, partial sequence	<i>Bacillus rossius</i>
R2	KJ958610	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reVIR1, partial sequence	<i>Bacillus rossius</i>
R2	KJ958609	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS17, partial sequence	<i>Bacillus rossius</i>
R2	KJ958608	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS16, partial sequence	<i>Bacillus rossius</i>
R2	KJ958607	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS15, partial sequence	<i>Bacillus rossius</i>
R2	KJ958606	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS14, partial sequence	<i>Bacillus rossius</i>
R2	KJ958605	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS13, partial sequence	<i>Bacillus rossius</i>
R2	KJ958604	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS12, partial sequence	<i>Bacillus rossius</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	KJ958603	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS7, partial sequence	<i>Bacillus rossius</i>
R2	KJ958602	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS6, partial sequence	<i>Bacillus rossius</i>
R2	KJ958601	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS5, partial sequence	<i>Bacillus rossius</i>
R2	KJ958600	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS4, partial sequence	<i>Bacillus rossius</i>
R2	KJ958599	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS3, partial sequence	<i>Bacillus rossius</i>
R2	KJ958598	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reTDS2, partial sequence	<i>Bacillus rossius</i>
R2	KJ958594	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reBER4, partial sequence	<i>Bacillus rossius</i>
R2	KJ958593	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reBER2, partial sequence	<i>Bacillus rossius</i>
R2	KJ958592	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reBER1, partial sequence	<i>Bacillus rossius</i>
R2	KJ958591	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roFOL7, partial sequence	<i>Bacillus rossius</i>
R2	KJ958590	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roFOL6, partial sequence	<i>Bacillus rossius</i>
R2	KJ958589	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roFOL5, partial sequence	<i>Bacillus rossius</i>
R2	KJ958588	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roFOL4, partial sequence	<i>Bacillus rossius</i>
R2	KJ958587	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roFOL3, partial sequence	<i>Bacillus rossius</i>
R2	KJ958586	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roFOL2, partial sequence	<i>Bacillus rossius</i>
R2	KJ958585	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roFOL1, partial sequence	<i>Bacillus rossius</i>
R2	KJ958584	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ10, partial sequence	<i>Bacillus rossius</i>
R2	KJ958583	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ9, partial sequence	<i>Bacillus rossius</i>
R2	KJ958582	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ8, partial sequence	<i>Bacillus rossius</i>
R2	KJ958581	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ7, partial sequence	<i>Bacillus rossius</i>
R2	KJ958580	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ6, partial sequence	<i>Bacillus rossius</i>
R2	KJ958579	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ5, partial sequence	<i>Bacillus rossius</i>
R2	KJ958578	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ4, partial sequence	<i>Bacillus rossius</i>
R2	KJ958577	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ3, partial sequence	<i>Bacillus rossius</i>
R2	KJ958576	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ2, partial sequence	<i>Bacillus rossius</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	KJ958575	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roANZ1, partial sequence	<i>Bacillus rossius</i>
R2	KJ958574	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP9, partial sequence	<i>Bacillus rossius</i>
R2	KJ958573	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP8, partial sequence	<i>Bacillus rossius</i>
R2	KJ958572	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP7, partial sequence	<i>Bacillus rossius</i>
R2	KJ958571	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP6, partial sequence	<i>Bacillus rossius</i>
R2	KJ958570	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP5, partial sequence	<i>Bacillus rossius</i>
R2	KJ958569	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP4, partial sequence	<i>Bacillus rossius</i>
R2	KJ958568	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP3, partial sequence	<i>Bacillus rossius</i>
R2	KJ958567	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP2, partial sequence	<i>Bacillus rossius</i>
R2	KJ958566	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP1, partial sequence	<i>Bacillus rossius</i>
R2	KJ958565	R2	<i>Bacillus rossius</i> non-LTR retrotransposon roCAP10, partial sequence	<i>Bacillus rossius</i>
R2	JN937654	R2	<i>Lepidurus apus lubbocki</i> isolate lu8a 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937653	R2	<i>Lepidurus apus lubbocki</i> isolate lu7a 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937652	R2	<i>Lepidurus apus lubbocki</i> isolate lu2a 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937651	R2	<i>Lepidurus apus lubbocki</i> isolate b7c7 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937650	R2	<i>Lepidurus apus lubbocki</i> isolate b6c4 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937649	R2	<i>Lepidurus apus lubbocki</i> isolate lu5a 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937648	R2	<i>Lepidurus apus lubbocki</i> isolate lu1a 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937647	R2	<i>Lepidurus apus lubbocki</i> isolate b6c5 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	JN937646	R2	<i>Lepidurus apus lubbocki</i> isolate b6c6 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937645	R2	<i>Lepidurus apus lubbocki</i> isolate lu4a 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937644	R2	<i>Lepidurus apus lubbocki</i> isolate lu3a 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937643	R2	<i>Lepidurus apus lubbocki</i> isolate b6c3 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937642	R2	<i>Lepidurus apus lubbocki</i> isolate lu6a 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon psR2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937641	R2	<i>Lepidurus apus lubbocki</i> isolate LM5h2 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937640	R2	<i>Lepidurus apus lubbocki</i> isolate LM2h5 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937639	R2	<i>Lepidurus apus lubbocki</i> isolate LM2h4 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937638	R2	<i>Lepidurus apus lubbocki</i> isolate LM5h5 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937637	R2	<i>Lepidurus apus lubbocki</i> isolate LM5h4 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937636	R2	<i>Lepidurus apus lubbocki</i> isolate LM5h3 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937635	R2	<i>Lepidurus apus lubbocki</i> isolate LM5h1 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937634	R2	<i>Lepidurus apus lubbocki</i> isolate LM2h3 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937633	R2	<i>Lepidurus apus lubbocki</i> isolate LM2h2 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>
R2	JN937632	R2	<i>Lepidurus apus lubbocki</i> isolate LM2h1 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2L1, complete sequence	<i>Lepidurus apus lubbocki</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	JN937631	R2	<i>Lepidurus arcticus</i> isolate T6 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937630	R2	<i>Lepidurus arcticus</i> isolate T5 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937629	R2	<i>Lepidurus arcticus</i> isolate T4 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937628	R2	<i>Lepidurus arcticus</i> isolate T3 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937627	R2	<i>Lepidurus arcticus</i> isolate T2 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937626	R2	<i>Lepidurus arcticus</i> isolate T1 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937625	R2	<i>Lepidurus arcticus</i> isolate V4 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937624	R2	<i>Lepidurus arcticus</i> isolate V3 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937623	R2	<i>Lepidurus arcticus</i> isolate V2 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937622	R2	<i>Lepidurus arcticus</i> isolate V1 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2La, complete sequence	<i>Lepidurus arcticus</i>
R2	JN937615	R2	<i>Lepidurus couesii</i> isolate D3a7f 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937614	R2	<i>Lepidurus couesii</i> isolate D3a5f 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937613	R2	<i>Lepidurus couesii</i> isolate D3a4f 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937612	R2	<i>Lepidurus couesii</i> isolate D3a3f 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937611	R2	<i>Lepidurus couesii</i> isolate D3a2f 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	JN937610	R2	<i>Lepidurus couesii</i> isolate D3_8 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937609	R2	<i>Lepidurus couesii</i> isolate D3_7 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937608	R2	<i>Lepidurus couesii</i> isolate D3_6 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937607	R2	<i>Lepidurus couesii</i> isolate D3_5 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937606	R2	<i>Lepidurus couesii</i> isolate D3_4 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937605	R2	<i>Lepidurus couesii</i> isolate D3_3 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937604	R2	<i>Lepidurus couesii</i> isolate D3_2 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937603	R2	<i>Lepidurus couesii</i> isolate D3_1 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcB, complete sequence	<i>Lepidurus couesii</i>
R2	JN937602	R2	<i>Lepidurus couesii</i> isolate C2_5 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937601	R2	<i>Lepidurus couesii</i> isolate LcoC2r1_5 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937600	R2	<i>Lepidurus couesii</i> isolate LcoC2r1_6 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937599	R2	<i>Lepidurus couesii</i> isolate C2_8 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937598	R2	<i>Lepidurus couesii</i> isolate C2_4 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937597	R2	<i>Lepidurus couesii</i> isolate C2_9 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937596	R2	<i>Lepidurus couesii</i> isolate C2_7 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	JN937595	R2	<i>Lepidurus couesii</i> isolate C2_6 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937594	R2	<i>Lepidurus couesii</i> isolate C2_3 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937593	R2	<i>Lepidurus couesii</i> isolate C2_2 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	JN937592	R2	<i>Lepidurus couesii</i> isolate C2_1 28S ribosomal RNA gene, partial sequence; and non-LTR retrotransposon R2LcA, complete sequence	<i>Lepidurus couesii</i>
R2	AF015822	R2 ORF	<i>Tenebrio molitor</i> retrotransposon R2 reverse transcriptase gene, partial cds	<i>Tenebrio molitor</i>
R2	AF015817	R2 ORF	<i>Tenebrio molitor</i> retrotransposon R2 reverse transcriptase gene, partial cds	<i>Tenebrio molitor</i>
R2	AF015816	R2 ORF	<i>Hippodamia convergens</i> retrotransposon R2 reverse transcriptase gene, partial cds	<i>Hippodamia convergens</i>
R2	KP657866	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-714).5 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657865	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-714).4 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657863	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-714).2 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657862	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-714).1 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657861	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).9 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657860	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).8 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657859	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).7 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657858	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).6 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657857	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).5 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657856	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).4 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657855	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).3 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	KP657854	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).2 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657853	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).10 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657852	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1297).1 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657851	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).9 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657850	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).8 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657849	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).7 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657848	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).6 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657847	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).5 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657846	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).4 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657845	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).3 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657844	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).2 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657843	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).10 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657842	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roCAP(-1172).1 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657824	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roANZ(-1062).5 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657823	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roANZ(-1062).4 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657822	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roANZ(-1062).3 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657820	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roANZ(-1062).2 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657816	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roANZ(-1062).16 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657814	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roANZ(-1062).14 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	KP657810	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roANZ(-1062).10 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657809	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate roANZ(-1062).1 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657759	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate rePAT(-1297).8 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657757	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate rePAT(-1297).6 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	KP657751	retrotransposon: R2Br	<i>Bacillus rossius</i> isolate rePAT(-1297).1 retrotransposon R2Br reverse transcriptase gene, partial cds	<i>Bacillus rossius</i>
R2	AB097125	rt	<i>Ciona savignyi</i> retrotransposon R2Cs-D DNA, partial sequence	<i>Ciona savignyi</i>
R2	AB097124	rt	<i>Ciona intestinalis</i> retrotransposon R2Ci-D DNA, partial sequence	<i>Ciona intestinalis</i>
R2	AB097123	rt	<i>Ciona intestinalis</i> retrotransposon R2Ci-C DNA, partial sequence	<i>Ciona intestinalis</i>
R2	AB097121	rt	<i>Ciona intestinalis</i> retrotransposon R2Ci-A DNA, complete sequence	<i>Ciona intestinalis</i>
R2	AB201417	rt	<i>Triops longicaudatus</i> non-LTR retrotransposon R2Tl gene for reverse transcriptase, partial cds	<i>Triops longicaudatus</i>
R2	AB201416	rt	<i>Procambarus clarkii</i> non-LTR retrotransposon R2Pc gene for reverse transcriptase, partial cds	<i>Procambarus clarkii</i>
R2	AB201415	rt	<i>Hasarius adansoni</i> non-LTR retrotransposon R2Ha gene for reverse transcriptase, partial cds	<i>Hasarius adansoni</i>
R2	AB201414	rt	<i>Metacrinus rotundus</i> non-LTR retrotransposon R2Mr gene for reverse transcriptase, partial cds	<i>Metacrinus rotundus</i>
R2	AB201413	rt	<i>Mauremys reevesii</i> non-LTR retrotransposon R2Cr-B2 gene for reverse transcriptase, partial cds	<i>Mauremys reevesii</i>
R2	AB201412	rt	<i>Mauremys reevesii</i> non-LTR retrotransposon R2Cr-B1 gene for reverse transcriptase, partial cds	<i>Mauremys reevesii</i>
R2	AB201411	rt	<i>Mauremys reevesii</i> non-LTR retrotransposon R2Cr-A gene for reverse transcriptase, partial cds	<i>Mauremys reevesii</i>
R2	AB201410	rt	<i>Oryzias latipes</i> non-LTR retrotransposon R2Ol-A gene for reverse transcriptase, partial cds	<i>Oryzias latipes</i>
R2	AB201409	rt	<i>Tanichthys albonubes</i> non-LTR retrotransposon R2Ta gene for reverse transcriptase, partial cds	<i>Tanichthys albonubes</i>
R2	AB201408	rt	<i>Eptatretus burgeri</i> non-LTR retrotransposon R2Eb gene for reverse transcriptase, partial cds	<i>Eptatretus burgeri</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	DQ099732	transposon: R2-like non-LTR retrotransposon R2Ag	<i>Aedes aegypti</i> transposon R2-like non-LTR retrotransposon R2Ag reverse transcriptase (pol) gene, partial cds	<i>Aedes aegypti</i>
R2	DQ099728	transposon: R2-like non-LTR retrotransposon R2Ag_B	<i>Aedes aegypti</i> transposon R2-like non-LTR retrotransposon R2Ag_B reverse transcriptase (pol) gene, partial cds	<i>Aedes aegypti</i>
R2	GU949559		<i>Kaloterme flavicollis</i> isolate Livorno non-LTR retrotransposon R2, complete sequence; and R2 protein gene, complete cds	<i>Kaloterme flavicollis</i>
R2	GU949557		<i>Reticulitermes balkanensis</i> non-LTR retrotransposon R2, partial sequence; and R2 protein gene, partial cds	<i>Reticulitermes balkanensis</i>
R2	GU949556		<i>Reticulitermes grassei</i> non-LTR retrotransposon R2, partial sequence; and R2 protein gene, partial cds	<i>Reticulitermes grassei</i>
R2	GU949554		<i>Reticulitermes urbis</i> non-LTR retrotransposon R2, complete sequence; and R2 protein gene, complete cds	<i>Reticulitermes urbis</i>
R2	AF412214		<i>Schistosoma japonicum</i> clone S10A non-LTR retrotransposon Sjr2-like, partial sequence	<i>Schistosoma japonicum</i>
R2	AF015685		<i>Drosophila mercatorum</i> R2 retrotransposon reverse transcriptase domain protein gene, complete cds	<i>Drosophila mercatorum</i>
R2	KJ958674		<i>Bacillus rossius</i> retrotransposon R2Br, complete sequence	<i>Bacillus rossius</i>
R2	AF015814	R2	<i>Limulus polyphemus</i> retrotransposon R2, complete sequence	<i>Limulus polyphemus</i>
R2	M16558	R2	<i>Bombyx mori</i> rDNA insertion element R2 (typeII), complete cds.	<i>Bombyx mori</i>
R2	GQ398057	R9Av	<i>Adineta vaga</i> copy 1 non-LTR retrotransposon R9, complete sequence; and disrupted 28S ribosomal RNA gene, partial sequence	<i>Adineta vaga</i>
R2Bm	AB076841	R2	<i>Bombyx mori</i> non-LTR retrotransposon R2Bm gene for reverse transcriptase, complete cds and 28S rRNA	<i>Bombyx mori</i>
R2Ci-B	AB097122	rt	<i>Ciona intestinalis</i> retrotransposon R2Ci-B DNA, complete sequence	<i>Ciona intestinalis</i>
R2Dr	AB097126	rt	<i>Danio rerio</i> retrotransposon R2Dr DNA, complete sequence	<i>Danio rerio</i>
R4	AH003588		<i>Parascaris equorum</i> transposon non-LTR retrotransposable element R4 reverse transcriptase gene, partial cds	<i>Parascaris equorum</i>
R4	ALU29445	R4	<i>Ascaris lumbricoides</i> site-specific non-LTR retrotransposable element R4 in 26S rDNA, complete sequence	<i>Ascaris lumbricoides</i>
R4	L08889	R4 Dong	<i>Bombyx mori</i> reverse transcriptase gene, complete cds	<i>Bombyx mori</i>
R4	DQ836390	MalR4-5	<i>Maculinea alcon</i> R4-like non-LTR retrotransposon R4-5 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris alcon</i>
R4	DQ836385	MnaR4-3	<i>Maculinea nausithous</i> R4-like non-LTR retrotransposon R4-3 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris nausithous</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R4	DQ836386	MnaR4-4	<i>Maculinea nausithous</i> R4-like non-LTR retrotransposon R4-4 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris nausithous</i>
R4	DQ836387	MnaR4-7	<i>Maculinea nausithous</i> R4-like non-LTR retrotransposon R4-7 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris nausithous</i>
R4	DQ836388	MnaR4-8	<i>Maculinea nausithous</i> R4-like non-LTR retrotransposon R4-8 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris nausithous</i>
R4	DQ836389	MnaR4-9	<i>Maculinea nausithous</i> R4-like non-LTR retrotransposon R4-9 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris nausithous</i>
R4	DQ836379	MteR4-1	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-1 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris teleius</i>
R4	DQ836384	MteR4-10	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-10 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris teleius</i>
R4	DQ836367	MteR4-2	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-2 reverse transcriptase (RT) gene, partial cds	<i>Phengaris teleius</i>
R4	DQ836380	MteR4-3	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-3 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris teleius</i>
R4	DQ836381	MteR4-4	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-4 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris teleius</i>
R4	DQ836382	MteR4-6	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-6 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris teleius</i>
R4	DQ836383	MteR4-8	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-8 reverse transcriptase (RT) pseudogene, partial sequence	<i>Phengaris teleius</i>
R4	DQ836374	transposon: R4-like non-LTR retrotransposon R4-1	<i>Maculinea alcon</i> R4-like non-LTR retrotransposon R4-1 reverse transcriptase (RT) gene, partial cds	<i>Phengaris alcon</i>
R4	DQ836373	transposon: R4-like non-LTR retrotransposon R4-1	<i>Maculinea nausithous</i> R4-like non-LTR retrotransposon R4-1 reverse transcriptase (RT) gene, partial cds	<i>Phengaris nausithous</i>
R4	DQ836375	transposon: R4-like non-LTR retrotransposon R4-2	<i>Maculinea alcon</i> R4-like non-LTR retrotransposon R4-2 reverse transcriptase (RT) gene, partial cds	<i>Phengaris alcon</i>
R4	DQ836376	transposon: R4-like non-LTR retrotransposon R4-3	<i>Maculinea alcon</i> R4-like non-LTR retrotransposon R4-3 reverse transcriptase (RT) gene, partial cds	<i>Phengaris alcon</i>
R4	DQ836377	transposon: R4-like non-LTR retrotransposon R4-4	<i>Maculinea alcon</i> R4-like non-LTR retrotransposon R4-4 reverse transcriptase (RT) gene, partial cds	<i>Phengaris alcon</i>
R4	DQ836371	transposon: R4-like non-LTR retrotransposon R4-5	<i>Maculinea nausithous</i> R4-like non-LTR retrotransposon R4-5 reverse transcriptase (RT) gene, partial cds	<i>Phengaris nausithous</i>
R4	DQ836368	transposon: R4-like	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-5	<i>Phengaris teleius</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R4	DQ836378	non-LTR retrotransposon R4-5 transposon: R4-like non-LTR retrotransposon R4-6	reverse transcriptase (RT) gene, partial cds <i>Maculinea alcon</i> R4-like non-LTR retrotransposon R4-6 reverse transcriptase (RT) gene, partial cds	<i>Phengaris alcon</i>
R4	DQ836372	transposon: R4-like non-LTR retrotransposon R4-6	<i>Maculinea nausithous</i> R4-like non-LTR retrotransposon R4-6 reverse transcriptase (RT) gene, partial cds	<i>Phengaris nausithous</i>
R4	DQ836369	transposon: R4-like non-LTR retrotransposon R4-7	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-7 reverse transcriptase (RT) gene, partial cds	<i>Phengaris teleius</i>
R4	DQ836370	transposon: R4-like non-LTR retrotransposon R4-9	<i>Maculinea teleius</i> R4-like non-LTR retrotransposon R4-9 reverse transcriptase (RT) gene, partial cds	<i>Phengaris teleius</i>
R4	AF286191	transposon: retrotransposon Rex6	<i>Xiphophorus maculatus</i> retrotransposon Rex6 reverse transcriptase pseudogene, partial sequence	<i>Xiphophorus maculatus</i>
R5	KJ958673	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reCUR5_deg, partial sequence	<i>Bacillus rossius</i>
R5	KJ958620	R2	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR5, partial sequence	<i>Bacillus rossius</i>
R5	KJ958614	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reVIR5, partial sequence	<i>Bacillus rossius</i>
R5	KJ958595	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reBER5, partial sequence	<i>Bacillus rossius</i>
R5	AJ006560	transposon: Amer5 non-LTR retrotransposon	<i>Anopheles merus</i> Amer5 non-LTR retrotransposon encoding reverse transcriptase, partial	<i>Anopheles merus</i>
R5	AF352479	transposon: NLRCth1-like non-LTR retrotransposon	<i>Chironomus circumdatus</i> clone cir5 transposon NLRCth1-like non-LTR retrotransposon reverse transcriptase gene, partial cds	<i>Chironomus circumdatus</i>
R5	AF352454	transposon: NLRCth1-like non-LTR retrotransposon	<i>Chironomus alpestris</i> clone dor50 transposon NLRCth1-like non-LTR retrotransposon reverse transcriptase gene, partial cds	<i>Chironomus alpestris</i>
R5	AF352404	transposon: NLRCth1-like non-LTR retrotransposon	<i>Chironomus luridus</i> clone lur5 transposon NLRCth1-like non-LTR retrotransposon reverse transcriptase gene, partial cds	<i>Chironomus luridus</i>
R8	FR852798	poly(A) tail	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline2_3	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	KJ958630	R2	<i>Bacillus rossius</i> non-LTR retrotransposon reVIR8_deg, partial sequence	<i>Bacillus rossius</i>
R8	KJ958621	R2	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR8, partial sequence	<i>Bacillus rossius</i>
R8	KP001560	rex3-RT pseudogene__Contig ILU_TR8	<i>Iberochondrostoma lusitanicum</i> clone tr8a non-LTR retrotransposon Rex3, complete sequence	<i>Iberochondrostoma lusitanicum</i>
R8	FR852885	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon BNR114 (Beline1_114)	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R8	FR852856	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon BNR45 (Belline1_45)	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852844	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon BNR22 (Belline1_22)	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852836	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline17_6	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852834	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline17_4	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852831	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline17_1	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852829	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline16_2	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852827	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline15_3	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852819	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline12_2	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852813	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline9_5	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852807	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline8_1	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852806	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline7_18	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852799	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon Belline2_4	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	FR852795	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon BNR19 (Belline1_19)	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8	AF352481	transposon: NLRCth1-like non-LTR retrotransposon	<i>Chironomus circumdatus</i> clone cir8 transposon NLRCth1-like non-LTR retrotransposon reverse transcriptase gene, partial cds	<i>Chironomus circumdatus</i>
R8; R5	FR852861	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon BNR59 (Belline1_59)	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8; R5	FR852857	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon BNR51 (Belline1_51)	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8; R7	FR852866	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon BNR76 (Belline1_76)	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8; R7	FR852838	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i> LINE-type retrotransposon BNR7 (Belline1_7)	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R8; R7	AF352455	transposon: NLRCth1-like non-LTR	<i>Chironomus alpestris</i> clone dor70 note identical sequence found in dor80 transposon	<i>Chironomus alpestris</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
		retrotransposon	NLRCh1-like non-LTR retrotransposon reverse transcriptase gene, partial cds	
R8; R9	FR852878	right terminal repeat	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>	<i>Beta vulgaris</i> subsp. <i>vulgaris</i>
R9	KJ958625	R2	LINE-type retrotransposon BNR96 (Belline1_96)	<i>Bacillus rossius</i>
Rex6	AJ293547	en	<i>Bacillus rossius</i> non-LTR retrotransposon trKOR9, partial sequence	<i>Oreochromis niloticus</i>
Rex6	AJ293546	en	<i>Oreochromis niloticus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Oni-3	<i>Oreochromis niloticus</i>
Rex6	AJ293545	en	<i>Oreochromis niloticus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Oni-2	<i>Oreochromis niloticus</i>
Rex6	AJ293517	en	<i>Oreochromis niloticus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Oni-1	<i>Oreochromis niloticus</i>
Rex6	AJ293516	en	<i>Xiphophorus maculatus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone Rex6-Xma-6	<i>Xiphophorus maculatus</i>
Rex6	AJ293515	en	<i>Xiphophorus maculatus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone Rex6-Xma-5	<i>Xiphophorus maculatus</i>
Rex6	AJ293514	en	<i>Xiphophorus maculatus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone Rex6-Xma-4	<i>Xiphophorus maculatus</i>
Rex6	AJ293513	en	<i>Xiphophorus maculatus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone Rex6-Xma-3	<i>Xiphophorus maculatus</i>
Rex6	AJ293512	en	<i>Xiphophorus maculatus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone Rex6-Xma-2	<i>Xiphophorus maculatus</i>
Rex6	AJ293538	en	<i>Xiphophorus maculatus</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone Rex6-Xma-1	<i>Poecilia formosa</i>
Rex6	AJ293537	en	<i>Poecilia formosa</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Pfo-6	<i>Poecilia formosa</i>
Rex6	AJ293536	en	<i>Poecilia formosa</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Pfo-5	<i>Poecilia formosa</i>
Rex6	AJ293535	en	<i>Poecilia formosa</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Pfo-4	<i>Poecilia formosa</i>
Rex6	AJ293534	en	<i>Poecilia formosa</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Pfo-3	<i>Poecilia formosa</i>
Rex6	AJ293533	en	<i>Poecilia formosa</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Pfo-2	<i>Poecilia formosa</i>
Rex6	AJ293526	en	<i>Poecilia formosa</i> Rex6 retrotransposon partial en pseudogene for endonuclease, clone rex6-Pfo-1	<i>Poeciliopsis gracilis</i>
Rex6	AJ293525	en	<i>Poeciliopsis gracilis</i> Rex6 retrotransposon partial en	<i>Poeciliopsis gracilis</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
Rex6	AJ293524	en	pseudogene for endonuclease, clone rex6-Pgr-3 <i>Poeciliopsis gracilis</i> Rex6 retrotransposon partial en	<i>Poeciliopsis gracilis</i>
Rex6	AJ293523	en	pseudogene for endonuclease, clone rex6-Pgr-2 <i>Poeciliopsis gracilis</i> Rex6 retrotransposon partial en	<i>Poeciliopsis gracilis</i>
Rex6	AJ293522	en	pseudogene for endonuclease, clone rex6-Pgr-1 <i>Oryzias latipes</i> Rex6 retrotransposon partial en	<i>Oryzias latipes</i>
Rex6	AJ293521	en	pseudogene for endonuclease, clone rex6-Ola-5 <i>Oryzias latipes</i> Rex6 retrotransposon partial en	<i>Oryzias latipes</i>
Rex6	AJ293520	en	pseudogene for endonuclease, clone Rex6-Ola-4 <i>Oryzias latipes</i> Rex6 retrotransposon partial en	<i>Oryzias latipes</i>
Rex6	AJ293519	en	pseudogene for endonuclease, clone Rex6-Ola-3 <i>Oryzias latipes</i> Rex6 retrotransposon partial en	<i>Oryzias latipes</i>
Rex6	AJ293518	en	pseudogene for endonuclease, clone Rex6-Ola-2 <i>Oryzias latipes</i> Rex6 retrotransposon partial en	<i>Oryzias latipes</i>
Rex6	AJ293549	en	pseudogene for endonuclease, clone Rex6-Ola-1 <i>Cichlasoma labridens</i> Rex6 retrotransposon partial en	<i>Herichthys labridens</i>
Rex6	AJ293548	en	pseudogene for endonuclease, clone rex6-Cla-2 <i>Cichlasoma labridens</i> Rex6 retrotransposon partial en	<i>Herichthys labridens</i>
Rex6	AJ293544	en	pseudogene for endonuclease, clone rex6-Cla-1 <i>Heterandria bimaculata</i> Rex6 retrotransposon partial en	<i>Pseudoxiphophorus bimaculatus</i>
Rex6	AJ293543	en	pseudogene for endonuclease, clone rex6-Hbi-6 <i>Heterandria bimaculata</i> Rex6 retrotransposon partial en	<i>Pseudoxiphophorus bimaculatus</i>
Rex6	AJ293542	en	pseudogene for endonuclease, clone rex6-Hbi-5 <i>Heterandria bimaculata</i> Rex6 retrotransposon partial en	<i>Pseudoxiphophorus bimaculatus</i>
Rex6	AJ293541	en	pseudogene for endonuclease, clone rex6-Hbi-4 <i>Heterandria bimaculata</i> Rex6 retrotransposon partial en	<i>Pseudoxiphophorus bimaculatus</i>
Rex6	AJ293540	en	pseudogene for endonuclease, clone rex6-Hbi-3 <i>Heterandria bimaculata</i> Rex6 retrotransposon partial en	<i>Pseudoxiphophorus bimaculatus</i>
Rex6	AJ293539	en	pseudogene for endonuclease, clone rex6-Hbi-2 <i>Heterandria bimaculata</i> Rex6 retrotransposon partial en	<i>Pseudoxiphophorus bimaculatus</i>
Rex6	AJ293532	en	pseudogene for endonuclease, clone rex6-Hbi-1 <i>Gambusia affinis</i> Rex6 retrotransposon partial en	<i>Gambusia affinis</i>
Rex6	AJ293531	en	pseudogene for endonuclease, clone rex6-Gaf-5 <i>Gambusia affinis</i> Rex6 retrotransposon partial en	<i>Gambusia affinis</i>
Rex6	AJ293530	en	pseudogene for endonuclease, clone rex6-Gaf-5 <i>Gambusia affinis</i> Rex6 retrotransposon partial en	<i>Gambusia affinis</i>
Rex6	AJ293529	en	pseudogene for endonuclease, clone rex6-Gaf-4 <i>Gambusia affinis</i> Rex6 retrotransposon partial en	<i>Gambusia affinis</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
Rex6	AJ293528	en	pseudogene for endonuclease, clone rex6-Gaf-3 <i>Gambusia affinis</i> Rex6 retrotransposon partial en	<i>Gambusia affinis</i>
Rex6	AJ293527	en	pseudogene for endonuclease, clone rex6-Gaf-2 <i>Gambusia affinis</i> Rex6 retrotransposon partial en	<i>Gambusia affinis</i>
Rex6	JX576459	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate i non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576458	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate h non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576457	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate g non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576456	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate f non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576455	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate e non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576454	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate d non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576453	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate c non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576452	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate b non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576451	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate a non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576450	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z8 non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576449	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z7 non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576448	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z6 non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576447	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z5 non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576446	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z4 non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576445	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z3 non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576444	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z2 non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576443	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z1 non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576442	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate z non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576441	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate x non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576440	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate v non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576439	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate u non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576438	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate t non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
Rex6	JX576437	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate s non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576436	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate r non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576435	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate q non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576434	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate p non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576433	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate o non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576432	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate n non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576431	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate m non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576430	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate l non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576429	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate k non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576428	non-LTR retrotransposon: Rex6	<i>Symphysodon discus</i> isolate j non-LTR retrotransposon Rex6, partial sequence	<i>Symphysodon discus</i>
Rex6	JX576427	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone e non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	JX576426	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone d non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	JX576425	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone c non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	JX576424	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone b non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	JX576423	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone a non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	JX576422	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone g non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	JX576421	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone f non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	JX576420	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone e non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	JX576419	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone d non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	JX576418	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone c non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	JX576417	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone b non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	JX576416	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone a non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	JX576415	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone g non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	JX576414	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone f non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	JX576413	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone e non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
Rex6	JX576412	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone d non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	JX576411	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone c non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	JX576410	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone b non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	JX576409	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone a non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	JX576408	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone h non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	JX576407	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone g non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	JX576406	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone f non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	JX576405	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone e non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	JX576404	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone d non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	JX576403	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone c non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	JX576402	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone b non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	JX576401	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone a non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131853	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone z7 non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131852	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone z6 non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131851	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone z5 non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131850	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone z4 non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131849	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone z3 non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131848	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone z2 non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131847	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone z1 non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131846	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone z non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131845	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone x non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131844	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone v non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131843	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone u non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131842	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone t non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131841	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone s non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
Rex6	KF131840	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone r non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131839	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone q non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131838	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone p non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131837	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone n non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131836	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone m non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131835	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone l non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131834	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone k non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131833	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone j non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131832	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone i non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131831	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone h non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131830	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone g non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131829	non-LTR retrotransposon: Rex6	<i>Pterophyllum scalare</i> clone f non-LTR retrotransposon Rex6, partial sequence	<i>Pterophyllum scalare</i>
Rex6	KF131828	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone z6 non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131827	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone z5 non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131826	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone z4 non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131825	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone z3 non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131824	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone z2 non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131823	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone z1 non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131822	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone z non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131821	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone x non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131820	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone v non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131819	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone u non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131818	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone t non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131817	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone s non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131816	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone r non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
Rex6	KF131815	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone q non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131814	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone p non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131813	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone o non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131812	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone n non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131811	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone m non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131810	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone l non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131809	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone k non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131808	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone j non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131807	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone i non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131806	non-LTR retrotransposon: Rex6	<i>Geophagus proximus</i> clone h non-LTR retrotransposon Rex6, partial sequence	<i>Geophagus proximus</i>
Rex6	KF131805	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z10 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131804	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z9 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131803	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z8 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131802	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z7 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131801	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z6 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131800	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z5 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131799	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z4 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131798	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z3 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131797	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z2 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131796	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z1 non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131795	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone z non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131794	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone x non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131793	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone v non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131792	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone u non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131791	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone t non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
Rex6	KF131790	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone s non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131789	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone r non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131788	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone q non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131787	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone p non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131786	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone o non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131785	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone n non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131784	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone m non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131783	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone l non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131782	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone k non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131781	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone j non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131780	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone i non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131779	non-LTR retrotransposon: Rex6	<i>Astronotus ocellatus</i> clone h non-LTR retrotransposon Rex6, partial sequence	<i>Astronotus ocellatus</i>
Rex6	KF131778	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone z6 non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131777	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone z5 non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131776	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone z4 non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131775	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone z3 non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131774	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone z2 non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131773	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone z1 non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131772	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone z non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131771	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone x non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131770	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone v non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131769	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone u non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131768	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone t non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131767	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone s non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131766	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone r non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
Rex6	KF131765	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone q non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131764	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone p non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131763	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone o non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131762	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone n non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131761	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone m non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131760	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone l non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131759	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone k non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131758	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone j non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
Rex6	KF131757	non-LTR retrotransposon: Rex6	<i>Cichla monoculus</i> clone i non-LTR retrotransposon Rex6, partial sequence	<i>Cichla monoculus</i>
SLACS	JN608782	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-46 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608781	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-45 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608780	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-41 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608779	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-40 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608778	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-38 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608777	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-37 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608776	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-36 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608775	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-35 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608774	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-34 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608773	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-33 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608772	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-32 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608771	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-30 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608770	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-29 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608769	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-28 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608768	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-27 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608767	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate Y-26 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>

TABLE 2: RLE-type non-LTR retrotransposon elements

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TABLE 2: RLE-type non-LTR retrotransposon elements

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TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
SLACS	JN608616	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-14 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608615	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-13 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608614	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-12 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608613	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-11 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608612	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-10 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608611	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-09 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608610	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-08 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608609	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-06 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608608	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-05 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608607	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-03 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608606	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate A-01 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608236	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR40 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608235	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR39 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608234	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR38 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608233	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR35 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608232	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR32 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608231	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR29 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608230	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR26 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608229	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR25 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608228	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR18 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608227	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR17 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608226	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR14 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608225	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR13 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608224	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR12 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608223	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate mR11 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>

TABLE 2: RLE-type non-LTR retrotransposon elements

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TABLE 2: RLE-type non-LTR retrotransposon elements

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TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
SLACS	JN608047	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate ff08 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608046	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate ff06 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608045	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate ff05 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608044	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate ff04 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608043	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate ff03 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
SLACS	JN608042	non-LTR retrotransposon: SLACS-like	<i>Silene latifolia</i> isolate ff02 non-LTR retrotransposon SLACS-like, partial sequence	<i>Silene latifolia</i>
YURECi	AB097133	rt	<i>Ciona intestinalis</i> retrotransposon YURECi DNA, complete sequence	<i>Ciona intestinalis</i>
CRE	.	Cnl1	<i>C. neoformans</i> non-LTR retrotransposon - consensus.	<i>Cryptococcus neoformans</i>
CRE	.	CRE-1_ACas	CRE non-LTR retrotransposon: consensus.	<i>Acanthamoeba castellanii</i>
CRE	.	Cre-1_BM	Cre-1_BM non-LTR retrotransposon - consensus.	<i>Bombyx mori</i>
CRE	.	CRE-1_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	Cre-1_FCy	Cre-1_FCy non-LTR retrotransposon - conceptual consensus.	<i>Fragilariopsis cylindrus</i>
CRE	.	Cre-1_HM	Cre-1_HM non-LTR retrotransposon - consensus.	<i>Hydra vulgaris</i>
CRE	.	CRE-1_HRo	Cre-like non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
CRE	.	CRE-1_LSa	CRE non-LTR retrotransposon: consensus.	<i>Lactuca sativa</i>
CRE	.	Cre-1_MB	Cre-1_MB non-LTR retrotransposon - consensus.	<i>Monosiga brevicollis</i>
CRE	.	Cre-1_NV	Cre-1_NV non-LTR retrotransposon - consensus.	<i>Nematostella vectensis</i>
CRE	.	CRE-1_PXu	Non-LTR retrotransposon from <i>Papilio xuthus</i> : consensus.	<i>Papilio xuthus</i>
CRE	.	CRE-10_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-11_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-12_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-13_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-14_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-15_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-16_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-17_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	Cre-2_BM	Cre-2_BM non-LTR retrotransposon - consensus.	<i>Bombyx mori</i>
CRE	.	CRE-2_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-2_HMa	CRE non-LTR retrotransposon: consensus.	<i>Hydra vulgaris</i>
CRE	.	CRE-2_HRo	Cre-like non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
CRE	.	CRE-2_NV	CRE non-LTR retrotransposon: consensus.	<i>Nematostella vectensis</i>
CRE	.	CRE-2_PXu	Non-LTR retrotransposon from <i>Papilio xuthus</i> : consensus.	<i>Papilio xuthus</i>
CRE	.	CRE-3_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
CRE	.	CRE-3_HRo	CRE-like non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
CRE	.	CRE-3_NV	CRE non-LTR retrotransposon: consensus.	<i>Nematostella vectensis</i>
CRE	.	CRE-4_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-4_HRo	CRE-like non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
CRE	.	CRE-5_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-5_HRo	CRE-like non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
CRE	.	CRE-6_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-6_HRo	CRE-like non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
CRE	.	CRE-7_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-8_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	.	CRE-9_CCri	Non-LTR retrotransposon from the red seaweed: consensus.	<i>Chondrus crispus</i>
CRE	M33009	CRE1	<i>C. fasciculata</i> retrotransposable element (CRE1).	<i>Crithidia fasciculata</i>
CRE	U19151	CRE2	<i>C. fasciculata</i> retrotransposable element (CRE2).	<i>Crithidia fasciculata</i>
CRE	M62862	CZAR	<i>T. cruzi</i> SL-RNA-associated non-LTR retrotransposon.	<i>Trypanosoma cruzi</i>
R4	.	Dong	<i>Bombyx mori</i> non-LTR retrotransposable element.	<i>Bombyx mori</i>
R4	.	DONG_FR2	Non-LTR retrotransposon; site-specific LINE; R4/Dong superfamily; DONG_FR2.	<i>Takifugu rubripes</i>
R4	.	Dong-1_AFC	Dong/R4-type non-LTR retrotransposon - consensus.	<i>Cichlidae</i>
R4	.	Dong-1_HMM	Non-LTR retrotransposon family from <i>Heliconius melpomene melpomene</i> .	<i>Heliconius melpomene melpomene</i>
R4	.	Dong-1_NVe	A Dong non-LTR retrotransposon family from <i>Nematostella vectensis</i> .	<i>Nematostella vectensis</i>
R4	.	Dong-1_PPo	Non-LTR retrotransposon from <i>Papilio polytes</i> : consensus.	<i>Papilio polytes</i>
R4	.	Dong-1_PXu	Non-LTR retrotransposon from <i>Papilio xuthus</i> : consensus.	<i>Papilio xuthus</i>
R4	.	Dong-2_BM	Non-LTR retrotransposon - a consensus.	<i>Bombyx mori</i>
R4	.	Dong-2_HMM	Non-LTR retrotransposon family from <i>Heliconius melpomene melpomene</i> .	<i>Heliconius melpomene melpomene</i>
R4	.	Dong-2_Lch	Dong-like non-LTR retrotransposon - consensus.	<i>Latimeria chalumnae</i>
R4	.	Dong-2_PPo	Non-LTR retrotransposon from <i>Papilio polytes</i> : consensus.	<i>Papilio polytes</i>
R4	.	DongAa	A Dong non-LTR retrotransposon family from <i>Aedes aegypti</i> .	<i>Aedes aegypti</i>
R4	AB097127	DongAG	<i>Anopheles gambiae</i> non-LTR retrotransposon DongAg - a partial sequence.	<i>Anopheles gambiae</i>
R4	AB097128	EhRLE2	<i>Entamoeba histolytica</i> retrotransposon EhRLE2, complete sequence.	<i>Entamoeba histolytica</i>
R4	AB097129	EhRLE3	<i>Entamoeba histolytica</i> retrotransposon EhRLE3, complete sequence.	<i>Entamoeba histolytica</i>
HERO	.	HERO-1_AFC	Hero-type non-LTR retrotransposon - consensus.	<i>Cichlidae</i>
HERO	.	HERO-1_BF	Amphioxus HERO-1_BF autonomous non-LTR Retrotransposon - consensus.	<i>Branchiostoma floridae</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
HERO	.	HERO-1_HR	A family of HERO non-LTR retrotransposons - a consensus sequence.	<i>Helobdella robusta</i>
HERO	.	HERO-1_PP	A family of HERO non-LTR retrotransposons - a consensus sequence.	<i>Physarum polycephalum</i>
HERO	AAGJ02121261	HERO-1_SP	Sea urchin HERO-1_SP autonomous non-LTR	<i>Strongylocentrotus purpuratus</i>
HERO	.	HERO-2_BF	Retrotransposon - consensus. Amphioxus HERO-2_BF autonomous non-LTR	<i>Branchiostoma floridae</i>
HERO	.	HERO-2_DR	Retrotransposon - consensus. HERO-2_DR is a family of HERO non-LTR retrotransposons - a consensus.	<i>Danio rerio</i>
HERO	.	HERO-2_HR	Non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
HERO	048B05	Hero-2_SPur	HERO-type non-ltr retrotransposon from sea urchin.	<i>Strongylocentrotus purpuratus</i>
HERO	.	HERO-3_BF	HERO-3_BF is a family of HERO non-LTR retrotransposons - a consensus.	<i>Branchiostoma floridae</i>
HERO	.	HERO-3_DR	HERO-3_DR is a family of HERO non-LTR retrotransposons - a consensus.	<i>Danio rerio</i>
HERO	.	HERO-3_HR	Non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
HERO	.	Hero-3_SPur	HERO-type non-LTR retrotransposon from sea urchin.	<i>Strongylocentrotus purpuratus</i>
HERO	.	HERO-4_DR	HERO-4_DR is a family of HERO non-LTR retrotransposons - a consensus.	<i>Danio rerio</i>
HERO	.	HERO-4_HR	Non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
HERO	.	HERO-5_HR	Non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
HERO	.	HERO-6_HR	Non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
HERO	.	HERO-7_HR	Non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
HERO	.	HERO-8_HR	Non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
HERO	.	HERO-9_HR	Non-LTR retrotransposon: consensus sequence.	<i>Helobdella robusta</i>
HERO	.	HERODr	HERODr is a family of HERO non-LTR retrotransposons - a consensus.	<i>Danio rerio</i>
HERO	.	HEROFr	A HERO clade non-LTR Retrotransposon family - consensus.	<i>Takifugu rubripes</i>
HERO	.	HEROTn	HEROTn or Zebulon non-LTR retrotransposon - a consensus sequence.	<i>Tetraodon nigroviridis</i>
NeSL	.	LIN10B_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN11_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN13_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN14_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN15_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN2_SM	Non-LTR retrotransposon (consensus).	<i>Schmidtea mediterranea</i>
NeSL	.	LIN21_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN23_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN24_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN24B_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN25_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
NeSL	.	LIN26_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN3_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN4_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN4b_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN5_SM	Non-LTR retrotransposon from <i>Schmidtea mediterranea</i> ; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN6_SM	Non-LTR retrotransposon from <i>Schmidtea mediterranea</i> ; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN7_SM	Non-LTR retrotransposon from <i>Schmidtea mediterranea</i> ; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN7B_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	LIN9_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
CRE	JQ747487	MoTeR1	Telomere-specific non-LTR retrotransposon MoTeR1 from <i>Magnaporthe oryzae</i> .	<i>Magnaporthe oryzae</i>
CRE	JQ747488	MoTeR2	Telomere-specific non-LTR retrotransposon MoTeR2 from <i>Magnaporthe oryzae</i> .	<i>Magnaporthe oryzae</i>
NeSL	Z82058	NeSL-1	NeSL-1 is a non-LTR retrotransposon, complete sequence.	<i>Caenorhabditis elegans</i>
NeSL	.	NeSL-1_C11	A family of NeSL non-LTR retrotransposons.	<i>Caenorhabditis tropicalis</i>
NeSL	.	NeSL-1_CA	A family of NeSL non-LTR retrotransposons.	<i>Caenorhabditis angaria</i>
NeSL	.	NeSL-1_CBre	A family of NeSL non-LTR retrotransposons - consensus.	<i>Caenorhabditis bremeri</i>
NeSL	.	NeSL-1_CBri	A family of NeSL non-LTR retrotransposons.	<i>Caenorhabditis briggsae</i>
NeSL	.	NeSL-1_CJap	A family of NeSL non-LTR retrotransposons - consensus.	<i>Caenorhabditis japonica</i>
NeSL	.	NeSL-1_CRem	A family of NeSL non-LTR retrotransposons - consensus.	<i>Caenorhabditis remanei</i>
NeSL	.	NeSL-1_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	NeSL-1_TV	A family of NeSL non-LTR retrotransposons - consensus.	<i>Trichomonas vaginalis</i>
NeSL	.	NeSL-2_CBre	A family of NeSL non-LTR retrotransposons - consensus.	<i>Caenorhabditis bremeri</i>
NeSL	.	NeSL-2_CRem	A family of NeSL non-LTR retrotransposons - consensus.	<i>Caenorhabditis remanei</i>
NeSL	.	NeSL-2_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	NeSL-3_CBre	A family of NeSL non-LTR retrotransposons - consensus.	<i>Caenorhabditis bremeri</i>
NeSL	.	NeSL-3_CRem	A family of NeSL non-LTR retrotransposons - consensus.	<i>Caenorhabditis remanei</i>
NeSL	chrUn	NeSL-4_CRem	A family of NeSL non-LTR retrotransposons.	<i>Caenorhabditis remanei</i>
NeSL	.	NeSL-4_SM	Non-LTR retrotransposon; consensus.	<i>Schmidtea mediterranea</i>
R2	BN000800	PERERE-9	<i>Schistosoma mansoni</i> Perere-9 non-LTR retrotransposon (EST).	<i>Schistosoma mansoni</i>
R4	.	Plat_R4	R4 Non-LTR Retrotransposon from <i>Ornithorhynchus</i> .	<i>Ornithorhynchus</i>
R2	AF015815	R2_AM	<i>Anurida maritima</i> retrotransposon R2, complete sequence.	<i>Anurida maritima</i>
R2	M16558	R2_BM	<i>Bombyx mori</i> rDNA insertion element R2 (type II), complete cds.	<i>Bombyx mori</i>
R2	AB097121	R2_CI	R2-type LINE.	<i>Ciona intestinalis</i>
R2	.	R2_CPB	Non-LTR retrotransposon; consensus.	<i>Chrysemys picta bellii</i>
R2	.	R2_DAn	28S rDNA-specific non-LTR retrotransposon R2 in <i>Drosophila ananassae</i> .	<i>Drosophila ananassae</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	X51967	R2_DM	LINE-like retrotransposable element R2DM.	<i>Drosophila melanogaster</i>
R2	.	R2_DPe	28S rDNA-specific non-LTR retrotransposon R2 in <i>Drosophila persimilis</i> .	<i>Drosophila persimilis</i>
R2	.	R2_DPs	28S rDNA-specific non-LTR retrotransposon R2 in <i>Drosophila pseudoobscura</i> .	<i>Drosophila pseudoobscura</i>
R2	.	R2_DSe	28S rDNA-specific non-LTR retrotransposon R2 in <i>Drosophila sechellia</i> .	<i>Drosophila sechellia</i>
R2	.	R2_DSi	28S rDNA-specific non-LTR retrotransposon R2 in <i>Drosophila simulans</i> .	<i>Drosophila simulans</i>
R2	.	R2_DYa	28S rDNA-specific non-LTR retrotransposon R2 in <i>Drosophila yakuba</i> .	<i>Drosophila yakuba</i>
R2	AF015819	R2_FA	<i>Forficula auricularia</i> retrotransposon R2, complete sequence.	<i>Forficula auricularia</i>
R2	AF015816	R2_HC	<i>Hippodamia convergens</i> retrotransposon R2 reverse transcriptase gene, partial cds.	<i>Hippodamia convergens</i>
R2	GU949558	R2_KF	28S rDNA-specific non-LTR retrotransposon R2 from <i>Kaloterms flavicollis</i> .	<i>Kaloterms flavicollis</i>
R2	AF015814	R2_LP	<i>Limulus polyphemus</i> retrotransposon R2, complete sequence.	<i>Limulus polyphemus</i>
R2	AF015818	R2_PS	<i>Porcellio scaber</i> retrotransposon R2, complete sequence.	<i>Porcellio scaber</i>
R2	GU949555	R2_RL	28S rDNA-specific non-LTR retrotransposon R2 from <i>Reticulitermes lucifugus</i> .	<i>Reticulitermes lucifugus</i>
R2	GU949554	R2_RU	28S rDNA-specific non-LTR retrotransposon R2 from <i>Reticulitermes urbis</i> .	<i>Reticulitermes urbis</i>
R2	.	R2-1_AAam	R2 non-LTR retrotransposon from lone star tick.	<i>Amblyomma americanum</i>
R2	.	R2-1_ACC	R2 non-LTR retrotransposon from golden eagle.	<i>Aquila chrysaetos canadensis</i>
R2	.	R2-1_ACh	R2 non-LTR retrotransposon from rifleman.	<i>Acanthisitta chloris</i>
R2	.	R2-1_AFo	R2 non-LTR retrotransposon from emperor penguin.	<i>Aptenodytes forsteri</i>
R2	.	R2-1_AMi	R2-type non-LTR retrotransposon.	<i>Alligator mississippiensis</i>
R2	.	R2-1_AOM	R2 non-LTR retrotransposon from kiwi.	<i>Apteryx spp.</i>
R2	.	R2-1_ApA	R2 non-LTR retrotransposon from north island brown kiwi.	<i>Apteryx australis mantelli</i>
R2	.	R2-1_APi	R2 non-LTR retrotransposon from pea aphid.	<i>Acyrtosiphon pisum</i>
R2	.	R2-1_BRG	R2 non-LTR retrotransposon from East African grey crowned crane.	<i>Balearica regulorum gibbericeps</i>
R2	.	R2-1_BTe	R2 non-LTR retrotransposon from buff-tailed bumblebee.	<i>Bombus terrestris</i>
R2	.	R2-1_CAnn	R2 non-LTR retrotransposon from Anna's hummingbird.	<i>Calypte anna</i>
R2	.	R2-1_CAu	R2 non-LTR retrotransposon from turkey vulture.	<i>Cathartes aura</i>
R2	.	R2-1_CBr	R2 non-LTR retrotransposon from American crow.	<i>Corvus brachyrhynchos</i>
R2	.	R2-1_CCa	R2 non-LTR retrotransposon from chuck-will's-widow.	<i>Antrostomus carolinensis</i>
R2	.	R2-1_CCan	R2 non-LTR retrotransposon from common cuckoo.	<i>Cuculus canorus</i>
R2	.	R2-1_CPu	R2 non-LTR retrotransposon from ruff.	<i>Calidris pugnax</i>
R2	.	R2-1_Crp	Non-LTR retrotransposon.	<i>Crocodylus porosus</i>
R2	.	R2-1_CSt	R2 non-LTR retrotransposon from speckled mousebird.	<i>Colius striatus</i>
R2	.	R2-1_CU	R2 non-LTR retrotransposon from MacQueen's bustard.	<i>Chlamydotis macqueenii</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	.	R2-1__CVo	R2 non-LTR retrotransposon from killdeer.	<i>Charadrius vociferus</i>
R2	.	R2-1__DWi	28S rDNA-specific non-LTR retrotransposon R2 in <i>Drosophila willistoni</i> .	<i>Drosophila willistoni</i>
R2	.	R2-1__EGa	R2 non-LTR retrotransposon from little egret.	<i>Egretta garzetta</i>
R2	.	R2-1__FAI	R2 non-LTR retrotransposon from collared flycatcher.	<i>Ficedula albicollis</i>
R2	.	R2-1__FCh	R2 non-LTR retrotransposon from Saker falcon.	<i>Falco cherrug</i>
R2	.	R2-1__FPe	R2 non-LTR retrotransposon from peregrine falcon.	<i>Falco peregrinus</i>
R2	.	R2-1__GA	R2 non-LTR retrotransposon from three-spined stickleback.	<i>Gasterosteus aculeatus</i>
R2	.	R2-1__Gav	Non-LTR retrotransposon.	<i>Gavialis gangeticus</i>
R2	.	R2-1__GFO	R2 non-LTR retrotransposon from medium ground finch.	<i>Geospiza fortis</i>
R2	.	R2-1__GSt	R2 non-LTR retrotransposon from red-throated loon.	<i>Gavia stellata</i>
R2	.	R2-1__HAI	R2 non-LTR retrotransposon from white-tailed eagle.	<i>Haliaeetus albicilla</i>
R2	.	R2-1__IS	R2 non-LTR retrotransposon from deer tick.	<i>Ixodes scapularis</i>
R2	.	R2-1__LCh	R2-type non-LTR retrotransposon - consensus.	<i>Latimeria chalumnae</i>
R2	.	R2-1__LDi	R2 non-LTR retrotransposon from cuckoo roller.	<i>Leptosomus discolor</i>
R2	.	R2-1__LSal	non-LTR retrotransposon, consensus.	<i>Lepeophtheirus salmonis</i>
R2	.	R2-1__LV	R2 non-LTR retrotransposon from green sea urchin.	<i>Lytechinus variegatus</i>
R2	.	R2-1__MDe	R2 non-LTR retrotransposon from Hessian fly.	<i>Mayetiola destructor</i>
R2	.	R2-1__MLE	R2 non-LTR retrotransposon - consensus.	<i>Mnemopsis leidyi</i>
R2	.	R2-1__MR	R2 non-LTR retrotransposon from alfalfa leafcutter bee.	<i>Megachile rotundata</i>
R2	.	R2-1__MUn	R2 non-LTR retrotransposon fragment from budgerigar.	<i>Melopsittacus undulatus</i>
R2	.	R2-1__MUni	R2 non-LTR retrotransposon from brown mesite.	<i>Mesitornis unicolor</i>
R2	.	R2-1__MVi	R2 non-LTR retrotransposon from golden-collared manakin.	<i>Manacus vitellinus</i>
R2	.	R2-1__NNi	R2 non-LTR retrotransposon from created ibis.	<i>Nipponia nippon</i>
R2	.	R2-1__NV	Starlet sea anemone R2-1__NV autonomous Non-LTR Retrotransposon - consensus.	<i>Nematostella vectensis</i>
R2	.	R2-1__OHO	R2 non-LTR retrotransposon from hoatzin.	<i>Opisthocomus hoazin</i>
R2	.	R2-1__PAd	R2 non-LTR retrotransposon from Adelie penguin.	<i>Pygoscelis adeliae</i>
R2	.	R2-1__PBa	R2 non-LTR retrotransposon from red harvester ant.	<i>Pogonomyrmex barbatus</i>
R2	.	R2-1__PCar	R2 non-LTR retrotransposon from great cormorant.	<i>Phalacrocorax carbo</i>
R2	.	R2-1__PCau	R2 non-LTR retrotransposon sequence.	<i>Priapulus caudatus</i>
R2	.	R2-1__PCr	R2 non-LTR retrotransposon from great crested grebe.	<i>Podiceps cristatus</i>
R2	.	R2-1__PCri	R2 non-LTR retrotransposon from Dalmatian pelican.	<i>Pelecanus crispus</i>
R2	.	R2-1__PGu	R2 non-LTR retrotransposon from sandgrouse.	<i>Pterocles gutturalis</i>
R2	.	R2-1__PLe	R2 non-LTR retrotransposon from tropicbird.	<i>Phaethon lepturus</i>
R2	.	R2-1__PM	R2-1__PM is a family of R2 non-LTR retrotransposons - consensus.	<i>Petromyzon marinus</i>
R2	.	R2-1__PPap	R2 non-LTR retrotransposon from sand fly.	<i>Phlebotomus papatasi</i>
R2	.	R2-1__PPu	R2 non-LTR retrotransposon from downy woodpecker.	<i>Picoides pubescens</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	.	R2-1_PRR	R2 non-LTR retrotransposon from American flamingo.	<i>Phoenicopterus ruber ruber</i>
R2	.	R2-1_PSi	R2 non-LTR retrotransposon from Chinese soft-shelled turtle.	<i>Pelodiscus sinensis</i>
R2	.	R2-1_RMi	R2 non-LTR retrotransposon from brown tick.	<i>Rhipicephalus microplus</i>
R2	.	R2-1_RPr	R2 non-LTR retrotransposon sequence.	<i>Rhodnius prolixus</i>
R2	.	R2-1_RPu	R2 non-LTR retrotransposon cDNA sequence from brown tick.	<i>Rhipicephalus pulchellus</i>
R2	.	R2-1_SCa	R2 non-LTR retrotransposon from Atlantic canary.	<i>Serinus canaria</i>
R2	.	R2-1_SK	R2 non-LTR retrotransposon from acorn worm.	<i>Saccoglossus kowalevskii</i>
R2	.	R2-1_SM	R2-type retrotransposon from <i>Schmidtea mediterranea</i> : consensus.	<i>Schmidtea mediterranea</i>
R2	.	R2-1_SP	R2 non-LTR retrotransposon from purple sea urchin.	<i>Strongylocentrotus purpuratus</i>
R2	AGKD01072455	R2-1_SSa	R2-type non-LTR retrotransposon.	<i>Salmo salar</i>
R2	.	R2-1_StC	R2 non-LTR retrotransposon from ostrich.	<i>Struthiocamelus australis</i>
R2	.	R2-1_TAI	R2 non-LTR retrotransposon from barn owl.	<i>Tyto alba</i>
R2	.	R2-1_TCas	R2 non-LTR retrotransposon from red flour beetle - consensus	<i>Tribolium castaneum</i>
R2	.	R2-1_TG	A family of R2 non-LTR retrotransposons - consensus sequence.	<i>Taeniopygia guttata</i>
R2	.	R2-1_TGut	R2 non-LTR retrotransposon from white-throated tinamou.	<i>Tinamus guttatus</i>
R2	.	R2-1_TSP	A family of R2 non-LTR retrotransposons in the <i>Trichinella spiralis</i> genome - a consensus.	<i>Trichinella spiralis</i>
R2	scaffold_6	R2-1_TUr	R2 non-LTR retrotransposon from twospotted spider mite.	<i>Tetranychus urticae</i>
R2	.	R2-1_XM	R2 non-LTR retrotransposon fragment from Southern platyfish.	<i>Xiphophorus maculatus</i>
R2	.	R2-1_ZA	R2 non-LTR retrotransposon from white-throated sparrow.	<i>Zonotrichia albicollis</i>
R2	.	R2-1_ZLM	R2 non-LTR retrotransposon from silveryeye.	<i>Zosterops lateralis melanops</i>
R2	.	R2-2_API	R2 non-LTR retrotransposon from pea aphid.	<i>Acyrtosiphon pisum</i>
R2	.	R2-2_CCan	R2 non-LTR retrotransposon from common cuckoo.	<i>Cuculus canorus</i>
R2	.	R2-2_CMa	R2 non-LTR retrotransposon from Macqueen's bustard.	<i>Chlamydotis macqueenii</i>
R2	.	R2-2_DWi	28S rDNA-specific non-LTR retrotransposon R2 in <i>Drosophila willistoni</i> .	<i>Drosophila willistoni</i>
R2	.	R2-2_HAI	R2 non-LTR retrotransposon from white-tailed eagle.	<i>Haliaeetus albicilla</i>
R2	.	R2-2_IS	R2 non-LTR retrotransposon from deer tick.	<i>Ixodes scapularis</i>
R2	.	R2-2_MR	R2 non-LTR retrotransposon from alfalfa leafcutter bee.	<i>Megachile rotundata</i>
R2	.	R2-2_MUn	R2 non-LTR retrotransposon fragment from budgerigar.	<i>Melopsittacus undulatus</i>
R2	.	R2-2_MUni	R2 non-LTR retrotransposon from brown mesite.	<i>Mesitornis unicolor</i>
R2	.	R2-2_NNi	R2 non-LTR retrotransposon from created ibis.	<i>Nipponia nippon</i>
R2	.	R2-2_NV	Starlet sea anemone R2-2_NV autonomous Non-LTR Retrotransposon - consensus.	<i>Nematostella vectensis</i>
R2	.	R2-2_PBa	R2 non-LTR retrotransposon from red harvester ant.	<i>Pogonomyrmex barbatus</i>
R2	.	R2-2_PM	R2-2_PM is a family of R2 non-LTR retrotransposons - a consensus.	<i>Petromyzon marinus</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	.	R2-2__RPr	R2 non-LTR retrotransposon sequence.	<i>Rhodnius prolixus</i>
R2	.	R2-2__SMed	R2 non-LTR retrotransposon from <i>Schmidtea mediterranea</i> : consensus.	<i>Schmidtea mediterranea</i>
R2	.	R2-2__TCas	R2 non-LTR retrotransposon from red flour beetle.	<i>Tribolium castaneum</i>
R2	scaffold_37	R2-2__TUr	R2 non-LTR retrotransposon from twospotted spider mite.	<i>Tetranychus urticae</i>
R2	ABJB010555169	R2-3__IS	R2 non-LTR retrotransposon from deer tick.	<i>Ixodes scapularis</i>
R2	.	R2-3__MR	R2 non-LTR retrotransposon from alfalfa leafcutter bee.	<i>Megachile rotundata</i>
R2	.	R2-4__MR	R2 non-LTR retrotransposon from alfalfa leafcutter bee.	<i>Megachile rotundata</i>
R2	.	R2-5__MR	R2 non-LTR retrotransposon from alfalfa leafcutter bee.	<i>Megachile rotundata</i>
R2	.	R2-6__MR	R2 non-LTR retrotransposon from alfalfa leafcutter bee.	<i>Megachile rotundata</i>
R2	.	R2-7__MR	R2 non-LTR retrotransposon from alfalfa leafcutter bee.	<i>Megachile rotundata</i>
R2	.	R2-8__MR	R2 non-LTR retrotransposon from alfalfa leafcutter bee.	<i>Megachile rotundata</i>
R2	.	R2-N1__Gav	Non-LTR retrotransposon.	<i>Gavialis gangeticus</i>
R2	.	R2-N2__Gav	Non-LTR retrotransposon.	<i>Gavialis gangeticus</i>
R2	.	R2-N2B__Gav	Non-LTR retrotransposon.	<i>Gavialis gangeticus</i>
R2	.	R2A__NVi	28S rDNA-specific non-LTR retrotransposon R2 in <i>Nasonia vitripennis</i> .	<i>Nasonia vitripennis</i>
R2	AF015817	R2A__TM	<i>Tenebrio molitor</i> retrotransposon R2 reverse transcriptase gene, partial cds.	<i>Tenebrio molitor</i>
R2	.	R2Amel	R2Amel - R2 non-LTR retrotransposon from the honeybee <i>Apis mellifera</i> .	<i>Apis mellifera</i>
R2	AF015685	R2B__DM	<i>Drosophila mercatorum</i> R2 retrotransposon reverse transcriptase domain protein gene, complete cds.	<i>Drosophila mercatorum</i>
R2	.	R2B__NVi	28S rDNA-specific non-LTR retrotransposon R2 in <i>Nasonia vitripennis</i> .	<i>Nasonia vitripennis</i>
R2	AF015822	R2B__TM	<i>Tenebrio molitor</i> retrotransposon R2 reverse transcriptase gene, partial cds.	<i>Tenebrio molitor</i>
R2	.	R2C__NGi	28S rDNA-specific non-LTR retrotransposon R2 in <i>Nasonia giraulti</i> .	<i>Nasonia giraulti</i>
R2	AB097122	R2Ci-B	<i>Ciona intestinalis</i> retrotransposon R2Ci-B, complete sequence.	<i>Ciona intestinalis</i>
R2	.	R2Ci-D	<i>Ciona intestinalis</i> retrotransposon R2CiD, complete sequence.	<i>Ciona intestinalis</i>
R2	AB097121	R2CIA__CI	<i>Ciona intestinalis</i> retrotransposon R2Ci-A, complete sequence.	<i>Ciona intestinalis</i>
R2	AB097125	R2Cs-D	<i>Ciona intestinalis</i> retrotransposon R2CsD, partial sequence.	<i>Ciona savignyi</i>
R2	.	R2D__NGi	28S rDNA-specific non-LTR retrotransposon R2 in <i>Nasonia giraulti</i> .	<i>Nasonia giraulti</i>
R2	NM__001030097	R2Dr	R2 non-LTR retrotransposon in the <i>Danio rerio</i> genome - a single copy.	<i>Danio rerio</i>
R2	.	R2E__NLo	28S rDNA-specific non-LTR retrotransposon R2 in <i>Nasonia longicornis</i> .	<i>Nasonia longicornis</i>
R2	AB201408	R2Eb	R2 non-LTR retrotransposon from <i>Eptatretus burgeri</i> .	<i>Eptatretus burgeri</i>
R2	AB201415	R2Ha	R2 non-LTR retrotransposon from <i>Hasarius adansoni</i> .	<i>Hasarius adansoni</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	JN937617	R2La	R2-type non-LTR retrotransposon.	<i>Lepidurus arcticus</i>
R2	.	R2LcA	R2-type non-LTR retrotransposon.	<i>Lepidurus couesii</i>
R2	JN937619	R2LcB	R2-type non-LTR retrotransposon.	<i>Lepidurus couesii</i>
R2	.	R2LcC	R2-type non-LTR retrotransposon.	<i>Lepidurus couesii</i>
R2	JN937616	R2LI	R2-type non-LTR retrotransposon.	<i>Lepidurus apus lubbocki</i>
R2	AB201414	R2Mr	R2 non-LTR retrotransposon from <i>Metacrinus rotundus</i> .	<i>Metacrinus rotundus</i>
R2	.	R2NS-1_CGi	R2-type retrotransposon from <i>Crassostrea gigas</i> .	<i>Crassostrea gigas</i>
R2	.	R2NS-1_CSi	R2-type retrotransposon from <i>Clonorchis sinensis</i> : consensus.	<i>Clonorchis sinensis</i>
R2	.	R2NS-1_PMi	R2-like non-LTR retrotransposon from bat star.	<i>Patiria miniata</i>
R2	.	R2NS-1_SMed	R2-type retrotransposon from <i>Schmidtea mediterranea</i> : consensus.	<i>Schmidtea mediterranea</i>
R2	.	R2Nvec-A	R2Nvec-A - R2 non-LTR retrotransposon from the starlet sea anemone <i>Nematostella vectensis</i> .	<i>Nematostella vectensis</i>
R2	.	R2OI-A	R2 non-LTR retrotransposon from the medaka <i>Oryzias latipes</i> - consensus.	<i>Oryzias latipes</i>
R2	AB201416	R2Pc	R2 non-LTR retrotransposon from <i>Procambarus clarkii</i> .	<i>Procambarus clarkii</i>
R2	.	R2Sm-A	R2Sm-A - R2 non-LTR retrotransposon from the bloodfluke <i>Schistosoma mansoni</i> .	<i>Schistosoma mansoni</i>
R2	AB201409	R2Ta	R2 non-LTR retrotransposon from <i>Tanichthys albonubes</i> .	<i>Tanichthys albonubes</i>
R2	EU854578	R2Tc	R2-type non-LTR retrotransposon.	<i>Triops cancriformis</i>
R2	JN937621	R2Tc_it	R2-type non-LTR retrotransposon.	<i>Triops cancriformis</i>
R2	AB201417	R2TI	R2 non-LTR retrotransposon from <i>Triops longicaudatus</i> .	<i>Triops longicaudatus</i>
R4	U29445	R4_AL	<i>Ascaris lumbricoides</i> site-specific non-LTR retrotransposable element R4 in 26S rDNA, complete sequence.	<i>Ascaris lumbricoides</i>
R4	U29590	R4_HC	<i>Haemonchus contortus</i> non-LTR retrotransposon specific to the large subunit rRNA genes of nematodes.	<i>Haemonchus contortus</i>
R4	.	R4_Hmel	a R4 element from <i>Heliconius melpomene</i> .	<i>Heliconius melpomene</i>
R4	.	R4-1_AC	A family of R4 non-LTR retrotransposons - consensus sequence.	<i>Anolis carolinensis</i>
R4	.	R4-1_ADi	R4-type retrotransposon: consensus.	<i>Acropora digitifera</i>
R4	.	R4-1_BM	Non-LTR retrotransposon - a consensus.	<i>Bombyx mori</i>
R4	CADV01008175	R4-1_BX	An R4 non-LTR retrotransposon family from <i>Bursaphelenchus xylophilus</i> .	<i>Bursaphelenchus xylophilus</i>
R4	ABLE03011482	R4-1_CJap	An R4 non-LTR retrotransposon family from <i>Caenorhabditis japonica</i> .	<i>Caenorhabditis japonica</i>
R4	.	R4-1_CM	Non-LTR retrotransposon from the elephant shark - consensus.	<i>Callorhynchus milii</i>
R4	.	R4-1_CPB	Non-LTR retrotransposon: consensus.	<i>Chrysomyspicta bellii</i>
R4	.	R4-1_ED	Autonomous non-LTR retrotransposon from the R4 clade - a consensus sequence.	<i>Entamoeba dispar</i>
R4	.	R4-1_HG	An R4 non-LTR retrotransposon family from <i>Heterodera glycines</i> .	<i>Heterodera glycines</i>
R4	.	R4-1_HMe	Non-LTR retrotransposon family from <i>Heliconius melpomene melpomene</i> .	<i>Heliconius melpomene melpomene</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R4	CABB01003843	R4-1_MI	An R4 non-LTR retrotransposon family from <i>Meloidogyne incognita</i> .	<i>Meloidogyne incognita</i>
R4	.	R4-1_PH	Non-LTR Retrotransposon, consensus.	<i>Parhyale hawaiiensis</i>
R4	CACX01002001	R4-1_SRa	An R4 non-LTR retrotransposon family from <i>Strongyloides ratti</i> .	<i>Strongyloides ratti</i>
R4	.	R4-1_TCa	R4-type retrotransposon: consensus.	<i>Tribolium castaneum</i>
R4	.	R4-1B_AC	Dong-type non-LTR retrotransposons - a consensus sequence.	<i>Anolis carolinensis</i>
R4	.	R4-2_AS	An R4 non-LTR retrotransposon family from <i>Ascaris suum</i> .	<i>Ascaris suum</i>
R4	CADV01009048	R4-2_BX	An R4 non-LTR retrotransposon family from <i>Bursaphelenchus xylophilus</i> .	<i>Bursaphelenchus xylophilus</i>
R4	ABLA01000389	R4-2_HG	An R4 non-LTR retrotransposon family from <i>Heterodera glycines</i> .	<i>Heterodera glycines</i>
R4	CACX01002006	R4-2_SRa	An R4 non-LTR retrotransposon family from <i>Strongyloides ratti</i> .	<i>Strongyloides ratti</i>
R4	CADV01008832	R4-3_BX	An R4 non-LTR retrotransposon family from <i>Bursaphelenchus xylophilus</i> .	<i>Bursaphelenchus xylophilus</i>
R4	.	R4-3_SRa	An R4 non-LTR retrotransposon family from <i>Strongyloides ratti</i> .	<i>Strongyloides ratti</i>
R4	.	R4-4_BX	An R4 non-LTR retrotransposon family from <i>Bursaphelenchus xylophilus</i> .	<i>Bursaphelenchus xylophilus</i>
R4	.	R4-4_SRa	An R4 non-LTR retrotransposon family from <i>Strongyloides ratti</i> .	<i>Strongyloides ratti</i>
R4	.	R4-5_BX	An R4 non-LTR retrotransposon family from <i>Bursaphelenchus xylophilus</i> .	<i>Bursaphelenchus xylophilus</i>
NeSL	AY216701	R5	<i>Girardia tigrina</i> R5 retrotransposon, complete sequence.	<i>Girardia tigrina</i>
NeSL	.	R5-1_SM	A family of planarian NeSL non-LTR retrotransposons - consensus.	<i>Schmidtea mediterranea</i>
NeSL	.	R5-2_SM	A family of planarian NeSL non-LTR retrotransposons - consensus.	<i>Schmidtea mediterranea</i>
R2	.	R8Hm-A	R8Hm-A - 18S rDNA-specific non-LTR retrotransposon from <i>Hydra magnipapillata</i> .	<i>Hydra vulgaris</i>
R2	.	R8Hm-B	R8Hm-B - 18S rDNA-specific non-LTR retrotransposon from <i>Hydra magnipapillata</i> .	<i>Hydra vulgaris</i>
R2	.	R9Av	R9Av, an rDNA-specific non-LTR retrotransposon family from rotifer.	<i>Adineta vaga</i>
R2	FJ461304	RaR2	28S rDNA-specific non-LTR retrotransposon R2 from <i>Rhynchosciara americana</i> .	<i>Rhynchosciara americana</i>
R4	.	Rex6	Non-LTR retrotransposon; site-specific LINE; R4/Dong superfamily; REX6; DONG_FR.	<i>Takifugu rubripes</i>
R4	.	Rex6-1_OL	A Rex6 non-LTR retrotransposon family from <i>Oryzias latipes</i> .	<i>Oryzias latipes</i>
CRE	X17078	SLACS	<i>Trypanosoma brucei</i> DNA for retrotransposable element SLACS.	<i>Trypanosoma brucei</i>
NeSL	.	Utopia-1_ACa	Utopia-1_ACa is a protozoan Utopia non-LTR retrotransposon - a complete sequence.	<i>Acanthamoeba castellanii</i>
NeSL	scaffold_474	Utopia-1_ACar	A family of NeSL non-LTR retrotransposons.	<i>Anolis carolinensis</i>
NeSL	.	Utopia-1_AEc	A family of Utopia non-LTR retrotransposons - consensus.	<i>Acromyrmex echinator</i>
NeSL	.	Utopia-1_AMi	A family of NeSL non-LTR retrotransposons - consensus.	<i>Alligator mississippiensis</i>
NeSL	.	Utopia-1_API	A family of Utopia non-LTR retrotransposons - consensus.	<i>Acyrtosiphon pisum</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
NeSL	.	Utopia-1__API	A family of Utopia non-LTR retrotransposons.	<i>Agrilus planipennis</i>
NeSL	.	Utopia-1__CFI	A family of Utopia non-LTR retrotransposons - consensus.	<i>Camponotus floridanus</i>
NeSL	.	Utopia-1__CMY	A family of Utopia non-LTR retrotransposons - consensus.	<i>Chelonia mydas</i>
NeSL	.	Utopia-1__CPB	A family of Utopia non-LTR retrotransposons - consensus.	<i>Chrysomyspicta bellii</i>
NeSL	.	Utopia-1__Cp	Non-LTR retrotransposon.	<i>Crocodylus porosus</i>
NeSL	.	Utopia-1__DPo	A family of Utopia non-LTR retrotransposons.	<i>Dendroctonus ponderosae</i>
NeSL	.	Utopia-1__DPu	A family of Utopia non-LTR retrotransposons - consensus.	<i>Daphnia pulex</i>
NeSL	.	Utopia-1__DYak	A family of Utopia non-LTR retrotransposons - consensus.	<i>Drosophila yakuba</i>
NeSL	.	Utopia-1__EBr	A family of Utopia non-LTR retrotransposons - consensus.	<i>Eimeria brunetti</i>
NeSL	.	Utopia-1__EMi	A family of Utopia non-LTR retrotransposons - consensus.	<i>Eimeria mitis</i>
NeSL	.	Utopia-1__ENe	A family of Utopia non-LTR retrotransposons - consensus.	<i>Eimeria necatrix</i>
NeSL	.	Utopia-1__Gav	Non-LTR retrotransposon.	<i>Gavialis gangeticus</i>
NeSL	.	Utopia-1__GG1	A family of Utopia non-LTR retrotransposons.	<i>Ganaspis</i>
NeSL	.	Utopia-1__HAr	A family of Utopia non-LTR retrotransposons.	<i>Hyaloperonospora arabidopsidis</i>
NeSL	.	Utopia-1__HG	A family of Utopia non-LTR retrotransposons.	<i>Heterodera glycines</i>
NeSL	.	Utopia-1__HMM	A family of Utopia non-LTR retrotransposons.	<i>Heliconius melpomene</i>
NeSL	.	Utopia-1__HSal	A family of Utopia non-LTR retrotransposons - consensus.	<i>Harpegnathos saltator</i>
NeSL	.	Utopia-1__IS	A family of Utopia non-LTR retrotransposons - consensus.	<i>Ixodes scapularis</i>
NeSL	.	Utopia-1__LAI	A family of Utopia non-LTR retrotransposons.	<i>Lasioglossum albipes</i>
NeSL	.	Utopia-1__LFu	A family of Utopia non-LTR retrotransposons.	<i>Ladona fulva</i>
NeSL	AGCV01358106	Utopia-1__LV	A family of Utopia non-LTR retrotransposons.	<i>Lytechinus variegatus</i>
NeSL	.	Utopia-1__MRo	A family of Utopia non-LTR retrotransposons - consensus.	<i>Megachile rotundata</i>
NeSL	.	Utopia-1__NVit	A family of Utopia non-LTR retrotransposons - consensus.	<i>Nasonia vitripennis</i>
NeSL	.	Utopia-1__PAIni	NeSL non-LTR retrotransposon from <i>Phytophthora alni</i> .	<i>Phytophthora alni</i>
NeSL	.	Utopia-1__PArrh	NeSL non-LTR retrotransposon from <i>Pythium arrhenomanes</i> .	<i>Pythium arrhenomanes</i>
NeSL	.	Utopia-1__PBa	A family of Utopia non-LTR retrotransposons - consensus.	<i>Pogonomyrmex barbatus</i>
NeSL	.	Utopia-1__PCa	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora capsici</i>
NeSL	.	Utopia-1__PCinn	NeSL non-LTR retrotransposon from <i>Phytophthora cinnamomi</i> .	<i>Phytophthora cinnamomi</i>
NeSL	AHJF01004292	Utopia-1__PCu	A family of Utopia non-LTR retrotransposons - consensus.	<i>Pseudoperonospora cubensis</i>
NeSL	.	Utopia-1__PI	A family of NeSL non-LTR retrotransposons - consensus.	<i>Phytophthora infestans</i>
NeSL	.	Utopia-1__PInsi	NeSL non-LTR retrotransposon from <i>Pythium insidiosum</i> .	<i>Pythium insidiosum</i>
NeSL	.	Utopia-1__PKern	NeSL non-LTR retrotransposon from <i>Phytophthora kernoviae</i> .	<i>Phytophthora kernoviae</i>
NeSL	.	Utopia-1__PLate	NeSL non-LTR retrotransposon from <i>Phytophthora lateralis</i> .	<i>Phytophthora lateralis</i>
NeSL	.	Utopia-1__PMi	A family of Utopia non-LTR retrotransposons.	<i>Patiria miniata</i>
NeSL	.	Utopia-1__PPac	A family of Utopia non-LTR retrotransposons.	<i>Pristionchus pacificus</i>
NeSL	.	Utopia-1__PPini	NeSL non-LTR retrotransposon from <i>Phytophthora pinifolia</i> .	<i>Phytophthora pinifolia</i>
NeSL	.	Utopia-1__PR	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora ramorum</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
NeSL	.	Utopia-1_PRe	A family of Utopia non-LTR retrotransposons - consensus.	<i>Panagrellus redivivus</i>
NeSL	.	Utopia-1_PS	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora sojae</i>
NeSL	.	Utopia-1_PSi	A family of Utopia non-LTR retrotransposons - consensus.	<i>Pelodiscus sinensis</i>
NeSL	.	Utopia-1_PT	A family of Utopia non-LTR retrotransposons.	<i>Parasteatoda tepidariorum</i>
NeSL	ADOS01001321	Utopia-1_PU	A family of Utopia non-LTR retrotransposons.	<i>Pythium ultimum</i>
NeSL	.	Utopia-1_PVexa	NeSL non-LTR retrotransposon from <i>Phytophthora vexans</i> .	<i>Phytophthora</i> aff. <i>vexans</i>
NeSL	.	Utopia-1_SaPa	A family of Utopia non-LTR retrotransposons.	<i>Saprolegnia parasitica</i>
NeSL	.	Utopia-1_SDicl	NeSL non-LTR retrotransposon from <i>Saprolegnia diclina</i> .	<i>Saprolegnia diclina</i>
NeSL	.	Utopia-1_SM	A family of Utopia non-LTR retrotransposons.	<i>Strigamia maritima</i>
NeSL	AAGJ02140537	Utopia-1_SP	A family of Utopia non-LTR retrotransposons.	<i>Strongylocentrotus purpuratus</i>
NeSL	.	Utopia-1_TSP	A family of Utopia non-LTR retrotransposons.	<i>Trichinella spiralis</i>
NeSL	.	Utopia-1B_CPB	A family of Utopia non-LTR retrotransposons - consensus.	<i>Chrysemys picta bellii</i>
NeSL	.	Utopia-2_API	A family of Utopia non-LTR retrotransposons.	<i>Acyrtosiphon pisum</i>
NeSL	.	Utopia-2_CMy	A family of Utopia non-LTR retrotransposons - consensus.	<i>Chelonia mydas</i>
NeSL	.	Utopia-2_CPB	A family of Utopia non-LTR retrotransposons - consensus.	<i>Chrysemys picta bellii</i>
NeSL	.	Utopia-2_DPu	A family of Utopia non-LTR retrotransposons.	<i>Daphnia pulex</i>
NeSL	.	Utopia-2_LFu	A family of Utopia non-LTR retrotransposons.	<i>Ladona fulva</i>
NeSL	.	Utopia-2_PCa	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora capsici</i>
NeSL	.	Utopia-2_PI	A family of NeSL non-LTR retrotransposons - consensus.	<i>Phytophthora infestans</i>
NeSL	.	Utopia-2_PR	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora ramorum</i>
NeSL	.	Utopia-2_PS	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora sojae</i>
NeSL	.	Utopia-2_PU	A family of Utopia non-LTR retrotransposons.	<i>Pythium ultimum</i>
NeSL	.	Utopia-3_CPB	A family of Utopia non-LTR retrotransposons - consensus.	<i>Chrysemys picta bellii</i>
NeSL	.	Utopia-3_DPu	A family of Utopia non-LTR retrotransposons.	<i>Daphnia pulex</i>
NeSL	.	Utopia-3_LFu	A family of Utopia non-LTR retrotransposons.	<i>Ladona fulva</i>
NeSL	.	Utopia-3_PCa	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora capsici</i>
NeSL	.	Utopia-3_PI	A family of NeSL non-LTR retrotransposons - consensus.	<i>Phytophthora infestans</i>
NeSL	.	Utopia-3_PR	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora ramorum</i>
NeSL	.	Utopia-4_LFu	A family of Utopia non-LTR retrotransposons.	<i>Ladona fulva</i>
NeSL	AATU01001281.1	Utopia-4_PI	A family of NeSL non-LTR retrotransposons - a copy.	<i>Phytophthora infestans</i>
NeSL	.	Utopia-4_PR	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora ramorum</i>
NeSL	.	Utopia-5_LFu	A family of Utopia non-LTR retrotransposons.	<i>Ladona fulva</i>
NeSL	.	Utopia-5_PI	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora infestans</i>
NeSL	.	Utopia-5_PR	A family of Utopia non-LTR retrotransposons - consensus.	<i>Phytophthora ramorum</i>
NeSL	.	Utopia-6_LFu	A family of Utopia non-LTR retrotransposons.	<i>Ladona fulva</i>
R4	.	X4_LINE	Conserved LINE element reconstructed from the human genome - consensus.	<i>Vertebrata</i>
NeSL	.	YURE_CSa	A NeSL non-LTR retrotransposon from <i>Ciona savignyi</i> .	<i>Ciona savignyi</i>

TABLE 2-continued

TABLE 2: RLE-type non-LTR retrotransposon elements				
Family	Accession	Mobile Element	Name/Description	Organism
R2	.	YURE-2_Cis	YURE non-LTR retrotransposon from <i>Ciona savignyi</i> .	<i>Ciona savignyi</i>
NeSL	.	YURECi	<i>Ciona intestinalis</i> retrotransposon YURECi.	<i>Ciona intestinalis</i>

A skilled artisan can, based on the Accession numbers provided in Tables 1-3 determine the nucleic acid and corresponding polypeptide sequences of each retrotransposon and domains thereof, e.g., by using routine sequence analysis tools as Basic Local Alignment Search Tool (BLAST) or CD-Search for conserved domain analysis. Other sequence analysis tools are known and can be found, e.g., at molbiol-tools.ca, for example, at molbiol-tools.ca/Motifs.htm. SEQ ID NOs 1-112 align with each row in Table 1, and SEQ ID NOs 113-1015 align with the first 903 rows of Table 2.

Tables 1-3 herein provide the sequences of exemplary transposons, including the amino acid sequence of the retrotransposase, and sequences of 5' and 3' untranslated regions to allow the retrotransposase to bind the template RNA, and the full transposon nucleic acid sequence. In some embodiments, a 5' UTR of any of Tables 1-3 allows the retrotransposase to bind the template RNA. In some embodiments, a 3' UTR of any of Tables 1-3 allows the retrotransposase to bind the template RNA. Thus, in some embodiments, a polypeptide for use in any of the systems described herein can be a polypeptide of any of Tables 1-3 herein, or a sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto. In some embodiments, the system further comprises one or both of a 5' or 3' untranslated region of any of Tables 1-3 herein (or a sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto), e.g., from the same transposon as the polypeptide referred to in the preceding sentence, as indicated in the same row of the same table. In some embodiments, the system comprises one or both of a 5' or 3' untranslated region of any of Tables 1-3 herein, e.g., a segment of the full transposon sequence that encodes an RNA that is capable of binding a retrotransposase, and/or the sub-sequence provided in the column entitled Predicted 5' UTR or Predicted 3' UTR.

In some embodiments, a polypeptide for use in any of the systems described herein can be a molecular reconstruction or ancestral reconstruction based upon the aligned polypeptide sequence of multiple retrotransposons. In some embodiments, a 5' or 3' untranslated region for use in any of the systems described herein can be a molecular reconstruction based upon the aligned 5' or 3' untranslated region of multiple retrotransposons. A skilled artisan can, based on the Accession numbers provided herein, align polypeptides or nucleic acid sequences, e.g., by using routine sequence analysis tools as Basic Local Alignment Search Tool (BLAST) or CD-Search for conserved domain analysis. Molecular reconstructions can be created based upon sequence consensus, e.g. using approaches described in Ivics et al., *Cell* 1997, 501 510; Wagstaff et al., *Molecular Biology and Evolution* 2013, 88-99. In some embodiments, the retrotransposon from which the 5' or 3' untranslated region or polypeptide is derived is a young or a recently active mobile element, as assessed via phylogenetic methods

such as those described in Boissinot et al., *Molecular Biology and Evolution* 2000, 915-928.

Table 3 (below) shows exemplary GENE WRITER™ proteins and associated sequences from a variety of retrotransposases, identified using data mining. Column 1 indicates the family to which the retrotransposon belongs. Column 2 lists the element name. Column 3 indicates an accession number, if any. Column 4 lists an organism in which the retrotransposase is found. Column 5 lists the DNA sequence of the retrotransposon. Column 6 lists the predicted 5' untranslated region, and column 7 lists the predicted 3' untranslated region; both are segments of the sequence of column 5 that are predicted to allow the template RNA to bind the retrotransposase of column 8. (It is understood that columns 5-7 show the DNA sequence, and that an RNA sequence according to any of columns 5-7 would typically include uracil rather than thymidine.) Column 8 lists the predicted retrotransposase sequence encoded in the retrotransposon of column 5.

Lengthy table referenced here

US12398392-20250826-T00001

Please refer to the end of the specification for access instructions.

Gene Writers, e.g. Thermostable GENE WRITER™ Genome Editor Polypeptides

While not wishing to be bound by theory, in some embodiments, retrotransposases that evolved in cold environments may not function as well at human body temperature. This application provides a number of thermostable GENE WRITER™ genome editor polypeptides, including proteins derived from avian retrotransposases. Exemplary avian transposase sequences in Table 3 include those of *Taeniopygia guttata* (zebra finch; transposon name R2-1_TG), *Geospiza fortis* (medium ground finch; transposon name R2-1_Gfo), *Zonotrichia albicollis* (white-throated sparrow; transposon name R2-1_ZA), and *Tinamus guttatus* (white-throated tinamou; transposon name R2-1_TGut).

Thermostability may be measured, e.g., by testing the ability of a GENE WRITER™ to polymerize DNA in vitro at a high temperature (e.g., 37° C.) and a low temperature (e.g., 25° C.). Suitable conditions for assaying in vitro DNA polymerization activity (e.g., processivity) are described, e.g., in Bibillo and Eickbush, "High Processivity of the Reverse Transcriptase from a Non-long Terminal Repeat Retrotransposon" (2002) JBC 277, 34836-34845. In some embodiments, the thermostable GENE WRITER™ polypeptide has an activity, e.g., a DNA polymerization activity, at 37° C. that is no less than 70%, 75%, 80%, 85%, 90%, or 95% of its activity at 25° C. under otherwise similar conditions.

In some embodiments, a GENE WRITER™ polypeptide (e.g., a sequence of Table 1, 2, or 3 or a sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or

99% identity thereto) is stable in a subject chosen from a mammal (e.g., human) or a bird. In some embodiments, a GENE WRITER™ polypeptide described herein is functional at 37° C. In some embodiments, a GENE WRITER™ polypeptide described herein has greater activity at 37° C. than it does at a lower temperature, e.g., at 30° C., 25° C., or 20° C. In some embodiments, a GENE WRITER™ polypeptide described herein has greater activity in a human cell than in a zebrafish cell.

In some embodiments, a GENE WRITER™ polypeptide is active in a human cell cultured at 37° C., e.g., using an assay of Example 6 or Example 7 herein.

In some embodiments, the assay comprises steps of: (1) introducing HEK293T cells into one or more wells of 6.4 mm diameter, at 10,000 cells/well, (2) incubating the cells at 37° C. for 24 hr, (3) providing a transfection mixture comprising 0.5 µl of FuGENE® HD transfection reagent and 80 ng DNA (wherein the DNA is a plasmid comprising, in order, (a) CMV promoter, (b) 100 bp of sequence homologous to the 100 bp upstream of the target site, (c) sequence encoding a 5' untranslated region that binds the GENE WRITER™ protein, (d) sequence encoding the GENE WRITER™ protein, (e) sequence encoding a 3' untranslated region that binds the GENE WRITER™ protein (f) 100 bp of sequence homologous to the 100 bp downstream of the target site, and (g) BGH polyadenylation sequence) and 10 µl Opti-MEM and incubating for 15 min at room temperature, (4) adding the transfection mixture to the cells, (5) incubating the cells for 3 days, and (6) assaying integration of the exogenous sequence into a target locus (e.g., rDNA) in the cell genome, e.g., wherein one or more of the preceding steps are performed as described in Example 6 herein.

In some embodiments, the GENE WRITER™ polypeptide results in insertion of the heterologous object sequence (e.g., the GFP gene) into the target locus (e.g., rDNA) at an average copy number of at least 0.01, 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2, 2.5, 3, 4, or 5 copies per genome. In some embodiments, a cell described herein (e.g., a cell comprising a heterologous sequence at a target insertion site) comprises the heterologous object sequence at an average copy number of at least 0.01, 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2, 2.5, 3, 4, or 5 copies per genome. In some embodiments, a GENE WRITER™ causes integration of a sequence in a target RNA with relatively few truncation events at the terminus. For instance, in some embodiments, a GENE WRITER™ protein (e.g., of SEQ ID NO: 1016) results in about 25-100%, 50-100%, 60-100%, 70-100%, 75-95%, 80%-90%, or 86.17% of integrants into the target site being non-truncated, as measured by an assay described herein, e.g., an assay of Example 6 and FIG. 8. In some embodiments, a GENE WRITER™ protein (e.g., of SEQ ID NO: 1016) results in at least about 30%, 40%, 50%, 60%, 70%, 80%, or 90% of integrants into the target site being non-truncated, as measured by an assay described herein. In some embodiments, an integrant is classified as truncated versus non-truncated using an assay comprising amplification with a forward primer situated 565 bp from the end of the element (e.g., a wild-type transposon sequence, e.g., of *Taeniopygia guttata*) and a reverse primer situated in the genomic DNA of the target insertion site, e.g., rDNA. In some embodiments, the number of full-length integrants in the target insertion site is greater than the number of integrants truncated by 300-565 nucleotides in the target insertion site, e.g., the number of full-length integrants is at least 1.1x, 1.2x, 1.5x, 2x, 3x, 4x, 5x, 6x, 7x, 8x, 9x, or 10x

the number of the truncated integrants, or the number of full-length integrants is at least 1.1x-10x, 2x-10x, 3x-10x, or 5x-10x the number of the truncated integrants.

In some embodiments, a system or method described herein results in insertion of the heterologous object sequence only at one target site in the genome of the target cell. Insertion can be measured, e.g., using a threshold of above 1%, 1.5%, 2%, 2.5%, 3%, 3.5%, 4%, 4.5%, 5%, e.g., as described in Example 8. In some embodiments, a system or method described herein results in insertion of the heterologous object sequence wherein less than 1%, 1.5%, 2%, 2.5%, 3%, 3.5%, 4%, 4.5%, 5%, 10%, 20%, 30%, 40%, or 50% of insertions are at a site other than the target site, e.g., using an assay described herein, e.g., an assay of Example 8.

In some embodiments, a system or method described herein results in “scarless” insertion of the heterologous object sequence, while in some embodiments, the target site can show deletions or duplications of endogenous DNA as a result of insertion of the heterologous sequence. The mechanisms of different retrotransposons could result in different patterns of duplications or deletions in the host genome occurring during retrotransposition at the target site. In some embodiments, the system results in a scarless insertion, with no duplications or deletions in the surrounding genomic DNA. In some embodiments, the system results in a deletion of less than 1, 2, 3, 4, 5, 10, 50, or 100 bp of genomic DNA upstream of the insertion. In some embodiments, the system results in a deletion of less than 1, 2, 3, 4, 5, 10, 50, or 100 bp of genomic DNA downstream of the insertion. In some embodiments, the system results in a duplication of less than 1, 2, 3, 4, 5, 10, 50, or 100 bp of genomic DNA upstream of the insertion. In some embodiments, the system results in a duplication of less than 1, 2, 3, 4, 5, 10, 50, or 100 bp of genomic DNA downstream of the insertion.

In some embodiments, a GENE WRITER™ described herein, or a DNA-binding domain thereof, binds to its target site specifically, e.g., as measured using an assay of Example 21. In some embodiments, the GENE WRITER™ or DNA-binding domain thereof binds to its target site more strongly than to any other binding site in the human genome. For example, in some embodiments, in an assay of Example 21, the target site represents more than 50%, 60%, 70%, 80%, 90%, or 95% of binding events of the GENE WRITER™ or DNA-binding domain thereof to human genomic DNA. Genetically Engineered, e.g., Dimerized GENE WRITER™ Genome Editor Polypeptides

Some non-LTR retrotransposons utilize two subunits to complete retrotransposition (Christensen et al PNAS 2006). In some embodiments, a retrotransposase described herein comprises two connected subunits as a single polypeptide. For instance, two wild-type retrotransposases could be joined with a linker to form a covalently “dimerized” protein (see FIG. 17). In some embodiments, the nucleic acid coding for the retrotransposase codes for two retrotransposase subunits to be expressed as a single polypeptide. In some embodiments, the subunits are connected by a peptide linker, such as has been described herein in the section entitled “Linker” and, e.g., in Chen et al Adv Drug Deliv Rev 2013. In some embodiments, the two subunits in the polypeptide are connected by a rigid linker. In some embodiments, the rigid linker consists of the motif (EAAAK)_n (SEQ ID NO: 1534). In other embodiments, the two subunits in the polypeptide are connected by a flexible linker. In some embodiments, the flexible linker consists of the motif (Gly)_n. In some embodiments, the flexible linker consists of the motif

(GGGGS)_n (SEQ ID NO: 1535). In some embodiments, the rigid or flexible linker consists of 1, 2, 3, 4, 5, 10, 15, or more amino acids in length to enable retrotransposition. In some embodiments, the linker consists of a combination of rigid and flexible linker motifs.

Based on mechanism, not all functions are required from both retrotransposase subunits. In some embodiments, the fusion protein may consist of a fully functional subunit and a second subunit lacking one or more functional domains. In some embodiments, one subunit may lack reverse transcriptase functionality. In some embodiments, one subunit may lack the reverse transcriptase domain. In some embodiments, one subunit may possess only endonuclease activity. In some embodiments, one subunit may possess only an endonuclease domain. In some embodiments, the two subunits comprising the single polypeptide may provide complementary functions.

In some embodiments, one subunit may lack endonuclease functionality. In some embodiments, one subunit may lack the endonuclease domain. In some embodiments, one subunit may possess only reverse transcriptase activity. In some embodiments, one subunit may possess only a reverse transcriptase domain. In some embodiments, one subunit may possess only DNA-dependent DNA synthesis functionality.

Linkers:

In some embodiments, domains of the compositions and systems described herein (e.g., the endonuclease and reverse transcriptase domains of a polypeptide or the DNA binding domain and reverse transcriptase domains of a polypeptide) may be joined by a linker. A composition described herein comprising a linker element has the general form S1-L-S2, wherein S1 and S2 may be the same or different and represent two domain moieties (e.g., each a polypeptide or nucleic acid domain) associated with one another by the linker. In some embodiments, a linker may connect two polypeptides. In some embodiments, a linker may connect two nucleic acid molecules. In some embodiments, a linker may connect a polypeptide and a nucleic acid molecule. A linker may be a chemical bond, e.g., one or more covalent bonds or non-covalent bonds. A linker may be flexible, rigid, and/or cleavable. In some embodiments, the linker is a peptide linker. Generally, a peptide linker is at least 2, 3, 4, 5, 6, 7, 8, 9, 10 or more amino acids in length, e.g., 2-50 amino acids in length, 2-30 amino acids in length.

The most commonly used flexible linkers have sequences consisting primarily of stretches of Gly and Ser residues ("GS" linker). Flexible linkers may be useful for joining domains that require a certain degree of movement or interaction and may include small, non-polar (e.g. Gly) or polar (e.g. Ser or Thr) amino acids. Incorporation of Ser or Thr can also maintain the stability of the linker in aqueous solutions by forming hydrogen bonds with the water molecules, and therefore reduce unfavorable interactions between the linker and the other moieties. Examples of such linkers include those having the structure [GGS]^{≥1} or [GGGS]^{≥1} (SEQ ID NO: 1536). Rigid linkers are useful to keep a fixed distance between domains and to maintain their independent functions. Rigid linkers may also be useful when a spatial separation of the domains is critical to preserve the stability or bioactivity of one or more components in the agent. Rigid linkers may have an alpha helix-structure or Pro-rich sequence, (XP)_n, with X designating any amino acid, preferably Ala, Lys, or Glu. Cleavable linkers may release free functional domains in vivo. In some

In vivo cleavable linkers may utilize the reversible nature of a disulfide bond. One example includes a thrombin-sensitive sequence (e.g., PRS) between the two Cys residues. In vitro thrombin treatment of CPRSC (SEQ ID NO: 1537) results in the cleavage of the thrombin-sensitive sequence, while the reversible disulfide linkage remains intact. Such linkers are known and described, e.g., in Chen et al. 2013. Fusion Protein Linkers: Property, Design and Functionality. Adv Drug Deliv Rev. 65(10): 1357-1369. In vivo cleavage of linkers in compositions described herein may also be carried out by proteases that are expressed in vivo under pathological conditions (e.g. cancer or inflammation), in specific cells or tissues, or constrained within certain cellular compartments. The specificity of many proteases offers slower cleavage of the linker in constrained compartments.

In some embodiments the amino acid linkers are (or are homologous to) the endogenous amino acids that exist between such domains in a native polypeptide. In some embodiments the endogenous amino acids that exist between such domains are substituted but the length is unchanged from the natural length. In some embodiments, additional amino acid residues are added to the naturally existing amino acid residues between domains.

In some embodiments, the amino acid linkers are designed computationally or screened to maximize protein function (Anad et al., FEBS Letters, 587:19, 2013).

Template RNA Component of GENE WRITER™ Gene Editor System

The GENE WRITER™ systems described herein can transcribe an RNA sequence template into host target DNA sites by target-primed reverse transcription. By writing DNA sequence(s) via reverse transcription of the RNA sequence template directly into the host genome, the GENE WRITER™ system can insert an object sequence into a target genome without the need for exogenous DNA sequences to be introduced into the host cell (unlike, for example, CRISPR systems), as well as eliminate an exogenous DNA insertion step. Therefore, the GENE WRITER™ system provides a platform for the use of customized RNA sequence templates containing object sequences, e.g., sequences comprising heterologous gene coding and/or function information.

In some embodiments the template RNA encodes a GENE WRITER™ protein in cis with a heterologous object sequence. Various cis constructs were described, for example, in Kuroki-Kami et al (2019) *Mobile DNA* 10:23 (incorporated by reference herein in its entirety), and can be used in combination with any of the embodiments described herein. For instance, in some embodiments, the template RNA comprises a heterologous object sequence, a sequence encoding a GENE WRITER™ protein (e.g., a protein comprising (i) a reverse transcriptase domain and (ii) an endonuclease domain, e.g., as described herein), a 5' untranslated region, and a 3' untranslated region. The components may be included in various orders. In some embodiments, the GENE WRITER™ protein and heterologous object sequence are encoded in different directions (sense vs. anti-sense), e.g., using an arrangement shown in FIG. 3A of Kuroki-Kami et al, Id. In some embodiments the GENE WRITER™ protein and heterologous object sequence are encoded in the same direction. In some embodiments, the nucleic acid encoding the polypeptide and the template RNA or the nucleic acid encoding the template RNA are covalently linked, e.g., are part of a fusion nucleic acid and/or are part of the same transcript. In some embodiments, the fusion nucleic acid comprises RNA or DNA.

The nucleic acid encoding the GENE WRITER™ polypeptide may, in some instances, be 5' of the heterologous object sequence. For example, in some embodiments, the template RNA comprises, from 5' to 3', a 5' untranslated region, a sense-encoded GENE WRITER™ polypeptide, a sense-encoded heterologous object sequence, and 3' untranslated region. In some embodiments, the template RNA comprises, from 5' to 3', a 5' untranslated region, a sense-encoded GENE WRITER™ polypeptide, anti-sense-encoded heterologous object sequence, and 3' untranslated region.

In some embodiments, the RNA further comprises homology to the DNA target site.

It is understood that, when a template RNA is described as comprising an open reading frame or the reverse complement thereof, in some embodiments the template RNA must be converted into double stranded DNA (e.g., through reverse transcription) before the open reading frame can be transcribed and translated.

In certain embodiments, customized RNA sequence template can be identified, designed, engineered and constructed to contain sequences altering or specifying host genome function, for example by introducing a heterologous coding region into a genome; affecting or causing exon structure/alternative splicing; causing disruption of an endogenous gene; causing transcriptional activation of an endogenous gene; causing epigenetic regulation of an endogenous DNA; causing up- or down-regulation of operably linked genes, etc. In certain embodiments, a customized RNA sequence template can be engineered to contain sequences coding for exons and/or transgenes, provide for binding sites to transcription factor activators, repressors, enhancers, etc., and combinations of thereof. In other embodiments, the coding sequence can be further customized with splice acceptor sites, poly-A tails. In certain embodiments the RNA sequence can contain sequences coding for an RNA sequence template homologous to the RLE transposase, be engineered to contain heterologous coding sequences, or combinations thereof.

The template RNA may have some homology to the target DNA. In some embodiments the template RNA has at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 175, 200 or more bases of exact homology to the target DNA at the 3' end of the RNA. In some embodiments the template RNA has at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 175, 180, or 200 or more bases of at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% homology to the target DNA, e.g., at the 5' end of the template RNA. In some embodiments the template RNA has a 3' untranslated region derived from a non-LTR retrotransposon, e.g. a non-LTR retrotransposons described herein. In some embodiments the template RNA has a 3' region of at least 10, 15, 20, 25, 30, 40, 50, 60, 80, 100, 120, 140, 160, 180, 200 or more bases of at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% homology to the 3' sequence of a non-LTR retrotransposon, e.g., a non-LTR retrotransposon described herein, e.g. a non-LTR retrotransposon in Table 1, 2, or 3. In some embodiments the template RNA has a 5' untranslated region derived from a non-LTR retrotransposon, e.g. a non-LTR retrotransposons described herein. In some embodiments the template RNA has a 5' region of at least 10, 15, 20, 25, 30, 40, 50, 60, 80, 100, 120, 140, 160, 180, or 200 or more bases of at least 40%, 50%, 60%, 70%, 80%, 90%, 95% or greater homology to the 5' sequence of a non-LTR retrotransposon, e.g., a non-LTR

retrotransposon described herein, e.g. a non-LTR retrotransposon described in Table 2 or 3.

The template RNA component of a GENE WRITER™ genome editing system described herein typically is able to bind the GENE WRITER™ genome editing protein of the system. In some embodiments the template RNA has a 3' region that is capable of binding a GENE WRITER™ genome editing protein. The binding region, e.g., 3' region, may be a structured RNA region, e.g., having at least 1, 2 or 3 hairpin loops, capable of binding the GENE WRITER™ genome editing protein of the system.

The template RNA component of a GENE WRITER™ genome editing system described herein typically is able to bind the GENE WRITER™ genome editing protein of the system. In some embodiments the template RNA has a 5' region that is capable of binding a GENE WRITER™ genome editing protein. The binding region, e.g., 5' region, may be a structured RNA region, e.g., having at least 1, 2 or 3 hairpin loops, capable of binding the GENE WRITER™ genome editing protein of the system. In some embodiments, the 5' untranslated region comprises a pseudoknot, e.g., a pseudoknot that is capable of binding to the GENE WRITER™ protein.

In some embodiments, the template RNA (e.g., an untranslated region of the hairpin RNA, e.g., a 5' untranslated region) comprises a stem-loop sequence. In some embodiments, the template RNA (e.g., an untranslated region of the hairpin RNA, e.g., a 5' untranslated region) comprises a hairpin. In some embodiments, the template RNA (e.g., an untranslated region of the hairpin RNA, e.g., a 5' untranslated region) comprises a helix. In some embodiments, the template RNA (e.g., an untranslated region of the hairpin RNA, e.g., a 5' untranslated region) comprises a pseudoknot. In some embodiments the template RNA comprises a ribozyme. In some embodiments the ribozyme is similar to an hepatitis delta virus (HDV) ribozyme, e.g., has a secondary structure like that of the HDV ribozyme and/or has one or more activities of the HDV ribozyme, e.g., a self-cleavage activity. See, e.g., Eickbush et al., *Molecular and Cellular Biology*, 2010, 3142-3150.

In some embodiments, the template RNA (e.g., an untranslated region of the hairpin RNA, e.g., a 3' untranslated region) comprises one or more stem-loops or helices. Exemplary structures of R2 3' UTRs are shown, for example, in Ruschak et al. "Secondary structure models of the 3' untranslated regions of diverse R2 RNAs" *RNA*. 2004 June; 10(6): 978-987, e.g., at FIG. 3, therein, and in Eickbush and Eickbush, "R2 and R2/R1 hybrid non-autonomous retrotransposons derived by internal deletions of full-length elements" *Mobile DNA* (2012) 3:10; e.g., at FIG. 3 therein, which articles are hereby incorporated by reference in their entirety.

In some embodiments, a template RNA described herein comprises a sequence that is capable of binding to a GENE WRITER™ protein described herein. For instance, in some embodiments, the template RNA comprises an MS2 RNA sequence capable of binding to an MS2 coat protein sequence in the GENE WRITER™ protein. In some embodiments, the template RNA comprises an RNA sequence capable of binding to a B-box sequence. In some embodiments, the template RNA comprises an RNA sequence (e.g., a crRNA sequence and/or tracrRNA sequence) capable of binding to a dCas sequence in the GENE WRITER™ protein. In some embodiments, in addition to or in place of a UTR, the template RNA is linked (e.g., covalently) to a non-RNA UTR, e.g., a protein or small molecule.

In some embodiments the template RNA has a poly-A tail at the 3' end. In some embodiments the template RNA does not have a poly-A tail at the 3' end.

In some embodiments the template RNA has a 5' region of at least 10, 15, 20, 25, 30, 40, 50, 60, 80, 100, 120, 140, 160, 180, 200 or more bases of at least 40%, 50%, 60%, 70%, 80%, 90%, 95% or greater homology to the 5' sequence of a non-LTR retrotransposon, e.g., a non-LTR retrotransposon described herein.

The template RNA of the system typically comprises an object sequence for insertion into a target DNA. The object sequence may be coding or non-coding.

In some embodiments a system or method described herein comprises a single template RNA. In some embodiments a system or method described herein comprises a plurality of template RNAs.

In some embodiments, the object sequence may contain an open reading frame. In some embodiments the template RNA has a Kozak sequence. In some embodiments the template RNA has an internal ribosome entry site. In some embodiments the template RNA has a self-cleaving peptide such as a T2A or P2A site. In some embodiments the template RNA has a start codon. In some embodiments the template RNA has a splice acceptor site. In some embodiments the template RNA has a splice donor site. In some embodiments the template RNA has a microRNA binding site downstream of the stop codon. In some embodiments the template RNA has a polyA tail downstream of the stop codon of an open reading frame. In some embodiments the template RNA comprises one or more exons. In some embodiments the template RNA comprises one or more introns. In some embodiments the template RNA comprises a eukaryotic transcriptional terminator. In some embodiments the template RNA comprises an enhanced translation element or a translation enhancing element. In some embodiments the RNA comprises the human T-cell leukemia virus (HTLV-1) R region. In some embodiments the RNA comprises a posttranscriptional regulatory element that enhances nuclear export, such as that of Hepatitis B Virus (HPRE) or Woodchuck Hepatitis Virus (WPRE). In some embodiments, in the template RNA, the heterologous object sequence encodes a polypeptide and is coded in an antisense direction with respect to the 5' and 3' UTR. In some embodiments, in the template RNA, the heterologous object sequence encodes a polypeptide and is coded in a sense direction with respect to the 5' and 3' UTR.

In some embodiments, a nucleic acid described herein (e.g., a template RNA or a DNA encoding a template RNA) comprises a microRNA binding site. In some embodiments, the microRNA binding site is used to increase the target-cell specificity of a GENE WRITER™ system. For instance, the microRNA binding site can be chosen on the basis that is recognized by a miRNA that is present in a non-target cell type, but that is not present (or is present at a reduced level relative to the non-target cell) in a target cell type. Thus, when the template RNA is present in a non-target cell, it would be bound by the miRNA, and when the template RNA is present in a target cell, it would not be bound by the miRNA (or bound but at reduced levels relative to the non-target cell). While not wishing to be bound by theory, binding of the miRNA to the template RNA may interfere with insertion of the heterologous object sequence into the genome. Accordingly, the heterologous object sequence would be inserted into the genome of target cells more efficiently than into the genome of non-target cells. A system having a microRNA binding site in the template RNA (or DNA encoding it) may also be used in combination with a

nucleic acid encoding a GENE WRITER™ polypeptide, wherein expression of the GENE WRITER™ polypeptide is regulated by a second microRNA binding site, e.g., as described herein, e.g., in the section entitled "Polypeptide component of GENE WRITER™ gene editor system".

In some embodiments, the object sequence may contain a non-coding sequence. For example, the template RNA may comprise a promoter or enhancer sequence. In some embodiments the template RNA comprises a tissue specific promoter or enhancer, each of which may be unidirectional or bidirectional. In some embodiments the promoter is an RNA polymerase I promoter, RNA polymerase II promoter, or RNA polymerase III promoter. In some embodiments the promoter comprises a TATA element. In some embodiments the promoter comprises a B recognition element. In some embodiments the promoter has one or more binding sites for transcription factors. In some embodiments the non-coding sequence is transcribed in an antisense-direction with respect to the 5' and 3' UTR. In some the non-coding sequence is transcribed in a sense direction with respect to the 5' and 3' UTR.

In some embodiments, a nucleic acid described herein (e.g., a template RNA or a DNA encoding a template RNA) comprises a promoter sequence, e.g., a tissue specific promoter sequence. In some embodiments, the tissue-specific promoter is used to increase the target-cell specificity of a GENE WRITER™ system. For instance, the promoter can be chosen on the basis that it is active in a target cell type but not active in (or active at a lower level in) a non-target cell type. Thus, even if the promoter integrated into the genome of a non-target cell, it would not drive expression (or only drive low level expression) of an integrated gene. A system having a tissue-specific promoter sequence in the template RNA may also be used in combination with a microRNA binding site, e.g., in the template RNA or a nucleic acid encoding a GENE WRITER™ protein, e.g., as described herein. A system having a tissue-specific promoter sequence in the template RNA may also be used in combination with a DNA encoding a GENE WRITER™ polypeptide, driven by a tissue-specific promoter, e.g., to achieve higher levels of GENE WRITER™ protein in target cells than in non-target cells.

In some embodiments the template RNA comprises a microRNA sequence, a siRNA sequence, a guide RNA sequence, a piwi RNA sequence.

In some embodiments the template RNA comprises a site that coordinates epigenetic modification. In some embodiments the template RNA comprises an element that inhibits, e.g., prevents, epigenetic silencing. In some embodiments the template RNA comprises a chromatin insulator. For example, the template RNA comprises a CTCF site or a site targeted for DNA methylation.

In order to promote higher level or more stable gene expression, the template RNA may include features that prevent or inhibit gene silencing. In some embodiments, these features prevent or inhibit DNA methylation. In some embodiments, these features promote DNA demethylation. In some embodiments, these features prevent or inhibit histone deacetylation. In some embodiments, these features prevent or inhibit histone methylation. In some embodiments, these features promote histone acetylation. In some embodiments, these features promote histone demethylation. In some embodiments, multiple features may be incorporated into the template RNA to promote one or more of these modifications. CpG dinucleotides are subject to methylation by host methyl transferases. In some embodiments, the template RNA is depleted of CpG dinucleotides,

e.g., does not comprise CpG nucleotides or comprises a reduced number of CpG dinucleotides compared to a corresponding unaltered sequence. In some embodiments, the promoter driving transgene expression from integrated DNA is depleted of CpG dinucleotides.

In some embodiments the template RNA comprises a gene expression unit composed of at least one regulatory region operably linked to an effector sequence. The effector sequence may be a sequence that is transcribed into RNA (e.g., a coding sequence or a non-coding sequence such as a sequence encoding a micro RNA).

In some embodiments the object sequence of the template RNA is inserted into a target genome in an endogenous intron. In some embodiments the object sequence of the template RNA is inserted into a target genome and thereby acts as a new exon. In some embodiments the insertion of the object sequence into the target genome results in replacement of a natural exon or the skipping of a natural exon.

In some embodiments the object sequence of the template RNA is inserted into the target genome in a genomic safe harbor site, such as AAVS1, CCR5, or ROSA26. In some embodiments the object sequence of the template RNA is added to the genome in an intergenic or intragenic region. In some embodiments the object sequence of the template RNA is added to the genome 5' or 3' within 0.1 kb, 0.25 kb, 0.5 kb, 0.75 kb, 1 kb, 2 kb, 3 kb, 4 kb, 5 kb, 7.5 kb, 10 kb, 15 kb, 20 kb, 25 kb, 50, 75 kb, or 100 kb of an endogenous active gene. In some embodiments the object sequence of the template RNA is added to the genome 5' or 3' within 0.1 kb, 0.25 kb, 0.5 kb, 0.75 kb, 1 kb, 2 kb, 3 kb, 4 kb, 5 kb, 7.5 kb, 10 kb, 15 kb, 20 kb, 25 kb, 50, 75 kb, or 100 kb of an endogenous promoter or enhancer. In some embodiments the object sequence of the template RNA can be, e.g., 50-50,000 base pairs (e.g., between 50-40,000 bp, between 500-30,000 bp between 500-20,000 bp, between 100-15,000 bp, between 500-10,000 bp, between 50-10,000 bp, between 50-5,000 bp. In some embodiments, the heterologous object sequence is less than 1,000, 1,300, 1500, 2,000, 3,000, 4,000, 5,000, or 7,500 nucleotides in length.

In some embodiments the genomic safe harbor site is a NATURAL HARBOR™ site. In some embodiments the NATURAL HARBOR™ site is ribosomal DNA (rDNA). In some embodiments the NATURAL HARBOR™ site is 5S rDNA, 18S rDNA, 5.8S rDNA, or 28S rDNA. In some embodiments the NATURAL HARBOR™ site is the Mutsu site in 5S rDNA. In some embodiments the NATURAL HARBOR™ site is the R2 site, the R5 site, the R6 site, the

R4 site, the R1 site, the R9 site, or the RT site in 28S rDNA. In some embodiments the NATURAL HARBOR™ site is the R8 site or the R7 site in 18S rDNA. In some embodiments the NATURAL HARBOR™ site is DNA encoding transfer RNA (tRNA). In some embodiments the NATURAL HARBOR™ site is DNA encoding tRNA-Asp or tRNA-Glu. In some embodiments the NATURAL HARBOR™ site is DNA encoding spliceosomal RNA. In some embodiments the NATURAL HARBOR™ site is DNA encoding small nuclear RNA (snRNA) such as U2 snRNA.

Thus, in some aspects, the present disclosure provides a method of inserting a heterologous object sequence into a NATURAL HARBOR™ site. In some embodiments, the method comprises using a GENE WRITER™ system described herein, e.g., using a polypeptide of any of Tables 1-3 or a polypeptide having sequence similarity thereto, e.g., at least 80%, 85%, 90%, or 95% identity thereto. In some embodiments, the method comprises using an enzyme, e.g., a retrotransposase, to insert the heterologous object sequence into the NATURAL HARBOR™ site. In some aspects, the present disclosure provides a host human cell comprising a heterologous object sequence (e.g., a sequence encoding a therapeutic polypeptide) situated at a NATURAL HARBOR™ site in the genome of the cell. In some embodiments, the NATURAL HARBOR™ site is a site described in Table 4 below. In some embodiments, the heterologous object sequence is inserted within 20, 50, 100, 150, 200, 250, 500, or 1000 base pairs of a sequence shown in Table 4. In some embodiments, the heterologous object sequence is inserted within 0.1 kb, 0.25 kb, 0.5 kb, 0.75 kb, 1 kb, 2 kb, 3 kb, 4 kb, 5 kb, 7.5 kb, 10 kb, 15 kb, 20 kb, 25 kb, 50, 75 kb, or 100 kb of a sequence shown in Table 4. In some embodiments, the heterologous object sequence is inserted into a site having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity to a sequence shown in Table 4. In some embodiments, the heterologous object sequence is inserted within 20, 50, 100, 150, 200, 250, 500, or 1000 base pairs, or within 0.1 kb, 0.25 kb, 0.5 kb, 0.75 kb, 1 kb, 2 kb, 3 kb, 4 kb, 5 kb, 7.5 kb, 10 kb, 15 kb, 20 kb, 25 kb, 50, 75 kb, or 100 kb, of a site having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity to a sequence shown in Table 4. In some embodiments, the heterologous object sequence is inserted within a gene indicated in Column 5 of Table 4, or within 20, 50, 100, 150, 200, 250, 500, or 1000 base pairs, or within 0.1 kb, 0.25 kb, 0.5 kb, 0.75 kb, 1 kb, 2 kb, 3 kb, 4 kb, 5 kb, 7.5 kb, 10 kb, 15 kb, 20 kb, 25 kb, 50, 75 kb, or 100 kb, of the gene.

TABLE 4

NATURAL HARBOR™ sites.					
Column 1 indicates a retrotransposon that inserts into the NATURAL HARBOR™ site. Column 2 indicates the gene at the NATURAL HARBOR™ site. Columns 3 and 4 show exemplary human genome sequence 5' and 3' of the insertion site (for example, 250 bp). Columns 5 and 6 list the example gene symbol and corresponding Gene ID.					
Target Site	Target Gene	5' flanking sequence	3' flanking sequence	Example Gene Symbol	Example Gene ID
R2	28S rDNA	CCGGTCCCCCGCGGGTCC GCCCCCGGGCGCGTTCC GCGCGCGCCTCGCTCGGC CGGCGCCTAGCAGCCGACTT AGAACTGGTGGGACCGAGG GAATCCGACTGTTAATTAAA ACAAAGCATCGGAAGGCC GCGCGGGTGTGACGCGAT GTGATTCTGCCAGTGCTCT	GTAGCCAAATGCCTCGTCATC TAATTAGTGACGCGCATGAAT GGATGAACGAGATTCCCACT GTCCTACCTACTATCCAGCG AAACCACAGCCAAGGGAACG GGCTTGGCGGAATCAGCGGG GAAAGAAGACCCTGTTGAGC TTGACTCTAGTCTGGCACGGT GAAGAGACATGAGAGGTGTA	RNA28SN1	106632264

TABLE 4-continued

NATURAL HARBOR [™] sites.					
Column 1 indicates a retrotransposon that inserts into the NATURAL HARBOR [™] site. Column 2 indicates the gene at the NATURAL HARBOR [™] site. Columns 3 and 4 show exemplary human genome sequence 5' and 3' of the insertion site (for example, 250 bp). Columns 5 and 6 list the example gene symbol and corresponding Gene ID.					
Target Site	Target Gene	5' flanking sequence	3' flanking sequence	Example Gene Symbol	Example Gene ID
R4	28S rDNA	GAATGTCAAAGTGAAGAAAT TCAATGAAGCGCGGTAAAC GGCGGGAGTAACATGACTC TCTTAAG (SEQ ID NO: 1508)	GAATAAGTGGGAGGCCCCCG GCGCCCCCGGTGTCCTCCCGC GAGGGGCCCGGGCGGGGT CCGCCG (SEQ ID NO: 1513)	RNA28SN1	106632264
R5	28S rDNA	GCGGTTCGCGCGGCGCCTC GCCTCGGCCGCGCCTAGCA GCCGACTTAGAACTGGTCG GACCAGGGGAATCCGACTGT TTAATTAACAAAGCATCGC GAAGGCCCGCGCGGGTGT GACGCGATGTGATTTCTGCC AGTGCTCTGAATGTCAAAGT GAAGAAATTCAATGAAGCGC GGGTAAACGGCGGAGTAAC TATGACTCTCTTAAGGTAGCC AAATGCCTCGTCATCTAATTA GTGACG (SEQ ID NO: 1509)	CGCATGAATGGATGAACGAG ATTCCCCTGTCCTACCTACT ATCCAGCGAAACCACAGCCA AGGGAACGGGCTTGGCGGA ATCAGCGGGGAAAGAACCC CTGTTGAGCTTGACTCTAGTC TGGCACGGTGAAGAGACATG AGAGGTGTAGAATAAGTGGG AGGCCCGCGCGCCCCCGG GTGTCCCGCGAGGGGCCCCG GGGCGGGGTCCGCCGCCCCC GCGGGCGCGCGGTGAAATAC CACTACTC (SEQ ID NO: 1514)	RNA28SN1	106632264
R9	28S rDNA	TCCCCCGCGCGGGTCCGCC CCGGGGCGCGGTTCGCGCG GGCGCCTCGCTCGGCCGCG GCCTAGCAGCGCACTTAGAA CTGGTGCAGGACAGGGGAAT CCGACTGTTTAAATTAACAA AGCATCGGAAGGCCCGCGG CGGGTGTGACGCGATGTGA TTTCTGCCAGTGTCTGAAT GTCAAAGTGAAGAAATTCAG TGAAGCGCGGTAAACGCGC GGAGTAACTATGACTCTCTTA AGGTAG (SEQ ID NO: 1510)	CCAAATGCCTCGTCATCTAAT TAGTGACGCGCATGAATGGA TGAACGAGATTCCTACTGTCC CTACCTACTATCCAGCGAAAC CACAGCCAAGGGAACGGGCT TGGCGGAATCAGCGGGGAAA GAAGACCCTGTTGAGCTTGA CTCTAGTCTGGCACGGTGAA GAGACATGAGAGGTGTAGAA TAAGTGGGAGGCCCGCGCG CCCCCGCGGTGTCGCCGCGAG GGGCCCGGGCGGGGTCCG CCGGCCC (SEQ ID NO: 1515)	RNA28SN1	106632264
R8	18S rDNA	CGGCGCGCTCGCCGCGGAG GTGGGATCCCGAGGCTCTC CAGTCCCGCAGGGCGCACC ACCGGCCCGTCTCGCCGCGG CGCCGGGAGGTGGAGCAC GAGCGCACGTGTTAGGACCC GAAAGATGGTGAACATGACC TGGGCAGGGCGAAGCCAGA GGAAACTCTGGTGGAGGTCC GTAGCGGTCTGACGTGCAA ATCGGTCGTCCGACCTGGGT ATAGGGGCGAAGACTAATC GAACCATCTAG (SEQ ID NO: 1511)	TAGTGGTTCCTCCGAAGTT TCCCTCAGGATAGCTGGCGCT CTCGCAGACCCGACGACCCCC CGCCACGAGTTTATCCGGT AAAGCGAATGATTAGAGGTG TTGGGGCCGAAACGATCTCA ACCTATTCTCAAACTTTAAAT GGGTAGAAAGCCCGCTCGC TGGCGTGGAGCCGGCGGTGG AATGCGAGTGCTTAGTGGGC CACTTTTGGTAAGCAGAACTG GCGCTGCGGGATGAACCGAA CGCC (SEQ ID NO: 1516)	RNA18SN1	106631781
R4-2_SRA	tRNA-Asp	GCATTGCTATTGCGCGCTAG AGGTGAAATCTTGACCCGG CGCAAGACGACGACGAGCGA AAGCATTTGCCAAGAATGTTT TCATTAATCAAGAACGAAAGT CGGAGGTTCAAGACGATCA GATACCGTCGTAGTTCCGACC ATAAACGATCCGACCGGCG ATGCGGCGGCGTTATTTCCAT GACCCGCGGGCAGCTTCCG GGAAACCAAAGTCTTTGGGT TCCGGGGGAGTATGTTGC AAAGC (SEQ ID NO: 1512)	TGAAACTTAAAGGAATTGAC GGAAGGGCACCACCGAGGT GGAGCCTGCGGCTTAATTTG ACTCAACACGGGAAACCTCA CCCGGCCCGGACACGGACAG GATTGACAGATTGATAGCTCT TTCTCGATTCCGTGGGTGGTG GTGCATGGCCGTTCTTAGTTG GTGGAGCGATTTGTCTGGTT AATTCGATAACGAAACGAGA CTCTGGCATGCTAACTAGTTA CGCGACCCCGAGCGGTCCG CGTCCC (SEQ ID NO: 1517)	TRD-GTC1-1	100189207
LIN25_SM	tRNA-Glu			TRE-CTC1-1	100189384

TABLE 4-continued

NATURAL HARBOR [™] sites.					
Column 1 indicates a retrotransposon that inserts into the NATURAL HARBOR [™] site. Column 2 indicates the gene at the NATURAL HARBOR [™] site. Columns 3 and 4 show exemplary human genome sequence 5' and 3' of the insertion site (for example, 250 bp). Columns 5 and 6 list the example gene symbol and corresponding Gene ID.					
Target Site	Target Gene	5' flanking sequence	3' flanking sequence	Example Gene Symbol	Example Gene ID
R1	28S rDNA	TAGCAGCCGACTTAGAAGTCTG GTGCGGACCAGGGGAATCCG ACTGTTTAATTAAACAAAGC ATCGCGAAGGCCCGCGGCGG GTGTTGACGCGATGTGATTTC TGCCCAAGTGTCTGAATGTCA AAGTGAAGAAATTCATGAA GCGCGGGTAAACGCGCGGA GTAACATATGACTCTCTTAAGG TAGCCAAATGCCTCGTCATCT AATTAGTGACGCGCATGAAT GGATGAACGAGATTCCTACT GTCCCT (SEQ ID NO: 1518)	ACCTACTATCCAGCGAAACCA CAGCCAAGGGAACGGGCTTG GCGGAATCAGCGGGGAAAG AAGACCTCTGTTGAGCTTGACT CTAGTCTGGCACGGTGAAGA GACATGAGAGGTGTAGAATA AGTGGGAGGCCCCCGGCGCC CCCCCGGTGTCCCGCGAGG GGCCCGGGCGGGTCCGCC GGCCCTGCGGGCGCGCGGTG AAATACCACTACTCTGATCGT TTTTTCACTGACCGGTGAGG CGGGGG (SEQ ID NO: 1524)	RNA28SN1	106632264
R6	28S rDNA	CCCCCGCCGGTCCGCCCCC GGGGCCGCGTTCCGCGCGG CGCCTCGCCTCGGCGGCGC CTAGCAGCCGACTTAGAAGT GTGCGGACCAGGGGAATCC GACTGTTTAATTAAACAAAG CATCGCGAAGGCCCGCGCGG GGTGTGACGCGATGTGATT TCTGCCAGTGTCTGAATGT CAAAGTGAAGAAATTCATG AAGCGCGGTAAACGCGCGG GAGTAACTATGACTCTCTTAA GGTAGCC (SEQ ID NO: 1519)	AAATGCCTCGTCATCTAATTA GTGACGCGCATGAATGGATG AACGAGATTCCCACTGTCCCT ACCTACTATCCAGCGAAACCA CAGCCAAGGGAACGGGCTTG GCGGAATCAGCGGGGAAAG AAGACCTCTGTTGAGCTTGACT CTAGTCTGGCACGGTGAAGA GACATGAGAGGTGTAGAATA AGTGGGAGGCCCCCGGCGCC CCCCCGGTGTCCCGCGAGG GGCCCGGGCGGGTCCGCC GGCCCTG (SEQ ID NO: 1525)	RNA28SN1	106632264
R7	18S rDNA	GCGCAAGACGCGACGAGCG AAAGCATTGCAAGAATGTT TTCATTAATCAAGAACGAAAG TCGAGAGTTCGAAGACGATC AGATACCGTCGTAGTTCGAC CATAAACGATGCCGACCGGC GATGCGGCGGCGTTATCCC ATGACCCGCGGGCAGCTTC CGGAAACCAAAGTCTTTGG GTTCCGGGGGAGTATGGTT GCAAAGCTGAAACTTAAAGG AATTGACGGAAGGGACCCAC CAGGAGT (SEQ ID NO: 1520)	GGAGCCTGCGGCTTAATTTG ACTCAACACGGGAAACCTCA CCCGCCCGGACACGGACAG GATTGACAGATTGATAGCTCT TTCTCGATTCCGTGGGTGGTG GTGCATGGCCGTTCTTAGTTG GTGGAGCGATTGTCTGGTT AATTCGGATAACGAACGAGA CTCTGGCATGCTAACTAGTTA CGCGACCCCGAGCGGTCCG CGTCCCCCACTTCTTAGAGG GACAAGTGGCGTTACGCCAC CCGAG (SEQ ID NO: 1526)	RNA18SN1	106631781
RT	28S rDNA	GGCCGGGCGCGACCCGCTCC GGGGACAGTGCCAGGTGGG GAGTTTGACTGGGGCGGTAC ACCTGTCAAACGTAACGCA GGTGTCCTAAGGCGAGCTCA GGGAGGACAGAAACCTCCCG TGGAGCAGAAGGGCAAAAG CTCGCTTGATCTTGATTTTCA GTACGAATACAGACCGTGAA AGCGGGGCTCACGATCCTTC TGACCTTTTGGGTTTAAAGCA GGAGGTGTCAGAAAAGTTAC CACAGGAT (SEQ ID NO: 1521)	AACTGGCTTGTGGCGGCCAA GCGTTTCATAGCGACGTGCTT TTTGATCCTTCGATGTCGGCT CTTCTATCATTGTGAAGCAG AATTCAACCAAGCGTTGGATTG TTCACCCACTAATAGGGAACG TGAGCTGGGTTAGACCGTC GTGAGACAGGTAGTTTACC CTACTGATGATGTGTTGTTGC CATGGTAATCTGCTCAGTAC GAGAGGAACCGCAGGTTTCA ACATTTGGTGTATGTGCTTGG C (SEQ ID NO: 1527)	RNA28SN1	106632264
Mutsu	5S rDNA	GTCTACGGCCATACCAACC (SEQ ID NO: 1522)	TGAACGCGCCGATCTCGTCT GATCTCGGAAGCTAAGCAGG GTCGGGCTTGGTTAGTACTT GGATGGGAGACCGCTGGGA ATACCGGTGCTGTAGGCTTT (SEQ ID NO: 1528)	RNA5S1	100169751
Utopia/ Keno	U2 snRNA	ATCGCTTCTCGGCCTTTTGGC TAAGATCAAGTGTAGTA (SEQ ID NO: 1523)	TCTGTTCTTATCAGTTTAATAT CTGATACGTCCTCTATCCGAG GACAATATATTAATGGATT TTGGAGCAGGAGATGGAAT	RNU2-1	6066

TABLE 4-continued

NATURAL HARBOR [™] sites.					
Column 1 indicates a retrotransposon that inserts into the NATURAL HARBOR [™] site. Column 2 indicates the gene at the NATURAL HARBOR [™] site. Columns 3 and 4 show exemplary human genome sequence 5' and 3' of the insertion site (for example, 250 bp). Columns 5 and 6 list the example gene symbol and corresponding Gene ID.					
Target Site	Target Gene	5' flanking sequence	3' flanking sequence	Example Gene Symbol	Example Gene ID
			AGGAGCTTGCTCCGTCCACTC CACGCATCGACCTGGTATTGC AGTACCTCCAGGACGGTGC ACCC (SEQ ID NO: 1529)		

In some embodiments, a system or method described herein results in insertion of a heterologous sequence into a target site in the human genome. In some embodiments, the target site in the human genome has sequence similarity to the corresponding target site of the corresponding wild-type retrotransposase (e.g., the retrotransposase from which the GENE WRITER[™] was derived) in the genome of the organism to which it is native. For instance, in some embodiments, the identity between the 40 nucleotides of human genome sequence centered at the insertion site and the 40 nucleotides of native organism genome sequence centered at the insertion site is less than 99.5%, 99%, 98%, 97%, 96%, 95%, 90%, 85%, 80%, 75%, 70%, 60%, or 50%, or is between 50-60%, 60-70%, 70-80%, 80-90%, or 90-100%. In some embodiments, the identity between the 100 nucleotides of human genome sequence centered at the insertion site and the 100 nucleotides of native organism genome sequence centered at the insertion site is less than 99.5%, 99%, 98%, 97%, 96%, 95%, 90%, 85%, 80%, 75%, 70%, 60%, or 50%, or is between 50-60%, 60-70%, 70-80%, 80-90%, or 90-100%. In some embodiments, the identity between the 500 nucleotides of human genome sequence centered at the insertion site and the 500 nucleotides of native organism genome sequence centered at the insertion site is less than 99.5%, 99%, 98%, 97%, 96%, 95%, 90%, 85%, 80%, 75%, 70%, 60%, or 50%, or is between 50-60%, 60-70%, 70-80%, 80-90%, or 90-100%.

Production of Compositions and Systems

As will be appreciated by one of skill, methods of designing and constructing nucleic acid constructs and proteins or polypeptides (such as the systems, constructs and polypeptides described herein) are routine in the art. Generally, recombinant methods may be used. See, in general, Smales & James (Eds.), *Therapeutic Proteins: Methods and Protocols* (Methods in Molecular Biology), Humana Press (2005); and Crommelin, Sindelar & Meibohm (Eds.), *Pharmaceutical Biotechnology: Fundamentals and Applications*, Springer (2013). Methods of designing, preparing, evaluating, purifying and manipulating nucleic acid compositions are described in Green and Sambrook (Eds.), *Molecular Cloning: A Laboratory Manual* (Fourth Edition), Cold Spring Harbor Laboratory Press (2012).

Exemplary methods for producing a therapeutic pharmaceutical protein or polypeptide described herein involve expression in mammalian cells, although recombinant proteins can also be produced using insect cells, yeast, bacteria, or other cells under control of appropriate promoters. Mammalian expression vectors may comprise non-transcribed elements such as an origin of replication, a suitable promoter, and other 5' or 3' flanking non-transcribed sequences,

and 5' or 3' non-translated sequences such as necessary ribosome binding sites, a polyadenylation site, splice donor and acceptor sites, and termination sequences. DNA sequences derived from the SV40 viral genome, for example, SV40 origin, early promoter, splice, and polyadenylation sites may be used to provide other genetic elements required for expression of a heterologous DNA sequence. Appropriate cloning and expression vectors for use with bacterial, fungal, yeast, and mammalian cellular hosts are described in Green & Sambrook, *Molecular Cloning: A Laboratory Manual* (Fourth Edition), Cold Spring Harbor Laboratory Press (2012).

Various mammalian cell culture systems can be employed to express and manufacture recombinant protein. Examples of mammalian expression systems include CHO, COS, HEK293, HeLa, and BHK cell lines. Processes of host cell culture for production of protein therapeutics are described in Zhou and Kantardjieff (Eds.), *Mammalian Cell Cultures for Biologics Manufacturing (Advances in Biochemical Engineering/Biotechnology)*, Springer (2014). Compositions described herein may include a vector, such as a viral vector, e.g., a lentiviral vector, encoding a recombinant protein. In some embodiments, a vector, e.g., a viral vector, may comprise a nucleic acid encoding a recombinant protein.

Purification of protein therapeutics is described in Franks, *Protein Biotechnology: Isolation, Characterization, and Stabilization*, Humana Press (2013); and in Cutler, *Protein Purification Protocols (Methods in Molecular Biology)*, Humana Press (2010).

Applications

By integrating coding genes into a RNA sequence template, the GENE WRITER[™] system can address therapeutic needs, for example, by providing expression of a therapeutic transgene in individuals with loss-of-function mutations, by replacing gain-of-function mutations with normal transgenes, by providing regulatory sequences to eliminate gain-of-function mutation expression, and/or by controlling the expression of operably linked genes, transgenes and systems thereof. In certain embodiments, the RNA sequence template encodes a promoter region specific to the therapeutic needs of the host cell, for example a tissue specific promoter or enhancer. In still other embodiments, a promoter can be operably linked to a coding sequence.

In embodiments, the GENE WRITER[™] gene editor system can provide therapeutic transgenes expressing, e.g., replacement blood factors or replacement enzymes, e.g., lysosomal enzymes. For example, the compositions, systems and methods described herein are useful to express, in a target human genome, agalsidase alpha or beta for treatment

of Fabry Disease; imiglucerase, taliglucerase alfa, velaglucerase alfa, or alglucerase for Gaucher Disease; sebelipase alfa for lysosomal acid lipase deficiency (Wolman disease/CESD); laronidase, idursulfase, elosulfase alfa, or galsulfase for mucopolysaccharidoses; alglucosidase alfa for Pompe disease. For example, the compositions, systems and methods described herein are useful to express, in a target human genome factor I, II, V, VII, X, XI, XII or XIII for blood factor deficiencies.

In some embodiments, the heterologous object sequence encodes an intracellular protein (e.g., a cytoplasmic protein, a nuclear protein, an organellar protein such as a mitochondrial protein or lysosomal protein, or a membrane protein). In some embodiments, the heterologous object sequence encodes a membrane protein, e.g., a membrane protein other than a CAR, and/or an endogenous human membrane protein. In some embodiments, the heterologous object sequence encodes an extracellular protein. In some embodiments, the heterologous object sequence encodes an enzyme, a structural protein, a signaling protein, a regulatory protein, a transport protein, a sensory protein, a motor protein, a defense protein, or a storage protein.

Administration

The composition and systems described herein may be used in vitro or in vivo. In some embodiments the system or components of the system are delivered to cells (e.g., mammalian cells, e.g., human cells), e.g., in vitro or in vivo. In some embodiments, the cells are eukaryotic cells, e.g., cells of a multicellular organism, e.g., an animal, e.g., a mammal (e.g., human, swine, bovine) a bird (e.g., poultry, such as chicken, turkey, or duck), or a fish. In some embodiments, the cells are non-human animal cells (e.g., a laboratory animal, a livestock animal, or a companion animal). In some embodiments, the cell is a stem cell (e.g., a hematopoietic stem cell), a fibroblast, or a T cell. In some embodiments, the cell is a non-dividing cell, e.g., a non-dividing fibroblast or non-dividing T cell. In some embodiments, the cell is an HSC and p53 is not upregulated or is upregulated by less than 10%, 5%, 2%, or 1%, e.g., as determined according to the method described in Example 30. The skilled artisan will understand that the components of the GENE WRITER™ system may be delivered in the form of polypeptide, nucleic acid (e.g., DNA, RNA), and combinations thereof.

For instance, delivery can use any of the following combinations for delivering the retrotransposase (e.g., as DNA encoding the retrotransposase protein, as RNA encoding the retrotransposase protein, or as the protein itself) and the template RNA (e.g., as DNA encoding the RNA, or as RNA):

1. Retrotransposase DNA+template DNA
2. Retrotransposase RNA+template DNA
3. Retrotransposase DNA+template RNA
4. Retrotransposase RNA+template RNA
5. Retrotransposase protein+template DNA
6. Retrotransposase protein+template RNA
7. Retrotransposase virus+template virus
8. Retrotransposase virus+template DNA
9. Retrotransposase virus+template RNA
10. Retrotransposase DNA+template virus
11. Retrotransposase RNA+template virus
12. Retrotransposase protein+template virus

As indicated above, in some embodiments, the DNA or RNA that encodes the retrotransposase protein is delivered using a virus, and in some embodiments, the template RNA (or the DNA encoding the template RNA) is delivered using a virus.

In one embodiment the system and/or components of the system are delivered as nucleic acid. For example, the GENE WRITER™ polypeptide may be delivered in the form of a DNA or RNA encoding the polypeptide, and the template RNA may be delivered in the form of RNA or its complementary DNA to be transcribed into RNA. In some embodiments the system or components of the system are delivered on 1, 2, 3, 4, or more distinct nucleic acid molecules. In some embodiments the system or components of the system are delivered as a combination of DNA and RNA. In some embodiments the system or components of the system are delivered as a combination of DNA and protein. In some embodiments the system or components of the system are delivered as a combination of RNA and protein. In some embodiments the GENE WRITER™ genome editor polypeptide is delivered as a protein.

In some embodiments the system or components of the system are delivered to cells, e.g. mammalian cells or human cells, using a vector. The vector may be, e.g., a plasmid or a virus. In some embodiments delivery is in vivo, in vitro, ex vivo, or in situ. In some embodiments the virus is an adeno associated virus (AAV), a lentivirus, an adenovirus. In some embodiments the system or components of the system are delivered to cells with a viral-like particle or a virosome. In some embodiments the delivery uses more than one virus, viral-like particle or virosome.

In one embodiment, the compositions and systems described herein can be formulated in liposomes or other similar vesicles. Liposomes are spherical vesicle structures composed of a uni- or multilamellar lipid bilayer surrounding internal aqueous compartments and a relatively impermeable outer lipophilic phospholipid bilayer. Liposomes may be anionic, neutral or cationic. Liposomes are biocompatible, nontoxic, can deliver both hydrophilic and lipophilic drug molecules, protect their cargo from degradation by plasma enzymes, and transport their load across biological membranes and the blood brain barrier (BBB) (see, e.g., Spuch and Navarro, *Journal of Drug Delivery*, vol. 2011, Article ID 469679, 12 pages, 2011 for review).

Vesicles can be made from several different types of lipids; however, phospholipids are most commonly used to generate liposomes as drug carriers. Methods for preparation of multilamellar vesicle lipids are known in the art (see for example U.S. Pat. No. 6,693,086, the teachings of which relating to multilamellar vesicle lipid preparation are incorporated herein by reference). Although vesicle formation can be spontaneous when a lipid film is mixed with an aqueous solution, it can also be expedited by applying force in the form of shaking by using a homogenizer, sonicator, or an extrusion apparatus (see, e.g., Spuch and Navarro, *Journal of Drug Delivery*, vol. 2011, Article ID 469679, 12 pages, 2011 for review). Extruded lipids can be prepared by extruding through filters of decreasing size, as described in Templeton et al., *Nature Biotech*, 15:647-652, 1997, the teachings of which relating to extruded lipid preparation are incorporated herein by reference.

Lipid nanoparticles are another example of a carrier that provides a biocompatible and biodegradable delivery system for the pharmaceutical compositions described herein. Nanostructured lipid carriers (NLCs) are modified solid lipid nanoparticles (SLNs) that retain the characteristics of the SLN, improve drug stability and loading capacity, and prevent drug leakage. Polymer nanoparticles (PNPs) are an important component of drug delivery. These nanoparticles can effectively direct drug delivery to specific targets and improve drug stability and controlled drug release. Lipid-polymer nanoparticles (PLNs), a new type of carrier that

combines liposomes and polymers, may also be employed. These nanoparticles possess the complementary advantages of PNPs and liposomes. A PLN is composed of a core-shell structure; the polymer core provides a stable structure, and the phospholipid shell offers good biocompatibility. As such, the two components increase the drug encapsulation efficiency rate, facilitate surface modification, and prevent leakage of water-soluble drugs. For a review, see, e.g., Li et al. 2017, *Nanomaterials* 7, 122.

Exosomes can also be used as drug delivery vehicles for the compositions and systems described herein. For a review, see Ha et al. July 2016. *Acta Pharmaceutica Sinica B*. Volume 6, Issue 4, Pages 287-296.

A GENE WRITER™ system can be introduced into cells, tissues and multicellular organisms. In some embodiments the system or components of the system are delivered to the cells via mechanical means or physical means.

Formulation of protein therapeutics is described in Meyer (Ed.), *Therapeutic Protein Drug Products: Practical Approaches to formulation in the Laboratory, Manufacturing, and the Clinic*, Woodhead Publishing Series (2012).

All publications, patent applications, patents, and other publications and references (e.g., sequence database reference numbers) cited herein are incorporated by reference in their entirety. For example, all GENBANK™, UNIGENE™, and ENTREZ™ sequences referred to herein, e.g., in any Table herein, are incorporated by reference. Unless otherwise specified, the sequence accession numbers specified herein, including in any Table herein, refer to the database entries current as of Aug. 27, 2018. When one gene or protein references a plurality of sequence accession numbers, all of the sequence variants are encompassed.

EXAMPLES

The invention is further illustrated by the following examples. The examples are provided for illustrative purposes only and are not to be construed as limiting the scope or content of the invention in any way.

Example 1: Delivery of a GENE WRITER™ System to Mammalian Cells

This example describes a GENE WRITER™ genome editing system delivered to a mammalian cell for site-specific insertion of exogenous DNA into a mammalian cell genome.

In this example, the polypeptide component of the GENE WRITER™ system is the R2Bm protein from *Bombyx mori* and the template RNA component is RNA for the R2Bm retrotransposase from *Bombyx mori* containing a mutation in the reverse transcriptase domain that renders the retrotransposase inactive.

HEK293T cells are transfected with the following test agents:

1. Scrambled RNA control
2. RNA coding for the polypeptide described above
3. Template RNA described above
4. Combination of 2 and 3

After transfection, HEK293T cells are cultured for at least 4 days and then assayed for site-specific genome editing. Genomic DNA is isolated from each group of HEK293 cells. PCR is conducted with primers that flank the R2Bm integration site in 28s rRNA genes. The PCR product is run on an agarose gel to measure the length of the amplified DNA.

A PCR product of the expected length, indicative of a successful GENE WRITING™ genome editing event that

inserts the sequence for the mutated R2Bm retrotransposase into the target genome, is observed only in cells that were transfected with the complete GENE WRITER™ system of group 4 above.

Example 2: Site-Specific Targeted Delivery of a GENE WRITER™ System into Insect Cells

This example describes a GENE WRITER™ genome editing system delivered to an insect cell at a specific target site of the genome.

In this example, the polypeptide component of the GENE WRITER™ system is derived from R2Bm of *Bombyx mori*, which is modified by replacing its DNA binding domain in the amino terminus of the polypeptide with a heterologous zinc-finger DNA binding domain. The zinc finger DNA binding domain is known to bind to DNA in the BmBLOS2 loci of *B. mori* cells (Takasu et al., *insect Biochemistry and Molecular Biology* 40(10): 759-765, 2010). The template RNA is RNA for the R2Bm retrotransposase from *Bombyx mori* containing a mutation in the reverse transcriptase domain that renders the retrotransposase inactive. Furthermore, the template RNA is modified at the 5' end to have 180 bases of homology to the target DNA site.

B. mori insect cell lines are transfected with the following test agents:

1. Scrambled RNA control
2. RNA coding for polypeptide component described above
3. Template RNA described above
4. Combination of 2 and 3

After transfection, the cells are cultured for at least 4 days and assayed for site-specific GENE WRITING™ genome editing. Genomic DNA is isolated from the cells and PCR is conducted with primers that flank the target integration site in the genome. The PCR product is run on an agarose gel to measure the length of DNA. A PCR product of the expected length, indicative of a successful GENE WRITING™ genome editing event that inserts the sequence for the mutated R2Bm retrotransposase into the target insect cell genome, is observed only in cells that were transfected with the complete GENE WRITER™ system of group 4 above.

Example 3: Site-Specific Targeted Delivery of a GENE WRITER™ System into Mammalian Cells

This example describes a GENE WRITER™ genome editor system used to insert a heterologous sequence into a specific site of the mammalian genome.

In this example, the polypeptide of the system is the R2Bm protein from *Bombyx mori* and the template RNA component is RNA coding for the GFP protein and flanked at the 5' end by the 5' UTR and at the 3' end by the 3' UTR of the R2Bm retrotransposase from *Bombyx mori*. The GFP gene has an internal ribosomal entry site upstream of its start codon and a polyA tail downstream of its stop codon.

HEK293 cells are transfected with the following test agents:

1. Scrambled RNA control
2. RNA coding for the polypeptide described above
3. Template RNA coding for GFP described above
4. Combination of 2 and 3

After transfection, HEK293 cells are cultured for at least 4 days and then assayed for a site-specific GENE WRITING™ genome editing event. Genomic DNA is isolated from the HEK293 cells and PCR is conducted with primers that flank the R2Bm integration site in 28s rRNA genes. The

PCR product is run on an agarose gel to measure the length of DNA. A PCR product of the expected length, indicative of a successful GENE WRITING™ genome editing event, is detected in cells transfected with the test agent of group 4 (complete GENE WRITER™ system). This result demonstrates that a GENE WRITING™ genome editing system can insert a novel transgene into the mammalian cell genome.

The transfected cells are cultured for a further 10 days, and after multiple cell culture passages are assayed for GFP expression via flow cytometry. The percent of cells that are GFP positive from each cell population are calculated. GFP positive cells are detected in the population of HEK293 cells that were transfected with the test agent of group 4 (complete GENE WRITER™ system). This result demonstrates that the novel transgene written into the mammalian cell genome is expressed.

Example 4: Targeted Delivery of a Gene Expression Unit into Mammalian Cells Using a GENE WRITER™ System

This example describes the making and using of a GENE WRITER™ genome editor to insert a heterologous gene expression unit into the mammalian genome.

In this example, the polypeptide of the GENE WRITER™ system is derived from the R2Bm polypeptide of *Bombyx mori* as modified by replacing its DNA binding domain in the amino terminus of the polypeptide with a heterologous zinc-finger DNA binding domain. The zinc finger DNA binding domain is known to bind to DNA in the AAVS1 locus of human cells (Hockemeyer et al., Nature Biotechnology 27(9): 851-857, 2009). The template RNA comprises a gene expression unit. A gene expression unit comprises at least one regulatory sequence operably linked to at least one coding sequence. In this example, the regulatory sequences include the CMV promoter and enhancer, an enhanced translation element, and a WPRE. The coding sequence is the GFP open reading frame. The gene expression unit is flanked at the 5' end by 180 bases of homology to the target DNA site and at the 3' end by the 3' UTR of the R2Bm retrotransposase from *Bombyx mori*.

HEK293 cells are transfected with the following test agents:

1. Scrambled control RNA
2. RNA coding for the polypeptide component described above
3. Template RNA comprising the gene expression unit (as described above)
4. The complete GENE WRITER™ system comprising both (2) and (3)

After transfection, HEK293 cells are cultured for at least 4 days and assayed for site-specific GENE WRITING™ genome editing. Genomic DNA is isolated from the HEK293 cells and PCR is conducted with primers that flank the target integration site in the genome. The PCR product is run on an agarose gel to measure the length of DNA. A PCR product of the expected length, indicative of a successful GENE WRITING™ genome editing event, is detected in cells transfected with the test agent of group 4 (complete GENE WRITER™ system).

The transfected cells are cultured for a further 10 days, and after multiple cell culture passages are assayed for GFP expression via flow cytometry. The percent of cells that are GFP positive from each cell population are calculated. GFP positive cells are detected in the population of HEK293 cells that were transfected with group 4 test agent, demonstrating

that a gene expression unit added into the mammalian cell genome via GENE WRITING™ genome editing is expressed.

Example 5: Targeted Delivery of a Gene Expression Unit into an Intronic Region of Mammalian Cells Using a GENE WRITER™ System

This example describes the making and use of a GENE WRITING™ genome editing system to add a heterologous sequence into an intronic region to act as a splice acceptor for an upstream exon.

The target integration site is the first intron of the albumin locus. Splicing into the first intron a new exon containing a splice acceptor site at the 5' end and a polyA tail at the 3' end will result in a mature mRNA containing the first natural exon of the albumin locus spliced to the new exon. Because the first exon of albumin is removed upon protein processing, the cell expressing the newly formed gene unit will secrete a mature protein comprising only the new exon.

In this example, the GENE WRITER™ genome editor polypeptide is derived from the R2Bm GENE WRITER™ genome editor of *Bombyx mori* as modified by replacing the DNA binding domain in the amino terminus of the polypeptide with a heterologous zinc-finger DNA binding domain. The zinc finger DNA binding domain is known to bind tightly to the albumin locus in the first intron as described in Sarma et al., Blood 126, 15: 1777-1784, 2015. The template RNA is RNA coding for EPO with a splice acceptor site immediately 5' to the first amino acid of mature EPO (the start codon and signal peptide is removed) and a 3' polyA tail downstream of the stop codon. The EPO RNA is further flanked at the 5' end by 180 bases of homology to target DNA site and at the 3' end by the 3' UTR of the R2Bm retrotransposase from *Bombyx mori*.

HEK293 cells are transfected with the following test agents:

1. Scrambled control RNA
2. RNA coding for the polypeptide described above
3. Template RNA comprising the EPO splice acceptor described above
4. The complete GENE WRITER™ system comprising both (2) and (3)

After transfection, HEK293 cells are cultured for at least 4 days and assayed for site-specific GENE WRITING™ genome editing and appropriate mRNA processing. Genomic DNA is isolated from the HEK293 cells. Reverse transcription-PCR is conducted to measure the mature mRNA containing the first natural exon of the albumin locus and the new exon. The RT-PCR reaction is conducted with forward primers that bind to the first natural exon of the albumin locus and with reverse primers that bind to EPO. The RT-PCR product is run on an agarose gel to measure the length of DNA. A PCR product of the expected length is detected in cells transfected with the test agent of group 4, indicative of a successful GENE WRITING™ genome editing event and a successful splice event. This result demonstrates that a GENE WRITING™ genome editing system can add a heterologous sequence encoding a gene into an intronic region to act as a splice acceptor for the upstream exon.

The transfected cells are cultured for a further 10 days, and after multiple cell culture passages are assayed for EPO secretion in the cell supernatant. The amount of EPO in the supernatant is measured via an EPO ELISA kit. EPO is detected in HEK293 cells that were transfected with the test

agent of group 4, demonstrating that a heterologous sequence can be added into an intronic region via GENE WRITING™ genome editing, to act as a splice acceptor for the upstream exon and is actively expressed.

Example 6: Targeted Delivery of R2Tg Retrotransposon to Mammalian Cells

This example describes targeted integration of the R2Tg retrotransposon element (see first row of Table 3 herein) to mammalian cells via DNA or RNA delivery.

R2Tg is an endogenous retrotransposon from the zebra finch (*Taenopygia guttata*). Because non-LTR R2 elements are not present in the human genome and are thought to be highly site-specific, the ability of R2Tg to accurately and efficiently integrate itself into the human genome would demonstrate the capability to perform genomic targeted integration and possibly enable human gene therapy.

In the DNA delivery method, plasmid harboring R2Tg (PLV014) was designed and synthesized such that the R2Tg element was codon optimized and flanked by its native un-translated regions (UTRs), with or without further flanking by 100 bp homology to the rDNA target locus. The R2Tg element expression was driven by the mammalian CMV promoter. Further, a 1bp deletion mutant (678*) having a frameshift in the coding sequence of the retrotransposase was constructed as an inactivated control (“frameshift mutant”). Each plasmid was introduced into HEK393T cells via FuGENE® HD transfection reagent. HEK293T cells were seeded in 96-well plate, 10,000 cells/well 24 hr before transfection. On the transfection day, 0.5 µl transfection reagent and 80 ng DNA was mixed in 10 µl Opti-MEM and incubated for 15 min at room temperature. Then the transfection mixture was added to the medium of the seeded cells. 3 days after transfection, genomic DNA was extracted for retrotransposition assays.

Next, integration of the R2Tg transposase into the human genome was assessed. Based on homology to the finch genome, a putative integration site in human rDNA was tested. Advanced MISEQ™ and ddPCR assays were used to assess integration.

Bias in MISEQ™ library construction was eliminated by introducing random unique molecular indices (UMIs) into initial PCRs (FIG. 7). Nested PCR was performed by first amplifying the expected 3' junction of R2Tg and the rDNA locus for 30 cycles. One MISEQ™ adapter, a multiplexing barcode, and an 8 bp UMI were introduced at this step. A second PCR was used to further enrich for expected products and add the second MISEQ™ adapter. Samples were sequenced on the MISEQ™ for 300 cycles. After demultiplexing, the samples were analyzed via MATLAB™. First, the UMI on each sequence was located by searching for neighboring sequence. A database of UMIs was created and next collapsed by uniqueness. For each unique read, for a search was performed for the sequence of the expected rDNA integration site and isolated sequences of aligned human genomic DNA and exogenous DNA. Exogenous DNA was then aligned to the expected integration sequence. Results of the MISEQ™ analysis pipeline are shown in FIGS. 8A-8B. Extensive unique integrations into the predicted integration site were found in cells treated with the wildtype R2Tg construct flanked by 100 bp homology to the target rDNA locus, but not with the frameshift mutant controls. Most integration events have a complete template RNA sequence integrated in the 565 bp most proximal to the integration site as demonstrated by sequencing reads that align perfectly to the expected sequence. A subset of inte-

gration events with the experimental R2Tg have either a ~300 bp or ~450 truncation as based upon sequencing reads that align to the expected sequence after a gap directly adjacent to the target site (FIG. 8A). More specifically, 86.17% of integrants observed were non-truncated in the 565 bp most proximal to the integration site. In contrast, FIG. 8B shows no integration events detected. Constructs without flanking rDNA homology showed insignificant integration signals near noise.

ddPCR was next performed to confirm integration and assess integration efficiency. A Taqman probe was designed to the 3'UTR portion of the R2Tg element. A forward primer was synthesized to bind directly upstream of the probe, and a reverse primer was synthesized to bind the rDNA. Therefore, amplification of the expected product across the integration junction degrades the probe and creates a fluorescent signal. ddPCR was performed on several replicate experiments of the above plasmids to determine the average copy number of the R2Tg integration event. The results of ddPCR copy number analysis (in comparison to reference gene RPP30) are shown in FIG. 9. Across several plasmid transfection conditions, average integration of 5 or more copies of R2Tg per genome at the target site when delivered with homology was noted with significant increase above control constructs. In contrast, the average copy number per genome in the frameshift mutant negative control was typically lower than 1. Insignificant signal was seen when constructs without homology were delivered to cells. The experiments collectively suggest efficient integration of the R2Tg retrotransposon into human cells at the target site.

In the RNA delivery method, R2Tg RNA (RNAV019) was designed such that the R2Tg element was codon optimized and flanked by its native untranslated regions (UTRs). More specifically, the construct includes, in order: a T7 promoter, a 5' 28S target homology region 100 nucleotides in length, a R2Tg wild-type 5' UTR, the R2Tg codon-optimized coding sequence, a R2Tg wild type 3' UTR, and a 3' 28S target homology region 100 nucleotides in length. The 100 bp 28S homology sequences were added outside of the UTRs to enhance the integration. R2Tg RNA was synthesized, and cap and polyA tail were added. The R2Tg element transcription was driven by the T7 promoter. The RNA was introduced into HEK393T cells via Lipofectamine™ RNAiMAX or TransIT®-mRNA transfection reagent with series of RNA dosages. HEK293T cells were seeded in 96-well plate 24 hr before transfection. On the transfection day, transfection reagent and RNA were mixed in 10 µl Opti-MEM, and the transfection mixture was added to the medium of the seeded cells. 3 days after transfection, genomic DNA was extracted to measure retrotransposition efficiency using ddPCR with the same design as the DNA delivery.

The results of ddPCR copy number analysis (normalized to reference gene RPP30) are shown in FIG. 12. Across several transfection conditions, the average integration was measured to be 0.01 of R2Tg copies per genome, significantly above the limit of detection. The results indicate successful integration of the R2Tg retrotransposon into human cells using an RNA delivery method.

Example 7: Targeted Delivery of a Heterologous Object Sequence Using R2Tg Retrotransposon to Mammalian Cells

This example describes the delivery of a transgene to human cells by utilizing the R2Tg retrotransposon system with multiple delivery machineries, including RNA-medi-

ated delivery of a heterologous object sequence to human cells by utilizing the R2Tg retrotransposon system.

R2 proteins recognize their template RNA structure in untranslated regions (UTRs) of each element to form ribonucleoprotein particles, which serve as the intermediates of downstream integration into a host genome. Therefore, the decoupling of UTRs from their native context and the introduction of UTRs into alternate exogenous sequence was engineered to deliver into the genome a desired nucleic acid using R2Tg machinery.

Trans-transgene integration was tested by constructing 1) R2Tg coding sequence and 2) transgene cassette flanked by R2Tg UTR sequences and 100 bp homology to 28S rDNA into separate driver and transgene plasmids, respectively. FIG. 13 illustrates the dual plasmid system. The dual plasmids were introduced into HEK293T cells via FuGENE® HD transfection reagent at multiple driver to transgene molar ratios. In addition to the WT R2Tg driver, backbone plasmid was used as a control. HEK293T cells were seeded in 96-well plates at 10,000 cells/well 24 hr before transfection. On the transfection day, transfection reagent and plasmids were mixed in 10p Opti-MEM and incubated for 15 minutes at room temperature, then added to the medium of the seeded cells. 3 days after transfection, genomic DNA was extracted for ddPCR assays to investigate the trans-retrotransposition efficiency. FIG. 14 demonstrates the ddPCR results for conditions with excess of transgene relative to driver.

Similar to the trans-transgene delivery with plasmids, RNA delivery was performed by constructing an amplicon of the coding sequence of R2Tg preceded by the T7 promoter sequence. The constructed amplicons contained the experimental R2Tg element as well as the 1 bp deletion frameshift mutant control. Separately, an amplicon was constructed that contained exogenous sequence coding for GFP and an EGF1-alpha reporter that was flanked regions sufficient to drive integration into the genome by R2Tg. More specifically, the construct included: a T7 promoter driving transcription of the RNA, wherein the RNA comprises, from 5' to 3', (a) a 5' 28S homology region of 10 nt in length, (b) a 5' untranslated region, (c) an anti-sense TKpA polyA sequence, (d) an anti-sense heterologous object sequence that encodes GFP, (e) an anti-sense Kozak sequence, (f) an anti-sense EF1 alpha promoter, (g) a 3' untranslated region that binds the GENE WRITER™ protein, and (h) a 3' 28S homology region of 10 nt in length. Each RNA was transcribed via the New England Biolabs HiScribe T7 ARCA kit and purified via Zymo RNA clean and concentrator.

The resulting heterologous object RNA and R2Tg RNA (either the experiment R2Tg element or frameshift mutant) were introduced into human HEK293T cells via TransIT®-mRNA Transfection Kit at 1:1 molar ratio. HEK293T cells were seeded in 96-well plate, 40,000 cells/well 24 hr before transfection. On the transfection day, 1 µl transfection reagent and 500 ng total RNA was mixed in 10 µl Opti-MEM and incubated for 5 min at room temperature. Then the transfection mixture was added to the medium of the seeded cells. 3 days after transfection, genomic DNA was extracted for PCR assays.

Nested PCR was performed by with a first 30 rounds of PCR across the 3' end of the expected transgene-rDNA junction, followed by 20 additional rounds of PCR amplification using an inner primer set. One of three replicates of nested PCR performed on genomic DNA extracted from cells treated with the wild-type transposase reaction produced a PCR product of the expected size (approximately

596 bp). In contrast, no PCR product was observed in genomic DNA extracted from cells treated with the frameshift-inactivated R2Tg mutant control, or no-transfection control. The PCR product was gel-purified via Zero Blunt® TOPO® PCR Cloning Kit, and the resulting colonies were Sanger sequenced. Each individual PCR product sequence was then aligned to the expected integration sequence. The fraction of PCR product sequences that align to the expected integrated heterologous object sequence is shown in FIG. 10. The majority of PCR products had the expected integrant as demonstrated by the sequencing alignment directly adjacent to the expected integration site at the right side of the alignment figure. This demonstrates RNA-mediated integration of the exogenous sequence via R2Tg machinery into human cells.

Example 8: Targeted Delivery of R2Tg Retrotransposon to Mammalian Cells

This example describes targeted integration of the R2Tg retrotransposon element to mammalian cells via DNA delivery.

Plasmid harboring R2Tg (PLV014) and control plasmid were designed and synthesized as described above in Example 6. Each plasmid was introduced into HEK293T cells via FuGENE® HD transfection reagent. HEK293T cells were seeded in 96-well plate, 10,000 cells/well 24 hr before transfection. On the transfection day, 0.5 µl transfection reagent and 80 ng DNA was mixed in 10 µl Opti-MEM and incubated for 15 min at room temperature. Then the transfection mixture was added to the medium of the seeded cells. 3 days after transfection, genomic DNA was extracted for retrotransposition assays or cells were frozen and underwent targeted locus amplification.

Target locus amplification was performed against hg38 reference human genome and the rDNA locus sequence hsl3369 (GENBANK™: U13369.1). Two independent primer sets were used to perform targeted locus amplification. Analysis with both primer sets showed that the 28S rDNA locus sequence is the only integration site detected above a 1% threshold. Thus, integration of the R2Tg transposon in mammalian cells is specific to this target site.

Example 9: Improved Trans RNA-Templated Integration into Mammalian Cells by RNA Refolding or Ratio of Driver to Template RNA

RNA templates are designed as in previous examples. Two RNAs consisting of a driver and a transgene payload are delivered to mammalian cells. To better promote folding, denaturing the payload RNA by heating to 95 C and cooling to room temperature are performed to encourage proper secondary structure formation. In some embodiments, cooling the RNA to room temperature will increase integration efficiency.

The molar ratio of transgene to driver is also varied to evaluate suitable stoichiometry of components. Integration is analyzed via ddPCR and sequencing. In some embodiments, a higher ratio of driver to transgene is used. In some embodiments, a higher ratio of transgene to driver is used.

Previous examples with cis transgene integration are similarly assayed for stoichiometry of driver to payload. Integration is analyzed via ddPCR and sequencing. In some embodiments, a higher ratio of driver transcription or translation to transgene transcription will result in higher integration efficiency. In some embodiments, a higher ratio of

transgene transcription to driver transcription and translation will result in higher integration efficiency.

Example 10: Hybrid Capture Assay

A hybrid capture experiment was performed to obtain an unbiased view of the specificity of retrotransposon integration into a target site. Retrotransposon experiments were performed as in previous examples by integrating R2Tg flanked by its native UTRs and 100 bp of homology to either side of the expected R2 rDNA target. The rDNA target site had two flanking sets of 100 nucleotides identity to the corresponding native target site. The retrotransposon was delivered to human 293T cells via plasmid or mRNA. Genomic DNA was extracted after 72 hours. After extraction, each genomic DNA sample was subjected to hybrid capture according to protocol with a custom probe set (Twist). Biotinylated probes were designed such that ~120 bp probes spanned both strands of the R2Tg coding sequence and UTRs. First, a next-generation library was created by fragmentation of the genomic DNA and ligation of sequencing adapters according to a protocol from Twist (available on the world wide web at: twistbioscience.com/ngs_protocol_custompanel_hybridcap). Next, probes were hybridized to genomic DNA libraries and the enriched samples were amplified. Final libraries were sequenced on the MISEQ™ using 300 bp paired-end reads. Custom MATLAB™ scripts were used to analyze reads. The resulting analysis is shown in FIGS. 15A and 15B for RNA delivery. Hybrid capture indicated on-target integration of R2Tg to the expected locus. With RNA delivery, 1 possible off-target with a single read was identified at an unexpected 3' junction in the data, compared to more than 100 reads at the expected locus, indicating a specificity of greater than 100:1. At the 5' junction, all 50 reads were at the expected locus, indicating a specificity of greater than 50:1. This experiment indicates a high specificity of integration.

Example 11: Long-Read PacBio Analysis

Long-range PCR amplification can be performed to measure integration of the desired full-length sequence into the target site in the human genome and to measure whether mutations are introduced during insertion. Retrotransposon integration experiments are performed as described in previous examples. In one example, PCR amplification is used to generate amplicons by designing one primer targeting the genomic integration site and one primer targeting the integrant sequence. In this example, these primers are designed to maximize the length of the amplified genomic locus fused with the integrant sequence. By pooling amplicons spanning both ends of the integrant and performing long-read next-generation sequencing, the fidelity of each integration is evaluated.

In another example, hybrid capture is performed as described in a previous example but with a larger target library length during initial library generation. The resulting library is then subjected to long-read next-generation sequencing.

In some embodiments, long-read next generation sequencing will show that there are less than 10%, 5%, 2%, 1%, 0.5%, 0.2%, or 0.1% SNPs in the integrated DNA across samples. In some embodiments, long-read next generation sequencing will show that less than 10%, 5%, 2%, or 1% of integrated DNA has a SNP. In some embodiments, long-read next generation sequencing will show that less than 10%, 5%, 2%, or 1% of integrated DNA has an internal deletion.

In some embodiments, long-read next generation sequencing will show that less than 10%, 5%, 2%, 1%, 0.5%, 0.2%, or 0.1% of total integrated DNA across the population is deleted. In some embodiments, long-read next generation sequencing will show that less than 10%, 5%, 2%, or 1% of integrated DNA is truncated.

Example 12: Experiment with Different Homology Lengths and Point Mutations in Homology

In this example, experiments are designed to characterize suitable lengths and starting positions of homology to the target site for efficient retrotransposon integration. Also, the homology is used to support the mechanism of integration being reverse transcription-driven.

A series of SNPs were introduced within the 100 bp downstream homology of R2Tg plasmids by modifying plasmid PLV014. The design of the SNPs is listed in FIG. 16. After the transfection, nested PCR was applied to recover the 3' integration junction site, producing a PCR product with an expected amplicon size of about 738 bp, and the PCR product was Sanger sequenced to check whether any SNPs were incorporated. In this experiment, a lack of SNP genetic markers being incorporated into the junction sequences indicates that the integration was driven by reverse transcription. The SNP design and the sequencing result are illustrated in FIG. 16. No SNP introduction was observed for the 18 genetic markers designed, consistent with the integration of R2Tg being directed by reverse transcription.

This example also describes the evaluation different homology regions to the target site to identify shorter regions that promote efficient integration into the genome. This example describes two approaches. First, different windows of 100 bp of homology to the target site are tested, starting from bp 1-100 3' of the target site, then testing 2-101 3' of the target site, 3-103 3' of the target site, and so on, through bp 30-131 3' of the target site. Second, shorter lengths of homology to the target site sufficient for DNA integration are tested, starting with bp 0-100 3' of the target site, then testing 0-95 3' of the target site, 0-90 3' of the target site, etc. through bp 0-10 3' of the target site. After the transfection of each plasmid into 293T cells, ddPCR is used to measure the retrotransposition efficiency.

In this example, different UTR regions with different lengths are evaluated to identify shorter sequences for efficient integration into the genome. The 3'UTR is tested by dividing this 325 bp sequence into 3 regions, 1-100 bp, 101-200 bp, and 201-325 bp. Constructs of R2Tg containing each truncated 3'UTR are generated to test the integration efficiency respectively.

Example 13: Assess Whether p53 or Other Repair Pathways are Upregulated

This example describes an evaluation of the effect of exogenous R2Tg retrotransposon on gene expression, especially tumor suppressor and DNA repair genes. An R2Tg expressing plasmid is delivered to multiple cancer cell lines, including 293T, MCF-7, and T47D. After confirmation of integration in each cell line, RNA-seq is conducted to assess the effect on gene expression profile. Gene set enrichment analysis is then applied to evaluate whether any DNA repair pathways are upregulated after retrotransposition. MCF-7 and T47D are breast cancer cell lines with wild type and mutant p53 respectively, which are used to evaluate the relationship between p53 and retrotransposon specificity.

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cally. In some embodiments, p53 is not upregulated when a retrotransposon GENE WRITER™ integrates into the genome. In some embodiments, no DNA repair genes are upregulated when a retrotransposon GENE WRITER™ integrates into the genome. In some embodiments, no tumor suppressor genes are upregulated when a retrotransposon GENE WRITER™ integrates into the genome.

Example 14: Retrotransposition in Presence of DNA Repair Inhibitors

In this example, experiments will test the effect of different DNA repair pathways on R2Tg retrotransposition via the application of DNA repair pathway inhibitors or DNA repair pathway deficient cell lines. When applying DNA repair pathway inhibitors, PrestoBlue cell viability assay is performed first to determine the toxicity of the inhibitors and whether any normalization should be applied for following assays. SCR7 is an inhibitor for NHEJ, which is applied at a series of dilutions during R2Tg delivery. PARP protein is a nuclear enzyme that binds as homodimers to both single- and double-strand breaks. Thus, its inhibitors are used in the test of relevant DNA repair pathways, including homologous recombination repair pathway and base excision repair pathway. The experiment procedure is the same with that of SCR7. Cell lines with deficient core proteins of nucleotide excision repair (NER) pathway are used to test the effect of NER on R2Tg retrotransposition. After the delivery of R2Tg element into the cell, ddPCR is used to evaluate the retrotransposition in the context of inhibition of DNA repair pathways. Sequencing analysis is also performed to evaluate whether certain DNA repair pathway plays a role in the alteration of integration junction. In some embodiments, R2Tg integration into the genome will not be decreased by the knockdown of any DNA repair pathways, suggesting that R2Tg does not rely on the host cell pathways for DNA integration.

Example 15: Retrotransposition in Fibroblasts and in T Cells

In this example, the previously performed R2Tg retrotransposition analysis of 293T cells is repeated in non-dividing cells, including fibroblast and T cells. Compared to 293T cells, non-dividing cells are sometimes more difficult to transfect with lipid reagent. Thus, nucleofection is used for the delivery of R2Tg element. The subsequent retrotransposition assay for integrating efficiency and sequencing analysis will be performed as described herein for 293T cells. In some embodiments, R2Tg integrates into the genome of fibroblasts and T cells.

Example 16: Single Cell ddPCR

In this example, a quantitative assay is used to determine the frequency of targeted genome integration at single cell level, and that information can be compared to the copy number of targeted genome integration per genome quantified from genomic DNA.

Approximately 5000 transfected cells will be collected and mixed with ddPCR reaction mixture before distributing into about 20,000 droplets, with the aim of each droplet containing one cell or no cells. ddPCR assays including 5'UTR and 3'UTR assays will be performed as described above to determine the frequency of R2 or transgene integration at single cell level. A control experiment will be performed in parallel using genomic DNA harvested from

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the same number of cells to determine the targeted genome integration efficiency per genome. In some embodiments, the frequency of targeted genome integration at the single cell level is calculated to be 1-80%, e.g., 25%, wherein the indicated percentage of cells have one or more copies of the transgene integrated into the desired locus.

Example 17: Single Cell Analysis Via Colony Isolation

In this example, a quantitative assay is used to determine genome integration copy number in cell colonies derived from single cell.

Single cell colonies will be isolated by colony picking up or by limited dilution and cultured in a 96 well format. When the cells reach >80% confluency, half of the cells will be frozen for backup and genomic DNA from the other half of the cells will be harvested for ddPCR. Optimized ddPCR assays including 5'UTR and 3'UTR assays will be performed as described previously to determine the frequency of R2 or transgene integration. At least 96 colonies will be screened for each R2 element with appropriate controls. The total number of colonies to be screened will be determined by single cell ddPCR data if applicable or the first set of single cell colony screen data. In some embodiments, the frequency of targeted genome integration at the single cell level will be calculated to be 1-80%, e.g., 25%, wherein the indicated percentage of cells have a single copy of the transgene integrated into the desired locus. The assay can also be used to determine the percentage of colonies that have more than one copy of the transgene integrated into the desired locus.

Example 18: DNA Binding Affinity and/or Re-Targeting

The DNA targeting module of wild-type R2 is made of a cysteine-histidine zinc finger and c-Myb transcription factor binding motifs. This N-terminal module can be substituted with different DNA binding modules such as DNA binding protein(s) (e.g., transcription factors), zinc finger(s) (e.g., natural or designed motifs), and/or nucleic acid guided, catalytically inactive endonucleases (e.g., Cas9 bound with a guide RNA (e.g., sgRNA) to form a Cas9-RNP). This DNA binding module is swapped for the naturally occurring module and, in some cases placed with a flexible linker attaching it to the RNA binding/RT module. Additionally, in some constructions, this new DNA binding module is placed in tandem with the same and/or different DNA binding modules. Furthermore, some constructions may split the GENE WRITER™ protein where one protein molecule contains the RNA binding module and the other protein contains the RT and endonuclease modules. In some embodiments, swapping of the DNA module increases specificity and/or affinity to a genomic location and in some cases allows for the specific targeting of new genomic locations.

Example 19: Assays to Measure DNA Binding Affinity

DNA binding activity of GENE WRITER™ genome editor polypeptides described herein (and DNA binding domains for the same) can be tested, e.g., as described in this example. DNA binding modules are purified by recombinantly expressing them in cells (e.g., *E. coli*) or they are expressed in a cell-free reactions of transcription and trans-

lation (e.g., T7 RNA polymerase+wheat germ extract). The purified DNA binding module(s) is tested for binding affinity by measuring the K_d in a binding assay (e.g., EMSA, Fluorescence anisotropy, dual-filter binding, FRET, SPR, or thermophoresis (temperature related intensity change)). The protein (DNA binding module) is labeled and/or the DNA molecule is labeled with a molecule that is compatible with the above binding assays (e.g., dye, radioisotope (for example, Protein: ^{35}S -methionine, maleimide dye, DNA: ^{32}P end or internal label, DNA with a linked amine reacted with NHS-ester dye)). The molecules are measured by changing their concentrations and fitting to a binding curve which calculates the binding affinity. In some assays, the nucleic acid sequence specificity is tested by mutational analysis of the DNA sequence or mutation to the DNA binding module by amino acid changes or alterations to protein-nucleic acid complex (e.g., Cas9-RNP DNA binding module). In some embodiments, increasing the K_d of the DNA binding module will decrease off-target insertions and, in some cases, will increase the activity of on-target sites by increasing the dwell time of the R2-RNA complex at the specific genomic location.

Example 20: Assays to Determine Global Specificity De Novo

The DNA binding module is expressed in cells (e.g., animal cells, e.g., human cells) as the DNA binding module alone, in the context of the full-length retrotransposon R2, or a control without retrotransposase. The expression of the module or retrotransposon is delivered to cells using conventional methods of delivering DNA, RNA, or protein. The complex is crosslinked (e.g., using chemical or UV light) or is not crosslinked. The cells are lysed and treated with DNase I so that only the bound DNA is protected from degradation. DNA is extracted, NGS library preparation of DNA fragments and de novo binding sites are identified, analogous to ChIP-seq or DIG-seq. In some embodiments, potential off-target sites are identified that can be followed-up to remove false-positives. In other embodiments this assay confirms the in vitro assay on the specificity of the DNA binding module to bind at its intended site and not at others.

An orthogonal assay to identify DNA binding sites in high-throughput uses the method described by Boyle et al, PNAS 2017 where the DNA binding domain is tested in a cell-free setting to determine specificity along with systematic analysis of sequence mutants related to the new DNA binding module.

Example 21: Modularity of RNA Molecule

The RNA molecule binds to the R2 protein via interactions found in the reverse transcriptase module, designated as a sub-module "RNA binding". The protein recognizes specific structures in the 5' and/or 3' UTRs to interact with the RNA. In some embodiments, swapping of the UTR modules increases protein interactions, changes the protein specificity to bind the UTR, stabilizes against nucleases, and/or improves cellular tolerance (e.g., leads to a reduced innate immune response). In other embodiments, addition and/or swapping of the RNA binding module of the R2 protein is compatible with the use of different sequence or ligands that are linked to the transgene and/or element module of the RNA. In some embodiments, combinations of new ligands in place of the UTRs will have better affinity to the RNA binding domain of R2 and lead to better insertion

efficiency. In some embodiments, the changes to the sequence of the UTRs or changes to the base modifications of the UTRs will increase the secondary structure stability that leads to better interaction with the RNA binding module.

Example 22: Assays to Measure RNA Binding Affinity to New Sequences

New UTR modules are tested in a binding assay. In the case of new RNAs, they are synthesized either by cell-free in vitro transcription using a synthetic DNA template or by chemical synthesis of the RNA in full-length or chemical synthesis of pieces that are ligated together to form a single RNA molecule. The binding affinity of the purified UTRs are measured in a binding assay (e.g., EMSA, Fluorescence anisotropy, dual-filter binding, FRET, SPR, or thermophoresis (temperature related intensity change)). The UTR module and/or RNA binding module/RT module is detected with or without a label which is described above for labeling RNAs. Measurement of the molecules at different concentrations is performed to determine a binding affinity. In some embodiments, alterations to/swapping of the 5' and/or 3' UTR binding module and/or changes to the RNA binding/RT module will lead to better interactions than the wild-type R2 protein or UTR. In some embodiments, the increased interaction will lead to an increase in the efficiency of retro-transposition and in some cases increases specificity of the R2 protein to interact with the RNA.

Example 23: Alternative UTRs

While not wishing to be bound by theory, in some embodiments the UTRs act as a handle for the R2 protein to interact with the RNA which it uses as a template for RT in concert with it binding a genomic location, nicking the DNA with its endonuclease module, and then using the bound RNA as a template for RT insertion at the cleavage site in the DNA. For the UTR to keep the template in close proximity to the RT module, then the UTR modules can be substituted with different ligands that would bind to a specific RNA binding module engineered into the R2 protein. Thus, in some embodiments, the alternative non-RNA UTR is either a protein, small molecule, or other chemical entity that is attached covalently, through protein-protein interaction, small molecule-protein interaction, or through hybridization. In some embodiments the RNA binding module binds specifically to a ligand that is not RNA that is attached to the transgene module RNA that increases the efficiency, stability, and/or rate of retro-transposition.

Example 24: Assays to Measure the Activity of UTR Constructs

Binding assays to measure affinity of R2 protein with engineered UTRs are performed as described above, e.g., for a protein-nucleic acid interaction. In cases of protein-protein or protein-small molecule interactions the assay uses a label on the RNA transgene module where the UTR module is attached.

Example 25: Targeted Genomic Integration

In this example, GENE WRITING™ technology is delivered to target cells and to non-target cells, and new DNA is integrated into the genome in target cells at a higher frequency than in non-target cells. As described in more detail

below, this approach takes advantage of the non-target cell having an endogenous miRNA that the target cell does not have (or has at a lower level). The endogenous miRNA is used to reduce DNA integration in the non-target cell.

The polypeptide used is the R2Tg protein and the template RNA component is RNA coding for the GFP protein and flanked at the 5' end by the 5' UTR and at the 3' end by the 3' UTR of the R2Tg retrotransposase. The 5' UTR is flanked by 100 bp of homology to the 5' of R2Tg 28s rDNA target site and the 3' UTR is flanked by 100 bp of homology to the 3' of R2Tg 28s rDNA target site. The GFP gene is facing in the antisense direction with regard to the 5' and 3' UTRs and has its own promoter and polyadenylation signal.

The template RNA further comprises a microRNA recognition sequence. This microRNA recognition sequence is bound by microRNAs in the non-target cells, leading to the inhibition (e.g., degradation) of the template RNA prior to genomic integration.

In this example the target cells are hepatocytes and the non-target cells are macrophages from the hematopoietic lineage. The target cells and non-target cells are cultured separately. The template RNA and retrotransposase protein can be delivered to cells as described herein, e.g., as RNA or using viral vectors (e.g. adeno-associated viral vectors), wherein the template RNA is transcribed from viral vector DNA.

Three days after treating the cells, GFP expression and genomic integration are assayed.

GFP expression is assayed via flow cytometry. In some embodiments, GFP expression will be higher in the hepatocyte population than in the macrophage population.

Genomic integration (in terms of copy number per cell normalized to a reference gene) is assayed via droplet digital PCR using methods described herein. In some embodiments, genomic integration will be higher in the hepatocyte population than in the macrophage population.

Example 26: Testing Modularity of the DNA Binding Domain

In this example, a series of experiments were performed to test the activity of various mutant retrotransposases, as well as gaining structural knowledge about these proteins. This experiments tested flexible linkers in different locations and lengths, in order to determine if the DNA binding domain (DBD) was modular. These experiments also provide support for being able to separate the DBD from the rest of R2Tg and replacing it with any DNA targeting protein sequence. This example thus supports an understanding that the transposases described herein can withstand the tested levels of sequence divergence at a plurality of locations (e.g., in the predicted -1 RNA binding motif, in an alpha helix, and in a coil region located C-terminal to the predicted c-myb DNA binding motif, e.g., as described below) identified by structural modeling, while maintaining function.

Briefly, the two linkers (Linker A: SGSETPGTSSESAT-PES (SEQ ID NO: 1023), and Linker B: GGGG (SEQ ID NO: 1024)) were inserted into 3 locations, noted herein as versions v1, v2, and v3. v1 was located at the N-terminal side of an alpha helical region of R2Tg that preceded the predicted -1 RNA binding motif, v2 was located at the C-terminal side of an alpha helical region of R2Tg that preceded the predicted -1 RNA binding motif, and v3 was located C-terminal to a random coil region that came after the predicted c-myb DNA binding motif of R2Tg. For each of v1, v2, and v3, one of linkers A or B were added by PCR to a DNA plasmid that expressed R2Tg, thereby yielding sequences v1A (v1+linker A), v1B (v1+linker B), v1C (v1+linker C), v2A (v2+linker A), v2B (v2+linker B), and v2C (v2+linker C), as shown in Table 5 below. The insertion of the linkers was verified by Sanger sequencing and the DNA plasmids were purified for transfection.

TABLE 5

Amino acid sequences of R2Tg mutants with linkers in the DNA binding domain (DBD)		
R2Tg Mutant + Linker	Amino Acid Sequence	SEQ ID NO
R2Tg with DBD Linker v1A	MASCPKPGPPVSAGAMSLESGLTTHSVLAIERGPNS LANSGLDFGGGLGLPLRLLRVSVGTQTSRSDWVDL VSWSHPGPTSKSQQVDLVSLFPKHRVDLLSKNDQVD LVAQFLPSKFPNLAENDLALLVNLBFYRSDLHVEE CVHFAAHWEGLSGLEVEYQLAPQPCVGETLHSSLP RDELFVPEEGSSEKESDAPKTSPPTPGKHGLEQT GEEKVMVTVPDKNPPCPCGTRVNSVLNLIHKLVS HGKRGVCFRCAKCGKENSNYHVVCHFPKCRGPETE KAPAGEWICEVCNRDFTTKIGLQHKRLAHPVRNQ ERIVASQPKETSNRGAHKRCWTKEEEELLIRLEAQF EGNKNINKLIAEHITTKAKQISDKRRLLSRKPAAE PREEPGTCHHTRRAAASLRTEPEMSHHAQAEEDRNG PGRRLPLGRAAAGGRTMDEIRRHDPKGNQQRPRTKQ KSEELQAYYKKTLEERLSAGALNTFPRAFKQVMEG RDIKLVINQTAQDSGSETPGTSSESATPESCFGCLES ISQIRATRDKKDVTREKHPKPPQKWMKDRAIKK GNLRFQRLFYLDRGKLAKIILDDIECLSCDIPLSE IYSVFKTRWETGSKSLGDFKTYGKADNTAFRELI TAKEIEKNVQEMSKGSAPGPDGITLGDVVKMDPEFS RTMEIFNLWLTGKIPDMVRGCRTPVLIPKSSKPDRL KDINNWRPITIGSILLRFLSRIVTARLSKACPLNPR QRGFIRAAGCSENLKLLQTI IWSAKREHRPLGVVVF DIAKAFD TVSHQHI IHALQQREVDPHIVGLVSNMYE NISTYITTKRNTHTDKIQIRVGKQGDPMSPLELNL AMDPLLCKLEESGKGYHRGQSSITAMAFADDLVLLS DSWENMNTNISILETFCNLTGLKTQGGKCHGFYIKP TKDSYTINGCAAWTINGTPLNMIDPGESEKYLGLQF DPWIGIARSLSTKLDLFWLQRIQDAPLKLPLQKTDIL KTYTIPRLIYIADHSEVKTALLETLQKIRTAVKEW	1017

TABLE 5-continued

Amino acid sequences of R2Tg mutants with linkers in the DNA binding domain (DBD)		
R2Tg Mutant + Linker	Amino Acid Sequence	SEQ ID NO
	LHLPPCTCDAILYSTRDGGGLGITKLGLIPSVQAR RLHRIAQSSDDTMKCFMEKEKMEQLHKKLWIQAGGD RENIPSIWEAPPSSEPPNNVSTNSEWEAPTQDKDFP KPCNWRKNEFKKWTKLASQGRGIVNFERDKISNHWI QYRRIPHRKLLTALQLRANVYPTREFLARGRQDQY IKACRHCADIESCAHIIGNCPVTQDARIKRHNYIC ELLLLEEAKKDWVVFKEPHIRDSNKELYKPDILFVK DARALVVDVTVRYEAAKSSLEEEAAEKVRKYKHLET EVRHLTNAKDVTFVGFPPLGARGKWHQDNFKLLTELG LSKSRQVKMAETFTSTVALFSSVDIVHMFASRARKSM VM	
R2Tg with DBD Linker v1B	MASCPKPGPPVSAGAMSLESGLTTHSVLAIERGPNS LANSGSDFGGGGLGLPLRLLRVSVGTQTSRSDWVDL VSWSHPGPTSKSQQVDLVSLFPHKRVDDLKNDQVD LVAQFLPSKFPPNLAENDLALLVNLEFYRSDLHVYE CVHFAAHWEGLSGLPEVYEQAPQPCVGETLHSSLP RDSLEFVPEEGSSEKESDAPKTSPPTPKGHGLEQT GEEKVMVTVDPKNPPCPCGTRVNSVLNIEHLKVS HGKRGVCFRCAKCGKENSNYHSVCHFPKCRGPETE KAPAGEWICEVCNRDFTTKIGLGQHKRLAHPAVRNQ ERIVASQPKETSNRGAHKRCWTKEEEELLIRLEAQF EGNKNINKLIAEHITTKAKQISDKRLLSRKPAEE PREEPGTCHHTRRAAASLRTEPEMSHHAQAEDRDNG PGRRLPGRAAAGGRTMDEIRRHDPKNGGQORPTKQ KSEEQQLQAYYKKTLEERLSAGALNTFPRAFQVMEG RDIKLVINQTAQDGGGSCFGCLESISQIRTATRDKK DTVTRKHPKPPQKWMKDRAIKKGNLRFQRLFYL DRGKLAKIILDDIECLSCDIPLSEIYSVFKTRWETT GSFKSLGDFKTYGKADNTAFRELI TAKEIEKNVQEM SKGSAPGPDGITLGDVVKMDPEFSRTMEIFNLWLT GKIPDMVRGCRTVLIPKSSKPDRLKINNWRPITIG SILLRFLSRIVTARLSKACPLNPRQGRFIRAAGCSE NLKLLQTI IWSAKREHRPLGVVFDIAKAFDTVSHQ HI IHALQQREVDPHIVGLVSNMYENISTYITTKRNT HTDKIQIRVGVKQGDPMSPLLFNAMDPLCLKLEES GKGYHRGQSSI TAMAFADDLVLLSDSWENMNTNISI LETFCNLTKLTKQKCHGFYIKPTKDSYTINDCAA WTINGTPLNMIDPGESEKYLGLQFDPWIGIARSGLS TKLDFWLQRIDQAPLKPLQKTDILKTYTIPRLIYIA DHSEVKTALLETLDQKIRTAVKEWLHLPCTCDAIL YSTRDGGGLGITKLGLIPSVQARRLHRIAQSSDDT MKCFMEKEKMEQLHKKLWIQAGGDRENI PSIWEAPP SSEPPNNVSTNSEWEAPTQDKDFPKPCNWRKNEFKK WTKLASQGRGIVNFERDKISNHWI QYRRIPHRKLL TALQLRANVYPTREFLARGRQDQYIKACRHCADIE SCAHIIGNCPVTQDARIKRHNYICELLLLEEAKKDW VVFKEPHIRDSNKELYKPDILFVKDARALVVDVTVR YEAAKSSLEEEAAEKVRKYKHLETEVRHLTNAKDV TFVGFPPLGARGKWHQDNFKLLTELGLSKSRQVKMAET FSTVALFSSVDIVHMFASRARKSMVM	1018
R2Tg with DBD Linker v2A	MASCPKPGPPVSAGAMSLESGLTTHSVLAIERGPNS LANSGSDFGGGGLGLPLRLLRVSVGTQTSRSDWVDL VSWSHPGPTSKSQQVDLVSLFPHKRVDDLKNDQVD LVAQFLPSKFPPNLAENDLALLVNLEFYRSDLHVYE CVHFAAHWEGLSGLPEVYEQAPQPCVGETLHSSLP RDSLEFVPEEGSSEKESDAPKTSPPTPKGHGLEQT GEEKVMVTVDPKNPPCPCGTRVNSVLNIEHLKVS HGKRGVCFRCAKCGKENSNYHSVCHFPKCRGPETE KAPAGEWICEVCNRDFTTKIGLGQHKRLAHPAVRNQ ERIVASQPKETSNRGAHKRCWTKEEEELLIRLEAQF EGNKNINKLIAEHITTKAKQISDKRLLSRKPAEE PREEPGTCHHTRRAAASLRTEPEMSHHAQAEDRDNG PGRRLPGRAAAGGRTMDEIRRHDPKNGGQORPTKQ KSEEQQLQAYYKKTLEERLSAGALNTFPRAFQVMEG RDIKLVINQTAQDCFGCLLESISQIRSGSETPGTSES ATPESTATRDKDDTVTRKHPKPPQKWMKDRAIKK GNLRFQRLFYLDGRGLAKIILDDIECLSCDIPLSE IYSVFKTRWETTGSFKSLGDFKTYGKADNTAFRELI TAKEIEKNVQEMSKGSAPGPDGITLGDVVKMDPEFS RTMEIFNLWLTGKIPDMVRGCRTVLIPKSSKPDRL KINNWRPITIGSILLRFLSRIVTARLSKACPLNPR QGRFIRAAGCSENLKLLQTI IWSAKREHRPLGVVFDIAKAFDTVSHQHI IHALQQREVDPHIVGLVSNMYE	1019

TABLE 5-continued

Amino acid sequences of R2Tg mutants with linkers in the DNA binding domain (DBD)		
R2Tg Mutant + Linker	Amino Acid Sequence	SEQ ID NO
	NISTYITTKRNTHTDKIQIRVGVKQGDPMSPLLFNL AMDPLLCKLEESGKGYHRGQSSITAMAFADDLVLLS DSWENMNTNISILETFCNLTLGLKTQGGKCHGFYIKP TKDSYTINDCAAWTINGTPLNMDPGESEKYLGLQF DPWIGIARSGLS TKLDFWLQRIQAPLKLQKTDIL KTYTIPRLIYIADHSEVKLTALLETLDQKIRTAVKEW LHLPPCTCDAILYSTRDGGGLGITKLGLIPSVQAR RLHRIAQSSDDTMKCFMEKEKMEQLHKKLWIQAGGD RENIPSIWEAPPSSEPPNNVSTNSEWEAPTQDKKFP KPCNWRKNEFKKWTKLASQGRGIVNFERDKISNHWI QYRRIIPHRKLLTALQLRANVYPTREFLARGRQDQY IKACRHCADIESCAHIIGNCPVTQDARIKRHNYIC ELLLEEAKKDWVVFKEPHIRDSNKELYKPDLI FVK DARALVVDVTVRYEAAKSSLEEAAAEKVRKYKHLET EVRHLTNAKDVTFVGFPPLGARGKWHQDNFKLLTELG LSKSRQVKMAETFTSTVALFSSVDIVHMFASRARKSM VM	
R2Tg with DBD Linker v2B	MASCPKPGPPVSAGAMSLESGLTTHSVLAIERGPNS LANSGLDFGGGGLGLPLRLLRVSVGTQTSRSDWVDL VSWSHPGPTSKSQQVDLVSLFPKHRVDLLSKNDQVD LVAQFLPSKFPNLAENDLALLVNLEFYRSDLHVYE CVHFPAHWEGLSGLPEVYEQ LAPQPCVGETLHSSLP RDELSEFVPEEGSSEKESADPKTSPPTPGKHGLEQT GEEKVMVTVDPKNPPCPCGTRVNSVLNIEHLKVS HGKRGVCFRCAKCGKENS NYHSVCHFPKCRGPETE KAPAGEWICEVCNRDFTTKIGLQGHKRLAHPAVRNQ ERIVASQPKETSNRGAHKRCWTKEEEELLIRLEAQF EGNKNINKLIAEHITTKTAKQISDKRLLSRKPAEE PREPGTCHHTRRAASLRTEPEMSHHAQAEDRDNG PGRRLPGRAAAGGR TMDEIRRHDPKNGQQRPTKQ KSEEQQLQAYYKKTLEERLSAGALNTPFRAFQVMEG RDIKLVINQTAQDCFGCLESISQIRGGGSTATRDKK DTVTREKHPKPPQKWMKDRAIKKGNLYRQRLFYF DRGKLAKIILDDIECLSCDIPLESIYSVFKTRWETT GSFKSLGDFKTYGKADNTAFRELITAKEIEKNVQEM SKGSAPGPDGITLGDVVKMDPEFSRTMEIFNLWLT GKIPDMVRGCRTVLIPKSSKPDRLKINNWRPITIG SILRLFSRIVTARLSKACPLNPRQGRFIRAAGCSE NLKLLQTI IWSAKREHRPLGVVFVDIAKAFDTVSHQ HI IHALQOREVDPHIVGLVSNMYENISTYITTKRNT HTDKIQIRVGVKQGDPMSPLLFNLAMDPLLCKLEES GKGYHRGQSSITAMAFADDLVLLSDSWENMNTNIS I LETFCNLTLGLKTQGGKCHGFYIKPTKDSYTINDCAA WTINGTPLNMDPGESEKYLGLQFDPWIGIARSGLS TKLDFWLQRIQAPLKLQKTDILKTYTIPRLIYIA DHSEVKLTALLETLDQKIRTAVKEWLHLPPCTCDAIL YSSTRDGGGLGITKLGLIPSVQARRLHRIAQSSDDT MKCFMEKEKMEQLHKKLWIQAGGDRENIPSIWEAPP SSEPPNNVSTNSEWEAPTQDKKFPKPCNWRKNEFKK WTKLASQGRGIVNFERDKISNHWIQYRRIIPHRKLL TALQLRANVYPTREFLARGRQDQYIKACRHCADIE SCAHIIGNCPVTQDARIKRHNYICELLLEEAKKDW VVFKEPHIRDSNKELYKPDLI FVKDARALVVDVTVR YEAAKSSLEEAAAEKVRKYKHLET EVRHLTNAKDV FVGFPPLGARGKWHQDNFKLLTELGLSKSRQVKMAET FSTVALFSSVDIVHMFASRARKSMVM	1020
R2Tg with DBD Linker v3A	MASCPKPGPPVSAGAMSLESGLTTHSVLAIERGPNS LANSGLDFGGGGLGLPLRLLRVSVGTQTSRSDWVDL VSWSHPGPTSKSQQVDLVSLFPKHRVDLLSKNDQVD LVAQFLPSKFPNLAENDLALLVNLEFYRSDLHVYE CVHFPAHWEGLSGLPEVYEQ LAPQPCVGETLHSSLP RDELSEFVPEEGSSEKESADPKTSPPTPGKHGLEQT GEEKVMVTVDPKNPPCPCGTRVNSVLNIEHLKVS HGKRGVCFRCAKCGKENS NYHSVCHFPKCRGPETE KAPAGEWICEVCNRDFTTKIGLQGHKRLAHPAVRNQ ERIVASQPKETSNRGAHKRCWTKEEEELLIRLEAQF EGNKNINKLIAEHITTKTAKQISDKRLLSRKPAEE PREPGTCHHTRRAASGSETPGTSESATPESASLRT EPMSHHAQAEDRDNGPGRRLPGRAAAGGR TMDEI RRHDPKNGQQRPTKQKSEEQQLQAYYKKTLEERLSA GALNTPFRAFQVMEGRDIKLVINQTAQDCFGCLES	1021

TABLE 5-continued

Amino acid sequences of R2Tg mutants with linkers in the DNA binding domain (DBD)		
R2Tg Mutant + Linker	Amino Acid Sequence	SEQ ID NO
	ISQIRTATRDKKDVTREKHPKPPQKWMKDRAIKK GNVLRQRLFYLDGRGLAKI ILDDIECLSCDIPLSE IYSVFKTRWETTGSGFKSLGDFKTYGKADNTAFRELI TAKEIEKNVQEMSKGSAPGPDGITLGDVVKMDPEFS RTMEIFNLWLTGKI PDMVRGCRTVLIPKSSKPDRL KDINNWRPITIGSILLRLFSRIVTARLSKACPLNPR QRGFIRAAGCSENLLQTI IWSAKREHRPLGVVVFV DIKAFD TVSHQHIIHALQQREVDPHIVGLVSNMYE NISTYITTKRNTHTDKI QIRVGKQGDPMSPLEFNL AMDPLLCKLEESGKGYHRGQSSITAMAFADDLVLLS DSWENMNTNIS ILETFCNLTLGLKTQGGKCHGFYIKP TKDSYITINDCAAWTINGTPLNMDPGESEKYLGLQF DPWIGIARSGSLTKLDFWLQRIQAPLPLQKTDIL KTYTIPRLIYIADHSEVKTALLETLQKIRTAVKEW LHLPCTCDAILYSTRDGGGLGITKLGLIPSVQAR RLHRIAQSSDDTMKCFMEKEKMEQLHKKLWIQAGGD RENIPSIWEAPPSSEPPNNVSTNSEWEAPTQDKKFP KPCNWRKNEFKKWTKLASQGRGIVNFERDKI SNHWI QYRRIPHRKLLTALQLRANVYPTREFLARGRQDQY IKACRHCDAIESCAHII GNCPVTQDARIKRHNYIC ELLLEEAKKDWVVFKEPHIRDSNKELYKPDLI FVK DARALVVDVTVRYEAAKSSLEEAEEKVRKYKHLET EVRHLTNAKDVT FVGFP LGARGKWHQDNFKLLTELG LSKSRQVKMAETFTVALFSSVDIVHMFASRARKSM VM	
R2Tg with DBD Linker v3B	MASCPKPGPPVSAGAMSLESLTTHSVLAIERGPNS LANSGSDFGGGGLGLPLRLLRVSVGTQTSRSDWVDL VWSHPGPTSKSQVDLVSLFPKHRVDLLSKNDQVD LVAQFLPSKFPPNLAENDLALLVNLEFYRSDLHVYE CVHFAAHWEGLSGLPEVYEQLAPQPCVGETLHSSLP RDESELFVPEEGSSEKESEDAPKTSPPTPGHLEQT GEEKVMVTVDPKNPPCPCGTRVNSVLNLI EHLKVS HGKRGVCFRCAKCGKENSNYHVVCHFPKCRGPETE KAPAGEWICEVCNRDFTTKI GLGQHKRLAHPAVRNQ ERIVASQPKETSNRGAHKRCWTKEEEELLIRLEAQF EGNKININKLIAEHITTKTAKQISDKRRLLSRKPAEE PREPGTCHHTRRAAGGGSASLREPEMSSHAQAE RDNGPGRRPLPGRAAAGGRTMDEIRRHDPDKNGQQR PTKQKSEEQLQAYYKKTLEERLSAGALNTPPRAFKQ VMEGRDIKLVINQTAQDCFGCLESISQIRTATRDKK DTVREKHPKPPQKWMKDRAIKKGNVLRQRLFYLD DRGKLAKI ILDDIECLSCDIPLSEIYSVFKTRWETT GSFKSLGDFKTYGKADNTAFRELI TAKEIEKNVQEM SKGSAPGPDGITLGDVVKMDPEFSRTMEIFNLWLT GKIPDMVRGCRTVLIPKSSKPDRLKDINNWRPITIG SILLRLFSRIVTARLSKACPLNPRQGRGFIRAAGCSE NLKLLQTI IWSAKREHRPLGVVFDIAKAFD TVSHQ HIIHALQQREVDPHIVGLVSNMYENISTYITTKRNT HTDKI QIRVGKQGDPMSPLEFNLAMDPLLCKLEES GKGYHRGQSSI TAMAFADDLVLLSDSWENMNTNIS I LETFCNLTLGLKTQGGKCHGFYIKPTKDSYITINDCAA WTINGTPLNMDPGESEKYLGLQFDPWIGIARSGLS TKLDFWLQRIQAPLPLQKTDILKTYTIPRLIYIA DHSEVKTALLETLQKIRTAVKEWLHLPCTCDAIL YSSTRDGGGLGITKLGLIPSVQARRLHRIAQSSDDT MKCFMEKEKMEQLHKKLWIQAGGDRENIPSIWEAPP SSEPPNNVSTNSEWEAPTQDKKFPKPCNWRKNEFKK WTKLASQGRGIVNFERDKI SNHWI QYRRIPHRKLL TALQLRANVYPTREFLARGRQDQYIKACRHCDAIE SCAHII GNCPVTQDARIKRHNYICELLLEEAKKDW VVFKEPHIRDSNKELYKPDLI FVKDARALVVDVTVR YEAAKSSLEEAEEKVRKYKHLETEVRHLTNAKDVT FVGFP LGARGKWHQDNFKLLTELGLSKSRQVKMAET FTVALFSSVDIVHMFASRARKSMVM	1022

HEK293T cells were plated in 96-well plates and grown overnight at 37° C., 5% CO₂. The HEK293T cells were transfected with plasmids that expressed R2Tg (wild-type), R2 endonuclease mutant, and linker mutants. The transfection was carried out using the Fugene HD transfection reagent according to the manufacturer recommendations, where each well received 80 ng of plasmid DNA and 0.5 µL

of transfection reagent. All transfections were performed in duplicate and the cells were incubated for 72 h prior to genomic DNA extraction.

Activity of the mutants was measured by a ddPCR assay that quantified the copy number of R2Tg integration per genome. The 5' and 3' junctions were quantified by generating two different amplicons at each end.

v3 (near the c-myb binding motif in the DBD) decreased integration activity with either linker A or B. v1 (N-terminal to the alpha helix preceding the -1 RNA binding motif) had comparable activity to the wild-type when used with linker A (16 AA) versus the shorter linker B (4 AA). This could be related to amino acid selection, length, or three-dimensional structure. v2 (C-terminal to the alpha helix preceding the -1 RNA binding motif) did not tolerate linker A; however, linker B had activity that was comparable and slightly better than the wild-type. v1 and v2 may therefore be considered preferred locations to add a linker that can separate R2Tg's DNA binding domain and the rest of the protein.

Example 27: Long-Read Sequencing to Determine Integration Fidelity

Retrotransposon integration experiments were performed as described in previous examples. In one example, PCR amplification was used to generate amplicons by designing one primer targeting the genomic integration site and one primer targeting the integrant sequence. In this example, these primers were designed to maximize the length of the amplified genomic locus fused with the integrant sequence. By pooling amplicons spanning both ends of the integrant and performing long-read next-generation sequencing, the fidelity of each integration was evaluated.

A cis construct of R2Tg was integrated into 293T cells via plasmid transfection as described herein. Amplicons spanning each end of the integrations were generated with flanking randomized UMIs to control for PCR bias. These amplicons were sequenced with PacBio next-generation sequencing. The resulting sequences were collapsed to remove reads with identical UMIs. By aligning unique reads, a coverage plot was constructed as shown in FIGS. 20A-20B. Sequence coverage largely shows uniform coverage across amplicons, indicating significant fidelity of integration. An associated reverse-transcriptase deficient mutant control produced no signal. Internal deletions were also analyzed in FIGS. 21A-21B. Internal deletions were generally low relative to overall unique read counts, with some clustering at the 5' junction of rDNA-R2Tg.

In another example, hybrid capture may be performed as described in a previous example but with a larger target library length during initial library generation. The resulting library can then be subjected to long-read next-generation sequencing.

Example 28: Targeted Delivery of R2Gfo and R4Al Retrotransposon to Mammalian Cells

This example describes targeted integration of the R2Gfo and R4Al retrotransposon elements to mammalian cells via DNA delivery.

In one example, we assayed the full R2 element R2-1_GFo (Rebase; Kojima et al *PLoS One* 11, e0163496 (2015)) from the medium ground finch, *Geospiza fortis* ("R2Gfo"). In another example, we assayed the full R4 element R4_AL (Rebase; Burke et al *Nucleic Acids Res.* 23, 4628-34 (1995)) from the large roundworm, *Ascaris lumbricoides* ("R4Al"). Because non-LTR R2 and R4 elements are not present in the human genome and are thought to be highly site-specific, the ability of retrotransposons to accurately and efficiently integrate itself into the human genome would demonstrate the capability to perform genomic targeted integration.

Plasmids harboring R2Gfo (PLV033) or R4Al (PLV462) were designed for cis integration of the R2Gfo or R4Al

elements as in previous examples. Plasmids were synthesized such that the wildtype element was flanked by its native un-translated regions (UTRs) and 100 bp of homology to its rDNA target (FIG. 22). The element expression was driven by the mammalian CMV promoter. We introduced each plasmid into HEK393T cells using the FuGENE® HD transfection reagent. HEK293T cells were seeded in 96-well plates at 10,000 cells/well 24 hours before transfection. On the transfection day, 0.5 µl transfection reagent and 80 ng DNA was mixed in 10 µl Opti-MEM and incubated for 15 minutes at room temperature. The transfection mixture was then added to the medium of the seeded cells. Three days after transfection, genomic DNA was extracted for retrotransposition assays. R2Tg was also delivered in parallel in the same format to serve as a comparison.

ddPCR was performed to confirm integration and assess integration efficiency. A Tagman probe was designed to the 3'UTR portion of each element. A forward primer was synthesized to bind directly upstream of the probe, and a reverse primer was synthesized to bind the rDNA. Thus, amplification of the expected product across the integration junction would degrade the probe and create a fluorescent signal. The results of the ddPCR copy number analysis (in comparison to reference gene RPP30) are shown in FIG. 23. R2Gfo integration achieved a mean copy number of 0.21 integrants/genome in this experiment. R4Al achieved a mean copy number of 0.085 integrants/genome.

Example 29: Integration of Retrotransposons into Human Fibroblasts

This example describes the cis integration of R2Tg into human fibroblasts. Briefly, a plasmid designed to integrate R2Tg in cis was synthesized such that R2Tg was flanked by its native UTRs and homologous sequence to its rDNA target as in previous examples. 0.5 µg PLV014 (wild-type) and PLV072 (EN mutant) plasmids were transfected into 100,000 human dermal fibroblasts isolated from neonatal foreskin (HDFn, C0045C, ThermoFisher Scientific) respectively using the Neon transfection system. Two programs were performed, each in duplicate. The setting for Program 1 was 1700V pulse voltage, 20 ms pulse width, and 1 pulse number. The setting for Program 2 was 1400V pulse voltage, 20 ms pulse width, and 2 pulse number. Both programs achieved 95% transfection efficiency measured using plasmid encoding the EGFP. Three days post transfection, genomic DNA was extracted for the ddPCR assay. ddPCR was performed to confirm integration and assess integration efficiency. A Tagman probe was designed to the 3' UTR portion of the R2Tg element. A forward primer was synthesized to bind directly upstream of the probe, and a reverse primer was synthesized to bind the rDNA. Thus, amplification of the expected product across the integration junction would degrade the probe and create a fluorescent signal. The results of ddPCR copy number analysis (in comparison to reference gene RPP30) are shown in FIG. 24. Wild-type (WT) R2Tg integration achieved a mean copy number of 0.036 integrants/genome in this experiment, significantly higher than a control R2Tg plasmid with a point mutation abolishing endonuclease activity (EN).

Example 30: Evaluation of DNA Damage Response Upon Retrotransposon Transfection

DNA damage (e.g., resulting from DSB formation or replication fork collapse) leads to the activation of p53, which among many other transcriptional responses, leads to

the upregulation of p21, resulting in cell cycle arrest or apoptosis. Genome editing using CRISPR/Cas9 has been shown to activate p53 and p21, which is a potential safety and efficacy problem for CRISPR-based therapeutics. To establish whether R2Tg delivery to the cell leads to activation of p53 and p21, U2OS cells were seeded at a density of 4×10^4 cells/well and transfected 24 hours later using the Eugene HD and Lipofectamine reagents with either 500 ng of R2Tg-WT plasmid or 500 ng of R2Tg-EN (a variant of R2Tg with a mutation in the endonuclease (EN) domain, rendering R2Tg inactive). To control for transfection efficiency, U2OS cells were also transfected with a plasmid

expressing GFP. Lastly, as a positive control for p53 and p21 activation, U2OS cells were treated with one of the DNA damage-inducing agents etoposide (20 μ M) or bleomycin (10 μ g/ml). The U2OS cells were collected 24 hours after transfection/treatment. Protein lysates were prepared in RIPA buffer and run on an SDS-PAGE gel, followed by transfer to nitrocellulose, followed by probing with antibodies against p53 and p21, as well as Actin and Vinculin. As shown in FIG. 25, no R2Tg-induced upregulation of p53 or p21 above the GFP plasmid control was detected in either transfection condition.

LENGTHY TABLES

The patent contains a lengthy table section. A copy of the table is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12398392B2>). An electronic copy of the table will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12398392B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

What is claimed is:

1. A method of modifying a target DNA strand in a cell, tissue or subject, the method comprising administering a composition for modifying DNA to the cell, tissue or subject, thereby modifying the target DNA strand;

wherein the composition for modifying DNA comprises:

(a) a first engineered RNA encoding a polypeptide, wherein the polypeptide comprises (i) a reverse transcriptase domain and (ii) an endonuclease domain, wherein the polypeptide comprises the amino acid sequence SEQ ID NO: 1387, or a sequence having at least 99% identity thereto; and

(b) a second engineered RNA comprising a template RNA, wherein the template RNA comprises (i) the sequence of SEQ ID NO: 1142, or a sequence having at least 99% identity thereto, that binds the encoded polypeptide, and (ii) a heterologous object sequence; and

wherein (a) and (b) are separate RNAs.

2. The method of claim 1, wherein the template RNA further comprises a sequence comprising at least 20 nucleotides of at least 80% identity to a target DNA.

3. The method of claim 1, wherein the template RNA comprises:

(iii) at least 3 bases of identity to a target DNA at the 3' end of the template RNA, and (iv) at least 3 bases of identity to the target DNA at the 5' end of the template RNA.

4. The method of claim 1, wherein the heterologous object sequence encodes an enzyme, a membrane protein, a blood factor, an intracellular protein, an extracellular protein, a

structural protein, a signaling protein, a regulatory protein, a transport protein, or a motor protein.

5. The method of claim 1, wherein the polypeptide further comprises one or both of a nuclear localization signal and a nucleolar localization signal.

6. The method of claim 1, wherein the composition comprises only RNA, or comprises more RNA than DNA by an RNA:DNA ratio of at least 10:1, 20:1, 30:1, 40:1, 50:1, 60:1, 70:1, 80:1, 90:1, or 100:1.

7. The method of claim 1, wherein the template RNA does not encode a reverse transcriptase or an endonuclease.

8. The method of claim 1, wherein the template RNA comprises: at least 10 bases of identity to a target DNA at the 3' end of the template RNA, and (iv) at least 10 bases of identity to the target DNA at the 5' end of the template RNA.

9. The method of claim 1, wherein the template RNA further comprises the sequence of SEQ ID NO: 1265, or a sequence having at least 95% identity thereto.

10. The method of claim 1, wherein the template RNA further comprises the sequence of SEQ ID NO: 1265, or a sequence having at least 99% identity thereto.

11. The method of claim 1, wherein the template RNA further comprises the sequence of SEQ ID NO: 1265.

12. The method of claim 1, wherein the template RNA further comprises the sequence of SEQ ID NO: 1142.

13. The method of claim 1, wherein the target DNA is a genomic safe harbor (GSH) site.

14. The method of claim 1, wherein the heterologous object sequence encodes a therapeutic polypeptide or fragment thereof.

15. The method of claim 1, wherein the heterologous object sequence comprises a non-coding sequence or a regulatory sequence.

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16. The method of claim **1**, wherein the composition comprises no more than 1%, 0.5%, 0.2%, 0.1%, 0.05%, 0.02%, or 0.01% DNA by mass or by molar amount of nucleic acid.

17. The method of claim **1**, wherein the composition is 5 capable of modifying DNA in the absence of homologous recombination activity.

18. The method of claim **15**, wherein the regulatory sequence is a promoter.

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